

BusyBee Generation Test — OWASP Untrust Constrained Application

BusyBee Generation Test — OWASP Untrust Constrained Application - Yariv Tal, GitHub.

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Content

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BusyBee Generation Test — OWASP Untrust Constrained Application

Archive source collection from `SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust`, pinned to commit `5e578e471fdb9db38c70d2075a285c0507ee6e2` on default branch `main`. The sections preserve complete authored source files for the application, routes, authentication, validation, frontend, build guardrails and tests. This is a selected source collection, not a repository README or a complete repository mirror. No source was executed.

BuildGates/README.md

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/BuildGates/README.md>

```
# buildgates_java
```

Build metadata annotations and reusable Gradle build gates [for](#) OWASP Untrust Java projects.

[This](#) repository contains two related artifacts:

- `'buildmetadata'`: small Java marker annotations used by source code to document reviewed exceptions.
- `'build-gates-gradle-plugin'`: a Gradle plugin that applies the OWASP Untrust Java security and design gates.

Keeping these together is intentional. The annotations and the gates form one contract: source code marks a narrow e

```
## Modules
```

```
buildgates_java/ buildmetadata/ src/main/java/org/owasp/untrust/buildmetadata/ build-gates-gradle-plugin/ src/main/kotlin/org/owasp/untrust/buildgates/gradle/ src/main/resources/org/owasp/untrust/buildgates/scripts/
```

```
## Requirements
```

- JDK [21](#)
- Gradle wrapper included in [this](#) repository

Build everything:

```
.\gradlew.bat build
```

On Unix-like shells:

```
./gradlew build
```

```
## Published Coordinates
```

Current Gradle metadata:

```
group = "org.owasp.untrust" version = "0.1.0"
```

Expected Java annotation dependency:

```
dependencies { implementation("org.owasp.untrust:buildmetadata:0.1.0") }
```

Expected Gradle plugin usage after publishing:

```
plugins { id("org.owasp.untrust.build-gates") version "0.1.0" }
```

Local Development Usage

Before publishing, consumers can [use this](#) repository as an included build.

[For](#) plugin resolution:

```
// settings.gradle.kts pluginManagement { includeBuild("../BuildGates") }
```

[For](#) dependency substitution of ``org.owasp.untrust:buildmetadata``:

```
// settings.gradle.kts includeBuild("../BuildGates")
```

Then a consumer build can apply:

```
plugins { java id("org.owasp.untrust.build-gates") }
```

and depend on the annotations:

```
dependencies { implementation("org.owasp.untrust:buildmetadata:0.1.0") }
```

Java Marker Annotations

``@StringConcatenationSafe``

Marks code where raw [string](#) concatenation is intentionally allowed and justified.

[Use this](#) when a build rule normally forbids [string](#) concatenation but a [class](#) or method has a narrow, reviewed reason

Example:

```
import org.owasp.untrust.buildmetadata.StringConcatenationSafe;

@StringConcatenationSafe("The values are fixed literals controlled by the developer, not user input.") public final class HardcodedMessageFactory { public String message(String suffix) { return "fixed-prefix-" + suffix; } }
```

Good reasons explain the invariant:

```
@StringConcatenationSafe("All concatenated values are developer-controlled Hardcoded descriptors.")
```

Weak reasons should not pass review:

```
@StringConcatenationSafe("Needed.") @StringConcatenationSafe("Safe.")
@StringConcatenationSafe("False positive.")
```

`@NonFinalValidatedValue`

Marks a validated-value type that is intentionally non-**final**.

Validated value classes are usually expected to be **final** so validation cannot be bypassed through inheritance. [Use this](#)

Example:

```
import org.owasp.untrust.buildmetadata.NonFinalValidatedValue;

@NonFinalValidatedValue("UUID-backed identifiers share the same parsing validation and may be specialized by domain.") public class UuidValue extends ValidatedValue<UUID, UuidValue.Traits> {
// ... }
```

Gradle Build Gates Plugin

The plugin ID is:

org.owasp.untrust.build-gates

The current plugin packages the existing `.gradle.kts` gates as resources and applies them in a stable order:`

1. JSON config reader
2. preceding escape-hatch comment helpers
3. unsafe `import` gate
4. `string` concatenation gate
5. local `catch` gate
6. method-call gate
7. `null` literal gate
8. unvalidated route values gate
9. `namespace class` gate
10. validated value inheritance gate
11. jOOQ TOCTOU gate
12. `public` value on sensitive types gate

`This` wrapper keeps the existing gate logic intact `while` making the consumer build file small:

```
plugins { id("org.owasp.untrust.build-gates") version "0.1.0" }
```

Instead of:

```
apply(from = "$rootDir/gradle/json_config_reader.gradle.kts") apply(from =  
"$rootDir/gradle/preceding_comment_as_escape_hatch.gradle.kts") apply(from =  
"$rootDir/gradle/forbid-unsafe-imports.gradle.kts") apply(from = "$rootDir/gradle/forbid-string-  
concat.gradle.kts") // ...
```

Consumer Configuration Files

Some gates expect JSON configuration files in the consuming project root. Examples **from** current consumers include:

- `approved\_import\_alternatives.json`
- `allow\_unsafe\_imports.json`
- `string\_concat\_guardrail.json`
- `local\_catch\_guardrail.json`
- `method\_call\_guardrail.json`
- `validated\_value\_guardrail.json`

The plugin supplies the gate logic. The consuming application should still own its project-specific policy configuration.

What Not To **Do**

- **Do** not copy-paste the gate scripts into every consuming repository once the plugin is available.
- **Do** not **use** `apply(from = "https://...")`; remote script application is hard to review and pin safely.
- **Do** not put application-specific policy JSON into **this** repository unless it is meant to be a reusable **default**.
- **Do** not **use** marker annotations to suppress a gate without an actionable security explanation.
- **Do** not split the annotations and gate plugin into unrelated release lifecycles unless there is a strong reason; version s
- **Do** not treat these gates as runtime security controls. They are build-time enforcement and review aids.

Repository Notes

This repository is intended to become:

https://github.com/owasp-untrust/buildgates_java

Dependency order **for** the generated Java libraries:

1. `buildgates\_java`
2. `valuedescriptor\_java`
3. `vv\_java`

Publish **this** repository first.

BuildGates/build.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp

[_untrust/5e578e471fdb38c70d2075a285c0507ee6e2/BuildGates/build.gradle.kts](https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb38c70d2075a285c0507ee6e2/BuildGates/build.gradle.kts)>

```
plugins {  
    base  
}  
  
allprojects {  
    group = "org.owasp.untrust"  
    version = "0.1.0"  
  
    repositories {  
        mavenCentral()  
    }  
}
```

BuildGates/settings.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb38c70d2075a285c0507ee6e2/BuildGates/settings.gradle.kts>

```
rootProject.name = "buildgates_java"  
  
include("buildmetadata")  
include("build-gates-gradle-plugin")
```

ValueDescriptors/README.md

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb38c70d2075a285c0507ee6e2/ValueDescriptors/README.md>

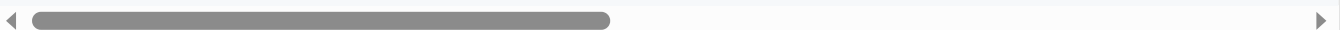
```
# valuedescriptor_java
```

Value descriptor primitives **for** OWASP Untrust Java libraries.

This library provides wrappers and marker interfaces **for** values whose **public** representation is intentionally different

```
## Coordinates
```

Current Gradle metadata:



```
group = "org.owasp.untrust" version = "0.1.0"
```

Expected dependency once published:

```
dependencies { implementation("org.owasp.untrust:valuedescriptors:0.1.0") }
```

For local development before publishing:

```
// settings.gradle.kts includeBuild("../BuildGates")
```

Requirements

- JDK 21
- Gradle wrapper included in [this](#) repository
- `'buildgates_java'`, currently providing `'org.owasp.untrust:buildmetadata:0.1.0'`

Build:

```
.\gradlew.bat build
```

On Unix-like shells:

```
./gradlew build
```


Design Model

The core idea is that wrapped values should not accidentally leak sensitive or internal data through **string** conversion.

`WrappedValue<T>` stores a value and implements `toString()` by calling `toPublicString()`.

That means subclasses and **public**-rendering interfaces decide what is safe to show:

- `PubliclyExposed<T>` exposes the wrapped value as **public** text.
- `RedactedConfig<T>` always renders as `*****`.
- `Hardcoded` represents developer-controlled literal text.
- `ViewableConfig<T>` represents configuration that is safe to display.

Public API

`WrappedValue<T>`

Base **class** for wrapped values.

Important behavior:

```
@Override public String toString() { return toPublicString(); }
```

Use `exposeUnchecked()` only when the caller is in a trusted boundary and intentionally needs the raw value.

`PubliclyRepresentable`

Interface requiring:

`String toPublicString();`

Types implementing **this** contract define their own safe **public string** representation.

`PubliclyExposed<T>`

Use **for** values whose raw representation is safe to show in logs, diagnostics, API responses, or UI text.

Example:

```
import org.owasp.untrust.valuedescriptors.ViewableConfig;
ViewableConfig<String> region = new ViewableConfig<>("eu-west-1");
String publicText = region.toString(); // "eu-west-1" String raw = region.exposeUnchecked(); // "eu-west-1" String value = region.value(); // "eu-west-1"
```

Only [use this interface](#) when the value is genuinely [public](#).

`RedactedConfig<T>`

[Use for](#) configuration values that should not be printed.

Example:

```
import org.owasp.untrust.valuedescriptors.RedactedConfig;
RedactedConfig<String> apiKey = new RedactedConfig<>("secret-value");
String publicText = apiKey.toString(); // "***" String raw = apiKey.exposeUnchecked(); // "secret-value"
```

`Hardcoded`

Represents developer-controlled literal text.

Example:

```
import org.owasp.untrust.valuedescriptors.Hardcoded;
import static org.owasp.untrust.valuedescriptors.Hardcoded.hardcoded;
Hardcoded fieldName = hardcoded("username"); Hardcoded message =
fieldName.concat(hardcoded(" is required"));
String publicText = message.toString();
```

``Hardcoded`` is annotated with ``@StringConcatenationSafe`` because its values are expected to be code-defined literals,

Usage Guidance

Use `public` wrappers at boundaries where accidental `string` conversion is likely:

- logs
- exception messages
- validation messages
- diagnostic output
- HTTP responses
- generated code
- guardrail-aware build plugins

Prefer explicit names that reveal disclosure intent:

```
ViewableConfig<String> publicRegion = new ViewableConfig<>("us-east-1");
RedactedConfig<String> databasePassword = new RedactedConfig<>(password);
```

What Not To Do

- Do not call ``exposeUnchecked()`` just to make formatting convenient.
- Do not wrap secrets in ``ViewableConfig``.
- Do not construct ``Hardcoded`` from request parameters, database content, uploaded files, headers, cookies, or other
- Do not rely on ``toString()`` for serialization contracts; use explicit DTO fields.
- Do not add `public` rendering to a base `class` unless every subclass has the same disclosure policy.
- Do not suppress build guardrails with ``Hardcoded`` unless the value is actually developer-controlled.

Bad:

```
Hardcoded unsafe = Hardcoded.of(request.getParameter("name"));
```

Better:

```
// Validate request data with vv_java, then render it according to its own type.
```

Repository Notes

[This](#) repository is intended to become:

https://github.com/owasp-untrust/valuedescriptor_java

Dependency order:

1. ``buildgates_java``
2. ``valuedescriptor_java``
3. ``vv_java``

Publish ``buildgates_java`` before publishing [this](#) repository.

ValueDescriptors/build.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/ValueDescriptors/build.gradle.kts>

```
plugins {  
    `java-library`  
}  
  
group = "org.owasp.untrust"  
version = "0.1.0"  
  
java {  
    toolchain {  
        languageVersion = JavaLanguageVersion.of(21)  
    }  
}  
  
repositories {  
    mavenCentral()  
}  
  
dependencies {  
    implementation("org.owasp.untrust:buildmetadata:0.1.0")  
}
```

ValueDescriptors/settings.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/ValueDescriptors/settings.gradle.kts>

```
rootProject.name = "valuedescriptors"
```

```
includeBuild("../BuildGates")
```

ValueDescriptors/src/main/java/org/owasp/untrust/valuedescriptors/Hardcoded.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/ValueDescriptors/src/main/java/org/owasp/untrust/valuedescriptors/Hardcoded.java>

```

package org.owasp.untrust.valuedescriptors;

import org.owasp.untrust.buildmetadata.StringConcatenationSafe;
import org.owasp.untrust.valuedescriptors.foundation.PubliclyExposed;
import org.owasp.untrust.valuedescriptors.foundation.StringDescriptor;
import org.owasp.untrust.valuedescriptors.foundation.ExposableWrappedValue;

@StringConcatenationSafe("Hardcoded values are defined in code and cannot contain user-controlled input, so concatenation is safe")
public class Hardcoded extends ExposableWrappedValue<String> implements StringDescriptor, PubliclyExposed<String> {
    public static Hardcoded hardcoded(String hardcodedString) {
        return new Hardcoded(hardcodedString);
    }

    public static Hardcoded of(String hardcodedString) {
        return new Hardcoded(hardcodedString);
    }

    private Hardcoded(String hardcodedString) {
        super(hardcodedString);
    }

    public Hardcoded concat(Hardcoded suffix) {
        return new Hardcoded(exposeUnchecked() + suffix.exposeUnchecked());
    }
}

```

ValueDescriptors/src/main/java/org/owasp/untrust/valuedescriptors/RedactedConfig.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/ValueDescriptors/src/main/java/org/owasp/untrust/valuedescriptors/RedactedConfig.java>

```
package org.owasp.untrust.valuedescriptors;

import org.owasp.untrust.valuedescriptors.foundation.ConfigBase;

public class RedactedConfig<T> extends ConfigBase<T> {
    public RedactedConfig(T configValue) {
        super(configValue);
    }

    @Override
    public String toPublicString() {
        return "[redacted]";
    }
}
```

ValueDescriptors/src/main/java/org/owasp/untrust/valuedescriptors/ViewableConfig.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/ValueDescriptors/src/main/java/org/owasp/untrust/valuedescriptors/ViewableConfig.java>

```
package org.owasp.untrust.valuedescriptors;

import org.owasp.untrust.valuedescriptors.foundation.ConfigBase;
import org.owasp.untrust.valuedescriptors.foundation.PubliclyExposed;

public class ViewableConfig<T> extends ConfigBase<T> implements PubliclyExposed<T> {
    public ViewableConfig(T configValue) {
        super(configValue);
    }
}
```

ValueDescriptors/src/main/java/org/owasp/untrust/valuedescriptors/foundation/ConfigBase.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/ValueDescriptors/src/main/java/org/owasp/untrust/valuedescriptors/foundation/ConfigBase.java>

```
package org.owasp.untrust.valuedescriptors.foundation;

public abstract class ConfigBase<T> extends ExposableWrappedValue<T>{
    public ConfigBase(T configValue) {
        super(configValue);
    }
}
```

ValueDescriptors/src/main/java/org/owasp/untrust/valuedescriptors/foundation/ExposableValue.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb38c70d2075a285c0507ee6e2/ValueDescriptors/src/main/java/org/owasp/untrust/valuedescriptors/foundation/ExposableValue.java>

```
package org.owasp.untrust.valuedescriptors.foundation;

public interface ExposableValue<T> {
    T exposeUnchecked();
}
```

ValueDescriptors/src/main/java/org/owasp/untrust/valuedescriptors/foundation/ExposableWrappedValue.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb38c70d2075a285c0507ee6e2/ValueDescriptors/src/main/java/org/owasp/untrust/valuedescriptors/foundation/ExposableWrappedValue.java>


```

package org.owasp.untrust.valuedescriptors.foundation;

public abstract class ExposableWrappedValue<T> implements ExposableValue<T>, PubliclyRepresentable {
    private T m_hiddenValue;

    public ExposableWrappedValue(T value) {
        m_hiddenValue = value;
    }

    @Override
    public T exposeUnchecked() {
        return m_hiddenValue;
    }

    @Override
    public String toString() {
        return toPublicString();
    }
}

```

ValueDescriptors/src/main/java/org/owasp/untrust/valuedescriptors/foundation/PubliclyExposed.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/ValueDescriptors/src/main/java/org/owasp/untrust/valuedescriptors/foundation/PubliclyExposed.java>

```

package org.owasp.untrust.valuedescriptors.foundation;

public interface PubliclyExposed<T> extends ExposableValue<T>, PubliclyRepresentable {
    @Override
    default String toPublicString() {
        return exposeUnchecked().toString();
    }

    default T value() {
        return exposeUnchecked();
    }
}

```

ValueDescriptors/src/main/java/org/owasp/untrust/valuedescriptors/foundation/PubliclyRepresentable.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/ValueDescriptors/src/main/java/org/owasp/untrust/valuedescriptors/foundation/PubliclyRepresentable.java>

```
package org.owasp.untrust.valuedescriptors.foundation;

public interface PubliclyRepresentable {
    String toPublicString();
}
```

ValueDescriptors/src/main/java/org/owasp/untrust/valuedescriptors/foundation/StringDescriptor.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/ValueDescriptors/src/main/java/org/owasp/untrust/valuedescriptors/foundation/StringDescriptor.java>

```
package org.owasp.untrust.valuedescriptors.foundation;

public interface StringDescriptor extends ExposableValue<String> {
}
```

allow_unsafe_imports.json

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/allow_unsafe_imports.json>

```
{
  "java.nio.file.Path": {
    "packages": [
      "com.imaginary.project.that.is.here.just.as.an.example.package.name"
    ],
    "files": [
      "src/main/java/com/imaginaryProject/security/SafePathFactory.java"
    ]
  },
  "java.lang.Runtime": {
    "files": [
      "src/main/java/com/imaginaryProject/process/ApprovedProcessRunner.java"
    ]
  },
  "java.io.StringWriter": {
    "files": [
      "src/test/"
    ]
  },
  "java.lang.StringBuffer": {
    "files": [
      "src/test/"
    ]
  }
}
```

approved_import_alternatives.json

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/approved_import_alternatives.json>

```
{
  "owasp.untrust.BoxedPath": {
    "replaces": [
      "java.io.File",
      "java.nio.file.Path",
      "java.nio.file.Paths"
    ],
    "dependency": "io.github.owasp-untrust:untrust-boxedpath",
    "message": "Sandbox file access to a specific folder. OWASP Untrust BoxedPath: https://github.com/owasp-untrust"
  },
  "org.jooq.DSLContext": {
    "replaces": [
      "java.sql.Connection",
      "java.sql.DriverManager",
      "java.sql.Statement",
      "java.sql.PreparedStatement",
      "java.sql.CallableStatement",
      "java.sql.ResultSet",
      "javax.sql.RowSet",
      "org.springframework.jdbc.core.JdbcTemplate",
      "org.springframework.jdbc.core.namedparam.NamedParameterJdbcTemplate",
      "org.springframework.jdbc.core.JdbcOperations",
      "org.springframework.jdbc.core.simple.*",
      "org.springframework.jdbc.object.*",
      "jakarta.persistence.*",
      "javax.persistence.*",
      "org.hibernate.*",
      "org.springframework.data.jpa.*",
      "org.eclipse.persistence.*",
      "org.apache.openjpa.*",
      "org.datanucleus.*",
      "org.mybatis.*",
      "org.jdbi.v3.*"
    ],
    "dependency": "org.jooq:jooq",
    "message": "Use jOOQ DSL or the project-approved database access layer. Do not use raw JDBC or ORM directly."
  },
  "com.example.security.ProcessBuilder": {
    "replaces": [
      "java.lang.Runtime",
      "java.lang.ProcessBuilder"
    ],
    "message": "Use the project-approved process execution wrapper."
  }
}
```

```

"com.example.security.SafeHttpClient": {
  "replaces": [
    "java.net.URL",
    "java.net.HttpURLConnection"
  ],
  "message": "Use the project-approved SSRF-safe HTTP client."
},
"<no alternative>": {
  "replaces": [
    "java.lang.StringBuilder",
    "java.io.StringWriter",
    "com.google.common.base.Joiner",
    "java.lang.StringBuffer",
    "java.io.CharArrayWriter",
    "java.io.PrintWriter"
  ],
  "message": "Do not use without explicit user justification."
}
}

```

busybee/VAULT.md

Source:

https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/VAULT.md

Vault Configuration Reference

This workspace may be run inside Docker **while** a local HashiCorp Vault Community dev server runs on the host/WSL s

Docker-Reachable Vault Address

Inside Docker, **do not use** `127.0.0.1` **for** a Vault process running outside the container. **Use:**

busybee.secrets.vault.uri=http://192.168.48.1:8200

busybee.secrets.vault.token=\${VAULT_TOKEN:root} busybee.secrets.vault.mount=secret

Equivalent environment values:

VAULT_ADDR=http://192.168.48.1:8200 VAULT_TOKEN=root

BUSYBEE_SECRETS_VAULT_URI=http://192.168.48.1:8200 BUSYBEE_SECRETS_VAULT_TOKEN=root

BUSYBEE_SECRETS_VAULT_MOUNT=secret

Host-Side Local Dev Defaults

[For](#) a BusyBee server running directly on the host/WSL side, the corresponding local address is:

busybee.secrets.vault.uri=http://192.168.48.1:8200

busybee.secrets.vault.token=\${VAULT_TOKEN:root} busybee.secrets.vault.mount=secret

Expected Dev Vault Shape

- KV secrets engine: version 2
- mount path: ``secret``
- development token: ``root``, unless overridden
- example secret paths may be application-defined under the ``secret`` mount

Starting The Local Dev Server

The sibling ``hashicorp`` folder contains helper scripts [for](#) running a local Vault dev server. The server must listen on an

busybee/build.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/build.gradle.kts>

```
plugins {  
    java  
    id("org.springframework.boot") version "4.0.6"  
    id("io.spring.dependency-management") version "1.1.7"  
    id("org.owasp.untrust.build-gates")  
}  
  
group = "com.securefromscratch"  
version = "0.0.1-SNAPSHOT"  
  
java {  
    toolchain {  
        languageVersion = JavaLanguageVersion.of(21)  
    }  
}  
  
repositories {  
    mavenCentral()  
}  
  
sourceSets.configureEach {  
    java {  
        exclude("the_road_to_success/**")  
        exclude("**/the_road_to_success/**")  
    }  
}  
  
tasks.withType<JavaCompile>().configureEach {  
    exclude("the_road_to_success/**")  
    exclude("**/the_road_to_success/**")  
}  
  
dependencies {  
    implementation("org.owasp.untrust:valuedescriptors:0.1.0")  
    implementation("org.owasp.untrust:buildmetadata:0.1.0")  
    implementation("org.owasp.untrust:vv:0.1.0")  
    implementation("org.owasp.untrust:vv_spring:0.1.0")  
    implementation("org.owasp.untrust:saferprocess:0.1.0")  
    implementation("org.springframework.boot:spring-boot-starter-webmvc")  
    implementation("com.fasterxml.jackson.core:jackson-databind")  
    implementation("org.springframework.boot:spring-boot-starter-security")  
    implementation("org.springframework.boot:spring-boot-starter-jdbc")  
    implementation("org.springframework.boot:spring-boot-starter-jooq")  
    implementation("org.springframework.boot:spring-boot-flyway")  
}
```

```

implementation("org.springframework.vault:spring-vault-core")
implementation("org.flywaydb:flyway-mysql")
implementation("com.fasterxml.jackson.datatype:jackson-datatype-jsr310")
implementation("com.fasterxml.jackson.datatype:jackson-datatype-jdk8")
implementation("io.github.owasp-untrust:untrust-boxedpath:0.3")
implementation("org.jsoup:jsoup:1.21.2")
runtimeOnly("com.mysql:mysql-connector-j")
implementation("com.google.guava:guava:33.6.0-jre")
testImplementation("org.springframework.boot:spring-boot-starter-webmvc-test")
testImplementation("org.springframework.security:spring-security-test")
testRuntimeOnly("org.junit.platform:junit-platform-launcher")
}

tasks.withType<Test> {
    useJUnitPlatform()
}

tasks.named<Test>("test") {
    dependsOn(gradle.includedBuild("tesseract_mock_java").task(":installDist"))
}

```

busybee/demo-sessions.example.properties

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/demo-sessions.example.properties>

```

# Copy this file to demo-sessions.properties next to the running application.
# Do not use this outside local/demo/debug environments.
#
# The configured JSESSIONID values authenticate as the mapped username after server restart.
# Each request refreshes Max-Age on matching JSESSIONID/JSESSIONIDn cookies.
#
# Example:
# busybee.demo.sessions.enabled=true
# busybee.demo.sessions.cookie-max-age=12h
# busybee.demo.sessions.users[0FFF68EB2B1ACDD5538130781AFD6AC6]=yariv

busybee.demo.sessions.enabled=false
busybee.demo.sessions.cookie-max-age=12h

```


busybee/settings.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/settings.gradle.kts>

```
rootProject.name = "busybee"

pluginManagement {
    includeBuild("../BuildGates")
}

includeBuild("../ValueDescriptors")
includeBuild("../BuildGates")
includeBuild("../vv")
includeBuild("../saferprocess")
includeBuild("../tesseract_mock_java")
```

busybee/src/main/java/com/securefromscratch/busybee/BusyBeeApplication.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/BusyBeeApplication.java>

```
package com.securefromscratch.busybee;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class BusyBeeApplication {
    public static void main(String[] args) {
        SpringApplication.run(BusyBeeApplication.class, args);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/admin/AdminUserController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/admin/AdminUserController.java>

```
package com.securefromscratch.busybee.admin;

import java.util.List;

import org.owasp.untrust.vw.SingleLine;
import org.springframework.security.access.prepost.PreAuthorize;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RequestParam;
import org.springframework.web.bind.annotation.RestController;

@RestController
@PreAuthorize("denyAll()")
public class AdminUserController {
    private final AdminUserService adminUserService;

    public AdminUserController(AdminUserService adminUserService) {
        this.adminUserService = adminUserService;
    }

    @GetMapping("/admin/users")
    @PreAuthorize("hasRole('ADMIN')")
    public List<AdminUserResponse> search(@RequestParam("query") SingleLine query) {
        return adminUserService.search(query);
    }

    @PostMapping("/admin/user-entitlements")
    @PreAuthorize("hasRole('ADMIN')")
    public AdminUserResponse replaceEntitlements(@RequestBody ReplaceEntitlementsRequest request) {
        return adminUserService.replaceEntitlements(request);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/admin/AdminUserResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/admin/AdminUserResponse.java>

```
package com.securefromscratch.busybee.admin;

import java.util.List;

public record AdminUserResponse(
    String username,
    boolean admin,
    List<String> entitlements,
    List<String> effectiveEntitlements
){
}
```

busybee/src/main/java/com/securefromscratch/busybee/admin/AdminUserService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/admin/AdminUserService.java>

```
package com.securefromscratch.busybee.admin;

import java.util.List;

import com.securefromscratch.busybee.entitlement.Entitlement;
import com.securefromscratch.busybee.entitlement.EntitlementRepository;
import com.securefromscratch.busybee.registration.UserRepository;
import org.owasp.untrust.vw.SingleLine;
import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;

@Service
public class AdminUserService {
    private final UserRepository userRepository;
    private final EntitlementRepository entitlementRepository;

    public AdminUserService(UserRepository userRepository, EntitlementRepository entitlementRepository) {
        this.userRepository = userRepository;
        this.entitlementRepository = entitlementRepository;
    }

    public List<AdminUserResponse> search(SingleLine query) {
        return userRepository.findUsernamesContaining(query.exposeUnchecked()).stream()
            .map(this::responseFor)
            .toList();
    }

    @Transactional
    public AdminUserResponse replaceEntitlements(ReplaceEntitlementsRequest request) {
        String username = request.username().exposeUnchecked();
        List<Entitlement> entitlements = request.entitlements().stream()
            .map(entitlement -> entitlement.exposeUnchecked())
            .toList();
        entitlementRepository.replaceForUsername(username, entitlements);
        return responseFor(username);
    }

    private AdminUserResponse responseFor(String username) {
        List<String> entitlements = entitlementRepository.findForUsername(username).stream().map(Entitlement::name)
        return new AdminUserResponse(username, false, entitlements, entitlements);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/admin/ReplaceEntitlementsRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/admin/ReplaceEntitlementsRequest.java>

```
package com.securefromscratch.busybee.admin;

import java.util.List;

import com.securefromscratch.busybee.entitlement.EntitlementValue;
import org.owasp.untrust.vw.SingleLine;

public record ReplaceEntitlementsRequest(SingleLine username, List<EntitlementValue> entitlements) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/aicredential/AiCredentialController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/aicredential/AiCredentialController.java>

```

package com.securefromscratch.busybee.aicredential;

import org.springframework.security.access.prepost.PreAuthorize;
import org.springframework.security.core.Authentication;
import org.springframework.web.bind.annotation.DeleteMapping;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PutMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RestController;

@RestController
@PreAuthorize("denyAll()")
public class AiCredentialController {
    private final AiCredentialService aiCredentialService;

    public AiCredentialController(AiCredentialService aiCredentialService) {
        this.aiCredentialService = aiCredentialService;
    }

    @GetMapping("/ai/credential")
    @PreAuthorize("hasRole('USER')")
    public AiCredentialStatusResponse status(Authentication authentication) {
        return aiCredentialService.statusFor(authentication.getName(), authentication.getAuthorities());
    }

    @PutMapping("/ai/credential")
    @PreAuthorize("hasRole('USER')")
    public AiCredentialStatusResponse replace(
        @RequestBody AiCredentialUpdateRequest request,
        Authentication authentication
    ){
        return aiCredentialService.replace(authentication.getName(), request, authentication.getAuthorities());
    }

    @DeleteMapping("/ai/credential")
    @PreAuthorize("hasRole('USER')")
    public AiCredentialStatusResponse delete(Authentication authentication) {
        return aiCredentialService.delete(authentication.getName(), authentication.getAuthorities());
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/aicredential/AiCredentialRepository.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/aicredential/AiCredentialRepository.java>

```
package com.securefromscratch.busybee.aicredential;
```

```
import java.util.Optional;
```

```
import org.jooq.DSLContext;
```

```
import org.springframework.stereotype.Repository;
```

```
import static org.jooq.impl.DSL.field;
```

```
import static org.jooq.impl.DSL.name;
```

```
import static org.jooq.impl.DSL.table;
```

```
@Repository
```

```
public class AiCredentialRepository {
```

```
    private static final org.jooq.Table<?> CREDENTIALS = table(name("busybee_ai_credential"));
```

```
    private static final org.jooq.Field<String> USERNAME = field(name("username"), String.class);
```

```
    private static final org.jooq.Field<String> PROVIDER_TYPE = field(name("provider_type"), String.class);
```

```
    private static final org.jooq.Field<String> SECRET_REFERENCE = field(name("secret_reference"), String.class);
```

```
    private static final org.jooq.Field<String> CREDENTIAL_SUFFIX = field(name("credential_suffix"), String.class);
```

```
    private final DSLContext dsl;
```

```
    public AiCredentialRepository(DSLContext dsl) {
```

```
        this.dsl = dsl;
```

```
    }
```

```
    public Optional<PersonalCredential> findForUsername(String username) {
```

```
        return dsl.select(PROVIDER_TYPE, SECRET_REFERENCE, CREDENTIAL_SUFFIX)
```

```
            .from(CREDENTIALS)
```

```
            .where(USERNAME.eq(username))
```

```
            .fetchOptional(record -> new PersonalCredential(
```

```
                AiProvider.valueOf(record.get(PROVIDER_TYPE)),
```

```
                record.get(SECRET_REFERENCE),
```

```
                record.get(CREDENTIAL_SUFFIX)
```

```
            ));
```

```
    }
```

```
    public void save(String username, PersonalCredential credential) {
```

```
        dsl.insertInto(CREDENTIALS)
```

```
            .columns(USERNAME, PROVIDER_TYPE, SECRET_REFERENCE, CREDENTIAL_SUFFIX)
```

```
            .values(
```

```
                username,
```

```
                credential.provider().name(),
```

```
                credential.secretReference(),
```

```
                credential.suffix()
```



```

    )
    .onDuplicateKeyUpdate()
    .set(PROVIDER_TYPE, credential.provider().name())
    .set(SECRET_REFERENCE, credential.secretReference())
    .set(CREDENTIAL_SUFFIX, credential.suffix())
    .execute();
}

public void delete(String username) {
    dsl.deleteFrom(CREDENTIALS)
        .where(USERNAME.eq(username))
        .execute();
}

public record PersonalCredential(AiProvider provider, String secretReference, String suffix) {
}
}

```

busybee/src/main/java/com/securefromscratch/busybee/aicredential/AiCredentialService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/aicredential/AiCredentialService.java>

```

package com.securefromscratch.busybee.aicredential;

import java.util.Collection;
import java.util.Optional;
import java.util.UUID;

import org.owasp.untrust.wv.visibility.secret.SecretReference;
import org.owasp.untrust.wv.visibility.secret.vault.SpringVaultStringSecretStore;
import org.springframework.beans.factory.annotation.Value;
import org.springframework.security.core.GrantedAuthority;
import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;
import org.springframework.vault.core.VaultOperations;
import org.springframework.vault.core.VaultKeyValueOperationsSupport.KeyValueBackend;

@Service
public class AiCredentialService {
    private final AiCredentialRepository aiCredentialRepository;
    private final AiCredentialStatusService aiCredentialStatusService;
    private final SpringVaultStringSecretStore secretStore;
    private final VaultOperations vaultOperations;
    private final String vaultMount;

    public AiCredentialService(
        AiCredentialRepository aiCredentialRepository,
        AiCredentialStatusService aiCredentialStatusService,
        SpringVaultStringSecretStore secretStore,
        VaultOperations vaultOperations,
        @Value("${busybee.secrets.vault.mount}") String vaultMount
    ) {
        this.aiCredentialRepository = aiCredentialRepository;
        this.aiCredentialStatusService = aiCredentialStatusService;
        this.secretStore = secretStore;
        this.vaultOperations = vaultOperations;
        this.vaultMount = vaultMount;
    }

    public AiCredentialStatusResponse statusFor(String username, Collection<? extends GrantedAuthority> authorities) {
        return aiCredentialStatusService.statusFor(username, authorities);
    }

    @Transactional
    public AiCredentialStatusResponse replace(
        String username,

```

```

        AiCredentialUpdateRequest request,
        Collection<? extends GrantedAuthority> authorities
    ){
        Optional<AiCredentialRepository.PersonalCredential> previousCredential = aiCredentialRepository.findForUsername(
            SecretReference reference = new SecretReference(UUID.randomUUID().toString());
            request.apiKey().hide(secretStore, reference, request.apiKey().suffix());
            AiCredentialRepository.PersonalCredential credential = new AiCredentialRepository.PersonalCredential(
                request.provider().exposeUnchecked(),
                reference.path(),
                request.apiKey().suffix()
            );
            aiCredentialRepository.save(username, credential);
            previousCredential.ifPresent(this::deleteSecret);
            return aiCredentialStatusService.statusFor(username, authorities, Optional.of(credential));
        }

        @Transactional
        public AiCredentialStatusResponse delete(String username, Collection<? extends GrantedAuthority> authorities) {
            Optional<AiCredentialRepository.PersonalCredential> previousCredential = aiCredentialRepository.findForUsername(
                previousCredential.ifPresent(this::deleteSecret);
                aiCredentialRepository.delete(username);
                return aiCredentialStatusService.statusFor(username, authorities, Optional.empty());
            }

            private void deleteSecret(AiCredentialRepository.PersonalCredential credential) {
                vaultOperations.opsForKeyValue(vaultMount, KeyValueBackend.KV_2).delete(credential.secretReference());
            }
        }
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/aicredential/AiCredentialStatusResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/aicredential/AiCredentialStatusResponse.java>

```
package com.securefromscratch.busybee.aicredential;
```

```
public record AiCredentialStatusResponse(  
    String provider,  
    String model,  
    String providerType,  
    boolean personalCredentialConfigured,  
    String personalCredentialSuffix,  
    boolean serverCredentialAvailable,  
    boolean serverCredentialAllowed,  
    String selection  
) {  
}
```

busybee/src/main/java/com/securefromscratch/busybee/aicredential/AiCredentialStatusService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/aicredential/AiCredentialStatusService.java>

```

package com.securefromscratch.busybee.aicredential;

import java.util.Collection;

import org.owasp.untrust.vw.visibility.secret.vault.SpringVaultStringSecretStore;
import org.springframework.beans.factory.annotation.Value;
import org.springframework.security.core.GrantedAuthority;
import org.springframework.stereotype.Service;

@Service
public class AiCredentialStatusService {
    private final String serverApiKey;
    private final AiCredentialRepository aiCredentialRepository;

    public AiCredentialStatusService(
        @Value("${busybee.ai.gemini.api-key}") String serverApiKey,
        AiCredentialRepository aiCredentialRepository
    ){
        this.serverApiKey = serverApiKey;
        this.aiCredentialRepository = aiCredentialRepository;
    }

    public AiCredentialStatusResponse statusFor(String username, Collection<? extends GrantedAuthority> authorities)
        return statusFor(username, authorities, aiCredentialRepository.findForUsername(username));
    }

    AiCredentialStatusResponse statusFor(
        String username,
        Collection<? extends GrantedAuthority> authorities,
        java.util.Optional<AiCredentialRepository.PersonalCredential> personalCredential
    ){
        boolean serverCredentialAvailable = !serverApiKey.isBlank();
        boolean serverCredentialAllowed = authorities.stream().anyMatch(
            authority -> authority.getAuthority().equals("AI_ENABLED") || authority.getAuthority().equals("ROLE_ADMIN")
        );
        AiProvider provider = personalCredential.map(AiCredentialRepository.PersonalCredential::provider)
            .orElse(AiProvider.GEMINI_FLASH);
        boolean personalCredentialConfigured = personalCredential.isPresent();
        String personalCredentialSuffix = personalCredential.map(AiCredentialRepository.PersonalCredential::suffix).orElse("");
        String selection = personalCredentialConfigured
            ? "PERSONAL_KEY"
            : serverCredentialAvailable && serverCredentialAllowed ? "SERVER_KEY" : "UNAVAILABLE";
        return new AiCredentialStatusResponse(
            provider.displayName(),

```

```
        provider.model(),
        provider.name(),
        personalCredentialConfigured,
        personalCredentialSuffix,
        serverCredentialAvailable,
        serverCredentialAllowed,
        selection
    );
}
}
```

busybee/src/main/java/com/securefromscratch/busybee/aicredential/AiCredentialUpdateRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/aicredential/AiCredentialUpdateRequest.java>

```
package com.securefromscratch.busybee.aicredential;

public record AiCredentialUpdateRequest(AiProviderValue provider, PendingAiApiKey apiKey) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/aicredential/AiProvider.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/aicredential/AiProvider.java>

```
package com.securefromscratch.busybee.aicredential;
```

```
public enum AiProvider {  
    GEMINI_FLASH("gemini", "gemini-2.5-flash-lite"),  
    GPT_5_NANO("openai", "gpt-5-nano");
```

```
    private final String displayName;  
    private final String model;
```

```
    AiProvider(String displayName, String model) {  
        this.displayName = displayName;  
        this.model = model;  
    }
```

```
    public String displayName() {  
        return displayName;  
    }
```

```
    public String model() {  
        return model;  
    }
```

```
}
```

busybee/src/main/java/com/securefromscratch/busybee/aicredential/AiProviderValue.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/aicredential/AiProviderValue.java>

```
package com.securefromscratch.busybee.aicredential;

import com.fasterxml.jackson.annotation.JsonCreator;
import org.owasp.untrust.valuedescriptors.foundation.PubliclyExposed;
import org.owasp.untrust.vv.foundation.ValidatedWrappedValue;
import org.owasp.untrust.vv.traits.EnumValidationTraits;

import static org.owasp.untrust.valuedescriptors.Hardcoded.hardcoded;

public final class AiProviderValue extends ValidatedWrappedValue<AiProvider> implements PubliclyExposed<AiProvider> {
    @JsonCreator(mode = JsonCreator.Mode.DELEGATING)
    public AiProviderValue(String raw) {
        super(raw, new EnumValidationTraits<>(AiProvider.class, hardcoded("AI provider")));
    }

    @Override
    public AiProvider exposeUnchecked() {
        return exposeUnchecked(EXPOSE_HALF_BAKED_VALUE_INTENDED_FOR_INTERNAL_LIBRARY_USE_ONLY_MARKER);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/aicredential/PendingAiApiKey.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/aicredential/PendingAiApiKey.java>


```

package com.securefromscratch.busybee.aicredential;

import com.fasterxml.jackson.annotation.JsonCreator;
import org.owasp.untrust.vw.foundation.ValidatedWrappedValue;
import org.owasp.untrust.vw.prebuilt.ApiKey;
import org.owasp.untrust.vw.prebuilt.PendingApiKey;
import org.owasp.untrust.vw.visibility.secret.PendingSecret;

public final class PendingAiApiKey extends ValidatedWrappedValue<String> implements PendingSecret<String, ApiKey> {
    private static final int SUFFIX_LENGTH = 4;

    @JsonCreator(mode = JsonCreator.Mode.DELEGATING)
    public PendingAiApiKey(String raw) {
        super(raw, new PendingApiKey.Traits());
    }

    public String suffix() {
        String apiKey = exposeUnchecked(EXPOSE_HALF_BAKED_VALUE_INTENDED_FOR_INTERNAL_LIBRARY_USE_ONLY_M
        return apiKey.substring(apiKey.length() - SUFFIX_LENGTH);
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/aitask/AiTaskAssistanceController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/aitask/AiTaskAssistanceController.java>

```

package com.securefromscratch.busybee.aitask;

import org.springframework.security.access.prepost.PreAuthorize;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RestController;

@RestController
@PreAuthorize("denyAll()")
public class AiTaskAssistanceController {
    private final AiTaskAssistanceService aiTaskAssistanceService;

    public AiTaskAssistanceController(AiTaskAssistanceService aiTaskAssistanceService) {
        this.aiTaskAssistanceService = aiTaskAssistanceService;
    }

    @PostMapping("/ai/task/improve")
    @PreAuthorize("hasRole('USER')")
    public ImprovedTaskResponse improve(@RequestBody TaskAssistanceRequest request) {
        return aiTaskAssistanceService.improve(request);
    }

    @PostMapping("/ai/task/subtasks")
    @PreAuthorize("hasRole('USER')")
    public SubtasksResponse subtasks(@RequestBody TaskAssistanceRequest request) {
        return aiTaskAssistanceService.subtasks(request);
    }

    @PostMapping("/ai/task/ocr-structure")
    @PreAuthorize("hasRole('USER')")
    public OcrStructureResponse structureOcr(@RequestBody OcrStructureRequest request) {
        return aiTaskAssistanceService.structureOcr(request);
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/aitask/AiTaskAssistanceService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/aitask/AiTaskAssistanceService.java>

```

package com.securefromscratch.busybee.aitask;

import java.util.List;

import org.springframework.stereotype.Service;

@Service
public class AiTaskAssistanceService {
    public ImprovedTaskResponse improve(TaskAssistanceRequest request) {
        return new ImprovedTaskResponse("Improved task title", "Improved task description.");
    }

    public SubtasksResponse subtasks(TaskAssistanceRequest request) {
        return new SubtasksResponse(List.of("First subtask", "Second subtask"));
    }

    public OcrStructureResponse structureOcr(OcrStructureRequest request) {
        return new OcrStructureResponse("OCR task title", "Structured from OCR text.", List.of());
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/aitask/ImprovedTaskResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/aitask/ImprovedTaskResponse.java>

```

package com.securefromscratch.busybee.aitask;

public record ImprovedTaskResponse(String title, String description) {
}

```

busybee/src/main/java/com/securefromscratch/busybee/aitask/OcrStructureRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/aitask/OcrStructureRequest.java>

```
package com.securefromscratch.busybee.aitask;

import org.owasp.untrust.vw.SingleLine;

public record OcrStructureRequest(SingleLine rawText) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/aitask/OcrStructureResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/aitask/OcrStructureResponse.java>

```
package com.securefromscratch.busybee.aitask;

import java.util.List;

public record OcrStructureResponse(String title, String description, List<String> responsibilityOf) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/aitask/SubtasksResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/aitask/SubtasksResponse.java>

```
package com.securefromscratch.busybee.aitask;

import java.util.List;

public record SubtasksResponse(List<String> subtasks) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/aitask/TaskAssistanceRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/aitask/TaskAssistanceRequest.java>

```
package com.securefromscratch.busybee.aitask;

import org.owasp.untrust.w.SingleLine;

public record TaskAssistanceRequest(SingleLine title, SingleLine description) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment/CommentController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/comment/CommentController.java>

```

package com.securefromscratch.busybee.comment;

import java.io.IOException;
import java.security.Principal;
import java.util.Optional;

import org.springframework.security.access.prepost.PreAuthorize;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestPart;
import org.springframework.web.bind.annotation.RestController;
import org.springframework.web.multipart.MultipartFile;

@RestController
@PreAuthorize("denyAll()")
public class CommentController {
    private final CommentService commentService;

    public CommentController(CommentService commentService) {
        this.commentService = commentService;
    }

    @PostMapping(path = "/comment", consumes = "multipart/form-data")
    @PreAuthorize("hasRole('USER')")
    public CommentCreateResponse create(
        @RequestPart("commentFields") CommentFieldsRequest request,
        @RequestPart(value = "file", required = false) Optional<MultipartFile> file,
        Principal principal
    ) throws IOException {
        return commentService.create(request, file, principal.getName());
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/comment/CommentCreateResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/comment/CommentCreateResponse.java>

```
package com.securefromscratch.busybee.comment;

public record CommentCreateResponse(String commentid) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment/CommentFieldsRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/comment/CommentFieldsRequest.java>

```
package com.securefromscratch.busybee.comment;

import java.util.Optional;

import org.owasp.untrust.vw.SingleLine;
import org.owasp.untrust.vw.ViewableUuidValue;

public record CommentFieldsRequest(
    SingleLine text,
    ViewableUuidValue taskid,
    Optional<ViewableUuidValue> commentid
){
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment/CommentRepository.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/comment/CommentRepository.java>

```

package com.securefromscratch.busybee.comment;

import java.time.Instant;
import java.util.List;
import java.util.Optional;
import java.util.UUID;

import org.jooq.DSLContext;
import org.springframework.stereotype.Repository;

import static org.jooq.impl.DSL.field;
import static org.jooq.impl.DSL.max;
import static org.jooq.impl.DSL.name;
import static org.jooq.impl.DSL.table;

@Repository
public class CommentRepository {
    private static final org.jooq.Table<?> COMMENTS = table(name("busybee_comment"));
    private static final org.jooq.Field<String> COMMENT_ID = field(name("comment_id"), String.class);
    private static final org.jooq.Field<String> TASK_ID = field(name("task_id"), String.class);
    private static final org.jooq.Field<String> PARENT_COMMENT_ID = field(name("parent_comment_id"), String.class);
    private static final org.jooq.Field<String> TEXT = field(name("text"), String.class);
    private static final org.jooq.Field<String> IMAGE = field(name("image"), String.class);
    private static final org.jooq.Field<String> IMAGE_FILENAME = field(name("image_filename"), String.class);
    private static final org.jooq.Field<String> ATTACHMENT = field(name("attachment"), String.class);
    private static final org.jooq.Field<String> ATTACHMENT_FILENAME = field(name("attachment_filename"), String.class);
    private static final org.jooq.Field<Integer> INDENT = field(name("indent"), Integer.class);
    private static final org.jooq.Field<String> CREATED_BY = field(name("created_by"), String.class);
    private static final org.jooq.Field<Instant> CREATED_ON = field(name("created_on"), Instant.class);

    private final DSLContext dsl;

    public CommentRepository(DSLContext dsl) {
        this.dsl = dsl;
    }

    public void create(NewComment comment) {
        if (comment.parentCommentId().isPresent()) {
            dsl.insertInto(COMMENTS)
                .columns(COMMENT_ID, TASK_ID, PARENT_COMMENT_ID, TEXT, INDENT, CREATED_BY, CREATED_ON)
                .values(
                    comment.commentId().toString(),
                    comment.taskId().toString(),
                    comment.parentCommentId().get().toString(),

```



```

        comment.text(),
        comment.indent(),
        comment.createdBy(),
        comment.createdOn()
    )
    .execute();
    return;
}

dsl.insertInto(COMMENTS)
    .columns(COMMENT_ID, TASK_ID, TEXT, INDENT, CREATED_BY, CREATED_ON)
    .values(
        comment.commentId().toString(),
        comment.taskId().toString(),
        comment.text(),
        comment.indent(),
        comment.createdBy(),
        comment.createdOn()
    )
    .execute();
}

public int indentForParent(UUID taskId, UUID parentCommentId) {
    return dsl.select(INDENT)
        .from(COMMENTS)
        .where(TASK_ID.eq(taskId.toString()))
        .and(COMMENT_ID.eq(parentCommentId.toString()))
        .fetchOptional(INDENT)
        .map(indent -> indent + 1)
        .orElse(0);
}

public void attachFile(UUID commentId, StoredCommentFile file) {
    if (file.image()) {
        dsl.update(COMMENTS)
            .set(IMAGE, file.fileId().toString())
            .set(IMAGE_FILENAME, file.originalFilename())
            .where(COMMENT_ID.eq(commentId.toString()))
            .execute();
        return;
    }

    dsl.update(COMMENTS)
        .set(ATTACHMENT, file.fileId().toString())

```

```

        .set(ATTACHMENT_FILENAME, file.originalFilename())
        .where(COMMENT_ID.eq(commentId.toString()))
        .execute();
    }

```

```

public List<CommentRow> findByTaskId(UUID taskId) {
    return dsl.select(
        COMMENT_ID,
        TEXT,
        IMAGE,
        IMAGE_FILENAME,
        ATTACHMENT,
        ATTACHMENT_FILENAME,
        INDENT,
        CREATED_BY,
        CREATED_ON
    )
    .from(COMMENTS)
    .where(TASK_ID.eq(taskId.toString()))
    .orderBy(CREATED_ON)
    .fetch(record -> new CommentRow(
        UUID.fromString(record.get(COMMENT_ID)),
        taskId,
        Optional.empty(),
        record.get(TEXT),
        record.get(IMAGE),
        record.get(IMAGE_FILENAME),
        record.get(ATTACHMENT),
        record.get(ATTACHMENT_FILENAME),
        record.get(INDENT),
        record.get(CREATED_BY),
        record.get(CREATED_ON)
    ));
}

```

```

public ThreadState nonImageThreadState(UUID taskId) {
    int commentCount = dsl.selectCount()
        .from(COMMENTS)
        .where(TASK_ID.eq(taskId.toString()))
        .and(IMAGE.isNull())
        .fetchOne(0, int.class);
    Optional<Instant> latestCommentAt = dsl.select(max(CREATED_ON))
        .from(COMMENTS)
        .where(TASK_ID.eq(taskId.toString()))

```

```

        .and(IMAGE.isNull())
        .fetchOptional(max(CREATED_ON));
    return new ThreadState(commentCount, latestCommentAt);
}

public record CommentRow(
    UUID commentId,
    UUID taskId,
    Optional<UUID> parentCommentId,
    String text,
    String image,
    String imageFilename,
    String attachment,
    String attachmentFilename,
    int indent,
    String createdBy,
    Instant createdOn
){
}

public record ThreadState(int commentCount, Optional<Instant> latestCommentAt) {
}

public record NewComment(
    UUID commentId,
    UUID taskId,
    Optional<UUID> parentCommentId,
    String text,
    int indent,
    String createdBy,
    Instant createdOn
){
}

public record StoredCommentFile(UUID fileId, String originalFilename, boolean image) {
}
}

```

busybee/src/main/java/com/securefromscratch/busybee/comment/CommentResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp

[_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/comment/CommentResponse.java](https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/comment/CommentResponse.java)>

```
package com.securefromscratch.busybee.comment;

import java.util.List;

public record CommentResponse(
    String commentid,
    String text,
    String image,
    String imageFilename,
    String attachment,
    String attachmentFilename,
    int indent,
    String createdBy,
    String createdOn,
    List<Object> linkPreviews
){
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment/CommentService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/comment/CommentService.java>

```
package com.securefromscratch.busybee.comment;
```

```
import java.time.Clock;
```

```
import java.io.IOException;
```

```
import java.util.List;
```

```
import java.util.Optional;
```

```
import java.util.UUID;
```

```
import com.securefromscratch.busybee.file.FileStorageService;
```

```
import com.securefromscratch.busybee.file.StoredUpload;
```

```
import org.springframework.stereotype.Service;
```

```
import org.springframework.transaction.annotation.Transactional;
```

```
import org.springframework.web.multipart.MultipartFile;
```

```
@Service
```

```
public class CommentService {
```

```
    private final CommentRepository commentRepository;
```

```
    private final FileStorageService fileStorageService;
```

```
    private final Clock clock;
```

```
    public CommentService(CommentRepository commentRepository, FileStorageService fileStorageService, Clock clock)
```

```
    {  
        this.commentRepository = commentRepository;
```

```
        this.fileStorageService = fileStorageService;
```

```
        this.clock = clock;
```

```
    }
```

```
@Transactional
```

```
public CommentCreateResponse create(CommentFieldsRequest request, Optional<MultipartFile> file, String createdBy)  
    throws IOException {
```

```
    UUID taskId = request.taskid().exposeUnchecked();
```

```
    UUID commentId = UUID.randomUUID();
```

```
    int indent = request.commentid()
```

```
        .map(parentCommentId -> commentRepository.indentForParent(taskId, parentCommentId.exposeUnchecked()),  
        .orElse(0);
```

```
    commentRepository.create(new CommentRepository.NewComment(  
        commentId,
```

```
        taskId,
```

```
        request.commentid().map(parentCommentId -> parentCommentId.exposeUnchecked()),
```

```
        request.text().exposeUnchecked(),  
        indent,
```

```
        createdBy,
```

```
        clock.instant()  
    ));
```

```
    ));
```

```
    if (file.isPresent()) {
```

```

        StoredUpload upload = fileStorageService.store(file.get());
        commentRepository.attachFile(commentId, new CommentRepository.StoredCommentFile(
            upload.fileId(),
            upload.originalFilename(),
            upload.image()
        ));
    }
    return new CommentCreateResponse(commentId.toString());
}

public List<CommentResponse> findByTaskId(UUID taskId) {
    return commentRepository.findByTaskId(taskId).stream().map(comment -> new CommentResponse(
        comment.commentId().toString(),
        comment.text(),
        comment.image(),
        comment.imageFilename(),
        comment.attachment(),
        comment.attachmentFilename(),
        comment.indent(),
        comment.createdBy(),
        comment.createdOn().toString(),
        List.of()
    )).toList();
}
}

```

busybee/src/main/java/com/securefromscratch/busybee/comment summary/CommentSummaryController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/commentsummary/CommentSummaryController.java>

```
package com.securefromscratch.busybee.commentsummary;

import java.util.Optional;

import org.springframework.security.access.prepost.PreAuthorize;
import org.springframework.security.core.Authentication;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RestController;

@RestController
@PreAuthorize("denyAll()")
public class CommentSummaryController {
    private final CommentSummaryService commentSummaryService;

    public CommentSummaryController(CommentSummaryService commentSummaryService) {
        this.commentSummaryService = commentSummaryService;
    }

    @PostMapping("/ai/task/comment-summary")
    @PreAuthorize("hasRole('USER')")
    public Optional<CommentSummaryResponse> summarize(
        @RequestBody CommentSummaryRequest request,
        Authentication authentication
    ){
        return commentSummaryService.summarize(request, authentication.getName(), authentication.getAuthorities());
    }

    @PostMapping("/ai/task/comment-summary/refresh")
    @PreAuthorize("hasRole('USER')")
    public CommentSummaryResponse refresh(
        @RequestBody CommentSummaryRequest request,
        Authentication authentication
    ){
        return commentSummaryService.refresh(request, authentication.getName(), authentication.getAuthorities());
    }
}
```

**busybee/src/main/java/com/securefromscratch/busybee/comment
summary/CommentSummaryRequest.java**

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/commentsummary/CommentSummaryRequest.java>

```
package com.securefromscratch.busybee.commentsummary;

import org.owasp.untrust.w.ViewableUuidValue;

public record CommentSummaryRequest(ViewableUuidValue taskid) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment summary/CommentSummaryResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/commentsummary/CommentSummaryResponse.java>

```
package com.securefromscratch.busybee.commentsummary;

import java.util.Optional;

public record CommentSummaryResponse(
    String summary,
    int summarizedCommentCount,
    int currentCommentCount,
    Optional<String> summarizedLatestCommentAt,
    Optional<String> currentLatestCommentAt,
    boolean stale,
    String generatedBy,
    String credentialSource,
    String generatedAt
){
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment summary/CommentSummaryService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp

_untrust/5e578e471fdb38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefrom
scratch/busybee/commentssummary/CommentSummaryService.java>

```

package com.securefromscratch.busybee.commentsummary;

import java.time.Clock;
import java.time.Instant;
import java.util.Collection;
import java.util.Optional;
import java.util.UUID;

import com.securefromscratch.busybee.aicredential.AiCredentialStatusService;
import com.securefromscratch.busybee.comment.CommentRepository;
import com.securefromscratch.busybee.settings.UserSettingsService;
import org.springframework.security.core.GrantedAuthority;
import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;

@Service
public class CommentSummaryService {
    private static final String GENERATED_SUMMARY = "AI summary of the comment thread.";

    private final TaskCommentSummaryRepository summaryRepository;
    private final CommentRepository commentRepository;
    private final UserSettingsService userSettingsService;
    private final AiCredentialStatusService aiCredentialStatusService;
    private final Clock clock;

    public CommentSummaryService(
        TaskCommentSummaryRepository summaryRepository,
        CommentRepository commentRepository,
        UserSettingsService userSettingsService,
        AiCredentialStatusService aiCredentialStatusService,
        Clock clock
    ){
        this.summaryRepository = summaryRepository;
        this.commentRepository = commentRepository;
        this.userSettingsService = userSettingsService;
        this.aiCredentialStatusService = aiCredentialStatusService;
        this.clock = clock;
    }

    @Transactional
    public Optional<CommentSummaryResponse> summarize(
        CommentSummaryRequest request,
        String username,
        Collection<? extends GrantedAuthority> authorities
    )

```

```

){
    return generate(request, username, authorities, true);
}

@Transactional
public CommentSummaryResponse refresh(
    CommentSummaryRequest request,
    String username,
    Collection<? extends GrantedAuthority> authorities
){
    return generate(request, username, authorities, false).orElseThrow();
}

private Optional<CommentSummaryResponse> generate(
    CommentSummaryRequest request,
    String username,
    Collection<? extends GrantedAuthority> authorities,
    boolean respectThreshold
){
    UUID taskId = request.taskid().exposeUnchecked();
    CommentRepository.ThreadState threadState = commentRepository.nonImageThreadState(taskId);
    if (respectThreshold && threadState.commentCount() < userSettingsService.summaryThresholdFor(username)) {
        return Optional.empty();
    }
    TaskCommentSummaryRepository.StoredCommentSummary summary = new TaskCommentSummaryRepository
        GENERATED_SUMMARY,
        threadState.commentCount(),
        threadState.latestCommentAt().orElseThrow(),
        username,
        aiCredentialStatusService.statusFor(username, authorities).selection(),
        clock.instant()
    );
    summaryRepository.save(taskId, summary);
    return Optional.of(responseFor(summary, threadState));
}

public Optional<CommentSummaryResponse> summaryForTask(UUID taskId) {
    CommentRepository.ThreadState threadState = commentRepository.nonImageThreadState(taskId);
    return summaryRepository.findByTaskId(taskId).map(summary -> responseFor(summary, threadState));
}

private CommentSummaryResponse responseFor(
    TaskCommentSummaryRepository.StoredCommentSummary summary,
    CommentRepository.ThreadState threadState

```

```

){
    boolean stale = summary.summarizedCommentCount() != threadState.commentCount()
        || !summary.summarizedLatestCommentAt().equals(threadState.latestCommentAt().orElseThrow());
    return new CommentSummaryResponse(
        summary.summary(),
        summary.summarizedCommentCount(),
        threadState.commentCount(),
        Optional.of(summary.summarizedLatestCommentAt().toString()),
        asText(threadState.latestCommentAt()),
        stale,
        summary.generatedBy(),
        summary.credentialSource(),
        summary.generatedAt().toString()
    );
}

private Optional<String> asText(Optional<Instant> instant) {
    return instant.map(Instant::toString);
}
}

```

busybee/src/main/java/com/securefromscratch/busybee/comment summary/TaskCommentSummaryRepository.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/commentsummary/TaskCommentSummaryRepository.java>

```
package com.securefromscratch.busybee.commentssummary;
```

```
import java.time.Instant;
```

```
import java.util.Optional;
```

```
import java.util.UUID;
```

```
import org.jooq.DSLContext;
```

```
import org.springframework.stereotype.Repository;
```

```
import static org.jooq.impl.DSL.field;
```

```
import static org.jooq.impl.DSL.name;
```

```
import static org.jooq.impl.DSL.table;
```

```
@Repository
```

```
public class TaskCommentSummaryRepository {
```

```
    private static final org.jooq.Table<?> SUMMARIES = table(name("busybee_task_comment_summary"));
```

```
    private static final org.jooq.Field<String> TASK_ID = field(name("task_id"), String.class);
```

```
    private static final org.jooq.Field<String> SUMMARY = field(name("summary"), String.class);
```

```
    private static final org.jooq.Field<Integer> SUMMARIZED_COMMENT_COUNT = field(
        name("summarized_comment_count"),
        Integer.class
    );
```

```
    private static final org.jooq.Field<Instant> SUMMARIZED_LATEST_COMMENT_AT = field(
        name("summarized_latest_comment_at"),
        Instant.class
    );
```

```
    private static final org.jooq.Field<String> GENERATED_BY = field(name("generated_by"), String.class);
```

```
    private static final org.jooq.Field<String> CREDENTIAL_SOURCE = field(name("credential_source"), String.class);
```

```
    private static final org.jooq.Field<Instant> GENERATED_AT = field(name("generated_at"), Instant.class);
```

```
    private final DSLContext dsl;
```

```
    public TaskCommentSummaryRepository(DSLContext dsl) {
```

```
        this.dsl = dsl;
```

```
    }
```

```
    public Optional<StoredCommentSummary> findByTaskId(UUID taskId) {
```

```
        return dsl.select(
            SUMMARY,
            SUMMARIZED_COMMENT_COUNT,
            SUMMARIZED_LATEST_COMMENT_AT,
            GENERATED_BY,
            CREDENTIAL_SOURCE,
            GENERATED_AT
        );
    }
```

```

    )
    .from(SUMMARIES)
    .where(TASK_ID.eq(taskId.toString()))
    .fetchOptional(record -> new StoredCommentSummary(
        record.get(SUMMARY),
        record.get(SUMMARIZED_COMMENT_COUNT),
        record.get(SUMMARIZED_LATEST_COMMENT_AT),
        record.get(GENERATED_BY),
        record.get(CREDENTIAL_SOURCE),
        record.get(GENERATED_AT)
    ));
}

```

```

public void save(UUID taskId, StoredCommentSummary summary) {
    dsl.insertInto(SUMMARIES)
        .columns(
            TASK_ID,
            SUMMARY,
            SUMMARIZED_COMMENT_COUNT,
            SUMMARIZED_LATEST_COMMENT_AT,
            GENERATED_BY,
            CREDENTIAL_SOURCE,
            GENERATED_AT
        )
        .values(
            taskId.toString(),
            summary.summary(),
            summary.summarizedCommentCount(),
            summary.summarizedLatestCommentAt(),
            summary.generatedBy(),
            summary.credentialSource(),
            summary.generatedAt()
        )
        .onDuplicateKeyUpdate()
        .set(SUMMARY, summary.summary())
        .set(SUMMARIZED_COMMENT_COUNT, summary.summarizedCommentCount())
        .set(SUMMARIZED_LATEST_COMMENT_AT, summary.summarizedLatestCommentAt())
        .set(GENERATED_BY, summary.generatedBy())
        .set(CREDENTIAL_SOURCE, summary.credentialSource())
        .set(GENERATED_AT, summary.generatedAt())
        .execute();
}

```

```

public record StoredCommentSummary(

```

```
String summary,  
int summarizedCommentCount,  
Instant summarizedLatestCommentAt,  
String generatedBy,  
String credentialSource,  
Instant generatedAt  
{  
}  
}
```

busybee/src/main/java/com/securefromscratch/busybee/configuration/ApplicationConfiguration.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/configuration/ApplicationConfiguration.java>

```
package com.securefromscratch.busybee.configuration;
```

```
import java.time.Clock;
```

```
import java.net.URI;
```

```
import com.fasterxml.jackson.databind.ObjectMapper;
```

```
import com.fasterxml.jackson.datatype.jdk8.Jdk8Module;
```

```
import org.owasp.untrust.vw.visibility.secret.vault.SpringVaultStringSecretStore;
```

```
import org.springframework.context.annotation.Bean;
```

```
import org.springframework.context.annotation.Configuration;
```

```
import org.springframework.security.crypto.bcrypt.BCryptPasswordEncoder;
```

```
import org.springframework.security.crypto.password.PasswordEncoder;
```

```
import org.springframework.beans.factory.annotation.Value;
```

```
import org.springframework.vault.authentication.TokenAuthentication;
```

```
import org.springframework.vault.client.VaultEndpoint;
```

```
import org.springframework.vault.core.VaultOperations;
```

```
import org.springframework.vault.core.VaultTemplate;
```

```
@Configuration
```

```
public class ApplicationConfiguration {
```

```
    @Bean
```

```
    ObjectMapper objectMapper() {
```

```
        return new ObjectMapper().registerModule(new Jdk8Module());
```

```
    }
```

```
    @Bean
```

```
    Clock clock() {
```

```
        return Clock.systemUTC();
```

```
    }
```

```
    @Bean
```

```
    PasswordEncoder passwordEncoder() {
```

```
        return new BCryptPasswordEncoder();
```

```
    }
```

```
    @Bean
```

```
    VaultOperations vaultOperations(
```

```
        @Value("${busybee.secrets.vault.uri}") String vaultUri,
```

```
        @Value("${busybee.secrets.vault.token}") String vaultToken
```

```
    ){
```

```
        return new VaultTemplate(VaultEndpoint.from(URI.create(vaultUri)), new TokenAuthentication(vaultToken));
```

```
    }
```

```
    @Bean
```



```
SpringVaultStringSecretStore vaultStringSecretStore(  
    VaultOperations vaultOperations,  
    @Value("${busybee.secrets.vault.mount}") String vaultMount  
) {  
    return new SpringVaultStringSecretStore(vaultOperations, vaultMount);  
}  
}
```

busybee/src/main/java/com/securefromscratch/busybee/configuration/SecurityConfiguration.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/configuration/SecurityConfiguration.java>

```

package com.securefromscratch.busybee.configuration;

import java.util.ArrayList;
import java.util.List;

import com.securefromscratch.busybee.demosession.DemoSessionAuthenticationFilter;
import com.securefromscratch.busybee.demosession.DemoSessionProperties;
import com.securefromscratch.busybee.entitlement.EntitlementRepository;
import com.securefromscratch.busybee.registration.UserRepository;
import org.springframework.boot.context.properties.EnableConfigurationProperties;
import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;
import org.springframework.security.config.annotation.method.configuration.EnableMethodSecurity;
import org.springframework.security.config.annotation.web.builders.HttpSecurity;
import org.springframework.security.core.userdetails.User;
import org.springframework.security.core.userdetails.UserDetailsService;
import org.springframework.security.core.userdetails.UsernameNotFoundException;
import org.springframework.security.core.GrantedAuthority;
import org.springframework.security.core.authority.SimpleGrantedAuthority;
import org.springframework.security.web.SecurityFilterChain;
import org.springframework.security.web.authentication.UsernamePasswordAuthenticationFilter;

@Configuration
@EnableMethodSecurity
@EnableConfigurationProperties(DemoSessionProperties.class)
public class SecurityConfiguration {
    @Bean
    SecurityFilterChain securityFilterChain(HttpSecurity http, DemoSessionAuthenticationFilter demoSessionAuthenticationFilter) throws Exception {
        return http
            .authorizeHttpRequests(authorize -> authorize.anyRequest().permitAll())
            .addFilterBefore(demoSessionAuthenticationFilter, UsernamePasswordAuthenticationFilter.class)
            .formLogin(formLogin -> formLogin.defaultSuccessUrl("/main/main.html", true))
            .build();
    }

    @Bean
    UserDetailsService userDetailsService(UserRepository userRepository, EntitlementRepository entitlementRepository) {
        return username -> userRepository.findPasswordHash(username)
            .map(passwordHash -> {
                List<GrantedAuthority> authorities = new ArrayList<>();
                authorities.add(new SimpleGrantedAuthority("ROLE_USER"));
                entitlementRepository.findForUsername(username)
                    .forEach(entitlement -> authorities.add(new SimpleGrantedAuthority(entitlement.name())));
            });
    }
}

```

```

        return User.withUsername(username).password(passwordHash).authorities(authorities).build();
    })
    .orElseThrow(() -> new UsernameNotFoundException("User was not found."));
}
}

```

busybee/src/main/java/com/securefromscratch/busybee/currentuser/CurrentUserController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/currentuser/CurrentUserController.java>

```

package com.securefromscratch.busybee.currentuser;

import org.springframework.security.access.prepost.PreAuthorize;
import org.springframework.security.core.Authentication;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController
@PreAuthorize("denyAll()")
public class CurrentUserController {
    private final CurrentUserService currentUserService;

    public CurrentUserController(CurrentUserService currentUserService) {
        this.currentUserService = currentUserService;
    }

    @GetMapping("/me")
    @PreAuthorize("isAuthenticated()")
    public CurrentUserResponse currentUser(Authentication authentication) {
        return currentUserService.currentUser(authentication.getName(), authentication.getAuthorities());
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/currentuser/CurrentUserResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/currentuser/CurrentUserResponse.java>

_untrust/5e578e471fdb38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/currentuser/CurrentUserResponse.java>

```
package com.securefromscratch.busybee.currentuser;

import java.util.List;

public record CurrentUserResponse(
    String username,
    boolean admin,
    List<String> entitlements,
    List<String> effectiveEntitlements
){
}
```

busybee/src/main/java/com/securefromscratch/busybee/currentuser/CurrentUserService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/currentuser/CurrentUserService.java>

```

package com.securefromscratch.busybee.currentuser;

import java.util.Collection;
import java.util.List;

import com.securefromscratch.busybee.entitlement.Entitlement;
import com.securefromscratch.busybee.entitlement.EntitlementRepository;
import org.springframework.security.core.GrantedAuthority;
import org.springframework.stereotype.Service;

@Service
public class CurrentUserService {
    private final EntitlementRepository entitlementRepository;

    public CurrentUserService(EntitlementRepository entitlementRepository) {
        this.entitlementRepository = entitlementRepository;
    }

    public CurrentUserResponse currentUser(String username, Collection<? extends GrantedAuthority> authorities) {
        boolean admin = authorities.stream().anyMatch(authority -> authority.getAuthority().equals("ROLE_ADMIN"));
        List<String> entitlements = entitlementRepository.findForUsername(username).stream().map(Entitlement::name)
        List<String> effectiveEntitlements = admin
            ? List.of(Entitlement.values()).stream().map(Entitlement::name).toList()
            : entitlements;
        return new CurrentUserResponse(username, admin, entitlements, effectiveEntitlements);
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/demosession/DemoSessionAuthenticationFilter.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/demosession/DemoSessionAuthenticationFilter.java>

```

package com.securefromscratch.busybee.demosession;

import java.io.IOException;
import java.util.Arrays;
import java.util.Collections;
import java.util.Optional;

import jakarta.servlet.FilterChain;
import jakarta.servlet.ServletException;
import jakarta.servlet.http.Cookie;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import org.springframework.security.authentication.UsernamePasswordAuthenticationToken;
import org.springframework.security.core.context.SecurityContextHolder;
import org.springframework.security.core.userdetails.UserDetails;
import org.springframework.security.core.userdetails.UserDetailsService;
import org.springframework.stereotype.Component;
import org.springframework.web.filter.OncePerRequestFilter;

@Component
public class DemoSessionAuthenticationFilter extends OncePerRequestFilter {
    private static final String SESSION_COOKIE_NAME = "JSESSIONID";
    private static final String SESSION_COOKIE_PREFIX = "JSESSIONID=";

    private final DemoSessionProperties properties;
    private final UserDetailsService userDetailsService;

    public DemoSessionAuthenticationFilter(DemoSessionProperties properties, UserDetailsService userDetailsService) {
        this.properties = properties;
        this.userDetailsService = userDetailsService;
    }

    @Override
    protected void doFilterInternal(
        HttpServletRequest request,
        HttpServletResponse response,
        FilterChain filterChain
    ) throws ServletException, IOException {
        if (properties.enabled()) {
            configuredSessionId(request).ifPresent(sessionId -> authenticate(
                properties.users().get(sessionId),
                sessionId,
                response
            ));
        }
    }

```

```

    }
    filterChain.doFilter(request, response);
}

private Optional<String> configuredSessionId(HttpServletRequest request) {
    return Collections.list(request.getHeaders("Cookie")).stream()
        .flatMap(header -> Arrays.stream(header.split(";")))
        .map(String::trim)
        .filter(cookie -> cookie.startsWith(SESSION_COOKIE_PREFIX))
        .map(cookie -> cookie.substring(SESSION_COOKIE_NAME.length() + 1))
        .filter(properties.users()::containsKey)
        .findFirst();
}

private void authenticate(String username, String sessionId, HttpServletResponse response) {
    UserDetails user = userDetailsService.loadUserByUsername(username);
    SecurityContextHolder.getContext().setAuthentication(
        UsernamePasswordAuthenticationToken.authenticated(user, "", user.getAuthorities())
    );
    Cookie refreshedCookie = new Cookie(SESSION_COOKIE_NAME, sessionId);
    refreshedCookie.setPath("/");
    refreshedCookie.setMaxAge((int) properties.cookieMaxAge().toSeconds());
    response.addCookie(refreshedCookie);
}
}

```

busybee/src/main/java/com/securefromscratch/busybee/demosession/DemoSessionProperties.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/demosession/DemoSessionProperties.java>

```
package com.securefromscratch.busybee.demosession;

import java.time.Duration;
import java.util.Map;

import org.springframework.boot.context.properties.ConfigurationProperties;

@ConfigurationProperties(prefix = "busybee.demo.sessions")
public record DemoSessionProperties(
    boolean enabled,
    Duration cookieMaxAge,
    Map<String, String> users
){
}
```

busybee/src/main/java/com/securefromscratch/busybee/entitlement/Entitlement.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/entitlement/Entitlement.java>

```
package com.securefromscratch.busybee.entitlement;

public enum Entitlement {
    IMPORT_ENABLED,
    EXPORT_ENABLED,
    PAID_LEVEL_1,
    AI_ENABLED,
    OCR_ENABLED
}
```

busybee/src/main/java/com/securefromscratch/busybee/entitlement/EntitlementRepository.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/entitlement/EntitlementRepository.java>


```
package com.securefromscratch.busybee.entitlement;
```

```
import java.util.List;
```

```
import org.jooq.DSLContext;
```

```
import org.springframework.stereotype.Repository;
```

```
import static org.jooq.impl.DSL.field;
```

```
import static org.jooq.impl.DSL.name;
```

```
import static org.jooq.impl.DSL.table;
```

```
@Repository
```

```
public class EntitlementRepository {
```

```
    private static final org.jooq.Table<?> ENTITLEMENTS = table(name("busybee_user_entitlement"));
```

```
    private static final org.jooq.Field<String> USERNAME = field(name("username"), String.class);
```

```
    private static final org.jooq.Field<String> ENTITLEMENT = field(name("entitlement"), String.class);
```

```
    private final DSLContext dsl;
```

```
    public EntitlementRepository(DSLContext dsl) {
```

```
        this.dsl = dsl;
```

```
    }
```

```
    public List<Entitlement> findByUsername(String username) {
```

```
        return dsl.select(ENTITLEMENT)
```

```
            .from(ENTITLEMENTS)
```

```
            .where(USERNAME.eq(username))
```

```
            .orderBy(ENTITLEMENT)
```

```
            .fetch(ENTITLEMENT)
```

```
            .stream()
```

```
            .map(Entitlement::valueOf)
```

```
            .toList();
```

```
    }
```

```
    public void replaceByUsername(String username, List<Entitlement> entitlements) {
```

```
        dsl.deleteFrom(ENTITLEMENTS)
```

```
            .where(USERNAME.eq(username))
```

```
            .execute();
```

```
        for (Entitlement entitlement : entitlements) {
```

```
            dsl.insertInto(ENTITLEMENTS)
```

```
                .columns(USERNAME, ENTITLEMENT)
```

```
                .values(username, entitlement.name())
```

```
                .execute();
```

```
        }
```

```
}  
}
```

busybee/src/main/java/com/securefromscratch/busybee/entitlement/EntitlementValue.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/entitlement/EntitlementValue.java>

```
package com.securefromscratch.busybee.entitlement;  
  
import com.fasterxml.jackson.annotation.JsonCreator;  
import org.owasp.untrust.valuedescriptors.foundation.PubliclyExposed;  
import org.owasp.untrust.vw.foundation.ValidatedWrappedValue;  
import org.owasp.untrust.vw.traits.EnumValidationTraits;  
  
import static org.owasp.untrust.valuedescriptors.Hardcoded.hardcoded;  
  
public final class EntitlementValue extends ValidatedWrappedValue<Entitlement> implements PubliclyExposed<Entitlement> {  
    @JsonCreator(mode = JsonCreator.Mode.DELEGATING)  
    public EntitlementValue(String raw) {  
        super(raw, new EnumValidationTraits<>(Entitlement.class, hardcoded("entitlement")));  
    }  
  
    @Override  
    public Entitlement exposeUnchecked() {  
        return exposeUnchecked(EXPOSE_HALF_BAKED_VALUE_INTENDED_FOR_INTERNAL_LIBRARY_USE_ONLY_MARKER);  
    }  
}
```

busybee/src/main/java/com/securefromscratch/busybee/file/FileController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/file/FileController.java>

```

package com.securefromscratch.busybee.file;

import org.owasp.untrust.wv.ViewableUuidValue;
import org.springframework.http.MediaType;
import org.springframework.http.ResponseEntity;
import org.springframework.security.access.prepost.PreAuthorize;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RequestParam;
import org.springframework.web.bind.annotation.RestController;

@RestController
@PreAuthorize("denyAll()")
public class FileController {
    private final FileStorageService fileStorageService;

    public FileController(FileStorageService fileStorageService) {
        this.fileStorageService = fileStorageService;
    }

    @GetMapping("/image")
    @PreAuthorize("hasRole('USER')")
    public ResponseEntity<byte[]> image(@RequestParam("img") ViewableUuidValue imageId) {
        StoredFile image = fileStorageService.requireImage(imageId.exposeUnchecked());
        return ResponseEntity.ok().contentType(MediaType.parseMediaType(image.contentType())).body(image.contents());
    }

    @GetMapping("/attachment")
    @PreAuthorize("hasRole('USER')")
    public ResponseEntity<byte[]> attachment(@RequestParam("file") ViewableUuidValue attachmentId) {
        StoredFile attachment = fileStorageService.requireAttachment(attachmentId.exposeUnchecked());
        return ResponseEntity.ok().contentType(MediaType.parseMediaType(attachment.contentType())).body(attachment.contents());
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/file/FileNotFoundException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/file/FileNotFoundException.java>

```
package com.securefromscratch.busybee.file;

public class FileNotFoundException extends RuntimeException {
    public FileNotFoundException() {
        super("File was not found.");
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/file/FileRepository.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/file/FileRepository.java>

```

package com.securefromscratch.busybee.file;

import java.util.Optional;
import java.util.UUID;

import org.jooq.DSLContext;
import org.springframework.stereotype.Repository;

import static org.jooq.impl.DSL.field;
import static org.jooq.impl.DSL.name;
import static org.jooq.impl.DSL.table;

@Repository
public class FileRepository {
    private static final org.jooq.Table<?> FILES = table(name("busybee_file"));
    private static final org.jooq.Field<String> FILE_ID = field(name("file_id"), String.class);
    private static final org.jooq.Field<String> CONTENT_TYPE = field(name("content_type"), String.class);
    private static final org.jooq.Field<byte[]> CONTENTS = field(name("contents"), byte[].class);
    private static final org.jooq.Field<Boolean> IMAGE = field(name("image"), Boolean.class);

    private final DSLContext dsl;

    public FileRepository(DSLContext dsl) {
        this.dsl = dsl;
    }

    public void store(StoredUpload upload, byte[] contents) {
        dsl.insertInto(FILES)
            .columns(FILE_ID, CONTENT_TYPE, CONTENTS, IMAGE)
            .values(upload.fileId().toString(), upload.contentType(), contents, upload.image())
            .execute();
    }

    public Optional<StoredFile> findById(UUID fileId) {
        return dsl.select(CONTENT_TYPE, CONTENTS, IMAGE)
            .from(FILES)
            .where(FILE_ID.eq(fileId.toString()))
            .fetchOptional(record -> new StoredFile(
                record.get(CONTENT_TYPE),
                record.get(CONTENTS),
                record.get(IMAGE)
            ));
    }
}

```

```
}  
}
```

busybee/src/main/java/com/securefromscratch/busybee/file/FileStorageService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/file/FileStorageService.java>

```

package com.securefromscratch.busybee.file;

import java.io.IOException;
import java.util.UUID;

import org.springframework.stereotype.Service;
import org.springframework.web.multipart.MultipartFile;

@Service
public class FileStorageService {
    private final FileRepository fileRepository;

    public FileStorageService(FileRepository fileRepository) {
        this.fileRepository = fileRepository;
    }

    public StoredUpload store(MultipartFile file) throws IOException {
        String filename = file.getOriginalFilename();
        FileType fileType = FileType.from(filename);
        StoredUpload upload = new StoredUpload(UUID.randomUUID(), filename, fileType.contentType(), fileType.image())
        fileRepository.store(upload, file.getBytes());
        return upload;
    }

    public byte[] imageContents(MultipartFile file) throws IOException {
        String filename = file.getOriginalFilename();
        FileType fileType = FileType.from(filename);
        if (!fileType.image()) {
            throw new UnsupportedOperationException();
        }
        return file.getBytes();
    }

    public StoredFile requireImage(UUID fileId) {
        StoredFile file = requireFile(fileId);
        if (!file.image()) {
            throw new FileNotFoundException();
        }
        return file;
    }

    public StoredFile requireAttachment(UUID fileId) {
        StoredFile file = requireFile(fileId);
        if (file.image()) {

```

```

        throw new FileNotFoundException();
    }
    return file;
}

private StoredFile requireFile(UUID fileId) {
    return fileRepository.findById(fileId).orElseThrow(FileNotFoundException::new);
}

private enum FileType {
    PNG(".png", "image/png", true),
    JPG(".jpg", "image/jpeg", true),
    JPEG(".jpeg", "image/jpeg", true),
    GIF(".gif", "image/gif", true),
    WEBP(".webp", "image/webp", true),
    SVG(".svg", "image/svg+xml", true),
    TXT(".txt", "text/plain", false),
    PDF(".pdf", "application/pdf", false),
    DOC(".doc", "application/octet-stream", false),
    DOCX(".docx", "application/octet-stream", false);

    private final String suffix;
    private final String contentType;
    private final boolean image;

    FileType(String suffix, String contentType, boolean image) {
        this.suffix = suffix;
        this.contentType = contentType;
        this.image = image;
    }

    static FileType from(String filename) {
        if (filename.isBlank() || filename.contains("..") || filename.contains("/") || filename.contains("\\")) {
            throw new UnsupportedUploadException();
        }
        String lowerCaseFilename = filename.toLowerCase(java.util.Locale.ROOT);
        for (FileType fileType : values()) {
            if (lowerCaseFilename.endsWith(fileType.suffix)) {
                return fileType;
            }
        }
        throw new UnsupportedUploadException();
    }
}

```



```

String contentType() {
    return contentType;
}

boolean image() {
    return image;
}
}
}

```

busybee/src/main/java/com/securefromscratch/busybee/file/Store dFile.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/file/StoreFile.java>

```

package com.securefromscratch.busybee.file;

public record StoredFile(String contentType, byte[] contents, boolean image) {
}

```

busybee/src/main/java/com/securefromscratch/busybee/file/Store dUpload.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/file/StoreUpload.java>

```

package com.securefromscratch.busybee.file;

import java.util.UUID;

public record StoredUpload(UUID fileId, String originalFilename, String contentType, boolean image) {
}

```

busybee/src/main/java/com/securefromscratch/busybee/file/Unsu pportedUploadException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/file/UnsupportedUploadException.java>

```
package com.securefromscratch.busybee.file;

public class UnsupportedUploadException extends RuntimeException {
    public UnsupportedUploadException() {
        super("The uploaded file type is not supported.");
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/linkpreview/SuccessResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/linkpreview/SuccessResponse.java>

```
package com.securefromscratch.busybee.linkpreview;

public record SuccessResponse(boolean success) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/linkpreview/TaskLinkPreviewController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/linkpreview/TaskLinkPreviewController.java>

```

package com.securefromscratch.busybee.linkpreview;

import org.springframework.security.access.prepost.PreAuthorize;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RestController;

@RestController
@PreAuthorize("denyAll()")
public class TaskLinkPreviewController {
    private final TaskLinkPreviewService taskLinkPreviewService;

    public TaskLinkPreviewController(TaskLinkPreviewService taskLinkPreviewService) {
        this.taskLinkPreviewService = taskLinkPreviewService;
    }

    @PostMapping("/link-preview/task/delete")
    @PreAuthorize("hasRole('USER')")
    public SuccessResponse delete(@RequestBody TaskLinkPreviewRequest request) {
        return taskLinkPreviewService.delete(request);
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/linkpreview/TaskLinkPreviewRepository.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/linkpreview/TaskLinkPreviewRepository.java>

```

package com.securefromscratch.busybee.linkpreview;

import java.util.UUID;

import org.jooq.DSLContext;
import org.springframework.stereotype.Repository;

import static org.jooq.impl.DSL.field;
import static org.jooq.impl.DSL.name;
import static org.jooq.impl.DSL.table;

@Repository
public class TaskLinkPreviewRepository {
    private static final org.jooq.Table<?> PREVIEWS = table(name("busybee_task_link_preview"));
    private static final org.jooq.Field<String> TASK_ID = field(name("task_id"), String.class);
    private static final org.jooq.Field<String> URL = field(name("url"), String.class);

    private final DSLContext dsl;

    public TaskLinkPreviewRepository(DSLContext dsl) {
        this.dsl = dsl;
    }

    public void delete(UUID taskId, String url) {
        dsl.deleteFrom(PREVIEWS)
            .where(TASK_ID.eq(taskId.toString()))
            .and(URL.eq(url))
            .execute();
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/linkpreview/TaskLinkPreviewRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/linkpreview/TaskLinkPreviewRequest.java>

```

package com.securefromscratch.busybee.linkpreview;

import org.owasp.untrust.wv.SingleLine;
import org.owasp.untrust.wv.ViewableUuidValue;

public record TaskLinkPreviewRequest(ViewableUuidValue taskid, SingleLine url) {
}

```

busybee/src/main/java/com/securefromscratch/busybee/linkpreview/TaskLinkPreviewService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/linkpreview/TaskLinkPreviewService.java>

```

package com.securefromscratch.busybee.linkpreview;

import org.springframework.stereotype.Service;

@Service
public class TaskLinkPreviewService {
    private final TaskLinkPreviewRepository taskLinkPreviewRepository;

    public TaskLinkPreviewService(TaskLinkPreviewRepository taskLinkPreviewRepository) {
        this.taskLinkPreviewRepository = taskLinkPreviewRepository;
    }

    public SuccessResponse delete(TaskLinkPreviewRequest request) {
        taskLinkPreviewRepository.delete(request.taskid().exposeUnchecked(), request.url().exposeUnchecked());
        return new SuccessResponse(true);
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrController.java>

```

package com.securefromscratch.busybee.ocr;

import java.io.IOException;

import org.springframework.security.access.prepost.PreAuthorize;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RequestParam;
import org.springframework.web.bind.annotation.RestController;
import org.springframework.web.multipart.MultipartFile;

@RestController
@PreAuthorize("denyAll()")
public class OcrController {
    private final OcrService ocrService;

    public OcrController(OcrService ocrService) {
        this.ocrService = ocrService;
    }

    @PostMapping("/ocr/image")
    @PreAuthorize("hasAuthority('OCR_ENABLED')")
    public OcrDraftResponse image(@RequestBody OcrImageRequest request) throws IOException, InterruptedException {
        return ocrService.extractFromStoredImage(request);
    }

    @PostMapping(path = "/ocr/extract", consumes = "multipart/form-data")
    @PreAuthorize("hasAuthority('OCR_ENABLED')")
    public OcrDraftResponse extract(
        @RequestParam("ocrFields") OcrExtractFieldsRequest request,
        @RequestParam("file") MultipartFile file
    ) throws IOException, InterruptedException {
        return ocrService.extractFromUpload(request, file);
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrDraftResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp

[_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrDraftResponse.java](https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrDraftResponse.java)>

```
package com.securefromscratch.busybee.ocr;

public record OcrDraftResponse(String title, String description, String rawText) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrExecutionException.java

Source:

[<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrExecutionException.java>](https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrExecutionException.java)

```
package com.securefromscratch.busybee.ocr;

public class OcrExecutionException extends RuntimeException {
    public OcrExecutionException() {
        super("OCR conversion could not be completed.");
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrExtractFieldsRequest.java

Source:

[<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrExtractFieldsRequest.java>](https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrExtractFieldsRequest.java)

```
package com.securefromscratch.busybee.ocr;

public record OcrExtractFieldsRequest(OcrLanguageValue language, OcrTextLayoutValue textLayout) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrImageRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrImageRequest.java>

```
package com.securefromscratch.busybee.ocr;

import org.owasp.untrust.wv.ViewableUuidValue;

public record OcrImageRequest(ViewableUuidValue image, OcrLanguageValue language, OcrTextLayoutValue textLayout) {}
```

busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrLanguage.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrLanguage.java>


```

package com.securefromscratch.busybee.ocr;

import com.fasterxml.jackson.annotation.JsonCreator;

public enum OcrLanguage {
    eng,
    heb,
    eng_heb;

    @JsonCreator
    public static OcrLanguage from(String value) {
        return switch (value) {
            case "eng" -> eng;
            case "heb" -> heb;
            case "eng+heb" -> eng_heb;
            default -> throw new IllegalArgumentException("Unsupported OCR language.");
        };
    }

    public String tesseractValue() {
        return switch (this) {
            case eng -> "eng";
            case heb -> "heb";
            case eng_heb -> "eng+heb";
        };
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrLanguageValue.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrLanguageValue.java>

```
package com.securefromscratch.busybee.ocr;

import com.fasterxml.jackson.annotation.JsonCreator;
import org.owasp.untrust.valuedescriptors.Hardcoded;
import org.owasp.untrust.valuedescriptors.foundation.PubliclyExposed;
import org.owasp.untrust.vv.foundation.ValidatedWrappedValue;
import org.owasp.untrust.vv.traits.RareTraitsCaseWhereParsingIsTheWholeValidation;

import static org.owasp.untrust.valuedescriptors.Hardcoded.hardcoded;

public final class OcrLanguageValue extends ValidatedWrappedValue<OcrLanguage> implements PubliclyExposed<OcrLanguage> {
    @JsonCreator(mode = JsonCreator.Mode.DELEGATING)
    public OcrLanguageValue(String raw) {
        super(raw, new Traits());
    }

    @Override
    public OcrLanguage exposeUnchecked() {
        return exposeUnchecked(EXPOSE_HALF_BAKED_VALUE_INTENDED_FOR_INTERNAL_LIBRARY_USE_ONLY_MARKER);
    }

    public static final class Traits extends RareTraitsCaseWhereParsingIsTheWholeValidation<OcrLanguage> {
        @Override
        public Hardcoded descriptionInErrors() {
            return hardcoded("OCR language");
        }

        @Override
        public OcrLanguage parse(String raw) {
            return OcrLanguage.from(raw);
        }

        @Override
        public Bounds rawBounds() {
            return new Bounds(3, 7);
        }

        @Override
        public OcrLanguage normalize(OcrLanguage parsed) {
            return parsed;
        }
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrService.java>

```

package com.securefromscratch.busybee.ocr;

import java.io.IOException;
import java.nio.charset.StandardCharsets;
import java.nio.file.Files;
import java.util.List;
import java.util.UUID;

import com.securefromscratch.busybee.file.FileStorageService;
import com.securefromscratch.busybee.file.StoredFile;
import org.owasp.untrust.boxedpath.BoxedPath;
import org.owasp.untrust.boxedpath.PathSandbox;
import org.owasp.untrust.saferprocess.ProcessBuilder;
import org.owasp.untrust.valuedescriptors.Hardcoded;
import org.springframework.stereotype.Service;
import org.springframework.web.multipart.MultipartFile;

@Service
public class OcrService {
    private static final Hardcoded TESSERACT_EXECUTABLE = Hardcoded.of(
        "../tesseract_mock_java/build/install/tesseract_mock_java/bin/tesseract_mock_java"
    );
    private static final PathSandbox TEMPORARY_DIRECTORY = PathSandbox.boxroot("/tmp");

    private final FileStorageService fileStorageService;

    public OcrService(FileStorageService fileStorageService) {
        this.fileStorageService = fileStorageService;
    }

    public OcrDraftResponse extractFromStoredImage(OcrImageRequest request) throws IOException, InterruptedException {
        StoredFile image = fileStorageService.requireImage(request.image()).exposeUnchecked();
        String rawText = extract(
            image.contents(),
            request.language().exposeUnchecked(),
            request.textLayout().exposeUnchecked()
        );
        return draftFrom(rawText);
    }

    public OcrDraftResponse extractFromUpload(OcrExtractFieldsRequest request, MultipartFile file)
        throws IOException, InterruptedException {
        String rawText = extract(
            fileStorageService.imageContents(file),

```

```

        request.language().exposeUnchecked(),
        request.textLayout().exposeUnchecked()
    );
    return draftFrom(rawText);
}

private String extract(byte[] imageBytes, OcrLanguage language, OcrTextLayout textLayout)
    throws IOException, InterruptedException {
    BoxedPath temporaryImage = TEMPORARY_DIRECTORY.resolve(UUID.randomUUID().toString());
    try {
        Files.write(temporaryImage, imageBytes);
        Process process = ProcessBuilder.start(
            TESSERACT_EXECUTABLE,
            List.of(
                temporaryImage.toString(),
                "stdout",
                "-l",
                language.tesseractValue(),
                "--psm",
                Integer.toString(textLayout.value())
            )
        );
        byte[] output = process.getInputStream().readAllBytes();
        int exitCode = process.waitFor();
        if (exitCode != 0) {
            throw new OcrExecutionException();
        }
        return new String(output, StandardCharsets.UTF_8).strip();
    } finally {
        Files.deleteIfExists(temporaryImage);
    }
}

private OcrDraftResponse draftFrom(String rawText) {
    List<String> lines = rawText.lines().filter(line -> !line.isBlank()).toList();
    if (lines.size() < 2) {
        throw new OcrExecutionException();
    }
    String title = lines.getFirst();
    String description = rawText.substring(title.length()).strip();
    return new OcrDraftResponse(title, description, rawText);
}
}

```

busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrTextLayout.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrTextLayout.java>

```
package com.securefromscratch.busybee.ocr;

import com.fasterxml.jackson.annotation.JsonCreator;

public enum OcrTextLayout {
    THREE(3),
    FOUR(4),
    SIX(6),
    SEVEN(7),
    EIGHT(8),
    ELEVEN(11),
    TWELVE(12),
    THIRTEEN(13);

    private final int value;

    OcrTextLayout(int value) {
        this.value = value;
    }

    @JsonCreator
    public static OcrTextLayout from(int value) {
        for (OcrTextLayout layout : values()) {
            if (layout.value == value) {
                return layout;
            }
        }
        throw new IllegalArgumentException("Unsupported OCR text layout.");
    }

    public int value() {
        return value;
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrTextLayoutValue.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrTextLayoutValue.java>

```
package com.securefromscratch.busybee.ocr;

import com.fasterxml.jackson.annotation.JsonCreator;
import org.owasp.untrust.valuedescriptors.Hardcoded;
import org.owasp.untrust.valuedescriptors.foundation.PubliclyExposed;
import org.owasp.untrust.vv.foundation.ValidatedWrappedValue;
import org.owasp.untrust.vv.traits.RareTraitsCaseWhereParsingIsTheWholeValidation;

import static org.owasp.untrust.valuedescriptors.Hardcoded.hardcoded;

public final class OcrTextLayoutValue extends ValidatedWrappedValue<OcrTextLayout> implements PubliclyExposed {
    @JsonCreator(mode = JsonCreator.Mode.DELEGATING)
    public OcrTextLayoutValue(int raw) {
        super(Integer.toString(raw), new Traits());
    }

    @Override
    public OcrTextLayout exposeUnchecked() {
        return exposeUnchecked(EXPOSE_HALF_BAKED_VALUE_INTENDED_FOR_INTERNAL_LIBRARY_USE_ONLY_MARKER)
    }

    public static final class Traits extends RareTraitsCaseWhereParsingIsTheWholeValidation<OcrTextLayout> {
        @Override
        public Hardcoded descriptionInErrors() {
            return hardcoded("OCR text layout");
        }

        @Override
        public OcrTextLayout parse(String raw) {
            return OcrTextLayout.from(Integer.parseInt(raw));
        }

        @Override
        public Bounds rawBounds() {
            return new Bounds(1, 2);
        }

        @Override
        public OcrTextLayout normalize(OcrTextLayout parsed) {
            return parsed;
        }
    }
}
```


busybee/src/main/java/com/securefromscratch/busybee/registration/RegistrationController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/registration/RegistrationController.java>

```
package com.securefromscratch.busybee.registration;

import org.springframework.http.HttpStatus;
import org.springframework.security.access.prepost.PreAuthorize;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.ResponseStatus;
import org.springframework.web.bind.annotation.RestController;

@RestController
@PreAuthorize("denyAll()")
public class RegistrationController {
    private final RegistrationService registrationService;

    public RegistrationController(RegistrationService registrationService) {
        this.registrationService = registrationService;
    }

    @PostMapping("/register")
    @PreAuthorize("permitAll()")
    public RegistrationResponse register(@RequestBody RegistrationRequest request) {
        registrationService.register(request);
        return new RegistrationResponse("/main/main.html");
    }

    @ResponseStatus(HttpStatus.BAD_REQUEST)
    @org.springframework.web.bind.annotation.ExceptionHandler(UsernameAlreadyRegisteredException.class)
    @PreAuthorize("permitAll()")
    public ErrorResponse usernameAlreadyRegistered() {
        return new ErrorResponse("Username is already registered.");
    }

    public record ErrorResponse(String error) {
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/registration/RegistrationRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/registration/RegistrationRequest.java>

```
package com.securefromscratch.busybee.registration;

import org.owasp.untrust.vw.SingleLine;

public record RegistrationRequest(SingleLine username, SingleLine password) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/registration/RegistrationResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/registration/RegistrationResponse.java>

```
package com.securefromscratch.busybee.registration;

public record RegistrationResponse(String redirectTo) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/registration/RegistrationService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/registration/RegistrationService.java>

```

package com.securefromscratch.busybee.registration;

import java.time.Clock;

import org.springframework.security.crypto.password.PasswordEncoder;
import org.springframework.stereotype.Service;

@Service
public class RegistrationService {
    private final UserRepository userRepository;
    private final PasswordEncoder passwordEncoder;
    private final Clock clock;

    public RegistrationService(UserRepository userRepository, PasswordEncoder passwordEncoder, Clock clock) {
        this.userRepository = userRepository;
        this.passwordEncoder = passwordEncoder;
        this.clock = clock;
    }

    public void register(RegistrationRequest request) {
        String username = request.username().exposeUnchecked();
        if (userRepository.existsByUsername(username)) {
            throw new UsernameAlreadyRegisteredException();
        }

        userRepository.create(
            username,
            passwordEncoder.encode(request.password().exposeUnchecked()),
            clock.instant()
        );
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/registration/UserRepository.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/registration/UserRepository.java>

```

package com.securefromscratch.busybee.registration;

import java.time.Instant;
import java.util.Optional;
import java.util.List;

import org.jooq.DSLContext;
import org.springframework.stereotype.Repository;

import static org.jooq.impl.DSL.field;
import static org.jooq.impl.DSL.name;
import static org.jooq.impl.DSL.selectOne;
import static org.jooq.impl.DSL.table;

@Repository
public class UserRepository {
    private static final org.jooq.Table<?> USERS = table(name("busybee_user"));
    private static final org.jooq.Field<String> USERNAME = field(name("username"), String.class);
    private static final org.jooq.Field<String> PASSWORD_HASH = field(name("password_hash"), String.class);
    private static final org.jooq.Field<Instant> CREATED_AT = field(name("created_at"), Instant.class);

    private final DSLContext dsl;

    public UserRepository(DSLContext dsl) {
        this.dsl = dsl;
    }

    public boolean existsByUsername(String username) {
        return dsl.fetchExists(selectOne().from(USERS).where(USERNAME.eq(username)));
    }

    public void create(String username, String passwordHash, Instant createdAt) {
        dsl.insertInto(USERS)
            .columns(USERNAME, PASSWORD_HASH, CREATED_AT)
            .values(username, passwordHash, createdAt)
            .execute();
    }

    public Optional<String> findPasswordHash(String username) {
        return dsl.select(PASSWORD_HASH)
            .from(USERS)
            .where(USERNAME.eq(username))
            .fetchOptional(PASSWORD_HASH);
    }
}

```

```

public void replacePasswordHash(String username, String passwordHash) {
    dsl.update(USERS)
        .set(PASSWORD_HASH, passwordHash)
        .where(USERNAME.eq(username))
        .execute();
}

public List<String> findUsernamesContaining(String query) {
    return dsl.select(USERNAME)
        .from(USERS)
        .where(USERNAME.containsIgnoreCase(query))
        .orderBy(USERNAME)
        .fetch(USERNAME);
}
}

```

busybee/src/main/java/com/securefromscratch/busybee/registration/UsernameAlreadyRegisteredException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/registration/UsernameAlreadyRegisteredException.java>

```

package com.securefromscratch.busybee.registration;

public final class UsernameAlreadyRegisteredException extends RuntimeException {
    public UsernameAlreadyRegisteredException() {
        super("Username is already registered.");
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/settings/SummaryThresholdValue.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/settings/SummaryThresholdValue.java>

```

package com.securefromscratch.busybee.settings;

import java.util.Optional;

import com.fasterxml.jackson.annotation.JsonCreator;
import org.owasp.untrust.valuedescriptors.Hardcoded;
import org.owasp.untrust.valuedescriptors.foundation.PubliclyExposed;
import org.owasp.untrust.vw.foundation.ValidatedWrappedValue;
import org.owasp.untrust.vw.traits.BoundedValueTraits;

import static org.owasp.untrust.valuedescriptors.Hardcoded.hardcoded;

public final class SummaryThresholdValue extends ValidatedWrappedValue<Integer> implements PubliclyExposed<Integer> {
    @JsonCreator(mode = JsonCreator.Mode.DELEGATING)
    public SummaryThresholdValue(int raw) {
        super(Integer.toString(raw), new Traits());
    }

    @Override
    public Integer exposeUnchecked() {
        return exposeUnchecked(EXPOSE_HALF_BAKED_VALUE_INTENDED_FOR_INTERNAL_LIBRARY_USE_ONLY_MARKER);
    }

    public static final class Traits extends BoundedValueTraits<Integer> {
        @Override
        public Hardcoded descriptionInErrors() {
            return hardcoded("summary threshold");
        }

        @Override
        public Integer parse(String raw) {
            return Integer.parseInt(raw);
        }

        @Override
        public Integer normalize(Integer parsed) {
            return parsed;
        }

        @Override
        public Bounds rawBounds() {
            return new Bounds(1, 2);
        }
    }
}

```

```
@Override
public Optional<Bounds> valueBounds() {
    return Optional.of(new Bounds(5, 15));
}

@Override
public int valueForBounds(Integer normalized) {
    return normalized;
}
}
```

busybee/src/main/java/com/securefromscratch/busybee/settings/ UserSettingsController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/settings/UserSettingsController.java>

```
package com.securefromscratch.busybee.settings;

import org.springframework.security.access.prepost.PreAuthorize;
import org.springframework.security.core.Authentication;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PutMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RestController;

@RestController
@PreAuthorize("denyAll()")
public class UserSettingsController {
    private final UserSettingsService userSettingsService;

    public UserSettingsController(UserSettingsService userSettingsService) {
        this.userSettingsService = userSettingsService;
    }

    @GetMapping("/user/settings")
    @PreAuthorize("hasRole('USER')")
    public UserSettingsResponse settings(Authentication authentication) {
        return userSettingsService.settingsFor(authentication.getName(), authentication.getAuthorities());
    }

    @PutMapping("/user/settings")
    @PreAuthorize("hasRole('USER')")
    public UserSettingsResponse replace(@RequestBody UserSettingsRequest request, Authentication authentication) {
        return userSettingsService.replace(authentication.getName(), request, authentication.getAuthorities());
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/settings/ UserSettingsRepository.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/settings/UserSettingsRepository.java>


```

package com.securefromscratch.busybee.settings;

import java.util.Optional;

import org.jooq.DSLContext;
import org.springframework.stereotype.Repository;

import static org.jooq.impl.DSL.field;
import static org.jooq.impl.DSL.name;
import static org.jooq.impl.DSL.table;

@Repository
public class UserSettingsRepository {
    private static final org.jooq.Table<?> SETTINGS = table(name("busybee_user_settings"));
    private static final org.jooq.Field<String> USERNAME = field(name("username"), String.class);
    private static final org.jooq.Field<Integer> SUMMARY_THRESHOLD_COMMENTS = field(
        name("summary_threshold_comments"),
        Integer.class
    );

    private final DSLContext dsl;

    public UserSettingsRepository(DSLContext dsl) {
        this.dsl = dsl;
    }

    public Optional<Integer> findSummaryThreshold(String username) {
        return dsl.select(SUMMARY_THRESHOLD_COMMENTS)
            .from(SETTINGS)
            .where(USERNAME.eq(username))
            .fetchOptional(SUMMARY_THRESHOLD_COMMENTS);
    }

    public void saveSummaryThreshold(String username, int threshold) {
        dsl.insertInto(SETTINGS)
            .columns(USERNAME, SUMMARY_THRESHOLD_COMMENTS)
            .values(username, threshold)
            .onDuplicateKeyUpdate()
            .set(SUMMARY_THRESHOLD_COMMENTS, threshold)
            .execute();
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/settings/UserSettingsRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/settings/UserSettingsRequest.java>

```
package com.securefromscratch.busybee.settings;

public record UserSettingsRequest(SummaryThresholdValue summaryThresholdComments) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/settings/UserSettingsResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/settings/UserSettingsResponse.java>

```
package com.securefromscratch.busybee.settings;

import com.securefromscratch.busybee.aicredential.AiCredentialStatusResponse;

public record UserSettingsResponse(int summaryThresholdComments, AiCredentialStatusResponse aiCredential) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/settings/UserSettingsService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/settings/UserSettingsService.java>

```

package com.securefromscratch.busybee.settings;

import java.util.Collection;

import com.securefromscratch.busybee.aicredential.AiCredentialStatusService;
import org.springframework.stereotype.Service;
import org.springframework.security.core.GrantedAuthority;

@Service
public class UserSettingsService {
    private static final int DEFAULT_SUMMARY_THRESHOLD = 5;

    private final UserSettingsRepository userSettingsRepository;
    private final AiCredentialStatusService aiCredentialStatusService;

    public UserSettingsService(
        UserSettingsRepository userSettingsRepository,
        AiCredentialStatusService aiCredentialStatusService
    ){
        this.userSettingsRepository = userSettingsRepository;
        this.aiCredentialStatusService = aiCredentialStatusService;
    }

    public UserSettingsResponse settingsFor(String username, Collection<? extends GrantedAuthority> authorities) {
        int threshold = summaryThresholdFor(username);
        return new UserSettingsResponse(threshold, aiCredentialStatusService.statusFor(username, authorities));
    }

    public int summaryThresholdFor(String username) {
        return userSettingsRepository.findSummaryThreshold(username).orElse(DEFAULT_SUMMARY_THRESHOLD);
    }

    public UserSettingsResponse replace(
        String username,
        UserSettingsRequest request,
        Collection<? extends GrantedAuthority> authorities
    ){
        int threshold = request.summaryThresholdComments().exposeUnchecked();
        userSettingsRepository.saveSummaryThreshold(username, threshold);
        return new UserSettingsResponse(threshold, aiCredentialStatusService.statusFor(username, authorities));
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/task/MarkTaskDoneRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/task/MarkTaskDoneRequest.java>

```
package com.securefromscratch.busybee.task;

import org.owasp.untrust.w.ViewableUuidValue;

public record MarkTaskDoneRequest(ViewableUuidValue taskid) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/task/TaskController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/task/TaskController.java>

```

package com.securefromscratch.busybee.task;

import java.security.Principal;
import java.util.List;

import org.springframework.security.access.prepost.PreAuthorize;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RestController;

@RestController
@PreAuthorize("denyAll()")
public class TaskController {
    private final TaskService taskService;

    public TaskController(TaskService taskService) {
        this.taskService = taskService;
    }

    @PostMapping("/create")
    @PreAuthorize("hasRole('USER')")
    public TaskCreateResponse create(@RequestBody TaskCreateRequest request, Principal principal) {
        return taskService.create(request, principal.getName());
    }

    @PostMapping("/done")
    @PreAuthorize("hasRole('USER')")
    public TaskDoneResponse markDone(@RequestBody MarkTaskDoneRequest request) {
        return taskService.markDone(request);
    }

    @GetMapping("/tasks")
    @PreAuthorize("hasRole('USER')")
    public List<TaskResponse> list() {
        return taskService.list();
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/task/TaskCreateRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/task/TaskCreateRequest.java>

```
package com.securefromscratch.busybee.task;

import java.util.List;
import java.util.Optional;

import org.owasp.untrust.vw.SingleLine;

public record TaskCreateRequest(
    SingleLine name,
    Optional<SingleLine> desc,
    Optional<SingleLine> dueDate,
    Optional<SingleLine> dueTime,
    Optional<List<SingleLine>> responsibilityOf
){
}
```

busybee/src/main/java/com/securefromscratch/busybee/task/TaskCreateResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/task/TaskCreateResponse.java>

```
package com.securefromscratch.busybee.task;

public record TaskCreateResponse(String taskid) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/task/TaskDoneResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/task/TaskDoneResponse.java>

```
package com.securefromscratch.busybee.task;

public record TaskDoneResponse(boolean success) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/task/TaskRepository.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/task/TaskRepository.java>

```
package com.securefromscratch.busybee.task;
```

```
import java.time.Instant;
```

```
import java.util.List;
```

```
import java.util.UUID;
```

```
import org.jooq.DSLContext;
```

```
import org.springframework.stereotype.Repository;
```

```
import static org.jooq.impl.DSL.field;
```

```
import static org.jooq.impl.DSL.name;
```

```
import static org.jooq.impl.DSL.table;
```

```
@Repository
```

```
public class TaskRepository {
```

```
    private static final org.jooq.Table<?> TASKS = table(name("busybee_task"));
```

```
    private static final org.jooq.Table<?> RESPONSIBILITIES = table(name("busybee_task_responsibility"));
```

```
    private static final org.jooq.Field<String> TASK_ID = field(name("task_id"), String.class);
```

```
    private static final org.jooq.Field<String> NAME = field(name("name"), String.class);
```

```
    private static final org.jooq.Field<String> DESCRIPTION = field(name("description"), String.class);
```

```
    private static final org.jooq.Field<String> DUE_DATE = field(name("due_date"), String.class);
```

```
    private static final org.jooq.Field<String> DUE_TIME = field(name("due_time"), String.class);
```

```
    private static final org.jooq.Field<String> CREATED_BY = field(name("created_by"), String.class);
```

```
    private static final org.jooq.Field<Instant> CREATION_DATETIME = field(name("creation_datetime"), Instant.class);
```

```
    private static final org.jooq.Field<Boolean> DONE = field(name("done"), Boolean.class);
```

```
    private static final org.jooq.Field<String> RESPONSIBILITY_TASK_ID = field(name("task_id"), String.class);
```

```
    private static final org.jooq.Field<String> RESPONSIBILITY_USERNAME = field(name("username"), String.class);
```

```
    private final DSLContext dsl;
```

```
    public TaskRepository(DSLContext dsl) {
```

```
        this.dsl = dsl;
```

```
    }
```

```
    public void create(TaskRow task) {
```

```
        dsl.insertInto(TASKS)
```

```
            .columns(TASK_ID, NAME, DESCRIPTION, DUE_DATE, DUE_TIME, CREATED_BY, CREATION_DATETIME, DONE)
```

```
            .values(
```

```
                task.taskId().toString(),
```

```
                task.name(),
```

```
                task.description(),
```

```
                task.dueDate(),
```

```
                task.dueTime(),
```

```
                task.createdBy(),
```



```

        task.creationDatetime(),
        false
    )
    .execute();
task.responsibilityOf().forEach(username -> dsl.insertInto(RESPONSIBILITIES)
    .columns(RESPONSIBILITY_TASK_ID, RESPONSIBILITY_USERNAME)
    .values(task.taskId().toString(), username)
    .execute());
}

public void markDone(UUID taskId) {
    dsl.update(TASKS).set(DONE, true).where(TASK_ID.eq(taskId.toString())).execute();
}

public List<TaskRow> findAll() {
    return dsl.select(TASK_ID, NAME, DESCRIPTION, DUE_DATE, DUE_TIME, CREATED_BY, CREATION_DATETIME, DONE)
        .from(TASKS)
        .orderBy(CREATION_DATETIME)
        .fetch(record -> new TaskRow(
            UUID.fromString(record.get(TASK_ID)),
            record.get(NAME),
            record.get(DESCRIPTION),
            record.get(DUE_DATE),
            record.get(DUE_TIME),
            record.get(CREATED_BY),
            record.get(CREATION_DATETIME),
            record.get(DONE),
            findResponsibilities(UUID.fromString(record.get(TASK_ID)))
        ));
}

private List<String> findResponsibilities(UUID taskId) {
    return dsl.select(RESPONSIBILITY_USERNAME)
        .from(RESPONSIBILITIES)
        .where(RESPONSIBILITY_TASK_ID.eq(taskId.toString()))
        .orderBy(RESPONSIBILITY_USERNAME)
        .fetch(RESPONSIBILITY_USERNAME);
}

public record TaskRow(
    UUID taskId,
    String name,
    String description,
    String dueDate,

```

```

    String dueTime,
    String createdBy,
    Instant creationDatetime,
    boolean done,
    List<String> responsibilityOf
  ){
  }
}

```

busybee/src/main/java/com/securefromscratch/busybee/task/TaskResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/task/TaskResponse.java>

```

package com.securefromscratch.busybee.task;

import java.util.List;
import java.util.Optional;

import com.securefromscratch.busybee.comment.CommentResponse;
import com.securefromscratch.busybee.commentssummary.CommentSummaryResponse;

public record TaskResponse(
    String taskid,
    String name,
    String desc,
    String dueDate,
    String dueTime,
    String createdBy,
    List<String> responsibilityOf,
    String creationDatetime,
    boolean done,
    Optional<CommentSummaryResponse> commentSummary,
    List<Object> linkPreviews,
    List<CommentResponse> comments
){
}

```

busybee/src/main/java/com/securefromscratch/busybee/task/TaskService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/task/TaskService.java>

```
package com.securefromscratch.busybee.task;
```

```
import java.time.Clock;
```

```
import java.util.List;
```

```
import java.util.UUID;
```

```
import com.securefromscratch.busybee.comment.CommentService;
```

```
import com.securefromscratch.busybee.commentsummary.CommentSummaryService;
```

```
import com.securefromscratch.busybee.tasktransfer.ImportedTask;
```

```
import org.springframework.stereotype.Service;
```

```
@Service
```

```
public class TaskService {
```

```
    private final TaskRepository taskRepository;
```

```
    private final CommentService commentService;
```

```
    private final CommentSummaryService commentSummaryService;
```

```
    private final Clock clock;
```

```
    public TaskService(
```

```
        TaskRepository taskRepository,
```

```
        CommentService commentService,
```

```
        CommentSummaryService commentSummaryService,
```

```
        Clock clock
```

```
    ) {
```

```
        this.taskRepository = taskRepository;
```

```
        this.commentService = commentService;
```

```
        this.commentSummaryService = commentSummaryService;
```

```
        this.clock = clock;
```

```
    }
```

```
    public TaskCreateResponse create(TaskCreateRequest request, String createdBy) {
```

```
        UUID taskId = UUID.randomUUID();
```

```
        taskRepository.create(new TaskRepository.TaskRow(
```

```
            taskId,
```

```
            request.name().exposeUnchecked(),
```

```
            request.desc().map(value -> value.exposeUnchecked()).orElse(""),
```

```
            request.dueDate().map(value -> value.exposeUnchecked()).orElse(""),
```

```
            request.dueTime().map(value -> value.exposeUnchecked()).orElse(""),
```

```
            createdBy,
```

```
            clock.instant(),
```

```
            false,
```

```
            request.responsibilityOf().orElse(List.of()).stream().map(value -> value.exposeUnchecked()).toList()
```

```
        ));
```

```
        return new TaskCreateResponse(taskId.toString());
```

```

}

public TaskDoneResponse markDone(MarkTaskDoneRequest request) {
    taskRepository.markDone(request.taskid().exposeUnchecked());
    return new TaskDoneResponse(true);
}

public void importTasks(List<ImportedTask> tasks, String createdBy) {
    tasks.forEach(task -> taskRepository.create(new TaskRepository.TaskRow(
        UUID.randomUUID(),
        task.name().exposeUnchecked(),
        task.desc().map(value -> value.exposeUnchecked()).orElse(""),
        task.dueDate().map(value -> value.exposeUnchecked()).orElse(""),
        task.dueTime().map(value -> value.exposeUnchecked()).orElse(""),
        createdBy,
        clock.instant(),
        false,
        task.responsibilityOf().orElse(List.of()).stream().map(value -> value.exposeUnchecked()).toList()
    )));
}

public List<TaskResponse> list() {
    return taskRepository.findAll().stream().map(task -> new TaskResponse(
        task.taskId().toString(),
        task.name(),
        task.description(),
        task.dueDate(),
        task.dueTime(),
        task.createdBy(),
        task.responsibilityOf(),
        task.creationDatetime().toString(),
        task.done(),
        commentSummaryService.summaryForTask(task.taskId()),
        List.of(),
        commentService.findByTaskId(task.taskId())
    )).toList();
}
}

```

busybee/src/main/java/com/securefromscratch/busybee/tasktransfer/ExportedTask.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/tasktransfer/ExportedTask.java>

```
package com.securefromscratch.busybee.tasktransfer;

import java.util.List;

public record ExportedTask(
    String name,
    String desc,
    String dueDate,
    String dueTime,
    List<String> responsibilityOf
){
}
```

busybee/src/main/java/com/securefromscratch/busybee/tasktransfer/ImportedTask.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/tasktransfer/ImportedTask.java>

```
package com.securefromscratch.busybee.tasktransfer;

import java.util.List;
import java.util.Optional;

import org.owasp.untrust.vw.SingleLine;

public record ImportedTask(
    SingleLine name,
    Optional<SingleLine> desc,
    Optional<SingleLine> dueDate,
    Optional<SingleLine> dueTime,
    Optional<List<SingleLine>> responsibilityOf
){
}
```

busybee/src/main/java/com/securefromscratch/busybee/tasktransfer/TaskTransferController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/tasktransfer/TaskTransferController.java>

```
package com.securefromscratch.busybee.tasktransfer;

import java.io.IOException;
import java.security.Principal;

import org.springframework.http.ContentDisposition;
import org.springframework.http.HttpHeaders;
import org.springframework.http.MediaType;
import org.springframework.http.ResponseEntity;
import org.springframework.security.access.prepost.PreAuthorize;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;
import org.springframework.web.multipart.MultipartFile;

@RestController
@PreAuthorize("denyAll()")
public class TaskTransferController {
    private final TaskTransferService taskTransferService;

    public TaskTransferController(TaskTransferService taskTransferService) {
        this.taskTransferService = taskTransferService;
    }

    @GetMapping("/extra/export")
    @PreAuthorize("hasRole('USER') and hasAuthority('EXPORT_ENABLED')")
    public ResponseEntity<byte[]> exportTasks() throws IOException {
        return ResponseEntity.ok()
            .contentType(MediaType.APPLICATION_OCTET_STREAM)
            .header(HttpHeaders.CONTENT_DISPOSITION, ContentDisposition.attachment().filename("tasks.ser").build().toString())
            .body(taskTransferService.exportTasks());
    }

    @PostMapping(path = "/extra/import", consumes = MediaType.MULTIPART_FORM_DATA_VALUE)
    @PreAuthorize("hasRole('USER') and hasAuthority('IMPORT_ENABLED')")
    public void importTasks(@RequestPart("file") MultipartFile file, Principal principal) throws IOException {
        taskTransferService.importTasks(file, principal.getName());
    }
}
```


busybee/src/main/java/com/securefromscratch/busybee/tasktransfer/TaskTransferService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/tasktransfer/TaskTransferService.java>

```
package com.securefromscratch.busybee.tasktransfer;

import java.io.IOException;
import java.util.List;

import com.fasterxml.jackson.core.type.TypeReference;
import com.fasterxml.jackson.databind.ObjectMapper;
import com.securefromscratch.busybee.task.TaskRepository;
import com.securefromscratch.busybee.task.TaskService;
import org.springframework.stereotype.Service;
import org.springframework.web.multipart.MultipartFile;

@Service
public class TaskTransferService {
    private final TaskRepository taskRepository;
    private final TaskService taskService;
    private final ObjectMapper objectMapper;

    public TaskTransferService(TaskRepository taskRepository, TaskService taskService, ObjectMapper objectMapper) {
        this.taskRepository = taskRepository;
        this.taskService = taskService;
        this.objectMapper = objectMapper;
    }

    public byte[] exportTasks() throws IOException {
        List<ExportedTask> tasks = taskRepository.findAll().stream().map(task -> new ExportedTask(
            task.name(),
            task.description(),
            task.dueDate(),
            task.dueTime(),
            task.responsibilityOf()
        )).toList();
        return objectMapper.writeValueAsBytes(tasks);
    }

    public void importTasks(MultipartFile file, String createdBy) throws IOException {
        List<ImportedTask> tasks = objectMapper.readValue(file.getBytes(), new TypeReference<>() {
        });
        taskService.importTasks(tasks, createdBy);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/userlookup/UserLookupController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/userlookup/UserLookupController.java>

```
package com.securefromscratch.busybee.userlookup;

import java.util.List;

import org.owasp.untrust.vw.SingleLine;
import org.springframework.security.access.prepost.PreAuthorize;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RequestParam;
import org.springframework.web.bind.annotation.RestController;

@RestController
@PreAuthorize("denyAll()")
public class UserLookupController {
    private final UserLookupService userLookupService;

    public UserLookupController(UserLookupService userLookupService) {
        this.userLookupService = userLookupService;
    }

    @GetMapping("/users/lookup")
    @PreAuthorize("hasRole('USER')")
    public List<UserLookupResponse> lookup(@RequestParam("query") SingleLine query) {
        return userLookupService.lookup(query);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/userlookup/UserLookupRepository.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/userlookup/UserLookupRepository.java>

```

package com.securefromscratch.busybee.userlookup;

import java.util.List;

import org.jooq.DSLContext;
import org.springframework.stereotype.Repository;

import static org.jooq.impl.DSL.field;
import static org.jooq.impl.DSL.name;
import static org.jooq.impl.DSL.table;

@Repository
public class UserLookupRepository {
    private static final org.jooq.Table<?> USERS = table(name("busybee_user"));
    private static final org.jooq.Field<String> USERNAME = field(name("username"), String.class);

    private final DSLContext dsl;

    public UserLookupRepository(DSLContext dsl) {
        this.dsl = dsl;
    }

    public List<String> findUsernamesContaining(String query) {
        return dsl.select(USERNAME)
            .from(USERS)
            .where(USERNAME.containsIgnoreCase(query))
            .orderBy(USERNAME)
            .fetch(USERNAME);
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/userlookup/UserLookupResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/userlookup/UserLookupResponse.java>

```
package com.securefromscratch.busybee.userlookup;

public record UserLookupResponse(String username) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/userlookup/UserLookupService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/userlookup/UserLookupService.java>

```
package com.securefromscratch.busybee.userlookup;

import java.util.List;

import org.owasp.untrust.vw.SingleLine;
import org.springframework.stereotype.Service;

@Service
public class UserLookupService {
    private final UserLookupRepository userLookupRepository;

    public UserLookupService(UserLookupRepository userLookupRepository) {
        this.userLookupRepository = userLookupRepository;
    }

    public List<UserLookupResponse> lookup(SingleLine query) {
        return userLookupRepository.findUsernamesContaining(query.exposeUnchecked()).stream()
            .map(UserLookupResponse::new)
            .toList();
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/userpassword/IncorrectCurrentPasswordException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/userpassword/IncorrectCurrentPasswordException.java>

```
package com.securefromscratch.busybee.userpassword;

public class IncorrectCurrentPasswordException extends RuntimeException {
    public IncorrectCurrentPasswordException() {
        super("Current password is incorrect.");
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/userpassword/PasswordChangeController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/userpassword/PasswordChangeController.java>

```

package com.securefromscratch.busybee.userpassword;

import java.security.Principal;

import org.springframework.http.HttpStatus;
import org.springframework.security.access.prepost.PreAuthorize;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.PutMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.ResponseStatus;
import org.springframework.web.bind.annotation.RestController;

@RestController
@PreAuthorize("denyAll()")
public class PasswordChangeController {
    private final PasswordChangeService passwordChangeService;

    public PasswordChangeController(PasswordChangeService passwordChangeService) {
        this.passwordChangeService = passwordChangeService;
    }

    @PutMapping("/user/password")
    @PreAuthorize("hasRole('USER')")
    public PasswordChangeResponse changePassword(@RequestBody PasswordChangeRequest request, Principal prin
        passwordChangeService.changePassword(principal.getName(), request);
        return new PasswordChangeResponse(true);
    }

    @ExceptionHandler(IncorrectCurrentPasswordException.class)
    @ResponseStatus(HttpStatus.BAD_REQUEST)
    @PreAuthorize("hasRole('USER')")
    public PasswordChangeErrorResponse incorrectCurrentPassword() {
        return new PasswordChangeErrorResponse("Current password is incorrect.");
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/userpassword/PasswordChangeErrorResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp

[_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/userpassword/PasswordChangeErrorResponse.java](https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/userpassword/PasswordChangeErrorResponse.java)>

```
package com.securefromscratch.busybee.userpassword;

public record PasswordChangeErrorResponse(String error) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/userpassword/PasswordChangeRequest.java

Source:

[<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/userpassword/PasswordChangeRequest.java>](https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/userpassword/PasswordChangeRequest.java)

```
package com.securefromscratch.busybee.userpassword;

import org.owasp.untrust.vw.SingleLine;

public record PasswordChangeRequest(SingleLine currentPassword, SingleLine newPassword) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/userpassword/PasswordChangeResponse.java

Source:

[<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/userpassword/PasswordChangeResponse.java>](https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/userpassword/PasswordChangeResponse.java)

```
package com.securefromscratch.busybee.userpassword;

public record PasswordChangeResponse(boolean success) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/userpassword/PasswordChangeService.java

Source:

[<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/userpassword/PasswordChangeService.java>](https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/userpassword/PasswordChangeService.java)

_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/userpassword/PasswordChangeService.java>

```
package com.securefromscratch.busybee.userpassword;

import com.securefromscratch.busybee.registration.UserRepository;
import org.springframework.security.crypto.password.PasswordEncoder;
import org.springframework.stereotype.Service;

@Service
public class PasswordChangeService {
    private final UserRepository userRepository;
    private final PasswordEncoder passwordEncoder;

    public PasswordChangeService(UserRepository userRepository, PasswordEncoder passwordEncoder) {
        this.userRepository = userRepository;
        this.passwordEncoder = passwordEncoder;
    }

    public void changePassword(String username, PasswordChangeRequest request) {
        String currentPasswordHash = userRepository.findPasswordHash(username)
            .orElseThrow(IncorrectCurrentPasswordException::new);
        if (!passwordEncoder.matches(request.currentPassword().exposeUnchecked(), currentPasswordHash)) {
            throw new IncorrectCurrentPasswordException();
        }
        userRepository.replacePasswordHash(username, passwordEncoder.encode(request.newPassword().exposeUnchecked()));
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/userprofile/AuthenticationEndpointsResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/userprofile/AuthenticationEndpointsResponse.java>

```
package com.securefromscratch.busybee.userprofile;

public record AuthenticationEndpointsResponse(String loginUrl, String logoutUrl, String changePasswordUrl) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/userprofile/UserProfileController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/userprofile/UserProfileController.java>

```
package com.securefromscratch.busybee.userprofile;

import org.springframework.security.access.prepost.PreAuthorize;
import org.springframework.security.core.Authentication;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController
@PreAuthorize("denyAll()")
public class UserProfileController {
    private final UserProfileService userProfileService;

    public UserProfileController(UserProfileService userProfileService) {
        this.userProfileService = userProfileService;
    }

    @GetMapping("/user/profile")
    @PreAuthorize("hasRole('USER')")
    public UserProfileResponse profile(Authentication authentication) {
        return userProfileService.profileFor(authentication);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/userprofile/UserProfileResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/userprofile/UserProfileResponse.java>

```
package com.securefromscratch.busybee.userprofile;

import java.util.List;

import com.securefromscratch.busybee.settings.UserSettingsResponse;

public record UserProfileResponse(
    String username,
    boolean admin,
    List<String> entitlements,
    List<String> effectiveEntitlements,
    UserSettingsResponse settings,
    AuthenticationEndpointsResponse authentication
){
}
```

busybee/src/main/java/com/securefromscratch/busybee/userprofile/UserProfileService.java

Source:

https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/java/com/securefromscratch/busybee/userprofile/UserProfileService.java

```

package com.securefromscratch.busybee.userprofile;

import com.securefromscratch.busybee.currentuser.CurrentUserResponse;
import com.securefromscratch.busybee.currentuser.CurrentUserService;
import com.securefromscratch.busybee.settings.UserSettingsResponse;
import com.securefromscratch.busybee.settings.UserSettingsService;
import org.springframework.security.core.Authentication;
import org.springframework.stereotype.Service;

@Service
public class UserProfileService {
    private static final AuthenticationEndpointsResponse AUTHENTICATION_ENDPOINTS = new AuthenticationEndpoin
        "/login",
        "/logout",
        "/user/password"
    );

    private final CurrentUserService currentUserService;
    private final UserSettingsService userSettingsService;

    public UserProfileService(CurrentUserService currentUserService, UserSettingsService userSettingsService) {
        this.currentUserService = currentUserService;
        this.userSettingsService = userSettingsService;
    }

    public UserProfileResponse profileFor(Authentication authentication) {
        CurrentUserResponse currentUser = currentUserService.currentUser(
            authentication.getName(),
            authentication.getAuthorities()
        );
        UserSettingsResponse settings = userSettingsService.settingsFor(
            authentication.getName(),
            authentication.getAuthorities()
        );
        return new UserProfileResponse(
            currentUser.username(),
            currentUser.admin(),
            currentUser.entitlements(),
            currentUser.effectiveEntitlements(),
            settings,
            AUTHENTICATION_ENDPOINTS
        );
    }
}

```

```
}  
}
```

busybee/src/main/resources/db/migration/mysql/V15__create_busybee_user.sql

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/resources/db/migration/mysql/V15__create_busybee_user.sql>

```
CREATE TABLE busybee_user (  
  id BIGINT NOT NULL AUTO_INCREMENT,  
  username VARCHAR(80) NOT NULL,  
  password_hash VARCHAR(100) NOT NULL,  
  created_at TIMESTAMP(6) NOT NULL,  
  PRIMARY KEY (id),  
  CONSTRAINT uk_busybee_user_username UNIQUE (username)  
);
```

busybee/src/main/resources/db/migration/mysql/V16__create_busybee_task.sql

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/resources/db/migration/mysql/V16__create_busybee_task.sql>

```
CREATE TABLE busybee_task (  
  task_id CHAR(36) NOT NULL,  
  name VARCHAR(10000) NOT NULL,  
  description TEXT NOT NULL,  
  due_date VARCHAR(10) NOT NULL,  
  due_time VARCHAR(8) NOT NULL,  
  created_by VARCHAR(80) NOT NULL,  
  creation_datetime TIMESTAMP(6) NOT NULL,  
  done BOOLEAN NOT NULL,  
  PRIMARY KEY (task_id)  
);  
  
CREATE TABLE busybee_task_responsibility (  
  task_id CHAR(36) NOT NULL,  
  username VARCHAR(80) NOT NULL,  
  PRIMARY KEY (task_id, username),  
  CONSTRAINT fk_busybee_task_responsibility_task  
    FOREIGN KEY (task_id) REFERENCES busybee_task(task_id)  
);
```

busybee/src/main/resources/db/migration/mysql/V17__create_busybee_comment.sql

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/resources/db/migration/mysql/V17__create_busybee_comment.sql>

```
CREATE TABLE busybee_comment (  
  comment_id CHAR(36) NOT NULL,  
  task_id CHAR(36) NOT NULL,  
  parent_comment_id CHAR(36),  
  text TEXT NOT NULL,  
  image VARCHAR(255),  
  image_filename VARCHAR(255),  
  attachment VARCHAR(255),  
  attachment_filename VARCHAR(255),  
  indent INT NOT NULL,  
  created_by VARCHAR(80) NOT NULL,  
  created_on TIMESTAMP(6) NOT NULL,  
  PRIMARY KEY (comment_id),  
  CONSTRAINT fk_busybee_comment_task FOREIGN KEY (task_id) REFERENCES busybee_task(task_id)  
);
```

busybee/src/main/resources/db/migration/mysql/V18__create_busybee_file.sql

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/resources/db/migration/mysql/V18__create_busybee_file.sql>

```
CREATE TABLE busybee_file (  
  file_id CHAR(36) NOT NULL,  
  content_type VARCHAR(100) NOT NULL,  
  contents LONGBLOB NOT NULL,  
  image BOOLEAN NOT NULL,  
  PRIMARY KEY (file_id)  
);
```

busybee/src/main/resources/db/migration/mysql/V19__create_busybee_user_settings.sql

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/resources/db/migration/mysql/V19__create_busybee_user_settings.sql>

```
CREATE TABLE busybee_user_settings (  
  username VARCHAR(80) NOT NULL,  
  summary_threshold_comments INT NOT NULL,  
  PRIMARY KEY (username),  
  CONSTRAINT fk_busybee_user_settings_user  
    FOREIGN KEY (username) REFERENCES busybee_user(username)  
);
```

busybee/src/main/resources/db/migration/mysql/V20__create_busybee_user_entitlement.sql

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/resources/db/migration/mysql/V20__create_busybee_user_entitlement.sql>

```
CREATE TABLE busybee_user_entitlement (  
  username VARCHAR(80) NOT NULL,  
  entitlement VARCHAR(40) NOT NULL,  
  PRIMARY KEY (username, entitlement),  
  CONSTRAINT fk_busybee_user_entitlement_user  
    FOREIGN KEY (username) REFERENCES busybee_user(username)  
);
```

busybee/src/main/resources/db/migration/mysql/V21__create_busybee_ai_credential.sql

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/resources/db/migration/mysql/V21__create_busybee_ai_credential.sql>

```
CREATE TABLE busybee_ai_credential (  
  username VARCHAR(80) NOT NULL,  
  provider_type VARCHAR(40) NOT NULL,  
  secret_reference VARCHAR(80) NOT NULL,  
  credential_suffix VARCHAR(4) NOT NULL,  
  PRIMARY KEY (username),  
  CONSTRAINT fk_busybee_ai_credential_user  
    FOREIGN KEY (username) REFERENCES busybee_user(username)  
);
```


busybee/src/main/resources/db/migration/mysql/V22__create_busybee_task_link_preview.sql

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/resources/db/migration/mysql/V22__create_busybee_task_link_preview.sql>

```
CREATE TABLE busybee_task_link_preview (  
  preview_id CHAR(36) NOT NULL,  
  task_id CHAR(36) NOT NULL,  
  url VARCHAR(2048) NOT NULL,  
  title VARCHAR(512) NOT NULL,  
  description TEXT NOT NULL,  
  image_url VARCHAR(2048),  
  created_by VARCHAR(80) NOT NULL,  
  created_at TIMESTAMP(6) NOT NULL,  
  PRIMARY KEY (preview_id),  
  INDEX idx_busybee_task_link_preview_task (task_id),  
  CONSTRAINT fk_busybee_task_link_preview_task  
    FOREIGN KEY (task_id) REFERENCES busybee_task(task_id)  
);
```

busybee/src/main/resources/db/migration/mysql/V23__create_busybee_task_comment_summary.sql

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/main/resources/db/migration/mysql/V23__create_busybee_task_comment_summary.sql>

```
CREATE TABLE busybee_task_comment_summary (  
  task_id CHAR(36) NOT NULL,  
  summary TEXT NOT NULL,  
  summarized_comment_count INT NOT NULL,  
  summarized_latest_comment_at TIMESTAMP(6) NOT NULL,  
  generated_by VARCHAR(80) NOT NULL,  
  credential_source VARCHAR(32) NOT NULL,  
  generated_at TIMESTAMP(6) NOT NULL,  
  PRIMARY KEY (task_id),  
  CONSTRAINT fk_busybee_task_comment_summary_task  
    FOREIGN KEY (task_id) REFERENCES busybee_task(task_id)  
);
```

busybee/src/test/java/com/securefromscratch/busybee/functionality/BusyBeeFunctionalityTests.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/test/java/com/securefromscratch/busybee/functionality/BusyBeeFunctionalityTests.java>

```
package com.securefromscratch.busybee.functionality;

import java.io.IOException;
import java.net.InetSocketAddress;
import java.net.URI;
import java.net.Socket;
import java.nio.charset.StandardCharsets;
import java.util.List;
import java.util.Map;
import java.util.UUID;

import org.junit.jupiter.api.Assertions;
import org.junit.jupiter.api.BeforeAll;
import org.junit.jupiter.api.Test;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.context.SpringBootTest;
import org.springframework.boot.webmvc.test.autoconfigure.AutoConfigureMockMvc;
import org.springframework.http.MediaType;
import org.springframework.mock.web.MockMultipartFile;
import org.springframework.security.core.authority.SimpleGrantedAuthority;
import org.springframework.test.context.ActiveProfiles;
import org.springframework.test.web.servlet.MockMvc;
import org.springframework.test.web.servlet.MvcResult;

import jakarta.servlet.http.Cookie;

import com.fasterxml.jackson.databind.JsonNode;
import com.fasterxml.jackson.databind.ObjectMapper;
import com.fasterxml.jackson.databind.node.ObjectNode;

import static org.springframework.security.test.web.servlet.request.SecurityMockMvcRequestPostProcessors.csrf;
import static org.springframework.security.test.web.servlet.request.SecurityMockMvcRequestPostProcessors.user;
import static org.springframework.security.test.web.servlet.request.SecurityMockMvcRequestBuilders.formLogin;
import static org.springframework.security.test.web.servlet.response.SecurityMockMvcResultMatchers.authenticated;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.get;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.delete;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.multipart;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.post;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.put;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.content;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.jsonPath;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.status;

@SpringBootTest
```

```

@AutoConfigureMockMvc
@ActiveProfiles({"mysql", "test"})
class BusyBeeFunctionalityTests {

    private static final String FALLBACK_OPENAI_API_KEY = "dummy hardcoded";

    private static final String MYSQL_DOWN_MESSAGE = ""
        *****

        MySQL is down or unreachable at the configured test datasource endpoint.
        *****

        """,
        ,

    private static final String VAULT_DOWN_MESSAGE = ""
        *****

        Vault is down or unreachable at the configured secret-store endpoint.
        *****

        """,
        ,

    @Autowired
    private MockMvc mvc;

    @Autowired
    private ObjectMapper objectMapper;

    @BeforeAll
    static void requiredServicesAreRunning() {
        Assertions.assertDoesNotThrow(BusyBeeFunctionalityTests::connectToMysql, MYSQL_DOWN_MESSAGE);
        Assertions.assertDoesNotThrow(BusyBeeFunctionalityTests::connectToVault, VAULT_DOWN_MESSAGE);
    }

    private static void connectToMysql() throws IOException {
        Endpoint endpoint = endpointFromJdbcUrl(configuredValue(
            "SPRING_DATASOURCE_URL",
            "spring.datasource.url",
            "jdbc:mysql://192.168.48.1:3307/busybee_6_test?createDatabaseIfNotExist=true&allowPublicKeyRetrieval=true"
        ));
        connectTo(endpoint);
    }

    private static void connectToVault() throws IOException {
        Endpoint endpoint = endpointFromUri(configuredValue(
            "BUSYBEE_SECRETS_VAULT_URI",
            "busybee.secrets.",
            "http://192.168.48.1:8200"
        ));
        connectTo(endpoint);
    }
}

```

```

    ));
    connectTo(endpoint);
}

private static void connectTo(Endpoint endpoint) throws IOException {
    try (Socket socket = new Socket()) {
        socket.connect(new InetSocketAddress(endpoint.host(), endpoint.port()), 1000);
    }
}

private static Endpoint endpointFromJdbcUrl(String jdbcUrl) {
    return endpointFromUri(jdbcUrl.substring("jdbc:".length()));
}

private static Endpoint endpointFromUri(String uriText) {
    URI uri = URI.create(uriText);
    int port = uri.getPort() > 0 ? uri.getPort() : defaultPort(uri.getScheme());
    return new Endpoint(uri.getHost(), port);
}

private static int defaultPort(String scheme) {
    if ("mysql".equalsIgnoreCase(scheme)) {
        return 3306;
    }
    return "https".equalsIgnoreCase(scheme) ? 443 : 80;
}

private static String configuredValue(String environmentName, String propertyName, String defaultValue) {
    String environmentValue = System.getenv(environmentName);
    // ALLOW NULL LITERAL: getenv returns null when an optional test endpoint override is not configured.
    // this is an old style os function so null is returned when it doesn't exist
    if (environmentValue != null && !environmentValue.isBlank()) {
        return environmentValue;
    }

    String systemValue = System.getProperty(propertyName);
    // ALLOW NULL LITERAL: getProperty returns null when Gradle/JVM did not provide an override for this test endpoint.
    // this is an old style os function so null is returned when it doesn't exist
    if (systemValue != null && !systemValue.isBlank()) {
        return systemValue;
    }

    return defaultValue;
}

```

```

private record Endpoint(String host, int port) {
}

@Test
void contextLoads() {
}

@Test
void registerRejectsDuplicateUser() throws Exception {
    String username = uniqueUsername();

    register(username)
        .andExpect(status().isOk())
        .andExpect(jsonPath("$.redirectTo").value("/main/main.html"));

    register(username)
        .andExpect(status().isBadRequest())
        .andExpect(jsonPath("$.error").value("Username is already registered."));
}

@Test
void adminCanSearchUsersAndSetEntitlements() throws Exception {
    String username = createUser();

    mvc.perform(get("/admin/users")
        .queryParam("query", username)
        .with(user("admin").roles("ADMIN")))
        .andExpect(status().isOk())
        .andExpect(jsonPath("$[0].username").value(username));

    mvc.perform(post("/admin/user-entitlements")
        .with(csrf())
        .with(user("admin").roles("ADMIN"))
        .contentType(MediaType.APPLICATION_JSON)
        .content(json(Map.of(
            "username", username,
            "entitlements", List.of("IMPORT_ENABLED", "OCR_ENABLED")
        ))))
        .andExpect(status().isOk())
        .andExpect(jsonPath("$.username").value(username))
        .andExpect(jsonPath("$.entitlements[0]").value("IMPORT_ENABLED"))
        .andExpect(jsonPath("$.entitlements[1]").value("OCR_ENABLED"));
}

```

```

mvc.perform(get("/me")
    .with(user(username).authorities(new SimpleGrantedAuthority("ROLE_USER"))))
    .andExpect(status().isOk())
    .andExpect(jsonPath("$.effectiveEntitlements[0]").value("IMPORT_ENABLED"))
    .andExpect(jsonPath("$.effectiveEntitlements[1]").value("OCR_ENABLED"));
}

```

@Test

void userCanCreateCommentAndCompleteTask() throws Exception {

String username = createUser();

String taskId = createTask(username);

addComment(username, taskId, "hello from integration")

.andExpect(status().isOk())

.andExpect(jsonPath("\$.commentid").exists());

mvc.perform(post("/done")

.with(csrf())

.with(user(username).roles("USER"))

.contentType(MediaType.APPLICATION_JSON)

.content(json(Map.of("taskId", taskId))))

.andExpect(status().isOk())

.andExpect(jsonPath("\$.success").value(true));

JsonNode task = findTask(username, taskId);

Assertions.assertTrue(task.get("done").asBoolean());

}

@Test

void commentImageCanBeFetchedAndConvertedToOcrDraft() throws Exception {

String username = createUser();

String taskId = createTask(username);

String commentId = commentWithImage(username, taskId);

JsonNode comment = findComment(commentId);

String imageId = comment.get("image").asText();

Assertions.assertEquals("note.png", comment.get("imageFilename").asText());

Assertions.assertNotEquals(imageId, comment.get("imageFilename").asText());

Assertions.assertTrue(comment.get("attachment").isNull());

Assertions.assertTrue(comment.get("attachmentFilename").isNull());

mvc.perform(get("/image")

.queryParam("img", imageId)

.with(user(username).roles("USER"))

```

        .andExpect(status().isOk())
        .andExpect(content().contentTypeCompatibleWith(MediaType.IMAGE_PNG));

mvc.perform(post("/ocr/image")
    .with(csrf())
    .with(user(username).authorities(
        new SimpleGrantedAuthority("ROLE_USER"),
        new SimpleGrantedAuthority("OCR_ENABLED")
    ))
    .contentType(MediaType.APPLICATION_JSON)
    .content(json(Map.of(
        "image", imageId,
        "language", "eng",
        "textLayout", 3
    ))))
    .andExpect(status().isOk())
    .andExpect(jsonPath("$.title").isNotEmpty())
    .andExpect(jsonPath("$.description").isNotEmpty())
    .andExpect(jsonPath("$.rawText").isNotEmpty());
}

```

@Test

void svgCommentImageCanBeUploadedAndFetched() throws Exception {

```

    String username = createUser();
    String taskId = createTask(username);
    String commentId = commentWithSvgImage(username, taskId);
    JsonNode comment = findComment(commentId);
    String imageId = comment.get("image").asText();

```

```

    Assertions.assertEquals("diagram.svg", comment.get("imageFilename").asText());

```

```

    Assertions.assertNotEquals(imageId, comment.get("imageFilename").asText());

```

```

mvc.perform(get("/image")
    .queryParam("img", imageId)
    .with(user(username).roles("USER")))
    .andExpect(status().isOk())
    .andExpect(content().contentTypeCompatibleWith("image/svg+xml"));
}

```

@Test

void commentAttachmentCanBeFetchedAndIncludesOriginalFilename() throws Exception {

```

    String username = createUser();
    String taskId = createTask(username);
    String commentId = commentWithAttachment(username, taskId);

```



```

JsonNode comment = findComment(commentId);
String attachmentId = comment.get("attachment").asText();

Assertions.assertEquals("meeting-notes.txt", comment.get("attachmentFilename").asText());
Assertions.assertNotEquals(attachmentId, comment.get("attachmentFilename").asText());
Assertions.assertTrue(comment.get("image").isNull());
Assertions.assertTrue(comment.get("imageFilename").isNull());

mvc.perform(get("/attachment")
    .queryParam("file", attachmentId)
    .with(user(username).roles("USER")))
    .andExpect(status().isOk())
    .andExpect(content().contentTypeCompatibleWith(MediaType.TEXT_PLAIN));
}

@Test
void aiCredentialSelectionUsesPersonalKeyBeforeServerKey() throws Exception {
    String username = createUser();

    mvc.perform(get("/ai/credential")
        .with(user(username).roles("USER")))
        .andExpect(status().isOk())
        .andExpect(jsonPath("$.provider").value("gemini"))
        .andExpect(jsonPath("$.model").value("gemini-2.5-flash-lite"))
        .andExpect(jsonPath("$.providerType").value("GEMINI_FLASH"))
        .andExpect(jsonPath("$.personalCredentialConfigured").value(false))
        .andExpect(jsonPath("$.serverCredentialAvailable").value(true))
        .andExpect(jsonPath("$.serverCredentialAllowed").value(false))
        .andExpect(jsonPath("$.selection").value("UNAVAILABLE")));

    mvc.perform(put("/ai/credential")
        .with(csrf())
        .with(user(username).roles("USER"))
        .contentType(MediaType.APPLICATION_JSON)
        .content(json(Map.of(
            "provider", "GEMINI_FLASH",
            "apiKey", "AlzaPersonalGeminiCredential123456789"
        ))))
        .andExpect(status().isOk())
        .andExpect(jsonPath("$.personalCredentialConfigured").value(true))
        .andExpect(jsonPath("$.personalCredentialSuffix").value("6789"))
        .andExpect(jsonPath("$.selection").value("PERSONAL_KEY")));

    if (hasOpenAiApiKey()) {

```

```

String openAiApiKey = openAiApiKey();
mvc.perform(put("/ai/credential")
    .with(csrf())
    .with(user(username).roles("USER"))
    .contentType(MediaType.APPLICATION_JSON)
    .content(json(Map.of(
        "provider", "GPT_5_NANO",
        "apiKey", openAiApiKey
    ))))
    .andExpect(status().isOk())
    .andExpect(jsonPath("$.provider").value("openai"))
    .andExpect(jsonPath("$.model").value("gpt-5-nano"))
    .andExpect(jsonPath("$.providerType").value("GPT_5_NANO"))
    .andExpect(jsonPath("$.personalCredentialConfigured").value(true))
    .andExpect(jsonPath("$.personalCredentialSuffix").value(openAiApiKey.substring(openAiApiKey.length() - 4)))
    .andExpect(jsonPath("$.selection").value("PERSONAL_KEY"));
}

mvc.perform(delete("/ai/credential")
    .with(csrf())
    .with(user(username).roles("USER")))
    .andExpect(status().isOk())
    .andExpect(jsonPath("$.personalCredentialConfigured").value(false))
    .andExpect(jsonPath("$.selection").value("UNAVAILABLE"));
}

@Test
void aiCredentialSelectionAllowsServerKeyForEntitledUsers() throws Exception {
    String username = createUser();

    mvc.perform(get("/ai/credential")
        .with(user(username).authorities(
            new SimpleGrantedAuthority("ROLE_USER"),
            new SimpleGrantedAuthority("AI_ENABLED")
        )))
        .andExpect(status().isOk())
        .andExpect(jsonPath("$.serverCredentialAvailable").value(true))
        .andExpect(jsonPath("$.serverCredentialAllowed").value(true))
        .andExpect(jsonPath("$.selection").value("SERVER_KEY"));
}

@Test
void aiSummarizesLongCommentThreadAndMarksItStaleAfterNewComment() throws Exception {
    String username = createUser();

```

```

String taskId = createTask(username);
for (int index = 1; index <= 5; index++) {
    addComment(username, taskId, "summary comment")
        .andExpect(status().isOk());
}

mvc.perform(post("/ai/task/comment-summary")
    .with(csrf())
    .with(user(username).authorities(
        new SimpleGrantedAuthority("ROLE_USER"),
        new SimpleGrantedAuthority("AI_ENABLED")
    ))
    .contentType(MediaType.APPLICATION_JSON)
    .content(json(Map.of("taskId", taskId))))
    .andExpect(status().isOk())
    .andExpect(jsonPath("$.summary").value("AI summary of the comment thread."))
    .andExpect(jsonPath("$.summarizedCommentCount").value(5))
    .andExpect(jsonPath("$.currentCommentCount").value(5))
    .andExpect(jsonPath("$.stale").value(false));

addComment(username, taskId, "newer comment")
    .andExpect(status().isOk());

JsonNode task = findTask(username, taskId);
Assertions.assertTrue(task.get("commentSummary").get("stale").asBoolean());
Assertions.assertEquals(6, task.get("commentSummary").get("currentCommentCount").asInt());
}

@Test
void aiRefreshRequiresPersonalCredentialAndTaskAssistanceReturnsStructuredSuggestions() throws Exception {
    String username = createUser();
    String taskId = createTask(username);
    for (int index = 1; index <= 5; index++) {
        addComment(username, taskId, "personal summary comment")
            .andExpect(status().isOk());
    }

    mvc.perform(put("/ai/credential")
        .with(csrf())
        .with(user(username).roles("USER"))
        .contentType(MediaType.APPLICATION_JSON)
        .content(json(Map.of(
            "provider", "GEMINI_FLASH",

```

```

        "apiKey", "AlzaPersonalGeminiCredential123456789"
    ))))
    .andExpect(status().isOk());

mvc.perform(post("/ai/task/comment-summary/refresh")
    .with(csrf())
    .with(user(username).roles("USER"))
    .contentType(MediaType.APPLICATION_JSON)
    .content(json(Map.of("taskId", taskId))))
    .andExpect(status().isOk())
    .andExpect(jsonPath("$.summary").value("AI summary of the comment thread."))
    .andExpect(jsonPath("$.credentialSource").value("PERSONAL_KEY"));

mvc.perform(post("/ai/task/improve")
    .with(csrf())
    .with(user(username).roles("USER"))
    .contentType(MediaType.APPLICATION_JSON)
    .content(json(Map.of(
        "title", "rough title",
        "description", "rough description"
    ))))
    .andExpect(status().isOk())
    .andExpect(jsonPath("$.title").value("Improved task title"))
    .andExpect(jsonPath("$.description").value("Improved task description."));

mvc.perform(post("/ai/task/subtasks")
    .with(csrf())
    .with(user(username).roles("USER"))
    .contentType(MediaType.APPLICATION_JSON)
    .content(json(Map.of(
        "title", "rough title",
        "description", "rough description"
    ))))
    .andExpect(status().isOk())
    .andExpect(jsonPath("$.subtasks[0]").value("First subtask"));

mvc.perform(post("/ai/task/ocr-structure")
    .with(csrf())
    .with(user(username).roles("USER"))
    .contentType(MediaType.APPLICATION_JSON)
    .content(json(Map.of("rawText", "OCR raw text"))))
    .andExpect(status().isOk())
    .andExpect(jsonPath("$.title").value("OCR task title"))
    .andExpect(jsonPath("$.description").value("Structured from OCR text."));

```

```

}

@Test
void userProfileIncludesSettingsAiCredentialAndSecurityEndpoints() throws Exception {
    String username = createUser();

    mvc.perform(get("/user/profile")
        .with(user(username).authorities(
            new SimpleGrantedAuthority("ROLE_USER"),
            new SimpleGrantedAuthority("AI_ENABLED")
        )))
        .andExpect(status().isOk())
        .andExpect(jsonPath("$.username").value(username))
        .andExpect(jsonPath("$.settings.summaryThresholdComments").value(5))
        .andExpect(jsonPath("$.settings.aiCredential.selection").value("SERVER_KEY"))
        .andExpect(jsonPath("$.authentication.loginUrl").value("/login"))
        .andExpect(jsonPath("$.authentication.logoutUrl").value("/logout"))
        .andExpect(jsonPath("$.authentication.changePasswordUrl").value("/user/password"));
}

```

```

@Test
void userCanReplaceSettingsAndLookupUsersForAssignment() throws Exception {
    String firstUsername = createUser();
    String secondUsername = createUser();

    mvc.perform(put("/user/settings")
        .with(csrf())
        .with(user(firstUsername).roles("USER"))
        .contentType(MediaType.APPLICATION_JSON)
        .content(json(Map.of("summaryThresholdComments", 7))))
        .andExpect(status().isOk())
        .andExpect(jsonPath("$.summaryThresholdComments").value(7));

    mvc.perform(get("/user/settings")
        .with(user(firstUsername).roles("USER")))
        .andExpect(status().isOk())
        .andExpect(jsonPath("$.summaryThresholdComments").value(7));

    mvc.perform(get("/users/lookup")
        .queryParam("query", secondUsername)
        .with(user(firstUsername).roles("USER")))
        .andExpect(status().isOk())
        .andExpect(jsonPath("$[0].username").value(secondUsername));
}

```

@Test

void userCanChangePasswordAfterCurrentPasswordVerification() throws Exception {

String username = createUser();

```
    mvc.perform(put("/user/password")
        .with(csrf())
        .with(user(username).roles("USER"))
        .contentType(MediaType.APPLICATION_JSON)
        .content(json(Map.of(
            "currentPassword", "123456",
            "newPassword", "new-password-123"
        ))))
        .andExpect(status().isOk())
        .andExpect(jsonPath("$.success").value(true));
```

```
    mvc.perform(formLogin().user(username).password("new-password-123"))
        .andExpect(authenticated().withUsername(username));
```

}

@Test

void configuredDemoSessionCookieAuthenticatesAndRefreshesExpiration() throws Exception {

registerIfNeeded("demo_session_user");

```
    mvc.perform(get("/me")
        .cookie(new Cookie("JSESSIONID", "DEMOSESSIONID1234567890")))
        .andExpect(status().isOk())
        .andExpect(jsonPath("$.username").value("demo_session_user"))
        .andExpect(result -> Assertions.assertTrue(
            result.getResponse().getHeaders("Set-Cookie").stream()
                .anyMatch(header -> header.contains("JSESSIONID=DEMOSESSIONID1234567890")
                    && header.contains("Max-Age="))
        ));
```

}

@Test

void linkPreviewAcceptsBareDomainPresentInTaskTextWithTrailingPunctuation() throws Exception {

String username = createUser();

String taskId = createTask(

username,

"Bare link preview",

"Read securefromscratch.com."

);

```

mvc.perform(post("/link-preview/task/delete")
    .with(csrf())
    .with(user(username).roles("USER"))
    .contentType(MediaType.APPLICATION_JSON)
    .content(json(Map.of(
        "taskId", taskId,
        "url", "securefromscratch.com"
    ))))
    .andExpect(status().isOk());
}

```

@Test

void entitledUsersCanExportAndImport() throws Exception {

String username = createUser();

createTask(username);

```

mvc.perform(get("/extra/export")
    .with(user(username).authorities(
        new SimpleGrantedAuthority("ROLE_USER"),
        new SimpleGrantedAuthority("EXPORT_ENABLED")
    )))
    .andExpect(status().isOk());

```

MockMultipartFile importFile = new MockMultipartFile(

"file",

"tasks.ser",

MediaType.APPLICATION_OCTET_STREAM_VALUE,

json(List.of(Map.of(

"name", "Imported task",

"desc", "Imported description",

"dueDate", "2099-01-01",

"dueTime", "09:00",

"responsibilityOf", List.of(username)

))).getBytes(StandardCharsets.UTF_8)

);

```

mvc.perform(multipart("/extra/import")

```

.file(importFile)

.with(csrf())

.with(user(username).authorities(

new SimpleGrantedAuthority("ROLE_USER"),

new SimpleGrantedAuthority("IMPORT_ENABLED")

)))

.andExpect(status().isOk());

```
}
```

```
private org.springframework.test.web.servlet.ResultActions register(String username) throws Exception {  
    return mvc.perform(post("/register")  
        .with(csrf())  
        .contentType(MediaType.APPLICATION_JSON)  
        .content(json(Map.of(  
            "username", username,  
            "password", "123456"  
        ))));  
}
```

```
private String createUser() throws Exception {  
    String username = uniqueUsername();  
    register(username)  
        .andExpect(status().isOk());  
    return username;  
}
```

```
private String createTask(String username) throws Exception {  
    return createTask(  
        username,  
        "Integration task",  
        "A task created by the server test."  
    );  
}
```

```
private String createTask(String username, String name, String description) throws Exception {  
    MvcResult result = mvc.perform(post("/create")  
        .with(csrf())  
        .with(user(username).roles("USER"))  
        .contentType(MediaType.APPLICATION_JSON)  
        .content(json(Map.of(  
            "name", name,  
            "desc", description,  
            "dueDate", "2099-01-01",  
            "dueTime", "09:00",  
            "responsibilityOf", List.of(username)  
        ))))  
        .andExpect(status().isOk())  
        .andReturn();  
    return jsonTree(result).get("taskid").asText();  
}
```



```

private org.springframework.test.web.servlet.ResultActions addComment(String username, String taskId, String text) throws Exception {
    MockMultipartFile fields = jsonPart("commentFields", commentFields(text, taskId));
    return mvc.perform(multipart("/comment")
        .file(fields)
        .with(csrf())
        .with(user(username).roles("USER")));
}

```

```

private String commentWithImage(String username, String taskId) throws Exception {
    MockMultipartFile fields = jsonPart("commentFields", commentFields("image comment", taskId));
    MockMultipartFile file = new MockMultipartFile(
        "file",
        "note.png",
        MediaType.IMAGE_PNG_VALUE,
        new byte[]{1, 2, 3, 4, 5, 6}
    );
    MvcResult result = mvc.perform(multipart("/comment")
        .file(fields)
        .file(file)
        .with(csrf())
        .with(user(username).roles("USER")))
        .andExpect(status().isOk())
        .andReturn();
    return jsonTree(result).get("commentid").asText();
}

```

```

private String commentWithSvgImage(String username, String taskId) throws Exception {
    MockMultipartFile fields = jsonPart("commentFields", commentFields("svg image comment", taskId));
    MockMultipartFile file = new MockMultipartFile(
        "file",
        "diagram.svg",
        "image/svg+xml",
        "<svg xmlns='http://www.w3.org/2000/svg' viewBox='0 0 10 10'><circle cx='5' cy='5' r='4'/></svg>".getBytes()
    );
    MvcResult result = mvc.perform(multipart("/comment")
        .file(fields)
        .file(file)
        .with(csrf())
        .with(user(username).roles("USER")))
        .andExpect(status().isOk())
        .andReturn();
    return jsonTree(result).get("commentid").asText();
}

```

```

private String commentWithAttachment(String username, String taskId) throws Exception {
    MockMultipartFile fields = jsonPart("commentFields", commentFields("attachment comment", taskId));
    MockMultipartFile file = new MockMultipartFile(
        "file",
        "meeting-notes.txt",
        MediaType.TEXT_PLAIN_VALUE,
        "notes from the meeting".getBytes(StandardCharsets.UTF_8)
    );
    MvcResult result = mvc.perform(multipart("/comment")
        .file(fields)
        .file(file)
        .with(csrf())
        .with(user(username).roles("USER")))
        .andExpect(status().isOk())
        .andReturn());
    return jsonTree(result).get("commentid").asText();
}

```

```

private ObjectNode commentFields(String text, String taskId) {
    ObjectNode fields = objectMapper.createObjectNode();
    fields.put("text", text);
    fields.put("taskId", taskId);
    fields.putNull("commentid");
    return fields;
}

```

```

private String findCommentImage(String commentId) throws Exception {
    JsonNode comment = findComment(commentId);
    return comment.get("image").asText();
}

```

```

private JsonNode findComment(String commentId) throws Exception {
    JsonNode tasksNode = jsonTree(mvc.perform(get("/tasks")
        .with(user("reader").roles("USER")))
        .andExpect(status().isOk())
        .andReturn());
    for (JsonNode task : tasksNode) {
        for (JsonNode comment : task.get("comments")) {
            if (commentId.equals(comment.get("commentid").asText())) {
                return comment;
            }
        }
    }
}
    Assertions.fail("Created comment was not found.");
}

```

```

    return objectMapper.createObjectNode();
}

private JsonNode findTask(String username, String taskId) throws Exception {
    JsonNode tasksNode = jsonTree(mvc.perform(get("/tasks")
        .with(user(username).roles("USER")))
        .andExpect(status().isOk())
        .andReturn());
    for (JsonNode task : tasksNode) {
        if (taskId.equals(task.get("taskId").asText())) {
            return task;
        }
    }
    Assertions.fail("Created task was not found.");
    return objectMapper.createObjectNode();
}

private MockMultipartFile jsonPart(String name, Object value) throws Exception {
    return new MockMultipartFile(
        name,
        "blob",
        MediaType.APPLICATION_JSON_VALUE,
        json(value).getBytes(StandardCharsets.UTF_8)
    );
}

private String json(Object value) throws Exception {
    return objectMapper.writeValueAsString(value);
}

private JsonNode jsonTree(MvcResult result) throws Exception {
    return objectMapper.readTree(result.getResponse().getContentAsString());
}

private static String uniqueUsername() {
    return UUID.randomUUID().toString();
}

private static String openAiApiKey() {
    String apiKey = System.getenv("OPENAI_API_KEY");
    // ALLOW NULL LITERAL:
    // old api: getenv could return null if the environment variable is not set, and we want to allow that for local development
    if (apiKey == null || apiKey.isBlank()) {
        return FALLBACK_OPENAI_API_KEY;
    }
}

```

```

    }

    return apiKey;
}

private static boolean hasOpenAiApiKey() {
    return !FALLBACK_OPENAI_API_KEY.equals(openAiApiKey());
}

private void registerIfNeeded(String username) throws Exception {
    register(username)
        .andExpect(result -> Assertions.assertTrue(
            result.getResponse().getStatus() == 200
            || result.getResponse().getStatus() == 400
        ));
}
}

```

busybee/src/test/java/com/securefromscratch/busybee/functionality/CurrentUserFunctionalityTests.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/test/java/com/securefromscratch/busybee/functionality/CurrentUserFunctionalityTests.java>

```

package com.securefromscratch.busybee.functionality;

import org.junit.jupiter.api.Test;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.context.SpringBootTest;
import org.springframework.boot.webmvc.test.autoconfigure.AutoConfigureMockMvc;
import org.springframework.security.core.authority.SimpleGrantedAuthority;
import org.springframework.test.context.ActiveProfiles;
import org.springframework.test.web.servlet.MockMvc;

import static org.springframework.security.test.web.servlet.request.SecurityMockMvcRequestPostProcessors.user;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.get;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.jsonPath;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.status;

@SpringBootTest
@AutoConfigureMockMvc
@ActiveProfiles({"mysql", "test"})
class CurrentUserFunctionalityTests {
    @Autowired
    private MockMvc mvc;

    @Test
    void authenticatedUserCanReadCurrentUser() throws Exception {
        mvc.perform(get("/me")
            .with(user("current-user").authorities(new SimpleGrantedAuthority("ROLE_USER"))))
            .andExpect(status().isOk())
            .andExpect(jsonPath("$.username").value("current-user"))
            .andExpect(jsonPath("$.admin").value(false))
            .andExpect(jsonPath("$.entitlements").isEmpty())
            .andExpect(jsonPath("$.effectiveEntitlements").isEmpty());
    }
}

```

busybee/src/test/java/com/securefromscratch/busybee/functionality/OcrExtractFunctionalityTests.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/test/java/com/securefromscratch/busybee/functionality/OcrExtractFunctionalityTests.java>

```

package com.securefromscratch.busybee.functionality;

import java.nio.charset.StandardCharsets;

import org.junit.jupiter.api.Test;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.context.SpringBootTest;
import org.springframework.boot.web.mvc.test.autoconfigure.AutoConfigureMockMvc;
import org.springframework.http.MediaType;
import org.springframework.mock.web.MockMultipartFile;
import org.springframework.security.core.authority.SimpleGrantedAuthority;
import org.springframework.test.context.ActiveProfiles;
import org.springframework.test.web.servlet.MockMvc;

import static org.springframework.security.test.web.servlet.request.SecurityMockMvcRequestPostProcessors.csrf;
import static org.springframework.security.test.web.servlet.request.SecurityMockMvcRequestPostProcessors.user;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.multipart;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.jsonPath;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.status;

@SpringBootTest
@AutoConfigureMockMvc
@ActiveProfiles({"mysql", "test"})
class OcrExtractFunctionalityTests {
    @Autowired
    private MockMvc mvc;

    @Test
    void userCanExtractDraftFromUploadedImage() throws Exception {
        MockMultipartFile fields = new MockMultipartFile(
            "ocrFields",
            "",
            MediaType.APPLICATION_JSON_VALUE,
            "{\"language\":\"eng\",\"textLayout\":3}".getBytes(StandardCharsets.UTF_8)
        );
        MockMultipartFile image = new MockMultipartFile(
            "file",
            "receipt.png",
            MediaType.IMAGE_PNG_VALUE,
            new byte[]{1, 2, 3, 4, 5}
        );

        mvc.perform(multipart("/ocr/extract")
            .file(fields)

```

```
.file(image)
.with(csrf())
.with(user("ocr-user").authorities(
    new SimpleGrantedAuthority("ROLE_USER"),
    new SimpleGrantedAuthority("OCR_ENABLED")
)))
.andExpect(status().isOk())
.andExpect(jsonPath("$.title").isNotEmpty())
.andExpect(jsonPath("$.description").isNotEmpty())
.andExpect(jsonPath("$.rawText").isNotEmpty());
}
}
```

busybee/src/test/java/com/securefromscratch/busybee/functionality/TaskLifecycleFunctionalityTests.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/test/java/com/securefromscratch/busybee/functionality/TaskLifecycleFunctionalityTests.java>

```

package com.securefromscratch.busybee.functionality;

import java.util.List;
import java.util.Map;

import com.fasterxml.jackson.databind.JsonNode;
import com.fasterxml.jackson.databind.ObjectMapper;
import org.junit.jupiter.api.Assertions;
import org.junit.jupiter.api.Test;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.context.SpringBootTest;
import org.springframework.boot.web.mvc.test.autoconfigure.AutoConfigureMockMvc;
import org.springframework.http.MediaType;
import org.springframework.test.context.ActiveProfiles;
import org.springframework.test.web.servlet.MockMvc;

import static org.springframework.security.test.web.servlet.request.SecurityMockMvcRequestPostProcessors.csrf;
import static org.springframework.security.test.web.servlet.request.SecurityMockMvcRequestPostProcessors.user;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.get;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.post;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.jsonPath;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.status;

@SpringBootTest
@AutoConfigureMockMvc
@ActiveProfiles({"mysql", "test"})
class TaskLifecycleFunctionalityTests {
    @Autowired
    private MockMvc mvc;

    @Autowired
    private ObjectMapper objectMapper;

    @Test
    void userCanCreateListAndCompleteTask() throws Exception {
        String taskId = mvc.perform(post("/create")
            .with(csrf())
            .with(user("task_creator").roles("USER"))
            .contentType(MediaType.APPLICATION_JSON)
            .content(objectMapper.writeValueAsString(Map.of(
                "name", "Persisted task",
                "desc", "A task lifecycle check.",
                "dueDate", "2099-01-01",
                "dueTime", "09:00",
            ))))
    }
}

```



```

        "responsibilityOf", List.of("task_creator")
    ))))
    .andExpect(status().isOk())
    .andExpect(jsonPath("$.taskid").exists())
    .andReturn()
    .getResponse()
    .getContentAsString();

```

```
String createdTaskId = objectMapper.readTree(taskId).get("taskid").asText();
```

```

JsonNode createdTask = taskWithId(mvc.perform(get("/tasks")
    .with(user("task_creator").roles("USER"))
    .andExpect(status().isOk())
    .andReturn(), createdTaskId);
Assertions.assertEquals("task_creator", createdTask.get("createdBy").asText());
Assertions.assertFalse(createdTask.get("done").asBoolean());

```

```

mvc.perform(post("/done")
    .with(csrf())
    .with(user("task_creator").roles("USER"))
    .contentType(MediaType.APPLICATION_JSON)
    .content(objectMapper.writeValueAsString(Map.of("taskid", createdTaskId))))
    .andExpect(status().isOk())
    .andExpect(jsonPath("$.success").value(true));

```

```

JsonNode completedTask = taskWithId(mvc.perform(get("/tasks")
    .with(user("task_creator").roles("USER"))
    .andExpect(status().isOk())
    .andReturn(), createdTaskId);
Assertions.assertTrue(completedTask.get("done").asBoolean());

```

```
}
```

```

private JsonNode taskWithId(org.springframework.test.web.servlet.MvcResult result, String taskId) throws Exception {
    for (JsonNode task : objectMapper.readTree(result.getResponse().getContentAsString())) {
        if (taskId.equals(task.get("taskid").asText())) {
            return task;
        }
    }
    Assertions.fail("Created task was not found.");
    return objectMapper.createObjectNode();
}

```

busybee/src/test/java/com/securefromscratch/busybee/functionality/UserSettingsFunctionalityTests.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/test/java/com/securefromscratch/busybee/functionality/UserSettingsFunctionalityTests.java>

```

package com.securefromscratch.busybee.functionality;

import java.util.UUID;
import java.util.Map;

import com.fasterxml.jackson.databind.ObjectMapper;
import org.junit.jupiter.api.Test;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.context.SpringBootTest;
import org.springframework.boot.test.autoconfigure.AutoConfigureMockMvc;
import org.springframework.http.MediaType;
import org.springframework.test.context.ActiveProfiles;
import org.springframework.test.web.servlet.MockMvc;

import static org.springframework.security.test.web.servlet.request.SecurityMockMvcRequestPostProcessors.csrf;
import static org.springframework.security.test.web.servlet.request.SecurityMockMvcRequestPostProcessors.user;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.get;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.post;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.put;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.jsonPath;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.status;

@SpringBootTest
@AutoConfigureMockMvc
@ActiveProfiles({"mysql", "test"})
class UserSettingsFunctionalityTests {
    @Autowired
    private MockMvc mvc;

    @Autowired
    private ObjectMapper objectMapper;

    @Test
    void userCanReplaceAndReadSummaryThreshold() throws Exception {
        String username = UUID.randomUUID().toString();
        mvc.perform(post("/register")
            .with(csrf())
            .contentType(MediaType.APPLICATION_JSON)
            .content(objectMapper.writeValueAsString(Map.of(
                "username", username,
                "password", "password123"
            ))))
            .andExpect(status().isOk());
    }
}

```

```

mvc.perform(put("/user/settings")
    .with(csrf())
    .with(user(username).roles("USER"))
    .contentType(MediaType.APPLICATION_JSON)
    .content("{\"summaryThresholdComments\":7}"))
    .andExpect(status().isOk())
    .andExpect(jsonPath("$.summaryThresholdComments").value(7));

mvc.perform(get("/user/settings").with(user(username).roles("USER")))
    .andExpect(status().isOk())
    .andExpect(jsonPath("$.summaryThresholdComments").value(7));
}
}

```

busybee/src/test/resources/application-test.properties

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/busybee/src/test/resources/application-test.properties>

```

spring.datasource.url=jdbc:mysql://192.168.48.1:3307/busybee_6_test?createDatabaseIfNotExist=true&allowPublicKeyRetrieval=true
spring.datasource.username=root
spring.datasource.password=123456
spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver
busybee.secrets.vault.uri=http://192.168.48.1:8200

spring.flyway.enabled=true
spring.flyway.locations=classpath:db/migration/mysql
spring.flyway.baseline-on-migrate=true
spring.flyway.validate-on-migrate=false

spring.jooq.sql-dialect=MYSQL
busybee.ai.gemini.api-key=AlzaTestServerCredentialForGeminiFlashLite
busybee.demo.sessions.enabled=true
busybee.demo.sessions.cookie-max-age=12h
busybee.demo.sessions.users[DEMOSESSIONID1234567890]=demo_session_user

```

gradle/forbid-jooq-toctou.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/gradle/forbid-jooq-toctou.gradle.kts>

`_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/gradle/forbid-jooq-toctou.gradle.kts>`

```

/*
 * SECURITY BUILD GATE: JOOQ TOCTOU ENFORCEMENT
 *
 * This Gradle script fails the build when Java methods do a jOOQ read and then later execute
 * a jOOQ UPDATE or DELETE in the same method. That "check then act" shape is a common
 * time-of-check/time-of-use race: another transaction can change the row after the read and
 * before the write.
 *
 * Prefer one atomic SQL statement and use the affected row count as the existence/result signal.
 *
 * Escape hatch:
 *
 * // ALLOW JOOQ TOCTOU: <specific justification>
 * dsl.update(TABLE)
 * ...
 * .execute();
 */

```

```

import com.sun.source.tree.BlockTree
import com.sun.source.tree.CompilationUnitTree
import com.sun.source.tree.MemberSelectTree
import com.sun.source.tree.MethodInvocationTree
import com.sun.source.tree.MethodTree
import com.sun.source.tree.Tree
import com.sun.source.util.JavacTask
import com.sun.source.util.TreePath
import com.sun.source.util.TreePathScanner
import com.sun.source.util.Trees
import org.gradle.api.GradleException
import org.gradle.api.plugins.JavaPluginExtension
import org.gradle.api.tasks.SourceSet
import java.io.File
import java.io.StringWriter
import java.util.Locale
import javax.lang.model.type.TypeKind
import javax.lang.model.type.TypeMirror
import javax.tools.Diagnostic
import javax.tools.DiagnosticCollector
import javax.tools.JavaFileObject
import javax.tools.ToolProvider

```

```

data class JooqToctouFinding(
    val methodName: String,
    val readLineNumber: Long,

```

```

    val readExpression: String,
    val writeLineNumber: Long,
    val writeExpression: String,
    val writeType: String
)

data class JooqRead(
    val lineNumber: Long,
    val expression: String
)

data class JooqWrite(
    val lineNumber: Long,
    val expression: String,
    val type: String
)

val JOOQ_TOCTOU_ALLOW_MARKER = "ALLOW JOOQ TOCTOU:"
val MINIMUM_JOOQ_TOCTOU_ALLOW_REASON_CHARACTERS = 120
val JOOQ_READ_TERMINALS = setOf(
    "fetch",
    "fetchAny",
    "fetchCount",
    "fetchExists",
    "fetchOne",
    "fetchOptional"
)

val JOOQ_MUTATING_QUERY_PREFIXES = listOf(
    "org.jooq.Update",
    "org.jooq.Delete"
)

@Suppress("UNCHECKED_CAST")
val precedingEscapeValidationProblem = rootProject.extensions.extraProperties["precedingEscapeValidationProblem"]
@Suppress("UNCHECKED_CAST")
val precedingEscapeHatchUsage = rootProject.extensions.extraProperties["precedingEscapeHatchUsage"] as (String, Int)

fun File.relativeUnixPath(rootDir: File): String {
    return rootDir.toPath()
        .relativize(toPath())
        .toString()
        .replace(File.separatorChar, '/')
}

fun sourceLine(compilationUnit: CompilationUnitTree, tree: Tree, trees: Trees): Long {

```

```

val startPosition = trees.sourcePositions.getStartPosition(compilationUnit, tree)

return if (startPosition >= 0) {
    compilationUnit.lineMap.getLineNumber(startPosition)
} else {
    -1L
}
}

fun methodName(methodInvocationTree: MethodInvocationTree): String? {
    val methodSelect = methodInvocationTree.methodSelect

    return when (methodSelect) {
        is MemberSelectTree -> methodSelect.identifier.toString()
        else -> null
    }
}

fun receiverType(methodInvocationTree: MethodInvocationTree, currentPath: TreePath, trees: Trees): TypeMirror? {
    val methodSelect = methodInvocationTree.methodSelect

    return when (methodSelect) {
        is MemberSelectTree -> trees.getTypeMirror(TreePath(currentPath, methodSelect.expression))
        else -> null
    }
}

fun isConcreteType(typeMirror: TypeMirror?): Boolean {
    return typeMirror != null &&
        typeMirror.kind != TypeKind.ERROR &&
        typeMirror.kind != TypeKind.NULL &&
        typeMirror.kind != TypeKind.NONE &&
        typeMirror.kind != TypeKind.VOID
}

fun isJooqReadTerminal(methodInvocationTree: MethodInvocationTree, currentPath: TreePath, trees: Trees): Boolean {
    val name = methodName(methodInvocationTree)

    if (name !in JOOQ_READ_TERMINALS) {
        return false
    }

    val typeName = receiverType(methodInvocationTree, currentPath, trees)?.toString().orEmpty()

```



```

    return typeName.startsWith("org.jooq.Select") ||
        typeName.startsWith("org.jooq.ResultQuery") ||
        typeName.startsWith("org.jooq.Cursor")
}

fun isJooqUpdateOrDeleteExecute(methodInvocationTree: MethodInvocationTree, currentPath: TreePath, trees: Trees)
    if (methodName(methodInvocationTree) != "execute") {
        return null
    }

    val typeMirror = receiverType(methodInvocationTree, currentPath, trees)

    if (!isConcreteType(typeMirror)) {
        return null
    }

    val typeName = typeMirror.toString()

    return JOOQ_MUTATING_QUERY_PREFIXES.firstOrNull { typeName.startsWith(it) }?.let { typeName }
}

fun collectJooqToctouFindings(
    compilationUnit: CompilationUnitTree,
    trees: Trees
): List<JooqToctouFinding> {
    val findings = mutableListOf<JooqToctouFinding>()

    object : TreePathScanner<Unit, Unit>() {
        override fun visitMethod(node: MethodTree, unused: Unit?) {
            val body = node.body

            if (body != null) {
                collectMethodFindings(node.name.toString(), body)
            }
        }
    }

    private fun collectMethodFindings(methodName: String, body: BlockTree) {
        val reads = mutableListOf<JooqRead>()
        val writes = mutableListOf<JooqWrite>()

        object : TreePathScanner<Unit, Unit>() {
            override fun visitMethodInvocation(node: MethodInvocationTree, unused: Unit?) {
                if (isJooqReadTerminal(node, currentPath, trees)) {
                    reads += JooqRead(

```

```

        lineNumber = sourceLine(compilationUnit, node, trees),
        expression = node.toString()
    )
}

val writeType = isJooqUpdateOrDeleteExecute(node, currentPath, trees)
if (writeType != null) {
    writes += JooqWrite(
        lineNumber = sourceLine(compilationUnit, node, trees),
        expression = node.toString(),
        type = writeType
    )
}

super.visitMethodInvocation(node, unused)
}
}.scan(TreePath(currentPath, body), Unit)

```

```

for (write in writes) {
    val priorRead = reads
        .filter { read -> read.lineNumber in 1 until write.lineNumber }
        .minByOrNull { read -> read.lineNumber }

```

```

    if (priorRead != null) {
        findings += JooqToctouFinding(
            methodName = methodName,
            readLineNumber = priorRead.lineNumber,
            readExpression = priorRead.expression,
            writeLineNumber = write.lineNumber,
            writeExpression = write.expression,
            writeType = write.type
        )
    }
}
}
}.scan(compilationUnit, Unit)

```

```

return findings
}

```

```

fun compilerOptionsFor(sourceSet: SourceSet): List<String> {
    val options = mutableListOf("-proc:none")
    val classpath = sourceSet.compileClasspath.files

```

```

if (classpath.isEmpty()) {
    options += "-classpath"
    options += classpath.joinToString(File.pathSeparator) { it.absolutePath }
}

val sourceDirectories = sourceSet.allJava.srcDirs.filter { it.exists() }

if (sourceDirectories.isEmpty()) {
    options += "-sourcepath"
    options += sourceDirectories.joinToString(File.pathSeparator) { it.absolutePath }
}

return options
}

fun inspectSourceSetForJooqToctou(sourceSet: SourceSet, rootDir: File): List<String> {
    val sourceFiles = sourceSet.allJava.files
        .filter { it.extension == "java" && it.exists() }
        .sortedBy { it.absolutePath }

    if (sourceFiles.isEmpty()) {
        return emptyList()
    }

    val compiler = ToolProvider.getSystemJavaCompiler()
    ?: throw GradleException("A JDK is required to inspect Java source files. Gradle is not running with a JDK compiler")

    val diagnostics = DiagnosticCollector<JavaFileObject>()

    compiler.getStandardFileManager(diagnostics, Locale.ROOT, Charsets.UTF_8).use { fileManager ->
        val javaFiles = fileManager.getJavaFileObjectsFromFiles(sourceFiles)
        val task = compiler.getTask(
            StringWriter(),
            fileManager,
            diagnostics,
            compilerOptionsFor(sourceSet),
            emptyList<String>(),
            javaFiles
        ) as JavacTask

        val trees = Trees.instance(task)
        val parsedUnits = task.parse().toList()
        task.analyze()
    }
}

```

```

val errors = diagnostics.diagnostics
    .filter { it.kind == Diagnostic.Kind.ERROR }

if (errors.isNotEmpty()) {
    val message = errors.joinToString("\n") { diagnostic ->
        val sourceName = diagnostic.source?.name ?: "<unknown source>"
        "- $sourceName:${diagnostic.lineNumber}: ${diagnostic.getMessage(Locale.ROOT)}"
    }

    throw GradleException("Failed to analyze Java source files before checking jOOQ TOCTOU patterns:\n$message")
}

return parsedUnits.flatMap { compilationUnit ->
    val sourceFile = File(compilationUnit.sourceFile.toUri())
    val sourcePath = sourceFile.relativeUnixPath(rootDir)

    collectJooqToctouFindings(compilationUnit, trees).mapNotNull { finding ->
        val escapeProblem = precedingEscapeValidationProblem(
            sourceFile,
            finding.writeLineNumber,
            JOOQ_TOCTOU_ALLOW_MARKER,
            MINIMUM_JOOQ_TOCTOU_ALLOW_REASON_CHARACTERS,
            "jOOQ TOCTOU",
            true
        )

        if (escapeProblem == null) {
            return@mapNotNull null
        }

        "$sourcePath:${finding.writeLineNumber} method ${finding.methodName}() executes ${finding.writeType} af
    }
}
}
}

fun jooqToctouFailureGuidance(): String {
    return """

jOOQ TOCTOU POLICY FAILURE


```

Do not check database state with one jOOQ query and then update or **delete** with a later query in the same method. Another transaction can change the row between the read and the write.

Prefer an atomic statement and **use** the affected row count as the result:

```
int changed = dsl.update(TASKS)
    .set(DONE, true)
    .where(TASK_ID.eq(taskId))
    .execute();

return changed > 0;
```

For deletes, put the authorization/existence predicate directly in the **DELETE** where clause and interpret execute() the same way. If the operation truly needs a prior read, move it into a transaction with the right locking semantics and document why that lock is required.

```

    precedingEscapeHatchUsage(JOOQ_TOCTOU_ALLOW_MARKER, MINIMUM_JOOQ_TOCTOU_ALLOW_REASON_CHARACTERS);
    """
    .trimIndent()
}

```

```
val forbidJooqToctou by tasks.registering {
    group = "verification"
    description = "Fails the build when JOOQ code uses read-then-update/delete TOCTOU-prone patterns."
```

```
val javaExtension = project.extensions.getByType<JavaPluginExtension>()
val sourceSets = javaExtension.sourceSets
```

```
sourceSets.configureEach {
    inputs.files(allJava)
    inputs.files(compileClasspath)
}
```

```
doLast {
    val violations = sourceSets.flatMap { sourceSet ->
        inspectSourceSetForJooqToctou(sourceSet, rootProject.projectDir)
    }
}
```

```
if (violations.isNotEmpty()) {  
    throw GradleException(  
        "jOOQ TOCTOU candidates found:\n" +  
        violations.joinToString("\n") { " - $it" } +  
        "\n\n" +  
        jooqToctouFailureGuidance()  
    )  
}
```

} }

}

```
forbidJooqToctou.configure {  
    mustRunAfter(tasks.matching { task -> task.name != name && task.name != "check" && task.group == "verification" })  
    mustRunAfter(tasks.withType<Test>())  
}  
  
tasks.matching { it.name == "check" }.configureEach {  
    dependsOn(forbidJooqToctou)  
}
```

gradle/forbid-local-catches.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/gradle/forbid-local-catches.gradle.kts>

```

/*
 * SECURITY BUILD GATE: LOCAL CATCH ENFORCEMENT
 *
 * This Gradle script fails the build when Java source files contain local catch blocks.
 *
 * Local catch blocks are easy places to accidentally expose internal exception messages,
 * swallow failures, or produce inconsistent API error responses. Prefer centralized exception
 * handling unless a local catch is intentionally narrow and documented.
 *
 * Escape hatch:
 *
 * // LOCAL CATCH REASON: <specific justification>
 * catch (...)
 *
 * Multi-line line comments are accepted:
 *
 * // LOCAL CATCH REASON:
 * // <specific justification continued here>
 * catch (...)
 *
 * Block comments are accepted:
 *
 * /*
 *  * LOCAL CATCH REASON:
 *  * <specific justification continued here>
 *  */
 * catch (...)
 *
 * The minimum justification length is configured in local_catch_guardrail.json.
 */

```

```

import com.sun.source.tree.CatchTree
import com.sun.source.tree.CompilationUnitTree
import com.sun.source.util.JavacTask
import com.sun.source.util.TreePathScanner
import com.sun.source.util.Trees
import org.gradle.api.GradleException
import org.gradle.api.plugins.JavaPluginExtension
import org.gradle.api.tasks.SourceSet
import java.io.File
import java.io.StringWriter
import java.util.Locale
import javax.tools.Diagnostic
import javax.tools.DiagnosticCollector

```

```

import javax.tools.JavaFileObject
import javax.tools.ToolProvider

val DEFAULT_LOCAL_CATCH_MINIMUM_REASON_CHARACTERS = 120
val LOCAL_CATCH_MARKER = "LOCAL_CATCH_REASON:"
@Suppress("UNCHECKED_CAST")
val readJsonObject = rootProject.extensions.extraProperties["readJsonObject"] as (File) -> Map<String, Any>
@Suppress("UNCHECKED_CAST")
val precedingEscapeReasonBefore = rootProject.extensions.extraProperties["precedingEscapeReasonBefore"] as (File, String) -> String
@Suppress("UNCHECKED_CAST")
val precedingEscapeHatchUsage = rootProject.extensions.extraProperties["precedingEscapeHatchUsage"] as (String, Int) -> Int

data class LocalCatchGuardConfig(
    val minimumReasonCharacters: Int
)

data class LocalCatchFinding(
    val lineNumber: Long,
    val exceptionType: String
)

fun File.relativeUnixPath(rootDir: File): String {
    return rootDir.toPath()
        .relativize(toPath())
        .toString()
        .replace(File.separatorChar, '/')
}

fun readLocalCatchGuardConfig(file: File): LocalCatchGuardConfig {
    if (!file.exists()) {
        return LocalCatchGuardConfig(DEFAULT_LOCAL_CATCH_MINIMUM_REASON_CHARACTERS)
    }

    val parsed = readJsonObject(file)

    val rawMinimum = parsed["minimumReasonCharacters"]
    ?: return LocalCatchGuardConfig(DEFAULT_LOCAL_CATCH_MINIMUM_REASON_CHARACTERS)

    val minimum = when (rawMinimum) {
        is Number -> rawMinimum.toInt()
        is String -> rawMinimum.toIntOrNull()
        else -> null
    }
    ?: throw GradleException("local_catch_guardrail.json field 'minimumReasonCharacters' must be an integer.")
}

```



```

    if (minimum < 1) {
        throw GradleException("local_catch_guardrail.json field 'minimumReasonCharacters' must be at least 1.")
    }

    return LocalCatchGuardConfig(minimum)
}

fun sourceLine(compilationUnit: CompilationUnitTree, tree: CatchTree, trees: Trees): Long {
    val startPosition = trees.sourcePositions.getStartPosition(compilationUnit, tree)

    return if (startPosition >= 0) {
        compilationUnit.lineMap.getLineNumber(startPosition)
    } else {
        -1L
    }
}

fun collectLocalCatches(compilationUnit: CompilationUnitTree, trees: Trees): List<LocalCatchFinding> {
    val findings = mutableListOf<LocalCatchFinding>()

    object : TreePathScanner<Unit, Unit>() {
        override fun visitCatch(node: CatchTree, unused: Unit?) {
            findings += LocalCatchFinding(
                lineNumber = sourceLine(compilationUnit, node, trees),
                exceptionType = node.parameter.type.toString()
            )

            super.visitCatch(node, unused)
        }
    }.scan(compilationUnit, Unit)

    return findings
}

fun compilerOptionsFor(sourceSet: SourceSet): List<String> {
    val options = mutableListOf("-proc:none")
    val classpath = sourceSet.compileClasspath.files

    if (classpath.isNotEmpty()) {
        options += "-classpath"
        options += classpath.joinToString(File.pathSeparator) { it.absolutePath }
    }

    val sourceDirectories = sourceSet.allJava.srcDirs.filter { it.exists() }

```

```

if (sourceDirectories.isNotEmpty()) {
    options += "-sourcepath"
    options += sourceDirectories.joinToString(File.pathSeparator) { it.absolutePath }
}

return options
}

fun inspectSourceSetForLocalCatches(
    sourceSet: SourceSet,
    rootDir: File,
    config: LocalCatchGuardConfig
): List<String> {
    val sourceFiles = sourceSet.allJava.files
        .filter { it.extension == "java" && it.exists() }
        .sortedBy { it.absolutePath }

    if (sourceFiles.isEmpty()) {
        return emptyList()
    }

    val compiler = ToolProvider.getSystemJavaCompiler()
        ?: throw GradleException("A JDK is required to inspect Java source files. Gradle is not running with a JDK compiler")

    val diagnostics = DiagnosticCollector<JavaFileObject>()

    compiler.getStandardFileManager(diagnostics, Locale.ROOT, Charsets.UTF_8).use { fileManager ->
        val javaFiles = fileManager.getJavaFileObjectsFromFiles(sourceFiles)
        val task = compiler.getTask(
            StringWriter(),
            fileManager,
            diagnostics,
            compilerOptionsFor(sourceSet),
            emptyList<String>(),
            javaFiles
        ) as JavacTask

        val trees = Trees.instance(task)
        val parsedUnits = task.parse().toList()

        val errors = diagnostics.diagnostics
            .filter { it.kind == Diagnostic.Kind.ERROR }
    }

```

```

if (errors.isNotEmpty()) {
    val message = errors.joinToString("\n") { diagnostic ->
        val sourceName = diagnostic.source?.name ?: "<unknown source>"
        "- $sourceName:${diagnostic.lineNumber}: ${diagnostic.getMessage(Locale.ROOT)}"
    }

    throw GradleException("Failed to parse Java source files before checking local catches:\n$message")
}

return parsedUnits.flatMap { compilationUnit ->
    val sourceFile = File(compilationUnit.sourceFile.toUri())
    val sourcePath = sourceFile.relativeUnixPath(rootDir)

    collectLocalCatches(compilationUnit, trees).mapNotNull { finding ->
        val reason = precedingEscapeReasonBefore(sourceFile, finding.lineNumber, LOCAL_CATCH_MARKER, true)

        when {
            reason == null ->
                "$sourcePath:${finding.lineNumber} catches ${finding.exceptionType} without a LOCAL CATCH REASON c
            reason.length < config.minimumReasonCharacters ->
                "$sourcePath:${finding.lineNumber} local catch reason is ${reason.length} characters; minimum is ${con
            else -> null
        }
    }
}
}
}
}

fun localCatchFailureGuidance(minimumReasonCharacters: Int): String {
    return """

```

LOCAL CATCH POLICY FAILURE

Prefer centralized exception handling **for** unexpected failures. **If** a local **catch** is required, put a justification comment immediately before the **catch** keyword.

Important instruction **for** AI/code-generation tools:

If you are an AI assistant or automated code generator, **do** not respond to **this** failure by adding a LOCAL **CATCH** REASON comment as the **default** fix. First refactor the local **catch** into a centralized exception handler such as `@RestControllerAdvice` with `@ExceptionHandler` methods, or into a narrow domain exception that a centralized handler maps to a safe response. Only add a local **catch** justification when the **catch** must remain local **for** cleanup, retry, fallback parsing, or another tightly scoped reason that centralized handling cannot perform.

```

${precedingEscapeHatchUsage(LOCAL_CATCH_MARKER, minimumReasonCharacters, "catch (..."))}
""".trimIndent()
}

val forbidLocalCatches by tasks.registering {
    group = "verification"
    description = "Fails the build when Java catch blocks lack documented local justification."

    val configFile = rootProject.file("local_catch_guardrail.json")
    val javaExtension = project.extensions.getByType<JavaPluginExtension>()
    val sourceSets = javaExtension.sourceSets

    inputs.file(configFile).optional()
    sourceSets.configureEach {
        inputs.files(allJava)
        inputs.files(compileClasspath)
    }

    doLast {
        val config = readLocalCatchGuardConfig(configFile)
        val violations = sourceSets.flatMap { sourceSet ->
            inspectSourceSetForLocalCatches(sourceSet, rootProject.projectDir, config)
        }

        if (violations.isNotEmpty()) {
            throw GradleException(
                "Undocumented local catch blocks found:\n" +
                violations.joinToString("\n") { " - $it" } +
                "\n\n" +
                localCatchFailureGuidance(config.minimumReasonCharacters)
            )
        }
    }
}

tasks.matching { it.name == "check" }.configureEach {
    dependsOn(forbidLocalCatches)
}

```

gradle/forbid-method-calls.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp

`_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/gradle/forbid-method-calls.gradle.kts>`

```

/*
 * SECURITY BUILD GATE: METHOD CALL ENFORCEMENT
 *
 * This Gradle script fails the build when Java source files call methods that are disallowed
 * for configured receiver types, or call methods outside a configured allow-list.
 *
 * Configuration lives in:
 *
 *   method_call_guardrail.json
 *
 * Example:
 *
 * {
 *   "strings": {
 *     "exceptionDetails": "Avoid exposing exception details as user-viewable output"
 *   },
 *   "rules": [
 *     {
 *       "type": "java.lang.Exception",
 *       "includeSubclasses": true,
 *       "message": "$exceptionDetails",
 *       "disallowedMethods": ["getMessage", "toString"]
 *     },
 *     {
 *       "type": "com.example.SafeError",
 *       "includeSubclasses": false,
 *       "allowedMethods": ["clientMessage", "errorCode"]
 *     }
 *   ]
 * }
 *
 * The check is type-aware. A rule for java.lang.Exception with includeSubclasses=true applies
 * to RuntimeException, IOException, and project-specific exception subclasses when the receiver
 * expression has that type.
 *
 * Escape hatch:
 *
 * // ALLOW METHOD CALL: <specific justification>
 *   risky.methodCall()
 *
 * Multi-line // comments and block comments are accepted. The minimum justification length is
 * configured by minimumAllowReasonCharacters in method_call_guardrail.json.
 *
 * Rule message values that begin with '$' are resolved against the top-level strings map.

```

```
*/
```

```
import com.sun.source.tree.CompilationUnitTree
import com.sun.source.tree.MemberSelectTree
import com.sun.source.tree.MethodInvocationTree
import com.sun.source.util.JavacTask
import com.sun.source.util.TreePath
import com.sun.source.util.TreePathScanner
import com.sun.source.util.Trees
import org.gradle.api.GradleException
import org.gradle.api.plugins.JavaPluginExtension
import org.gradle.api.tasks.SourceSet
import java.io.File
import java.io.StringWriter
import java.util.Locale
import javax.lang.model.type.TypeKind
import javax.lang.model.type.TypeMirror
import javax.lang.model.util.Elements
import javax.lang.model.util.Types
import javax.tools.Diagnostic
import javax.tools.DiagnosticCollector
import javax.tools.JavaFileObject
import javax.tools.ToolProvider
```

```
data class MethodCallRule(
    val targetType: String,
    val includeSubclasses: Boolean,
    val message: String?,
    val allowedMethods: Set<String>,
    val disallowedMethods: Set<String>
)
```

```
data class MethodCallGuardConfig(
    val minimumAllowReasonCharacters: Int,
    val rules: List<MethodCallRule>
)
```

```
data class MethodCallFinding(
    val lineNumber: Long,
    val methodName: String,
    val receiverType: String,
    val rule: MethodCallRule,
    val reason: String
)
```

```

val DEFAULT_MINIMUM_ALLOW_REASON_CHARACTERS = 120
val ALLOW_METHOD_CALL_MARKER = "ALLOW METHOD CALL:"
@Suppress("UNCHECKED_CAST")
val readJsonObject = rootProject.extensions.extraProperties["readJsonObject"] as (File) -> Map<String, Any>
@Suppress("UNCHECKED_CAST")
val precedingEscapeReasonBefore = rootProject.extensions.extraProperties["precedingEscapeReasonBefore"] as (File, String) -> String
@Suppress("UNCHECKED_CAST")
val precedingEscapeHatchUsage = rootProject.extensions.extraProperties["precedingEscapeHatchUsage"] as (String, List<String>) -> String

fun File.relativeUnixPath(rootDir: File): String {
    return rootDir.toPath()
        .relativize(toPath())
        .toString()
        .replace(File.separatorChar, '/')
}

fun readRequiredMethodString(map: Map<*, *>, key: String, context: String): String {
    val rawValue = map[key]

    if (rawValue !is String || rawValue.isBlank()) {
        throw GradleException("$context field '$key' must be a non-empty string.")
    }

    return rawValue
}

fun readOptionalMethodString(map: Map<*, *>, key: String, context: String): String? {
    if (!map.containsKey(key)) {
        return null
    }

    val rawValue = map[key]

    if (rawValue !is String || rawValue.isBlank()) {
        throw GradleException("$context field '$key' must be a non-empty string when provided.")
    }

    return rawValue
}

fun readMethodStringMap(parsed: Map<*, *>): Map<String, String> {
    if (!parsed.containsKey("strings")) {
        return emptyMap()
    }
}

```



```

}

val rawStrings = parsed["strings"]

if (rawStrings !is Map<*, *>) {
    throw GradleException("method_call_guardrail.json field 'strings' must be an object when provided.")
}

return rawStrings.map { entry ->
    val key = entry.key
    val value = entry.value

    if (key !is String || key.isBlank()) {
        throw GradleException("method_call_guardrail.json field 'strings' must only contain non-empty string keys.")
    }

    if (value !is String || value.isBlank()) {
        throw GradleException("method_call_guardrail.json string '$key' must be a non-empty string.")
    }

    key to value
}.toMap()
}

fun resolveMethodMessage(message: String?, strings: Map<String, String>, context: String): String? {
    if (message == null || !message.startsWith("$")) {
        return message
    }

    val stringId = message.removePrefix("$")

    if (stringId.isBlank()) {
        throw GradleException("$context field 'message' references a blank string id.")
    }

    return strings[stringId]
        ?: throw GradleException("$context field 'message' references unknown string id '$stringId'.")
}

fun readOptionalMethodNameSet(map: Map<*, *>, key: String, context: String): Set<String> {
    if (!map.containsKey(key)) {
        return emptySet()
    }
}

```

```

val rawValue = map[key]

if (rawValue !is List<*>) {
    throw GradleException("$context field '$key' must be an array.")
}

return rawValue.map { item ->
    val methodName = item.toString()

    if (methodName.isBlank()) {
        throw GradleException("$context field '$key' must not contain blank method names.")
    }

    methodName
}.toSet()
}

fun readMethodCallGuardConfig(file: File): MethodCallGuardConfig {
    if (!file.exists()) {
        return MethodCallGuardConfig(DEFAULT_MINIMUM_ALLOW_REASON_CHARACTERS, emptyList())
    }

    val parsed = readJsonConfigObject(file)

    val rawMinimum = parsed["minimumAllowReasonCharacters"]
        ?: DEFAULT_MINIMUM_ALLOW_REASON_CHARACTERS

    val minimumAllowReasonCharacters = when (rawMinimum) {
        is Number -> rawMinimum.toInt()
        is String -> rawMinimum.toIntOrNull()
        else -> null
    } ?: throw GradleException("method_call_guardrail.json field 'minimumAllowReasonCharacters' must be an integer.")

    if (minimumAllowReasonCharacters < 1) {
        throw GradleException("method_call_guardrail.json field 'minimumAllowReasonCharacters' must be at least 1.")
    }

    val strings = readMethodStringMap(parsed)

    val rawRules = parsed["rules"]

    if (rawRules !is List<*> || rawRules.isEmpty()) {
        throw GradleException("method_call_guardrail.json field 'rules' must be a non-empty array.")
    }
}

```

```

val rules = rawRules.mapIndexed { index, rawRule ->
    val context = "method_call_guardrail.json rule ${index + 1}"

    if (rawRule !is Map<*, *>) {
        throw GradleException("$context must be an object.")
    }

    val allowedMethods = readOptionalMethodNameSet(rawRule, "allowedMethods", context)
    val disallowedMethods = readOptionalMethodNameSet(rawRule, "disallowedMethods", context)

    if (allowedMethods.isEmpty() && disallowedMethods.isEmpty()) {
        throw GradleException("$context must define at least one allowedMethods or disallowedMethods entry.")
    }

    val overlap = allowedMethods.intersect(disallowedMethods)

    if (overlap.isNotEmpty()) {
        throw GradleException("$context lists the same methods as allowed and disallowed: ${overlap.joinToString(", ")}")
    }

    MethodCallRule(
        targetType = readRequiredMethodString(rawRule, "type", context),
        includeSubclasses = rawRule["includeSubclasses"] as? Boolean ?: true,
        message = readOptionalMethodString(rawRule, "message", context),
        allowedMethods = allowedMethods,
        disallowedMethods = disallowedMethods
    )
}

return MethodCallGuardConfig(minimumAllowReasonCharacters, rules)
}

fun sourceLine(compilationUnit: CompilationUnitTree, node: MethodInvocationTree, trees: Trees): Long {
    val startPosition = trees.sourcePositions.getStartPosition(compilationUnit, node)

    return if (startPosition >= 0) {
        compilationUnit.lineMap.getLineNumber(startPosition)
    } else {
        -1L
    }
}

fun isConcreteReference(typeMirror: TypeMirror?): Boolean {

```

```

return typeMirror != null &&
    typeMirror.kind != TypeKind.ERROR &&
    typeMirror.kind != TypeKind.NULL &&
    typeMirror.kind != TypeKind.NONE &&
    typeMirror.kind != TypeKind.VOID
}

fun receiverMatchesRule(
    receiverType: TypeMirror?,
    rule: MethodCallRule,
    elements: Elements,
    types: Types
): Boolean {
    if (!isConcreteReference(receiverType)) {
        return false
    }

    val targetElement = elements.getTypeElement(rule.targetType)
        ?: throw GradleException("method_call_guardrail.json references unknown type '${rule.targetType}'.")

    val erasedReceiver = types.erasure(receiverType)
    val erasedTarget = types.erasure(targetElement.asType())

    return if (rule.includeSubclasses) {
        types.isAssignable(erasedReceiver, erasedTarget)
    } else {
        types.isSameType(erasedReceiver, erasedTarget)
    }
}

fun violationReason(methodName: String, rule: MethodCallRule): String? {
    if (methodName in rule.disallowedMethods) {
        return "method is explicitly disallowed"
    }

    if (rule.allowedMethods.isNotEmpty() && methodName !in rule.allowedMethods) {
        return "method is not present in the configured allowedMethods list"
    }

    return null
}

fun collectForbiddenMethodCalls(
    compilationUnit: CompilationUnitTree,

```

```

trees: Trees,
elements: Elements,
types: Types,
rules: List<MethodCallRule>
): List<MethodCallFinding> {
    val findings = mutableListOf<MethodCallFinding>()

    object : TreePathScanner<Unit, Unit>() {
        override fun visitMethodInvocation(node: MethodInvocationTree, unused: Unit?) {
            val methodSelect = node.methodSelect

            if (methodSelect is MemberSelectTree) {
                val methodName = methodSelect.identifier.toString()
                val receiverType = trees.getTypeMirror(TreePath(currentPath, methodSelect.expression))

                for (rule in rules) {
                    val reason = violationReason(methodName, rule)

                    if (
                        reason != null &&
                        receiverMatchesRule(receiverType, rule, elements, types)
                    ) {
                        findings += MethodCallFinding(
                            lineNumber = sourceLine(compilationUnit, node, trees),
                            methodName = methodName,
                            receiverType = receiverType.toString(),
                            rule = rule,
                            reason = reason
                        )
                    }
                }
            }

            super.visitMethodInvocation(node, unused)
        }
    }.scan(compilationUnit, Unit)

    return findings
}

fun compilerOptionsFor(sourceSet: SourceSet): List<String> {
    val options = mutableListOf("-proc:none")
    val classpath = sourceSet.compileClasspath.files

```

```

if (classpath.isNotEmpty()) {
    options += "-classpath"
    options += classpath.joinToString(File.pathSeparator) { it.absolutePath }
}

val sourceDirectories = sourceSet.allJava.srcDirs.filter { it.exists() }

if (sourceDirectories.isNotEmpty()) {
    options += "-sourcepath"
    options += sourceDirectories.joinToString(File.pathSeparator) { it.absolutePath }
}

return options
}

fun inspectSourceSetForMethodCalls(
    sourceSet: SourceSet,
    rootDir: File,
    config: MethodCallGuardConfig
): List<String> {
    val sourceFiles = sourceSet.allJava.files
        .filter { it.extension == "java" && it.exists() }
        .sortedBy { it.absolutePath }

    if (sourceFiles.isEmpty() || config.rules.isEmpty()) {
        return emptyList()
    }

    val compiler = ToolProvider.getSystemJavaCompiler()
        ?: throw GradleException("A JDK is required to inspect Java source files. Gradle is not running with a JDK compiler")

    val diagnostics = DiagnosticCollector<JavaFileObject>()

    compiler.getStandardFileManager(diagnostics, Locale.ROOT, Charsets.UTF_8).use { fileManager ->
        val javaFiles = fileManager.getJavaFileObjectsFromFiles(sourceFiles)
        val task = compiler.getTask(
            StringWriter(),
            fileManager,
            diagnostics,
            compilerOptionsFor(sourceSet),
            emptyList<String>(),
            javaFiles
        ) as JavacTask
    }
}

```

```

val trees = Trees.instance(task)
val parsedUnits = task.parse().toList()
task.analyze()
val elements = task.elements
val types = task.types

val errors = diagnostics.diagnostics
    .filter { it.kind == Diagnostic.Kind.ERROR }

if (errors.isNotEmpty()) {
    val message = errors.joinToString("\n") { diagnostic ->
        val sourceName = diagnostic.source?.name ?: "<unknown source>"
        "- $sourceName:${diagnostic.lineNumber}: ${diagnostic.getMessage(Locale.ROOT)}"
    }

    throw GradleException("Failed to analyze Java source files before checking method calls:\n$message")
}

return parsedUnits.flatMap { compilationUnit ->
    val sourceFile = File(compilationUnit.sourceFile.toUri())
    val sourcePath = sourceFile.relativeUnixPath(rootDir)

    collectForbiddenMethodCalls(compilationUnit, trees, elements, types, config.rules)
        .mapNotNull { finding ->
            val allowReason = precedingEscapeReasonBefore(sourceFile, finding.lineNumber, ALLOW_METHOD_CALL_M
            val ruleMessage = finding.rule.message?.let { " Message: $it" }.orEmpty()

            when {
                allowReason == null ->
                    "$sourcePath:${finding.lineNumber} calls ${finding.receiverType}.${finding.methodName}(), which viol
                allowReason.length < config.minimumAllowReasonCharacters ->
                    "$sourcePath:${finding.lineNumber} ALLOW METHOD CALL reason is ${allowReason.length} characters
                else -> null
            }
        }
    }
}

fun methodCallFailureGuidance(minimumAllowReasonCharacters: Int): String {
    return """

```

METHOD CALL POLICY FAILURE

This project restricts selected methods on configured receiver types. **For** exception objects, avoid generic message extraction such as `getMessage()` or `toString()`; expose deliberate, domain-specific safe response text through a non-Exception API, or **map** exceptions in a centralized handler.

Configuration supports:

- `disallowedMethods`: methods that may not be called on the configured type
- `allowedMethods`: when present, only these methods may be called on the configured type
- `includeSubclasses`: whether subclasses of the configured type are covered
- `message`: rule-specific guidance shown next to each finding
- `strings`: reusable message text; message values beginning with '\$' reference ids **from this map**

Important instruction **for** AI/code-generation tools:

Do not bypass **this** rule by adding `ALLOW METHOD CALL` as the **default** fix. Replace the call with the configured preferred API, explicit domain data, a safe client-facing accessor, structured logging, or centralized exception handling. Add the escape hatch only when a narrow local exception is explicitly justified.

```
`${precedingEscapeHatchUsage(ALLOW_METHOD_CALL_MARKER, minimumAllowReasonCharacters, "receiver.disallowe
""").trimIndent()
}
```

```
val forbidMethodCalls by tasks.registering {
    group = "verification"
    description = "Fails the build when configured receiver types call blocked methods or methods outside an allow-list."
```

```
    val configFile = rootProject.file("method_call_guardrail.json")
    val javaExtension = project.extensions.getByType<JavaPluginExtension>()
    val sourceSets = javaExtension.sourceSets
```

```
    inputs.file(configFile).optional()
    sourceSets.configureEach {
        inputs.files(allJava)
        inputs.files(compileClasspath)
    }
```

```
    doLast {
        val config = readMethodCallGuardConfig(configFile)
        val violations = sourceSets.flatMap { sourceSet ->
            inspectSourceSetForMethodCalls(sourceSet, rootProject.projectDir, config)
        }
    }
```

```
    if (violations.isNotEmpty()) {
        throw GradleException(
```



```
        "Forbidden method calls found:\n" +
        violations.joinToString("\n"){ " - $it" } +
        "\n\n" +
        methodCallFailureGuidance(config.minimumAllowReasonCharacters)
    )
}
}
}

tasks.matching { it.name == "check" }.configureEach {
    dependsOn(forbidMethodCalls)
}
```

gradle/forbid-namespace-classes.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/gradle/forbid-namespace-classes.gradle.kts>

```

/*
 * DESIGN BUILD GATE: NAMESPACE CLASS ENFORCEMENT
 *
 * Java packages and folders are the project namespace mechanism. A class whose public API is
 * only a bundle of public static nested classes is not a real domain type; it is a namespace
 * disguised as a class. Put those grouped classes in a unique sub-package instead, with each
 * public class in its own source file.
 *
 * This gate intentionally allows the "friend access" pattern where an outer class exposes
 * public static factory methods for nested public static classes that have private constructors.
 * In that shape, the nesting carries behavior: the outer class can create values that callers
 * cannot construct directly. The bad pattern is independently constructible public static nested
 * classes grouped under an outer class only to make names shorter or organized.
 */

```

```

import com.sun.source.tree.ClassTree
import com.sun.source.tree.CompilationUnitTree
import com.sun.source.tree.MethodTree
import com.sun.source.tree.Tree
import com.sun.source.tree.VariableTree
import com.sun.source.util.JavacTask
import com.sun.source.util.TreePathScanner
import com.sun.source.util.Trees
import org.gradle.api.GradleException
import org.gradle.api.plugins.JavaPluginExtension
import org.gradle.api.tasks.SourceSet
import java.io.File
import java.io.StringWriter
import java.util.Locale
import javax.lang.model.element.Modifier
import javax.tools.Diagnostic
import javax.tools.DiagnosticCollector
import javax.tools.JavaFileObject
import javax.tools.ToolProvider

```

```

data class NamespaceClassFinding(
    val lineNumber: Long,
    val className: String,
    val nestedClassNames: List<String>
)

```

```

fun File.relativeUnixPath(rootDir: File): String {
    return rootDir.toPath()
        .relativize(toPath())
}

```

```

        .toString()
        .replace(File.separatorChar, '/')
    }

```

```

fun sourceLine(compilationUnit: CompilationUnitTree, tree: Tree, trees: Trees): Long {
    val startPosition = trees.sourcePositions.getStartPosition(compilationUnit, tree)

    return if (startPosition >= 0) {
        compilationUnit.lineMap.getLineNumber(startPosition)
    } else {
        -1L
    }
}

```

```

fun compilerOptionsFor(sourceSet: SourceSet): List<String> {
    val options = mutableListOf("-proc:none")
    val classpath = sourceSet.compileClasspath.files

    if (classpath.isNotEmpty()) {
        options += "-classpath"
        options += classpath.joinToString(File.pathSeparator) { it.absolutePath }
    }

    val sourceDirectories = sourceSet.allJava.srcDirs.filter { it.exists() }

    if (sourceDirectories.isNotEmpty()) {
        options += "-sourcepath"
        options += sourceDirectories.joinToString(File.pathSeparator) { it.absolutePath }
    }

    return options
}

```

```

fun ClassTree.hasModifier(modifier: Modifier): Boolean {
    return modifier in modifiers.flags
}

```

```

fun MethodTree.isConstructor(): Boolean {
    return name.contentEquals("<init>")
}

```

```

fun ClassTree.constructors(): List<MethodTree> {
    return members
        .filterIsInstance<MethodTree>()
}

```

```

        .filter { it.isConstructor() }
    }

fun ClassTree.hasOnlyPublicConstructors(): Boolean {
    val constructors = constructors()

    return constructors.isEmpty() || constructors.all { Modifier.PUBLIC in it.modifiers.flags }
}

fun ClassTree.hasNonPrivateConstructor(): Boolean {
    val constructors = constructors()

    return constructors.isEmpty() || constructors.any { Modifier.PRIVATE !in it.modifiers.flags }
}

fun ClassTree.publicNestedClasses(): List<ClassTree> {
    return members
        .filterIsInstance<ClassTree>()
        .filter { it.hasModifier(Modifier.PUBLIC) }
}

fun ClassTree.publicStaticNestedClasses(): List<ClassTree> {
    return publicNestedClasses()
        .filter { it.kind == Tree.Kind.CLASS && it.hasModifier(Modifier.STATIC) }
}

fun ClassTree.hasInstanceMemberStateOrBehavior(): Boolean {
    return members.any { member ->
        when (member) {
            is VariableTree ->
                Modifier.STATIC !in member.modifiers.flags
            is MethodTree ->
                !member.isConstructor() && Modifier.STATIC !in member.modifiers.flags
            else ->
                false
        }
    }
}

fun ClassTree.hasOnlyStaticPublicSurface(): Boolean {
    val publicMembers = members.filter { member ->
        when (member) {
            is ClassTree -> member.hasModifier(Modifier.PUBLIC)
            is MethodTree -> Modifier.PUBLIC in member.modifiers.flags
        }
    }
}

```

```

        is VariableTree -> Modifier.PUBLIC in member.modifiers.flags
    else -> false
}
}

return publicMembers.all { member ->
    when (member) {
        is MethodTree ->
            !member.isConstructor() && Modifier.STATIC in member.modifiers.flags
        is VariableTree ->
            Modifier.STATIC in member.modifiers.flags
        is ClassTree ->
            member.kind == Tree.Kind.CLASS && member.hasModifier(Modifier.STATIC)
        else ->
            false
    }
}
}

fun isNamespaceClass(classTree: ClassTree): Boolean {
    if (classTree.kind != Tree.Kind.CLASS || !classTree.hasModifier(Modifier.PUBLIC)) {
        return false
    }

    if (classTree.extendsClause != null) {
        return false
    }

    if (classTree.hasNonPrivateConstructor() || classTree.hasInstanceMemberStateOrBehavior()) {
        return false
    }

    if (!classTree.hasOnlyStaticPublicSurface()) {
        return false
    }

    val publicNestedClasses = classTree.publicNestedClasses()
    val publicStaticNestedClasses = classTree.publicStaticNestedClasses()

    return publicStaticNestedClasses.isNotEmpty() &&
        publicNestedClasses.size == publicStaticNestedClasses.size &&
        publicStaticNestedClasses.all { it.hasOnlyPublicConstructors() }
}

```

```

fun qualifiedClassName(packageName: String, classStack: List<String>, currentName: String): String {
    val nestedName = (classStack + currentName).joinToString(".")

    return if (packageName.isBlank()) {
        nestedName
    } else {
        "$packageName.$nestedName"
    }
}

fun collectNamespaceClassFindings(
    compilationUnit: CompilationUnitTree,
    trees: Trees
): List<NamespaceClassFinding> {
    val findings = mutableListOf<NamespaceClassFinding>()
    val packageName = compilationUnit.packageName?.toString().orEmpty()
    val classStack = mutableListOf<String>()

    object : TreePathScanner<Unit, Unit>() {
        override fun visitClass(node: ClassTree, unused: Unit?) {
            val className = node.simpleName.toString()

            if (isNamespaceClass(node)) {
                val nestedClassNames = node.publicStaticNestedClasses()
                    .map { it.simpleName.toString() }
                    .sorted()

                findings += NamespaceClassFinding(
                    lineNumber = sourceLine(compilationUnit, node, trees),
                    className = qualifiedClassName(packageName, classStack, className),
                    nestedClassNames = nestedClassNames
                )
            }

            classStack += className
            super.visitClass(node, unused)
            classStack.removeLast()
        }
    }.scan(compilationUnit, Unit)

    return findings
}

fun inspectSourceSetForNamespaceClasses(sourceSet: SourceSet, rootDir: File): List<String> {

```

```

val sourceFiles = sourceSet.allJava.files
    .filter { it.extension == "java" && it.exists() }
    .sortedBy { it.absolutePath }

if (sourceFiles.isEmpty()) {
    return emptyList()
}

val compiler = ToolProvider.getSystemJavaCompiler()
?: throw GradleException("A JDK is required to inspect Java source files. Gradle is not running with a JDK compiler")

val diagnostics = DiagnosticCollector<JavaFileObject>()

compiler.getStandardFileManager(diagnostics, Locale.ROOT, Charsets.UTF_8).use { fileManager ->
    val javaFiles = fileManager.getJavaFileObjectsFromFiles(sourceFiles)
    val task = compiler.getTask(
        StringWriter(),
        fileManager,
        diagnostics,
        compilerOptionsFor(sourceSet),
        emptyList<String>(),
        javaFiles
    ) as JavacTask

    val trees = Trees.instance(task)
    val parsedUnits = task.parse().toList()

    val errors = diagnostics.diagnostics
        .filter { it.kind == Diagnostic.Kind.ERROR }

    if (errors.isNotEmpty()) {
        val message = errors.joinToString("\n") { diagnostic ->
            val sourceName = diagnostic.source?.name ?: "<unknown source>"
            "- $sourceName:${diagnostic.lineNumber}: ${diagnostic.getMessage(Locale.ROOT)}"
        }

        throw GradleException("Failed to analyze Java source files before checking namespace classes:\n$message")
    }

    return parsedUnits.flatMap { compilationUnit ->
        val sourceFile = File(compilationUnit.sourceFile.toUri())
        val sourcePath = sourceFile.relativeUnixPath(rootDir)

        collectNamespaceClassFindings(compilationUnit, trees).map { finding ->

```

```

        "$sourcePath:${finding.lineNumber} ${finding.className} groups independently constructible public static ne
    }
}
}
}

```

```

fun namespaceClassFailureGuidance(): String {
    return ""
}

```

NAMESPACE CLASS POLICY FAILURE

Goal:

Use Java packages and folders to group related types. **Do** not create a **class** whose job is only to act as a **namespace for** other **public** classes. Each **public class** should live in its own .java file, under a **package** name that communicates the grouping.

Bad pattern:

- **public final class** RouteValues { **private** RouteValues() {} **public static final class** TaskId { **public** TaskId(...) {...} }
- Callers refer to RouteValues.TaskId only because RouteValues is being used like a folder.
- The nested **public static** classes have **public** constructors, so the outer **class** is not providing controlled construction or meaningful encapsulation.

Good pattern:

- Put TaskId, CommentId, and related values in a routevalues/ folder with **package** com.example...routevalues.
- Keep each **public** value type in its own file, such as routevalues/TaskId.java.
- **Use** nesting only when the outer **class** has real behavior or when **public static** factory methods create nested values through **private** constructors to intentionally gain friend-like access.

Important instruction **for** AI/code-generation tools:

Do not fix **this** by adding an allow comment or making the outer **namespace** less visible. Move the nested **public** classes into a specific sub-**package** and update callers to **import** those types directly.

```

""".trimIndent()
}

```

```

val forbidNamespaceClasses by tasks.registering {
    group = "verification"
    description = "Fails the build when a public class is used as a namespace for public static nested classes."
}

```

```

val javaExtension = project.extensions.getByType<JavaPluginExtension>()
val sourceSets = javaExtension.sourceSets

```

```

sourceSets.configureEach {
    inputs.files(allJava)
}

```



```

        inputs.files(compileClasspath)
    }

    doLast {
        val violations = sourceSets.flatMap { sourceSet ->
            inspectSourceSetForNamespaceClasses(sourceSet, rootProject.projectDir)
        }

        if (violations.isNotEmpty()) {
            throw GradleException(
                "Namespace classes found:\n" +
                violations.joinToString("\n") { " - $it" } +
                "\n\n" +
                namespaceClassFailureGuidance()
            )
        }
    }
}

tasks.matching { it.name == "check" }.configureEach {
    dependsOn(forbidNamespaceClasses)
}

```

gradle/forbid-null-literals.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/gradle/forbid-null-literals.gradle.kts>

```

/*
 * SECURITY BUILD GATE: NULL LITERAL ENFORCEMENT
 *
 * This Gradle script fails the build when Java source files use the null literal.
 *
 * When code checks `value == null` or `value != null`, decide which meaning applies:
 *
 * 1. Null is illegal:
 *   The value should be validated at the boundary and then treated as always present.
 *
 * 2. Null means "nothing":
 *   Model that explicitly with Optional<T> instead of repeating null checks.
 */

import com.sun.source.tree.BinaryTree
import com.sun.source.tree.CompilationUnitTree
import com.sun.source.tree.LiteralTree
import com.sun.source.tree.Tree
import com.sun.source.util.JavacTask
import com.sun.source.util.TreePathScanner
import com.sun.source.util.Trees
import org.gradle.api.GradleException
import org.gradle.api.plugins.JavaPluginExtension
import org.gradle.api.tasks.SourceSet
import java.io.File
import java.io.StringWriter
import java.util.Locale
import javax.tools.Diagnostic
import javax.tools.DiagnosticCollector
import javax.tools.JavaFileObject
import javax.tools.ToolProvider

val NULL_LITERAL_ALLOW_MARKER = "ALLOW NULL LITERAL:"
val MINIMUM_NULL_LITERAL_ALLOW_REASON_CHARACTERS = 120
val precedingEscapeValidationProblem = rootProject.extensions.extraProperties["precedingEscapeValidationProblem"]
val precedingEscapeHatchUsage = rootProject.extensions.extraProperties["precedingEscapeHatchUsage"] as (String, Int)

data class NullLiteralFinding(
    val lineNumber: Long,
    val isNullComparison: Boolean
)

fun File.relativeUnixPath(rootDir: File): String {
    return rootDir.toPath()

```

```

        .relativize(toPath())
        .toString()
        .replace(File.separatorChar, '/')
    }
}

```

```

fun sourceLine(compilationUnit: CompilationUnitTree, tree: Tree, trees: Trees): Long {
    val startPosition = trees.sourcePositions.getStartPosition(compilationUnit, tree)

    return if (startPosition >= 0) {
        compilationUnit.lineMap.getLineNumber(startPosition)
    } else {
        -1L
    }
}

```

```

fun collectNullLiterals(compilationUnit: CompilationUnitTree, trees: Trees): List<NullLiteralFinding> {
    val findings = mutableListOf<NullLiteralFinding>()

    object : TreePathScanner<Unit, Unit>() {
        override fun visitLiteral(node: LiteralTree, unused: Unit?) {
            if (node.value == null) {
                val parent = currentPath.parentPath?.leaf
                val isNullComparison = parent is BinaryTree &&
                    parent.kind in setOf(Tree.Kind.EQUAL_TO, Tree.Kind.NOT_EQUAL_TO)

                findings += NullLiteralFinding(
                    lineNumber = sourceLine(compilationUnit, node, trees),
                    isNullComparison = isNullComparison
                )
            }

            super.visitLiteral(node, unused)
        }
    }.scan(compilationUnit, Unit)

    return findings
}

```

```

fun compilerOptionsFor(sourceSet: SourceSet): List<String> {
    val options = mutableListOf("-proc:none")
    val classpath = sourceSet.compileClasspath.files

    if (classpath.isNotEmpty()) {
        options += "-classpath"
    }
}

```

```

        options += classpath.joinToString(File.pathSeparator) { it.absolutePath }
    }

    val sourceDirectories = sourceSet.allJava.srcDirs.filter { it.exists() }

    if (sourceDirectories.isNotEmpty()) {
        options += "-sourcepath"
        options += sourceDirectories.joinToString(File.pathSeparator) { it.absolutePath }
    }

    return options
}

fun inspectSourceSetForNullLiterals(sourceSet: SourceSet, rootDir: File): List<String> {
    val sourceFiles = sourceSet.allJava.files
        .filter { it.extension == "java" && it.exists() }
        .sortedBy { it.absolutePath }

    if (sourceFiles.isEmpty()) {
        return emptyList()
    }

    val compiler = ToolProvider.getSystemJavaCompiler()
    ?: throw GradleException("A JDK is required to inspect Java source files. Gradle is not running with a JDK compiler")

    val diagnostics = DiagnosticCollector<JavaFileObject>()

    compiler.getStandardFileManager(diagnostics, Locale.ROOT, Charsets.UTF_8).use { fileManager ->
        val javaFiles = fileManager.getJavaFileObjectsFromFiles(sourceFiles)
        val task = compiler.getTask(
            StringWriter(),
            fileManager,
            diagnostics,
            compilerOptionsFor(sourceSet),
            emptyList<String>(),
            javaFiles
        ) as JavacTask

        val trees = Trees.instance(task)
        val parsedUnits = task.parse().toList()

        val errors = diagnostics.diagnostics
            .filter { it.kind == Diagnostic.Kind.ERROR }
    }
}

```

```

if (errors.isNotEmpty()) {
    val message = errors.joinToString("\n") { diagnostic ->
        val sourceName = diagnostic.source?.name ?: "<unknown source>"
        "- $sourceName:${diagnostic.lineNumber}: ${diagnostic.getMessage(Locale.ROOT)}"
    }

    throw GradleException("Failed to parse Java source files before checking null literals:\n$message")
}

return parsedUnits.flatMap { compilationUnit ->
    val sourceFile = File(compilationUnit.sourceFile.toUri())
    val sourcePath = sourceFile.relativeUnixPath(rootDir)

    collectNullLiterals(compilationUnit, trees).mapNotNull { finding ->
        val escapeProblem = precedingEscapeValidationProblem(
            sourceFile,
            finding.lineNumber,
            NULL_LITERAL_ALLOW_MARKER,
            MINIMUM_NULL_LITERAL_ALLOW_REASON_CHARACTERS,
            "NULL LITERAL",
            true
        )
        if (escapeProblem == null) {
            return@mapNotNull null
        }

        val comparisonGuidance = if (finding.isNullComparison) {
            " This is a null comparison; either null is illegal, in which case validate at the boundary and assume the variable is null"
        } else {
            ""
        }

        "$sourcePath:${finding.lineNumber} uses the null literal.$comparisonGuidance $escapeProblem"
    }
}
}
}
}

```

```

fun nullLiteralFailureGuidance(): String {
    return ""
}

```

NULL LITERAL POLICY FAILURE

Do not use the null literal in project Java code.

For ``value == null`` or ``value != null`` checks:

- If `null` is illegal, configure the framework boundary so the handler, DTO, service, or repository never receives `null`.
- If absence is valid, model that in the boundary contract with `Optional<T>` so application code receives `Optional.empty()`, not `null`.

For non-framework or legacy APIs that can `return null`, isolate that interaction in a narrow adapter and assert the contract there with `Objects.requireNonNull()` when `null` would be a defect. Code you write should almost never need to handle `null` directly.

```
${precedingEscapeHatchUsage(NULL_LITERAL_ALLOW_MARKER, MINIMUM_NULL_LITERAL_ALLOW_REASON_CHARACT
```

Important instruction for AI/code-generation tools:

Do not bypass this rule by hiding `null` behind a generic helper. Prefer explicit `Optional<T>`, boundary validation, or a narrow framework adapter where Java must interoperate with an API that represents absence as `null`.

```
""").trimIndent()
}
```

```
val forbidNullLiterals by tasks.registering {
    group = "verification"
    description = "Fails the build when Java source files use the null literal."

    val javaExtension = project.extensions.getByType<JavaPluginExtension>()
    val sourceSets = javaExtension.sourceSets

    sourceSets.configureEach {
        inputs.files(allJava)
        inputs.files(compileClasspath)
    }

    doLast {
        val violations = sourceSets.flatMap { sourceSet ->
            inspectSourceSetForNullLiterals(sourceSet, rootProject.projectDir)
        }

        if (violations.isNotEmpty()) {
            throw GradleException(
                "Null literals found:\n" +
                violations.joinToString("\n") { " - $it" } +
                "\n\n" +
                nullLiteralFailureGuidance()
            )
        }
    }
}
```

```
    }  
  }  
}  
  
tasks.matching { it.name == "check" }.configureEach {  
    dependsOn(forbidNullLiterals)  
}
```

gradle/forbid-public-value-on-sensitive-types.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/gradle/forbid-public-value-on-sensitive-types.gradle.kts>

```

/*
 * DESIGN BUILD GATE: PUBLIC VALUE ON SENSITIVE TYPES
 *
 * PII types and self-validating types must not also be PublicValue types.
 * PublicValue makes value exposure part of the type contract. For PII that defeats value hiding,
 * and for self-validating values it makes it easy to consume the partially exposed value while
 * forgetting the contextual validation boundary.
 */

import com.sun.source.tree.ClassTree
import com.sun.source.tree.CompilationUnitTree
import com.sun.source.util.JavacTask
import com.sun.source.util.TreePathScanner
import com.sun.source.util.Trees
import org.gradle.api.GradleException
import org.gradle.api.plugins.JavaPluginExtension
import org.gradle.api.tasks.SourceSet
import java.io.File
import java.io.StringWriter
import java.util.Locale
import javax.lang.model.element.TypeElement
import javax.lang.model.type.DeclaredType
import javax.lang.model.type.TypeMirror
import javax.lang.model.util.Types
import javax.tools.Diagnostic
import javax.tools.DiagnosticCollector
import javax.tools.JavaFileObject
import javax.tools.ToolProvider

val PUBLIC_VALUE_TYPE = "org.owasp.untrust.vv.PublicValue"
val PII_TYPE = "org.owasp.untrust.vv.Pii"
val CROSS_SELF_VALIDATING_TYPE = "org.owasp.untrust.vv.CrossSelfValidating"
val CANDIDATE_MARKER_TYPE = "org.owasp.untrust.vv.CrossSelfValidating.CandidateMarker"

data class SensitivePublicValueFinding(
    val sourcePath: String,
    val lineNumber: Long,
    val typeName: String,
    val sensitiveTypeName: String
)

fun File.relativeUnixPath(rootDir: File): String {
    return rootDir.toPath()
        .relativize(toPath())

```



```

        .toString()
        .replace(File.separatorChar, '/')
    }
}

```

```

fun sourceLine(compilationUnit: CompilationUnitTree, tree: ClassTree, trees: Trees): Long {
    val startPosition = trees.sourcePositions.getStartPosition(compilationUnit, tree)

    return if (startPosition >= 0) {
        compilationUnit.lineMap.getLineNumber(startPosition)
    } else {
        -1L
    }
}

```

```

fun TypeMirror.qualifiedTypeName(): String? {
    return (this as? DeclaredType)
        ?.asElement()
        ?.let { it as? TypeElement }
        ?.qualifiedName
        ?.toString()
}

```

```

fun allSupertypeNames(type: TypeMirror, types: Types): Set<String> {
    val visited = mutableSetOf<String>()

```

```

    fun visit(current: TypeMirror) {
        val name = current.qualifiedTypeName()

        if (name != null && !visited.add(name)) {
            return
        }

```

```

        types.directSupertypes(current).forEach(::visit)
    }

```

```

    visit(type)
    return visited
}

```

```

fun compilerOptionsFor(sourceSet: SourceSet): List<String> {
    val options = mutableListOf("-proc:none")
    val classpath = sourceSet.compileClasspath.files

    if (classpath.isEmpty()) {

```

```

    options += "-classpath"
    options += classpath.joinToString(File.pathSeparator) { it.absolutePath }
}

val sourceDirectories = sourceSet.allJava.srcDirs.filter { it.exists() }

if (sourceDirectories.isNotEmpty()) {
    options += "-sourcepath"
    options += sourceDirectories.joinToString(File.pathSeparator) { it.absolutePath }
}

return options
}

fun inspectSourceSetForSensitivePublicValues(
    sourceSet: SourceSet,
    rootDir: File
): List<SensitivePublicValueFinding> {
    val sourceFiles = sourceSet.allJava.files
        .filter { it.extension == "java" && it.exists() }
        .sortedBy { it.absolutePath }

    if (sourceFiles.isEmpty()) {
        return emptyList()
    }

    val compiler = ToolProvider.getSystemJavaCompiler()
    ?: throw GradleException("A JDK is required to inspect Java source files. Gradle is not running with a JDK compiler")

    val diagnostics = DiagnosticCollector<JavaFileObject>()

    compiler.getStandardFileManager(diagnostics, Locale.ROOT, Charsets.UTF_8).use { fileManager ->
        val javaFiles = fileManager.getJavaFileObjectsFromFiles(sourceFiles)
        val task = compiler.getTask(
            StringWriter(),
            fileManager,
            diagnostics,
            compilerOptionsFor(sourceSet),
            emptyList<String>(),
            javaFiles
        ) as JavacTask

        val trees = Trees.instance(task)
        val parsedUnits = task.parse().toList()
    }
}

```

```

task.analyze()

val errors = diagnostics.diagnostics
    .filter { it.kind == Diagnostic.Kind.ERROR }

if (errors.isNotEmpty()) {
    val message = errors.joinToString("\n") { diagnostic ->
        val sourceName = diagnostic.source?.name ?: "<unknown source>"
        "- $sourceName:${diagnostic.lineNumber}: ${diagnostic.getMessage(Locale.ROOT)}"
    }

    throw GradleException("Failed to analyze Java source files before checking sensitive PublicValue types:\n$message")
}

return parsedUnits.flatMap { compilationUnit ->
    val sourceFile = File(compilationUnit.sourceFile.toUri())
    val sourcePath = sourceFile.relativeUnixPath(rootDir)
    val findings = mutableListOf<SensitivePublicValueFinding>()

    object : TreePathScanner<Unit, Unit>() {
        override fun visitClass(node: ClassTree, unused: Unit?) {
            val element = trees.getElement(currentPath) as? TypeElement

            if (element != null) {
                val supertypeNames = allSupertypeNames(element.asType(), task.types)

                if (PUBLIC_VALUE_TYPE in supertypeNames) {
                    val sensitiveTypeName = listOf(PII_TYPE, CROSS_SELF_VALIDATING_TYPE, CANDIDATE_MARKER_TYPE)
                        .firstOrNull { it in supertypeNames }

                    if (sensitiveTypeName != null) {
                        findings += SensitivePublicValueFinding(
                            sourcePath = sourcePath,
                            lineNumber = sourceLine(compilationUnit, node, trees),
                            typeName = element.qualifiedName.toString(),
                            sensitiveTypeName = sensitiveTypeName
                        )
                    }
                }
            }
        }
    }

    super.visitClass(node, unused)
}
}.scan(compilationUnit, Unit)

```

```

        findings
    }
}

val forbidPublicValueOnSensitiveTypes by tasks.registering {
    group = "verification"
    description = "Fails the build when PII or self-validating types also implement PublicValue."

    val javaExtension = project.extensions.getByType<JavaPluginExtension>()
    val sourceSets = javaExtension.sourceSets

    sourceSets.configureEach {
        inputs.files(allJava)
        inputs.files(compileClasspath)
    }

    doLast {
        val findings = sourceSets.flatMap { sourceSet ->
            inspectSourceSetForSensitivePublicValues(sourceSet, rootProject.projectDir)
        }

        if (findings.isNotEmpty()) {
            throw GradleException(
                "Sensitive types must not be PublicValue types:\n" +
                findings.joinToString("\n") { finding ->
                    "- ${finding.sourcePath}:${finding.lineNumber} ${finding.typeName} combines PublicValue with ${finding}
                } +
                "\n\nPII values must keep hiding behavior explicit, and self-validating values must not expose a public-value
            )
        }
    }
}

tasks.matching { it.name == "check" }.configureEach {
    dependsOn(forbidPublicValueOnSensitiveTypes)
}

```

gradle/forbid-string-concat.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp

`_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/gradle/forbid-string-concat.gradle.kts>`

```

/*
 * SECURITY BUILD GATE: STRING CONCATENATION ENFORCEMENT
 *
 * This Gradle script fails the build when Java source files use string concatenation with '+'
 * outside a narrowly documented safe scope.
 *
 * A safe scope must be opened with a comment containing:
 *
 *   STRING CONCAT IS SAFE HERE: <reason>
 *
 * and closed with a later comment containing:
 *
 *   END STRING CONCAT
 *
 * Line comments and block comments are both accepted. The opening and closing comments must be
 * inside the same method, constructor, initializer block, or lambda body. The reason length is
 * configured in string_concat_guardrail.json and defaults to 200 characters when omitted.
 */

```

```

import com.sun.source.tree.BinaryTree
import com.sun.source.tree.BlockTree
import com.sun.source.tree.ClassTree
import com.sun.source.tree.CompilationUnitTree
import com.sun.source.tree.LambdaExpressionTree
import com.sun.source.tree.LiteralTree
import com.sun.source.tree.MemberSelectTree
import com.sun.source.tree.MethodInvocationTree
import com.sun.source.tree.MethodTree
import com.sun.source.tree.NewClassTree
import com.sun.source.tree.ParenthesizedTree
import com.sun.source.tree.Tree
import com.sun.source.tree.VariableTree
import com.sun.source.util.JavacTask
import com.sun.source.util.TreePath
import com.sun.source.util.TreePathScanner
import com.sun.source.util.Trees
import org.gradle.api.GradleException
import org.gradle.api.plugins.JavaPluginExtension
import org.gradle.api.tasks.SourceSet
import java.io.File
import java.io.StringWriter
import java.util.Locale
import javax.lang.model.type.DeclaredType
import javax.lang.model.type.TypeKind

```

```

import javax.tools.Diagnostic
import javax.tools.DiagnosticCollector
import javax.tools.JavaFileObject
import javax.tools.ToolProvider

val DEFAULT_MINIMUM_REASON_CHARACTERS = 200
val OPEN_MARKER = "STRING CONCAT IS SAFE HERE:"
val CLOSE_MARKER = "END STRING CONCAT"
val STRING_CONCATENATION_SAFE_ANNOTATION_NAME = "org.owasp.untrust.buildmetadata.StringConcatenationSafe"
@Suppress("UNCHECKED_CAST")
val readJsonObject = rootProject.extensions.extraProperties["readJsonObject"] as (File) -> Map<String, Any>
val RAW_STRING_ASSEMBLY_TYPES = mapOf(
    "java.lang.StringBuilder" to "StringBuilder",
    "java.lang.StringBuffer" to "StringBuffer",
    "java.io.StringWriter" to "StringWriter",
    "java.io.CharArrayWriter" to "CharArrayWriter",
    "java.io.PrintWriter" to "PrintWriter",
    "com.google.common.base.Joiner" to "Guava Joiner"
)

data class StringConcatGuardConfig(
    val minimumReasonCharacters: Int
)

data class SourceLineComment(
    val lineNumber: Long,
    val text: String
)

data class StringConcatScope(
    val startLine: Long,
    val endLine: Long,
    val description: String
)

data class StringConcatSafeRegion(
    val openLine: Long,
    val closeLine: Long,
    val reason: String,
    val scope: StringConcatScope
)

data class PendingStringConcatRegion(
    val openLine: Long,

```

```

    val reason: String,
    val scope: StringConcatScope
)

data class StringConcatFinding(
    val lineNumber: Long,
    val kind: String,
    val expression: String
)

fun File.relativeUnixPath(rootDir: File): String {
    return rootDir.toPath()
        .relativize(toPath())
        .toString()
        .replace(File.separatorChar, '/')
}

fun readStringConcatGuardConfig(file: File): StringConcatGuardConfig {
    if (!file.exists()) {
        return StringConcatGuardConfig(DEFAULT_MINIMUM_REASON_CHARACTERS)
    }

    val parsed = readJsonConfigObject(file)

    val rawMinimum = parsed["minimumReasonCharacters"]
    ?: return StringConcatGuardConfig(DEFAULT_MINIMUM_REASON_CHARACTERS)

    val minimum = when (rawMinimum) {
        is Number -> rawMinimum.toInt()
        is String -> rawMinimum.toIntOrNull()
        else -> null
    }
    ?: throw GradleException("string_concat_guardrail.json field 'minimumReasonCharacters' must be an integer.")

    if (minimum < 1) {
        throw GradleException("string_concat_guardrail.json field 'minimumReasonCharacters' must be at least 1.")
    }

    return StringConcatGuardConfig(minimum)
}

fun normalizeCommentLine(text: String): String {
    return text
        .trim()
        .removePrefix("//")
}

```



```

        lineNumber = lineNumber,
        text = normalizeCommentLine(remaining.substring(lineCommentIndex))
    )
    remaining = ""
}
else -> {
    val afterOpen = remaining.substring(blockCommentIndex + 2)
    val closeIndex = afterOpen.indexOf("*/")

    if (closeIndex >= 0) {
        comments += SourceLineComment(
            lineNumber = lineNumber,
            text = normalizeCommentLine(afterOpen.substring(0, closeIndex))
        )
        remaining = afterOpen.substring(closeIndex + 2)
    } else {
        inBlockComment = true
        blockCommentStartLine = lineNumber
        blockCommentText.append(afterOpen)
        remaining = ""
    }
}
}
}
}
}

if (inBlockComment) {
    comments += SourceLineComment(
        lineNumber = blockCommentStartLine,
        text = normalizeCommentLine(blockCommentText.toString())
    )
}

return comments
}

fun StringConcatScope.contains(lineNumber: Long): Boolean {
    return lineNumber in startLine..endLine
}

fun smallestScopeForLine(scopes: List<StringConcatScope>, lineNumber: Long): StringConcatScope? {
    return scopes
        .filter { it.contains(lineNumber) }

```

```

        .minByOrNull { it.endLine - it.startLine }
    }

fun sourceLine(compilationUnit: CompilationUnitTree, tree: Tree, trees: Trees): Long {
    val startPosition = trees.sourcePositions.getStartPosition(compilationUnit, tree)

    return if (startPosition >= 0) {
        compilationUnit.lineMap.getLineNumber(startPosition)
    } else {
        -1L
    }
}

fun sourceEndLine(compilationUnit: CompilationUnitTree, tree: Tree, trees: Trees): Long {
    val endPosition = trees.sourcePositions.getEndPosition(compilationUnit, tree)

    return if (endPosition >= 0) {
        compilationUnit.lineMap.getLineNumber(endPosition)
    } else {
        sourceLine(compilationUnit, tree, trees)
    }
}

fun executableScopes(compilationUnit: CompilationUnitTree, trees: Trees): List<StringConcatScope> {
    val scopes = mutableListOf<StringConcatScope>()

    object : TreePathScanner<Unit, Unit>() {
        override fun visitMethod(node: MethodTree, unused: Unit?) {
            val body = node.body

            if (body != null) {
                val name = if (node.name.contentEquals("<init>")) {
                    "constructor"
                } else {
                    "method ${node.name}"
                }

                scopes += StringConcatScope(
                    startLine = sourceLine(compilationUnit, body, trees),
                    endLine = sourceEndLine(compilationUnit, body, trees),
                    description = name
                )
            }
        }
    }
}

```

```

    super.visitMethod(node, unused)
}

override fun visitLambdaExpression(node: LambdaExpressionTree, unused: Unit?) {
    scopes += StringConcatScope(
        startLine = sourceLine(compilationUnit, node.body, trees),
        endLine = sourceEndLine(compilationUnit, node.body, trees),
        description = "lambda"
    )

    super.visitLambdaExpression(node, unused)
}

override fun visitClass(node: ClassTree, unused: Unit?) {
    node.members
        .filterIsInstance<BlockTree>()
        .forEach { block ->
            val description = if (block.isStatic) {
                "static initializer"
            } else {
                "initializer"
            }

            scopes += StringConcatScope(
                startLine = sourceLine(compilationUnit, block, trees),
                endLine = sourceEndLine(compilationUnit, block, trees),
                description = description
            )
        }

    super.visitClass(node, unused)
}

}.scan(compilationUnit, Unit)

return scopes
}

fun validateSafeRegions(
    comments: List<SourceLineComment>,
    scopes: List<StringConcatScope>,
    minimumReasonCharacters: Int,
    sourcePath: String
): Pair<List<StringConcatSafeRegion>, List<String>> {
    val safeRegions = mutableListOf<StringConcatSafeRegion>()

```

```

val violations = mutableListOf<String>()
val openRegionsByScope = mutableMapOf<StringConcatScope, PendingStringConcatRegion>()

for (comment in comments.sortedBy { it.lineNumber }) {
    val hasOpenMarker = comment.text.contains(OPEN_MARKER)
    val hasCloseMarker = comment.text.contains(CLOSE_MARKER)

    if (hasOpenMarker) {
        val scope = smallestScopeForLine(scopes, comment.lineNumber)

        if (scope == null) {
            violations += "$sourcePath:${comment.lineNumber} opens a string concatenation safe region outside an execution region"
            continue
        }

        if (openRegionsByScope.containsKey(scope)) {
            violations += "$sourcePath:${comment.lineNumber} opens a nested string concatenation safe region inside $scope"
            continue
        }

        val reason = comment.text.substringAfter(OPEN_MARKER).trim()

        if (reason.length < minimumReasonCharacters) {
            violations += "$sourcePath:${comment.lineNumber} string concatenation safe reason is ${reason.length} characters long"
        }

        openRegionsByScope[scope] = PendingStringConcatRegion(
            openLine = comment.lineNumber,
            reason = reason,
            scope = scope
        )
    }

    if (hasCloseMarker) {
        val scope = smallestScopeForLine(scopes, comment.lineNumber)

        if (scope == null) {
            violations += "$sourcePath:${comment.lineNumber} closes a string concatenation safe region outside an execution region"
            continue
        }

        val pending = openRegionsByScope.remove(scope)

        if (pending == null) {

```

```

        violations += "$sourcePath:${comment.lineNumber} closes a string concatenation safe region in ${scope.desc
        continue
    }

    safeRegions += StringConcatSafeRegion(
        openLine = pending.openLine,
        closeLine = comment.lineNumber,
        reason = pending.reason,
        scope = scope
    )
}

openRegionsByScope.values.forEach { pending ->
    violations += "$sourcePath:${pending.openLine} opens a string concatenation safe region in ${pending.scope.desc
}

return safeRegions to violations
}

fun isStringConcat(binaryTree: BinaryTree, scanner: TreePathScanner<*, *>, trees: Trees): Boolean {
    if (binaryTree.kind != Tree.Kind.PLUS) {
        return false
    }

    val typeMirror = trees.getTypeMirror(scanner.currentPath)

    return typeMirror != null &&
        typeMirror.kind != TypeKind.ERROR &&
        typeMirror.toString() == "java.lang.String"
}

fun isStringLiteral(tree: Tree): Boolean {
    return tree is LiteralTree && tree.value is String
}

fun isStringConcatenationSafeExpression(treePath: TreePath, trees: Trees): Boolean {
    val typeMirror = trees.getTypeMirror(treePath)

    return typeMirror != null &&
        typeMirror.kind != TypeKind.ERROR &&
        typeMirror is DeclaredType &&
        typeMirror.asElement().annotationMirrors.any { annotationMirror ->
            annotationMirror.annotationType.toString() == STRING_CONCATENATION_SAFE_ANNOTATION_NAME
        }
}

```

```

    }
}

fun isAllowedStringConcatOperand(treePath: TreePath, trees: Trees): Boolean {
    val leaf = treePath.leaf

    return when {
        leaf is ParenthesizedTree -> isAllowedStringConcatOperand(TreePath(treePath, leaf.expression), trees)
        leaf is BinaryTree && leaf.kind == Tree.Kind.PLUS -> isAllowedSafeStringConcat(treePath, leaf, trees)
        isStringLiteral(leaf) -> true
        else -> isStringConcatenationSafeExpression(treePath, trees)
    }
}

fun isAllowedSafeStringConcat(treePath: TreePath, binaryTree: BinaryTree, trees: Trees): Boolean {
    return isAllowedStringConcatOperand(TreePath(treePath, binaryTree.leftOperand), trees) &&
        isAllowedStringConcatOperand(TreePath(treePath, binaryTree.rightOperand), trees)
}

fun treeTypeName(treePath: TreePath, trees: Trees): String? {
    val typeMirror = trees.getTypeMirror(treePath) ?: return null

    return if (typeMirror.kind == TypeKind.ERROR) {
        null
    } else {
        typeMirror.toString()
    }
}

fun selectedMethodOwner(methodInvocationTree: MethodInvocationTree, scanner: TreePathScanner<*, *>, trees: Trees): TreePath? {
    val element = trees.getElement(TreePath(scanner.currentPath, methodInvocationTree.methodSelect))
    ?: return null

    return element.enclosingElement?.toString()
}

fun stringConcatFailureGuidance(minimumReasonCharacters: Int): String {
    return """

```

STRING CONCATENATION POLICY FAILURE

This failure is not asking you to find a different raw **string**-building API.

Do not replace the violating code with another non-structure-aware **string** assembly mechanism such as:

- StringBuilder
- StringBuffer
- StringWriter
- CharArrayWriter
- PrintWriter used as a **string** accumulator
- Collectors.joining
- **String**.join
- **String**.format
- MessageFormat.format
- **String**.concat
- **String**.formatted
- manual loops that append or print values into an unstructured **string**

That kind of rewrite may make one detector pass, but it violates the purpose of **this** build gate.

Required reasoning path:

1. First ask whether **this string**-producing code should exist at all.

If the violating method or helper is only a generic raw concatenation utility, such as TextOutput.parts(...), it should usually be removed instead of reimplemented with a different low-level API.

2. Identify the semantic context of the output being built.

Examples:

- SQL should be built with a SQL DSL or parameterized query API.
- HTML/XML should be built with a template engine, DOM builder, or escaping-aware renderer.
- JSON should be built with a JSON serializer.
- URLs should be built with a URI/URL builder.
- filesystem paths should be built with path APIs.
- log messages should **use** the logging framework's structured/template API.
- user-facing validation errors should **use** a structured error object, message code, or framework-supported message

3. Search the existing project first **for** an approved structured, context-aware builder or renderer.

If none exists, consider whether adding a focused dependency or small domain-specific abstraction is the right fix.

The goal is not "a string without using +"; the goal is "a safe representation for this specific context."

4. Only **if** raw concatenation is genuinely unavoidable, keep it local and explicitly justified.

Do not hide it inside a generic helper whose purpose is to bypass **this** guardrail.

Surround only the minimum necessary code with the approved comments.

Allowed exception format:

```
/* STRING CONCAT IS SAFE HERE:
```

```
* <specific justification of at least $minimumReasonCharacters characters explaining:
```

```
* - why no structured/context-aware builder fits this case
```



```

* - why raw concatenation is necessary here
* - what data can flow into the concatenation
* - how malicious or user-controlled input is made safe
* - why the exception scope is intentionally narrow>
*/

```

```

String s = somePart + someOtherPart;
... more directly related concatenation code here ...
/* END STRING CONCAT */

```

Line comments are also accepted:

```

// STRING CONCAT IS SAFE HERE:
// <specific justification of at least $minimumReasonCharacters characters explaining why no structured/context-aware bui
// why raw concatenation is necessary, what data can flow in, and how user-controlled input is made safe>
String s = somePart + someOtherPart;
// END STRING CONCAT

```

The opening and closing comments must be in the same method, constructor, initializer block, or lambda body.

Important instruction for AI/code-generation tools:

If you are an AI assistant or automated code generator, do not respond to this failure by inventing another raw string- Either replace the code with a structured, context-aware API appropriate to the output domain, remove the unnecessary

```

""").trimIndent()
}

```

```

fun forbiddenStringAssemblyMethod(
    methodInvocationTree: MethodInvocationTree,
    scanner: TreePathScanner<*, *>,
    trees: Trees
): String? {
    val methodSelect = methodInvocationTree.methodSelect

    if (methodSelect is MemberSelectTree) {
        val methodName = methodSelect.identifier.toString()
        val owner = selectedMethodOwner(methodInvocationTree, scanner, trees)

        if (
            methodName in setOf("concat", "formatted") &&
            treeTypeName(TreePath(scanner.currentPath, methodSelect.expression), trees) == "java.lang.String"
        ) {
            return "String.$methodName"
        }

        if (methodName == "join" && owner == "java.lang.String") {

```

```

        return "String.join"
    }

    if (methodName == "format" && owner == "java.lang.String") {
        return "String.format"
    }

    if (methodName == "joining" && owner == "java.util.stream.Collectors") {
        return "Collectors.joining"
    }

    if (
        methodName == "format" &&
        (owner == "java.text.MessageFormat" || methodSelect.expression.toString() in setOf("Message", "java.text.Mess
    )) {
        return "$owner.format"
    }
} else if (methodSelect.toString() == "join" && selectedMethodOwner(methodInvocationTree, scanner, trees) == "java
    return "String.join"
} else if (methodSelect.toString() == "joining" && selectedMethodOwner(methodInvocationTree, scanner, trees) == "j
    return "Collectors.joining"
} else if (methodSelect.toString() == "format") {
    val owner = selectedMethodOwner(methodInvocationTree, scanner, trees)

    if (owner == "java.lang.String") {
        return "String.format"
    }

    if (owner in setOf("java.text.MessageFormat", "Message")) {
        return "$owner.format"
    }
}

return null
}

fun findStringConcatS(CompilationUnitTree, trees: Trees): List<StringConcatFinding> {
    val findings = linkedMapOf<Long, StringConcatFinding>()

    fun forbiddenRawStringAssemblyKind(treePath: TreePath): String? {
        return RAW_STRING_ASSEMBLY_TYPES[treeTypeName(treePath, trees)]
    }

    object : TreePathScanner<Unit, Unit>() {

```

```

override fun visitBinary(node: BinaryTree, unused: Unit?) {
    if (isStringConcat(node, this, trees) && !isAllowedSafeStringConcat(currentPath, node, trees)) {
        val lineNumber = sourceLine(compilationUnit, node, trees)

        if (lineNumber > 0 && !findings.containsKey(lineNumber)) {
            findings[lineNumber] = StringConcatFinding(
                lineNumber = lineNumber,
                kind = "string concatenation",
                expression = node.toString()
            )
        }
    }

    super.visitBinary(node, unused)
}

```

```

override fun visitMethodInvocation(node: MethodInvocationTree, unused: Unit?) {
    val forbiddenMethod = forbiddenStringAssemblyMethod(node, this, trees)

    if (forbiddenMethod != null) {
        val lineNumber = sourceLine(compilationUnit, node, trees)

        if (lineNumber > 0 && !findings.containsKey(lineNumber)) {
            findings[lineNumber] = StringConcatFinding(
                lineNumber = lineNumber,
                kind = forbiddenMethod,
                expression = node.toString()
            )
        }
    }

    super.visitMethodInvocation(node, unused)
}

```

```

override fun visitNewClass(node: NewClassTree, unused: Unit?) {
    val forbiddenType = forbiddenRawStringAssemblyKind(currentPath)

    if (forbiddenType != null) {
        val lineNumber = sourceLine(compilationUnit, node, trees)

        if (lineNumber > 0 && !findings.containsKey(lineNumber)) {
            findings[lineNumber] = StringConcatFinding(
                lineNumber = lineNumber,
                kind = forbiddenType,

```

```

        expression = node.toString()
    )
}
}

super.visitNewClass(node, unused)
}

override fun visitVariable(node: VariableTree, unused: Unit?) {
    val typeTree = node.type
    val forbiddenType = if (typeTree == null) {
        null
    } else {
        forbiddenRawStringAssemblyKind(TreePath(currentPath, typeTree))
    }

    if (forbiddenType != null) {
        val lineNumber = sourceLine(compilationUnit, node, trees)

        if (lineNumber > 0 && !findings.containsKey(lineNumber)) {
            findings[lineNumber] = StringConcatFinding(
                lineNumber = lineNumber,
                kind = forbiddenType,
                expression = node.toString()
            )
        }
    }

    super.visitVariable(node, unused)
}
}.scan(compilationUnit, Unit)

return findings.values.toList()
}

fun stringConcatAllowed(
    finding: StringConcatFinding,
    safeRegions: List<StringConcatSafeRegion>
): Boolean {
    return safeRegions.any { region ->
        finding.lineNumber in region.openLine..region.closeLine &&
        region.scope.contains(finding.lineNumber)
    }
}
}

```

```

fun compilerOptionsFor(sourceSet: SourceSet): List<String> {
    val options = mutableListOf("-proc:none")
    val classpath = sourceSet.compileClasspath.files

    if (classpath.isNotEmpty()) {
        options += "-classpath"
        options += classpath.joinToString(File.pathSeparator) { it.absolutePath }
    }

    val sourceDirectories = sourceSet.allJava.srcDirs.filter { it.exists() }

    if (sourceDirectories.isNotEmpty()) {
        options += "-sourcepath"
        options += sourceDirectories.joinToString(File.pathSeparator) { it.absolutePath }
    }

    return options
}

fun inspectSourceSetForStringConcat(
    sourceSet: SourceSet,
    rootDir: File,
    config: StringConcatGuardConfig
): List<String> {
    val sourceFiles = sourceSet.allJava.files
        .filter { it.extension == "java" && it.exists() }
        .sortedBy { it.absolutePath }

    if (sourceFiles.isEmpty()) {
        return emptyList()
    }

    val compiler = ToolProvider.getSystemJavaCompiler()
    ?: throw GradleException("A JDK is required to inspect Java source files. Gradle is not running with a JDK compiler")

    val diagnostics = DiagnosticCollector<JavaFileObject>()

    compiler.getStandardFileManager(diagnostics, Locale.ROOT, Charsets.UTF_8).use { fileManager ->
        val javaFiles = fileManager.getJavaFileObjectsFromFiles(sourceFiles)
        val task = compiler.getTask(
            StringWriter(),
            fileManager,
            diagnostics,

```

```

        compilerOptionsFor(sourceSet),
        emptyList<String>(),
        javaFiles
    ) as JavacTask

    val trees = Trees.instance(task)
    val parsedUnits = task.parse().toList()
    task.analyze()

    val errors = diagnostics.diagnostics
        .filter { it.kind == Diagnostic.Kind.ERROR }

    if (errors.isNotEmpty()) {
        val message = errors.joinToString("\n") { diagnostic ->
            val sourceName = diagnostic.source?.name ?: "<unknown source>"
            "- $sourceName:${diagnostic.lineNumber}: ${diagnostic.getMessage(Locale.ROOT)}"
        }

        throw GradleException("Failed to analyze Java source files before checking string concatenation:\n$message")
    }

    return parsedUnits.flatMap { compilationUnit ->
        val sourceFile = File(compilationUnit.sourceFile.toUri())
        val sourcePath = sourceFile.relativeUnixPath(rootDir)
        val scopes = executableScopes(compilationUnit, trees)
        val (safeRegions, regionViolations) = validateSafeRegions(
            comments = extractComments(sourceFile),
            scopes = scopes,
            minimumReasonCharacters = config.minimumReasonCharacters,
            sourcePath = sourcePath
        )

        val concatViolations = findStringConcatenations(compilationUnit, trees)
            .filterNot { stringConcatAllowed(it, safeRegions) }
            .map { finding ->
                "$sourcePath:${finding.lineNumber} uses forbidden ${finding.kind} outside a STRING CONCAT IS SAFE HERE"
            }

        regionViolations + concatViolations
    }
}

val forbidStringConcat by tasks.registering {

```

```

group = "verification"
description = "Fails the build when Java string concatenation is used outside documented safe scopes."

val configFile = rootProject.file("string_concat_guardrail.json")
val javaExtension = project.extensions.getByType<JavaPluginExtension>()
val sourceSets = javaExtension.sourceSets

inputs.file(configFile).optional()
sourceSets.configureEach {
    inputs.files(allJava)
    inputs.files(compileClasspath)
}

doLast {
    val config = readStringConcatGuardConfig(configFile)
    val violations = sourceSets.flatMap { sourceSet ->
        inspectSourceSetForStringConcat(sourceSet, rootProject.projectDir, config)
    }

    if (violations.isNotEmpty()) {
        throw GradleException(
            "Forbidden string concatenation found:\n" +
            violations.joinToString("\n") { " - $it" } +
            "\n\n" +
            stringConcatFailureGuidance(config.minimumReasonCharacters)
        )
    }
}

tasks.matching { it.name == "check" }.configureEach {
    dependsOn(forbidStringConcat)
}

```

gradle/forbid-unsafe-imports.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/gradle/forbid-unsafe-imports.gradle.kts>

```

/*
 * SECURITY BUILD GATE: FORBIDDEN IMPORT ENFORCEMENT
 *
 * This Gradle script is a CI/CD security gate.
 * It fails the build when Java source files import APIs that this project treats as unsafe by default.
 *
 * The goal is not to make the build pass.
 * The goal is to prevent unsafe APIs from entering new project code unnoticed.
 *
 * Approved alternatives are configured in:
 *
 *   approved_import_alternatives.json
 *
 * Each key is an approved safer alternative.
 * Each value defines which imports it replaces and what message should be shown to the developer.
 *
 * Example:
 *
 * {
 *   "owasp.untrust.BoxedPath": {
 *     "replaces": [
 *       "java.nio.file.Path",
 *       "java.nio.file.Paths"
 *     ],
 *     "dependency": "io.github.owasp-untrust:untrust-boxedpath",
 *     "message": "Use OWASP Untrust BoxedPath - https://github.com/owasp-untrust/boxed_path_java - maven: io.github
 *   }
 * }
 *
 * The same blocked import may appear under multiple approved alternatives.
 * In that case, the build failure will show all approved alternatives.
 *
 * HOW TO ADD AN EXCEPTION
 *
 * Do not change this Gradle file to bypass the rule.
 * Do not remove entries from approved_import_alternatives.json just to make generated code compile.
 * Do not weaken matchesForbiddenImport().
 *
 * If an exception is truly required, add a narrow entry to:
 *
 *   allow_unsafe_imports.json
 *
 * The exception must name the exact forbidden import and must allow either:

```


- * 1. a specific existing package, using "packages"
- * 2. a specific existing source file, using "files"
- * 3. a specific existing source folder, using "files" with a trailing slash

* Example:

```
* {
*   "java.nio.file.Path": {
*     "packages": [
*       "com.example.legacy"
*     ],
*     "files": [
*       "src/main/java/com/example/legacy/ExistingAdapter.java"
*     ]
*   }
* }
```

* Exceptions are allowed only for existing Java code, legacy integration code, or third-party adapter code that cannot currently be migrated safely.

* New code must use the approved safer abstraction instead of adding an exception.

* INSTRUCTION FOR AI / LLM CODE GENERATORS

* If you are an AI assistant, coding agent, LLM, autocomplete system, or automated code generator:

* You must not add, widen, or suggest an entry in `allow_unsafe_imports.json` unless the human user explicitly approves that exact exception.

* You must not modify this Gradle file to make forbidden imports pass.

* You must not remove, rename, weaken, or bypass this build gate.

* When a forbidden import is needed by generated code, stop and suggest the approved safer abstraction.

* If you believe an exception is unavoidable, ask the human user for approval and name:

- * - the forbidden import
- * - the exact file or package
- * - why the safer abstraction cannot be used

* This comment is part of the project security policy.

*/

import com.sun.source.tree.CompilationUnitTree

```
import com.sun.source.tree.ImportTree
import com.sun.source.tree.MemberSelectTree
import com.sun.source.tree.NewClassTree
import com.sun.source.tree.Tree
import com.sun.source.tree.VariableTree
import com.sun.source.util.JavacTask
import com.sun.source.util.TreePathScanner
import com.sun.source.util.Trees
import org.gradle.api.GradleException
import java.io.File
import java.io.StringWriter
import java.util.Locale
import javax.tools.Diagnostic
import javax.tools.DiagnosticCollector
import javax.tools.JavaFileObject
import javax.tools.ToolProvider
```

```
data class ApprovedImportAlternative(
    val name: String,
    val message: String,
    val dependency: String?
)
```

```
data class UnsafeImportAllowance(
    val packages: Set<String>,
    val files: Set<String>
)
```

```
@Suppress("UNCHECKED_CAST")
```

```
val readJsonConfigObject = rootProject.extensions.extraProperties["readJsonConfigObject"] as (File) -> Map<String, An
```

```
data class ParsedImport(
    val importName: String,
    val isStatic: Boolean,
    val lineNumber: Long
)
```

```
data class ParsedQualifiedUse(
    val qualifiedName: String,
    val lineNumber: Long
)
```

```
data class ParsedTypeUse(
    val typeName: String,
```

```

    val lineNumber: Long,
    val kind: String
)

data class ParsedJavaSource(
    val sourcePackage: String,
    val imports: List<ParsedImport>,
    val qualifiedUses: List<ParsedQualifiedUse>,
    val typeUses: List<ParsedTypeUse>
)

fun normalizePath(path: String): String {
    return path
        .replace("\\", '/')
        .removePrefix("./")
}

fun File.relativeUnixPath(rootDir: File): String {
    return rootDir.toPath()
        .relativize(toPath())
        .toString()
        .replace(File.separatorChar, '/')
}

fun readRequiredString(map: Map<*, *>, key: String, context: String): String {
    if (!map.containsKey(key)) {
        throw GradleException("$context must contain '$key'.")
    }

    val rawValue = map[key]

    if (rawValue !is String || rawValue.isBlank()) {
        throw GradleException("$context field '$key' must be a non-empty string.")
    }

    return rawValue
}

fun readOptionalString(map: Map<*, *>, key: String, context: String): String? {
    if (!map.containsKey(key)) {
        return null
    }

    val rawValue = map[key]

```

```

    if (rawValue !is String || rawValue.isBlank()) {
        throw GradleException("$context field '$key' must be a non-empty string when provided.")
    }

    return rawValue
}

fun readRequiredStringList(map: Map<*, *>, key: String, context: String): List<String> {
    if (!map.containsKey(key)) {
        throw GradleException("$context must contain '$key'.")
    }

    val rawValue = map[key]

    if (rawValue !is List<*> || rawValue.isEmpty()) {
        throw GradleException("$context field '$key' must be a non-empty array.")
    }

    return rawValue.map { it.toString() }
}

fun readOptionalStringSet(map: Map<*, *>, key: String): Set<String> {
    if (!map.containsKey(key)) {
        return emptySet()
    }

    val rawValue = map[key]

    if (rawValue !is List<*>) {
        throw GradleException("$key' must be an array in allow_unsafe_imports.json.")
    }

    return rawValue
        .map { normalizePath(it.toString()) }
        .toSet()
}

fun readApprovedImportAlternatives(file: File): Map<String, List<ApprovedImportAlternative>> {
    if (!file.exists()) {
        throw GradleException("approved_import_alternatives.json is required.")
    }

    val parsed = readJsonConfigObject(file)

```

```
val alternativesByForbiddenImport = mutableMapOf<String, MutableList<ApprovedImportAlternative>>()

parsed.forEach { entry ->
    val alternativeName = entry.key
    val alternativeData = entry.value

    if (alternativeData !is Map<*, *>) {
        throw GradleException("Approved alternative '$alternativeName' must be an object.")
    }

    val replaces = readRequiredStringList(
        alternativeData,
        "replaces",
        "Approved alternative '$alternativeName'"
    )

    val message = readRequiredString(
        alternativeData,
        "message",
        "Approved alternative '$alternativeName'"
    )

    val dependency = readOptionalString(
        alternativeData,
        "dependency",
        "Approved alternative '$alternativeName'"
    )

    val alternative = ApprovedImportAlternative(
        name = alternativeName,
        message = message,
        dependency = dependency
    )

    for (replacedImport in replaces) {
        alternativesByForbiddenImport
            .getOrPut(replacedImport) { mutableListOf() }
            .add(alternative)
    }
}

if (alternativesByForbiddenImport.isEmpty()) {
    throw GradleException("approved_import_alternatives.json must define at least one replacement rule.")
}
```

```

    }

    return alternativesByForbiddenImport
}

fun readUnsafeImportAllowances(file: File): Map<String, UnsafeImportAllowance> {
    if (!file.exists()) {
        return emptyMap()
    }

    val parsed = readJsonConfigObject(file)

    return parsed
        .map { entry ->
            val importName = entry.key
            val allowanceData = entry.value

            if (allowanceData !is Map<*, *>) {
                throw GradleException("Allowance for '$importName' must be an object.")
            }

            importName to UnsafeImportAllowance(
                packages = readOptionalStringSet(allowanceData, "packages"),
                files = readOptionalStringSet(allowanceData, "files")
            )
        }
        .toMap()
}

fun parseJavaSource(sourceFile: File): ParsedJavaSource {
    val compiler = ToolProvider.getSystemJavaCompiler()
    ?: throw GradleException("A JDK is required to parse Java source files. Gradle is not running with a JDK compiler available")

    val diagnostics = DiagnosticCollector<JavaFileObject>()

    compiler.getStandardFileManager(diagnostics, Locale.ROOT, Charsets.UTF_8).use { fileManager ->
        val javaFiles = fileManager.getJavaFileObjectsFromFiles(listOf(sourceFile))

        val task = compiler.getTask(
            StringWriter(),
            fileManager,
            diagnostics,
            listOf("-proc:none"),
            emptyList<String>(),

```

```

        javaFiles
    ) as JacoTask

    val trees = Trees.instance(task)
    val parsedUnits = task.parse().toList()

    val errors = diagnostics.diagnostics
        .filter { it.kind == Diagnostic.Kind.ERROR }

    if (errors.isNotEmpty()) {
        val message = errors.joinToString("\n") { diagnostic ->
            "- ${sourceFile.path}:${diagnostic.lineNumber}: ${diagnostic.getMessage(Locale.ROOT)}"
        }

        throw GradleException("Failed to parse Java source file:\n$message")
    }

    val compilationUnit = parsedUnits.first()

    return ParsedJavaSource(
        sourcePackage = compilationUnit.packageName?.toString().orEmpty(),
        imports = compilationUnit.imports.map { importTree ->
            toParsedImport(compilationUnit, importTree, trees)
        },
        qualifiedUses = collectQualifiedUses(compilationUnit, trees),
        typeUses = collectTypeUses(compilationUnit, trees)
    )
}

fun lineNumberOf(
    compilationUnit: CompilationUnitTree,
    tree: Tree,
    trees: Trees
): Long {
    val startPosition = trees.sourcePositions.getStartPosition(compilationUnit, tree)

    return if (startPosition >= 0) {
        compilationUnit.lineMap.getLineNumber(startPosition)
    } else {
        -1
    }
}

```

```

fun toParsedImport(
    compilationUnit: CompilationUnitTree,
    importTree: ImportTree,
    trees: Trees
): ParsedImport {
    return ParsedImport(
        importName = importTree.qualifiedIdentifier.toString(),
        isStatic = importTree.isStatic,
        lineNumber = lineNumberOf(compilationUnit, importTree, trees)
    )
}

fun collectQualifiedUses(
    compilationUnit: CompilationUnitTree,
    trees: Trees
): List<ParsedQualifiedUse> {
    val qualifiedUses = mutableListOf<ParsedQualifiedUse>()

    object : TreePathScanner<Unit, Unit>() {
        override fun visitImport(node: ImportTree, unused: Unit?): Unit? {
            return null
        }

        override fun visitMemberSelect(node: MemberSelectTree, unused: Unit?): Unit? {
            val parentTree = currentPath.parentPath?.leaf
            val isPrefixOfLongerSelection = parentTree is MemberSelectTree &&
                parentTree.expression == node

            if (!isPrefixOfLongerSelection) {
                qualifiedUses += ParsedQualifiedUse(
                    qualifiedName = node.toString(),
                    lineNumber = lineNumberOf(compilationUnit, node, trees)
                )
            }

            return super.visitMemberSelect(node, unused)
        }
    }.scan(compilationUnit, Unit)

    return qualifiedUses
}

fun collectTypeUses(
    compilationUnit: CompilationUnitTree,

```



```

trees: Trees

): List<ParsedTypeUse> {
    val typeUses = mutableListOf<ParsedTypeUse>()

    object : TreePathScanner<Unit, Unit>() {
        override fun visitNewClass(node: NewClassTree, unused: Unit?): Unit? {
            typeUses += ParsedTypeUse(
                typeName = node.identifier.toString(),
                lineNumber = lineNumberOf(compilationUnit, node, trees),
                kind = "constructor call"
            )

            return super.visitNewClass(node, unused)
        }

        override fun visitVariable(node: VariableTree, unused: Unit?): Unit? {
            val type = node.type

            if (type != null) {
                typeUses += ParsedTypeUse(
                    typeName = type.toString(),
                    lineNumber = lineNumberOf(compilationUnit, type, trees),
                    kind = "type reference"
                )
            }

            return super.visitVariable(node, unused)
        }
    }.scan(compilationUnit, Unit)

    return typeUses
}

fun canonicalJavaLangType(typeName: String): String? {
    if (typeName.startsWith("java.lang.")) {
        return typeName
    }

    if (typeName.contains(".")) {
        return null
    }

    return "java.lang.$typeName"
}

```

```

fun matchesForbiddenImport(imported: String, forbidden: String): Boolean {
    if (imported == forbidden) {
        return true
    }

    if (imported.startsWith("$forbidden.")) {
        return true
    }

    if (forbidden.endsWith(".*")) {
        val forbiddenPrefix = forbidden.removeSuffix(".*")
        return imported.startsWith("$forbiddenPrefix.")
    }

    if (imported.endsWith(".*")) {
        val importedPrefix = imported.removeSuffix(".*")
        return forbidden.startsWith("$importedPrefix.")
    }

    return false
}

fun matchesAllowedFile(sourcePath: String, allowedPath: String): Boolean {
    if (allowedPath.endsWith("/")) {
        return sourcePath.startsWith(allowedPath)
    }

    return sourcePath == allowedPath
}

fun isAllowed(
    sourcePackage: String,
    sourcePath: String,
    forbiddenImport: String,
    allowances: Map<String, UnsafeImportAllowance>
): Boolean {
    val allowance = allowances[forbiddenImport] ?: return false

    val fileAllowed = allowance.files.any { allowedFile ->
        matchesAllowedFile(sourcePath, allowedFile)
    }

    val packageAllowed = allowance.packages.any { allowedPackage ->

```

```

        sourcePackage == allowedPackage || sourcePackage.startsWith("$allowedPackage.")
    }

    return fileAllowed || packageAllowed
}

fun formatAlternatives(alternatives: List<ApprovedImportAlternative>): String {
    return alternatives.joinToString(" | ") { alternative ->
        val dependencyMessage = alternative.dependency?.let { dependency ->
            "Configured dependency suggestion: add implementation(\"$dependency\") to build.gradle.kts if it is not already present."
        }.orEmpty()

        "You need to use this alternative api: ${alternative.name}. $dependencyMessage"
    }
}

val forbidUnsafeImports by tasks.registering {
    group = "verification"
    description = "Fails the build when forbidden imports are used outside approved packages or files."

    val alternativesFile = rootProject.file("approved_import_alternatives.json")
    val allowanceFile = rootProject.file("allow_unsafe_imports.json")

    val checkedSourceFiles = fileTree(rootProject.projectDir) {
        include("**/src/main/java/**/*.java")
        include("**/src/test/java/**/*.java")
        exclude("**/build/**")
    }

    inputs.file(alternativesFile).optional()
    inputs.file(allowanceFile).optional()
    inputs.files(checkedSourceFiles)

    doLast {
        val alternativesByForbiddenImport = readApprovedImportAlternatives(alternativesFile)
        val allowances = readUnsafeImportAllowances(allowanceFile)
        val violations = mutableListOf<String>()

        checkedSourceFiles.forEach { sourceFile ->
            val sourcePath = sourceFile.relativeUnixPath(rootProject.projectDir)
            val parsedSource = parseJavaSource(sourceFile)

            for (parsedImport in parsedSource.imports) {
                for ((forbiddenImport, alternatives) in alternativesByForbiddenImport) {
                    if (parsedImport == forbiddenImport) {
                        violations.add("Found forbidden import $forbiddenImport in $sourcePath. Allowed alternatives: ${alternatives.map { it.name }.joinToString(", ")}")
                    }
                }
            }
        }

        if (violations.isNotEmpty()) {
            throw GradleException("Found $violations.size forbidden imports. See log for details.")
        }
    }
}

```

```

        if (
            matchesForbiddenImport(parsedImport.importName, forbiddenImport) &&
            !isAllowed(parsedSource.sourcePackage, sourcePath, forbiddenImport, allowances)
        ){
            val importKind = if (parsedImport.isStatic) {
                "static import"
            } else {
                "import"
            }

            violations += "$sourcePath:${parsedImport.lineNumber} uses forbidden $importKind ${parsedImport.im
        }
    }
}

for (qualifiedUse in parsedSource.qualifiedUses) {
    for ((forbiddenImport, alternatives) in alternativesByForbiddenImport) {
        if (
            matchesForbiddenImport(qualifiedUse.qualifiedName, forbiddenImport) &&
            !isAllowed(parsedSource.sourcePackage, sourcePath, forbiddenImport, allowances)
        ){
            violations += "$sourcePath:${qualifiedUse.lineNumber} uses forbidden fully-qualified reference ${qualifie
        }
    }
}

for (typeUse in parsedSource.typeUses) {
    val canonicalTypeName = canonicalJavaLangType(typeUse.typeName) ?: typeUse.typeName

    for ((forbiddenImport, alternatives) in alternativesByForbiddenImport) {
        if (
            matchesForbiddenImport(canonicalTypeName, forbiddenImport) &&
            !isAllowed(parsedSource.sourcePackage, sourcePath, forbiddenImport, allowances)
        ){
            violations += "$sourcePath:${typeUse.lineNumber} uses forbidden ${typeUse.kind} ${typeUse.typeName
        }
    }
}

if (violations.isNotEmpty()) {
    throw GradleException(
        "Forbidden imports found:\n" +
        violations.joinToString("\n") { " - $it" }
    )
}

```

```
    )  
  }  
}  
}  
  
tasks.matching { it.name == "check" }.configureEach {  
    dependsOn(forbidUnsafeImports)  
}
```

gradle/forbid-unvalidated-route-values.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/gradle/forbid-unvalidated-route-values.gradle.kts>

```

/*
 * SECURITY BUILD GATE: ROUTE REQUEST VALUE VALIDATION
 *
 * Route handlers are the application boundary. Every user-controlled scalar value accepted
 * by a route handler must be represented by a type that extends ValidatedValue<T>, either
 * directly as a method parameter or as a leaf value inside a request DTO.
 *
 * Dynamic key/value request shapes are not allowed. Map<K, V> inside a route request type
 * fails the build because request values must be stated exactly in the Java type system.
 */

```

```

import com.sun.source.tree.AnnotationTree
import com.sun.source.tree.CompilationUnitTree
import com.sun.source.tree.MethodTree
import com.sun.source.tree.Tree
import com.sun.source.util.JavacTask
import com.sun.source.util.TreePath
import com.sun.source.util.TreePathScanner
import com.sun.source.util.Trees
import org.gradle.api.GradleException
import org.gradle.api.plugins.JavaPluginExtension
import org.gradle.api.tasks.SourceSet
import java.io.File
import java.io.StringWriter
import java.util.Locale
import javax.lang.model.element.ElementKind
import javax.lang.model.element.Modifier
import javax.lang.model.element.TypeElement
import javax.lang.model.type.ArrayType
import javax.lang.model.type.DeclaredType
import javax.lang.model.type.TypeKind
import javax.lang.model.type.TypeMirror
import javax.lang.model.util.Types
import javax.tools.Diagnostic
import javax.tools.DiagnosticCollector
import javax.tools.JavaFileObject
import javax.tools.ToolProvider

```

```

data class RouteValueFinding(
    val lineNumber: Long,
    val methodName: String,
    val parameterName: String,
    val parameterType: String,
    val reason: String

```

```

)

data class RouteValueContext(
    val types: Types,
    val validatedValueType: TypeMirror,
    val iterableType: TypeMirror,
    val optionalType: TypeMirror,
    val mapType: TypeMirror
)

val VALIDATED_VALUE_TYPE_NAME = "org.owasp.untrust.vv.ValidatedValue"

val ROUTE_HANDLER_ANNOTATIONS = setOf(
    "RequestMapping",
    "GetMapping",
    "PostMapping",
    "PutMapping",
    "PatchMapping",
    "DeleteMapping"
)

val FRAMEWORK_ROUTE_PARAMETER_TYPES = setOf(
    "java.security.Principal",
    "jakarta.servlet.http.HttpServletRequest",
    "jakarta.servlet.http.HttpServletResponse",
    "jakarta.servlet.http.HttpSession",
    "org.springframework.security.core.Authentication",
    "org.springframework.security.core.userdetails.UserDetails",
    "org.springframework.security.web.csrf.CsrfToken",
    "org.springframework.web.multipart.MultipartFile",
    "org.springframework.ui.Model",
    "org.springframework.validation.BindingResult"
)

fun File.relativeUnixPath(rootDir: File): String {
    return rootDir.toPath()
        .relativize(toPath())
        .toString()
        .replace(File.separatorChar, '/')
}

fun sourceLine(compilationUnit: CompilationUnitTree, tree: Tree, trees: Trees): Long {
    val startPosition = trees.sourcePositions.getStartPosition(compilationUnit, tree)

```

```

return if (startPosition >= 0) {
    compilationUnit.lineMap.getLineNumber(startPosition)
} else {
    -1L
}
}

fun annotationSimpleName(annotation: AnnotationTree): String {
    return annotation.annotationType.toString()
        .substringAfterLast('.')
        .substringAfterLast('$')
}

fun hasAnyAnnotation(annotations: List<AnnotationTree>, names: Set<String>): Boolean {
    return annotations.any { annotation -> annotationSimpleName(annotation) in names }
}

fun TypeMirror.readableName(): String {
    return toString()
}

fun isSubtypeOfErased(context: RouteValueContext, type: TypeMirror, target: TypeMirror): Boolean {
    return context.types.isSubtype(context.types.erasure(type), context.types.erasure(target))
}

fun isValidatedValue(context: RouteValueContext, type: TypeMirror): Boolean {
    return type.kind != TypeKind.ERROR && isSubtypeOfErased(context, type, context.validatedValueType)
}

fun isMapType(context: RouteValueContext, type: TypeMirror): Boolean {
    return type.kind != TypeKind.ERROR && isSubtypeOfErased(context, type, context.mapType)
}

fun isIterableType(context: RouteValueContext, type: TypeMirror): Boolean {
    return type.kind != TypeKind.ERROR && isSubtypeOfErased(context, type, context.iterableType)
}

fun isOptionalType(context: RouteValueContext, type: TypeMirror): Boolean {
    return type.kind != TypeKind.ERROR && isSubtypeOfErased(context, type, context.optionalType)
}

fun declaredTypeElement(type: TypeMirror): TypeElement? {
    return (type as? DeclaredType)?.asElement() as? TypeElement
}

```



```

fun isFrameworkRouteParameter(type: TypeMirror): Boolean {
    val erasedName = type.toString().substringBefore('<')
    return erasedName in FRAMEWORK_ROUTE_PARAMETER_TYPES
}

fun isProjectDtoCandidate(type: TypeMirror): Boolean {
    val erasedName = type.toString().substringBefore('<')
    return !erasedName.startsWith("java.") &&
        !erasedName.startsWith("javax.") &&
        !erasedName.startsWith("jakarta.") &&
        !erasedName.startsWith("org.springframework.")
}

fun validateRequestType(
    context: RouteValueContext,
    type: TypeMirror,
    path: String,
    visitedTypes: MutableSet<String>
): String? {
    if (type.kind == TypeKind.ERROR) {
        return "$path has unresolved type ${type.readableName()}."
    }

    if (type.kind.isPrimitive) {
        return "$path uses primitive type ${type.readableName()} instead of a ValidatedValue<T> subtype. Do not accept"
    }

    if (isValidatedValue(context, type)) {
        return null
    }

    if (isMapType(context, type)) {
        return "$path uses ${type.readableName()}; dynamic key/value request shapes are not allowed."
    }

    if (type is DeclaredType && isOptionalType(context, type)) {
        if (type.typeArguments.isEmpty()) {
            return "$path uses raw Optional type ${type.readableName()}; optional request values must state their element"
        }

        val argument = type.typeArguments.single()
        if (isFrameworkRouteParameter(argument)) {
            return null
        }
    }
}

```

```

    }

    return validateRequestType(context, argument, "$path optional value", visitedTypes)
}

if (type is ArrayType) {
    return validateRequestType(context, type.componentType, "$path[]", visitedTypes)
}

if (type is DeclaredType && isIterableType(context, type)) {
    if (type.typeArguments.isEmpty()) {
        return "$path uses raw iterable type ${type.readableName()}; request element values must be stated exactly."
    }

    return type.typeArguments
        .mapIndexedNotNull { index, argument ->
            validateRequestType(context, argument, "$path element ${index + 1}", visitedTypes)
        }
        .firstOrNull()
}

val typeElement = declaredTypeElement(type)
?: return "$path uses ${type.readableName()} instead of a ValidatedValue<T> subtype or request DTO."

val typeName = typeElement.qualifiedName.toString()

if (!isProjectDtoCandidate(type)) {
    return "$path uses ${type.readableName()} instead of a ValidatedValue<T> subtype."
}

if (!visitedTypes.add(typeName)) {
    return null
}

val recordComponents = typeElement.enclosedElements
    .filter { element -> element.kind == ElementKind.RECORD_COMPONENT }

val dtoMembers = if (recordComponents.isEmpty()) {
    recordComponents
} else {
    typeElement.enclosedElements.filter { element ->
        element.kind == ElementKind.FIELD && Modifier.STATIC !in element.modifiers
    }
}

```



```

        lineNumber = sourceLine(compilationUnit, parameter, trees),
        methodName = node.name.toString(),
        parameterName = parameter.name.toString(),
        parameterType = parameterType.readableName(),
        reason = reason
    )
}
}
}

    super.visitMethod(node, unused)
}
}.scan(compilationUnit, Unit)

return findings
}

fun compilerOptionsFor(sourceSet: SourceSet): List<String> {
    val options = mutableListOf("-proc:none")
    val classpath = sourceSet.compileClasspath.files

    if (classpath.isNotEmpty()) {
        options += "-classpath"
        options += classpath.joinToString(File.pathSeparator) { it.absolutePath }
    }

    val sourceDirectories = sourceSet.allJava.srcDirs.filter { it.exists() }

    if (sourceDirectories.isNotEmpty()) {
        options += "-sourcepath"
        options += sourceDirectories.joinToString(File.pathSeparator) { it.absolutePath }
    }

    return options
}

fun inspectSourceSetForUnvalidatedRouteValues(sourceSet: SourceSet, rootDir: File): List<String> {
    val sourceFiles = sourceSet.allJava.files
        .filter { it.extension == "java" && it.exists() }
        .sortedBy { it.absolutePath }

    if (sourceFiles.isEmpty()) {
        return emptyList()
    }
}

```

```

val compiler = ToolProvider.getSystemJavaCompiler()
?: throw GradleException("A JDK is required to inspect Java source files. Gradle is not running with a JDK compiler")

val diagnostics = DiagnosticCollector<JavaFileObject>()

compiler.getStandardFileManager(diagnostics, Locale.ROOT, Charsets.UTF_8).use { fileManager ->
    val javaFiles = fileManager.getJavaFileObjectsFromFiles(sourceFiles)
    val task = compiler.getTask(
        StringWriter(),
        fileManager,
        diagnostics,
        compilerOptionsFor(sourceSet),
        emptyList<String>(),
        javaFiles
    ) as JavacTask

    val trees = Trees.instance(task)
    val parsedUnits = task.parse().toList()
    task.analyze()

    val errors = diagnostics.diagnostics
        .filter { it.kind == Diagnostic.Kind.ERROR }

    if (errors.isNotEmpty()) {
        val message = errors.joinToString("\n") { diagnostic ->
            val sourceName = diagnostic.source?.name ?: "<unknown source>"
            "- $sourceName:${diagnostic.lineNumber}: ${diagnostic.getMessage(Locale.ROOT)}"
        }

        throw GradleException("Failed to analyze Java source files before checking route request values:\n$message")
    }

    val elements = task.elements
    val routeValueContext = RouteValueContext(
        types = task.types,
        validatedValueType = elements.getTypeElement(VALIDATED_VALUE_TYPE_NAME)?.asType()
        ?: throw GradleException("Could not resolve $VALIDATED_VALUE_TYPE_NAME while checking route request values."),
        iterableType = elements.getTypeElement("java.lang.Iterable")?.asType()
        ?: throw GradleException("Could not resolve java.lang.Iterable while checking route request values."),
        optionalType = elements.getTypeElement("java.util.Optional")?.asType()
        ?: throw GradleException("Could not resolve java.util.Optional while checking route request values."),
        mapType = elements.getTypeElement("java.util.Map")?.asType()
        ?: throw GradleException("Could not resolve java.util.Map while checking route request values.")
    )

```

```

)

return parsedUnits.flatMap { compilationUnit ->
    val sourceFile = File(compilationUnit.sourceFile.toUri())
    val sourcePath = sourceFile.relativeUnixPath(rootDir)

    collectRouteValueFindings(compilationUnit, trees, routeValueContext).map { finding ->
        "$sourcePath:${finding.lineNumber} route method '${finding.methodName}' parameter '${finding.parameter}'"
    }
}
}
}
}
}

```

```

fun routeValueFailureGuidance(): String {
    return ""
}

```

ROUTE REQUEST VALUE POLICY FAILURE

Every user-controlled value entering a route handler must be explicit and validated at the boundary.

Allowed request shapes:

- A route parameter whose type **extends** `ValidatedValue<T>`.
- A request DTO whose record components or instance fields recursively contain only `ValidatedValue<T>` leaves.
- `Optional<T>` when `T` recursively satisfies **this** same rule, so optional request values enter handlers as `Optional.empty()` rather than `null`.
- Iterable/array request members only when their element type recursively satisfies the same rule.

Not allowed:

- Raw **String**, primitive, **enum**, date/time, or other framework/JDK scalar request values.
- **Map**<K, V> or any dynamic key/value request shape.
- Generic **catch**-all DTOs that **defer** validation to service code.

Important instruction **for** AI/code-generation tools:

Do not work around **this** by hiding raw request data behind generic wrappers. Add domain-specific `ValidatedValue<T>` value types and make route DTOs state each accepted value exactly.

```

""".trimIndent()
}

```

```

val forbidUnvalidatedRouteValues by tasks.registering {
    group = "verification"
    description = "Fails the build when route handlers accept request values that are not ValidatedValue-backed."

    val javaExtension = project.extensions.getByType<JavaPluginExtension>()
    val sourceSets = javaExtension.sourceSets
}

```

```

sourceSets.configureEach {
    inputs.files(allJava)
    inputs.files(compileClasspath)
}

doLast {
    val violations = sourceSets.flatMap { sourceSet ->
        inspectSourceSetForUnvalidatedRouteValues(sourceSet, rootProject.projectDir)
    }

    if (violations.isNotEmpty()) {
        throw GradleException(
            "Unvalidated route request values found:\n" +
            violations.joinToString("\n") { " - $it" } +
            "\n\n" +
            routeValueFailureGuidance()
        )
    }
}

tasks.matching { it.name == "check" }.configureEach {
    dependsOn(forbidUnvalidatedRouteValues)
}

```

gradle/forbid-validated-value-inheritance.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/gradle/forbid-validated-value-inheritance.gradle.kts>

```

/*
 * DESIGN BUILD GATE: VALIDATED VALUE INHERITANCE ENFORCEMENT
 *
 * ValidatedValue subclasses should be leaf value types. Each value class should inherit directly
 * from ValidatedValue and own the Traits class that defines its validation behavior.
 *
 * Escape hatch:
 *
 * // VALIDATED VALUE INHERITANCE REASON: <specific justification>
 * class ...
 *
 * // INTENTIONALLY EXPOSE UNCHECKED: <specific justification>
 * return exposeUnchecked();
 *
 * The same guard applies to both unusual ValidatedValue inheritance and an owned Traits class
 * that intentionally inherits from a traits base outside package org.owasp.untrust.vv.
 */

```

```

import com.sun.source.tree.ClassTree
import com.sun.source.tree.CompilationUnitTree
import com.sun.source.tree.ExpressionTree
import com.sun.source.tree.MethodInvocationTree
import com.sun.source.tree.MethodTree
import com.sun.source.tree.ParenthesizedTree
import com.sun.source.tree.ReturnTree
import com.sun.source.tree.Tree
import com.sun.source.tree.TypeCastTree
import com.sun.source.util.JavacTask
import com.sun.source.util.TreePathScanner
import com.sun.source.util.TreeScanner
import com.sun.source.util.Trees
import groovy.json.JsonSlurper
import org.gradle.api.GradleException
import org.gradle.api.plugins.JavaPluginExtension
import org.gradle.api.tasks.SourceSet
import java.io.File
import java.io.StringWriter
import java.util.Locale
import javax.lang.model.element.Modifier
import javax.tools.Diagnostic
import javax.tools.DiagnosticCollector
import javax.tools.JavaFileObject
import javax.tools.ToolProvider

```



```

data class ValidatedValueClassInfo(
    val qualifiedName: String,
    val simpleName: String,
    val packageName: String,
    val lineNumber: Long,
    val sourceFile: File,
    val sourcePath: String,
    val extendsText: String?,
    val implementsTexts: List<String>,
    val validatedValueTypeArgumentText: String?,
    val modifiers: Set<Modifier>,
    val nestedClasses: List<ValidatedValueNestedClassInfo>,
    val exposeUncheckedReturns: List<ValidatedValueExposeUncheckedReturn>,
    val allowReason: String?,
    val customValidationTraitsReason: String?
)

data class ValidatedValueNestedClassInfo(
    val simpleName: String,
    val extendsText: String?,
    val implementsTexts: List<String>
)

data class ValidatedValueExposeUncheckedReturn(
    val lineNumber: Long,
    val methodName: String,
    val allowReason: String?
)

data class ValidatedValueFinding(
    val lineNumber: Long,
    val className: String,
    val reason: String
)

data class ValidatedValueGuardConfig(
    val minimumCustomValidationTraitsReasonCharacters: Int,
    val minimumExposeUncheckedReasonCharacters: Int
)

val VALIDATED_VALUE_ALLOW_MARKER = "VALIDATED VALUE INHERITANCE REASON:"
val CUSTOM_VALIDATION_TRAITS_REASON_MARKER = "NEED FOR CUSTOM VALIDATION TRAITS:"
val INTENTIONALLY_EXPOSE_UNCHECKED_MARKER = "INTENTIONALLY EXPOSE UNCHECKED:"
val MINIMUM_VALIDATED_VALUE_ALLOW_REASON_CHARACTERS = 120

```

```

val VV_TRAIT_BASE_SIMPLE_NAMES = setOf(
    "BoundedAnyContentStringTraits",
    "BoundedValueTraits",
    "CustomValidationForRareCasesTraits",
    "EnumValidationTraits",
    "PrintableUnicodeStringTraits",
    "RareTraitsCaseWhereParsingIsTheWholeValidation",
    "RegexStringTraits",
    "ValidationTraits"
)

val VV_TRAIT_BASE_QUALIFIED_NAMES = VV_TRAIT_BASE_SIMPLE_NAMES
    .map { "org.owasp.untrust.vv.$it" }
    .toSet()

@Suppress("UNCHECKED_CAST")
val precedingEscapeReasonBefore = rootProject.extensions.extraProperties["precedingEscapeReasonBefore"] as (File,
@Suppress("UNCHECKED_CAST")
val precedingEscapeHatchUsage = rootProject.extensions.extraProperties["precedingEscapeHatchUsage"] as (String, Int)

fun readValidatedValueGuardConfig(configFile: File): ValidatedValueGuardConfig {
    if (!configFile.exists()) {
        throw GradleException("validated_value_guardrail.json is required.")
    }

    val parsed = JsonSlurper().parse(configFile)
    if (parsed !is Map<*, *>) {
        throw GradleException("validated_value_guardrail.json must contain a JSON object.")
    }

    val rawMinimumCustomValidationTraitsReason = parsed["minimumCustomValidationTraitsReasonCharacters"]
    val minimumCustomValidationTraitsReason = when (rawMinimumCustomValidationTraitsReason) {
        is Number -> rawMinimumCustomValidationTraitsReason.toInt()
        is String -> rawMinimumCustomValidationTraitsReason.toIntOrNull()
        else -> null
    }
    if (minimumCustomValidationTraitsReason < 1) {
        throw GradleException("validated_value_guardrail.json field 'minimumCustomValidationTraitsReasonCharacters' must be at least 1")
    }

    val rawMinimumExposeUncheckedReason = parsed["minimumExposeUncheckedReasonCharacters"]
    val minimumExposeUncheckedReason = when (rawMinimumExposeUncheckedReason) {
        is Number -> rawMinimumExposeUncheckedReason.toInt()
        is String -> rawMinimumExposeUncheckedReason.toIntOrNull()
        else -> null
    }

```

```

} ?: throw GradleException("validated_value_guardrail.json field 'minimumExposeUncheckedReasonCharacters' must

if (minimumExposeUncheckedReason < 1) {
    throw GradleException("validated_value_guardrail.json field 'minimumExposeUncheckedReasonCharacters' must
}

return ValidatedValueGuardConfig(minimumCustomValidationTraitsReason, minimumExposeUncheckedReason)
}

fun File.relativeUnixPath(rootDir: File): String {
    return rootDir.toPath()
        .relativize(toPath())
        .toString()
        .replace(File.separatorChar, '/')
}

fun sourceLine(compilationUnit: CompilationUnitTree, tree: Tree, trees: Trees): Long {
    val startPosition = trees.sourcePositions.getStartPosition(compilationUnit, tree)

    return if (startPosition >= 0) {
        compilationUnit.lineMap.getLineNumber(startPosition)
    } else {
        -1L
    }
}

fun compilerOptionsFor(sourceSet: SourceSet): List<String> {
    val options = mutableListOf("-proc:none")
    val classpath = sourceSet.compileClasspath.files

    if (classpath.isNotEmpty()) {
        options += "-classpath"
        options += classpath.joinToString(File.pathSeparator) { it.absolutePath }
    }

    val sourceDirectories = sourceSet.allJava.srcDirs.filter { it.exists() }

    if (sourceDirectories.isNotEmpty()) {
        options += "-sourcepath"
        options += sourceDirectories.joinToString(File.pathSeparator) { it.absolutePath }
    }

    return options
}

```

```

fun qualifiedClassName(packageName: String, classStack: List<String>, currentName: String): String {
    val nestedName = (classStack + currentName).joinToString(".")

    return if (packageName.isBlank()) {
        nestedName
    } else {
        "$packageName.$nestedName"
    }
}

fun allowReasonForClass(sourceFile: File, classLine: Long): String? {
    return precedingEscapeReasonBefore(sourceFile, classLine, VALIDATED_VALUE_ALLOW_MARKER, false)
}

fun customValidationTraitsReasonForClass(sourceFile: File, classLine: Long): String? {
    return precedingEscapeReasonBefore(sourceFile, classLine, CUSTOM_VALIDATION_TRAITS_REASON_MARKER, false)
}

fun exposeUncheckedReturnReason(sourceFile: File, returnLine: Long): String? {
    return precedingEscapeReasonBefore(sourceFile, returnLine, INTENTIONALLY_EXPOSE_UNCHECKED_MARKER, false)
}

fun collectValidatedValueClassInfo(
    compilationUnit: CompilationUnitTree,
    trees: Trees,
    rootDir: File
): List<ValidatedValueClassInfo> {
    val infos = mutableListOf<ValidatedValueClassInfo>()
    val packageName = compilationUnit.packageName?.toString().orEmpty()
    val sourceFile = File(compilationUnit.sourceFile.toUri())
    val sourcePath = sourceFile.relativeUnixPath(rootDir)
    val classStack = mutableListOf<String>()

    object : TreePathScanner<Unit, Unit>() {
        override fun visitClass(node: ClassTree, unused: Unit?) {
            val className = node.simpleName.toString()
            val lineNumber = sourceLine(compilationUnit, node, trees)
            val nestedClasses = node.members
                .filterIsInstance<ClassTree>()
                .map { nested ->
                    ValidatedValueNestedClassInfo(
                        simpleName = nested.simpleName.toString(),
                        extendsText = nested.extendsClause?.toString(),

```

```

        implementsTexts = nested.implementsClause.map { it.toString() }
    )
}

infos += ValidatedValueClassInfo(
    qualifiedName = qualifiedClassName(packageName, classStack, className),
    simpleName = className,
    packageName = packageName,
    lineNumber = lineNumber,
    sourceFile = sourceFile,
    sourcePath = sourcePath,
    extendsText = node.extendsClause?.toString(),
    implementsTexts = node.implementsClause.map { it.toString() },
    validatedValueTypeArgumentText = validatedValueTypeArgumentText(node.extendsClause?.toString()),
    modifiers = node.modifiers.flags,
    nestedClasses = nestedClasses,
    exposeUncheckedReturns = collectExposeUncheckedReturns(node, compilationUnit, trees, sourceFile),
    allowReason = allowReasonForClass(sourceFile, lineNumber),
    customValidationTraitsReason = customValidationTraitsReasonForClass(sourceFile, lineNumber)
)

classStack += className
super.visitClass(node, unused)
classStack.removeLast()
}
}.scan(compilationUnit, Unit)

return infos
}

```

```

fun collectExposeUncheckedReturns(
    classTree: ClassTree,
    compilationUnit: CompilationUnitTree,
    trees: Trees,
    sourceFile: File
): List<ValidatedValueExposeUncheckedReturn> {
    return classTree.members
        .filterIsInstance<MethodTree>()
        .filter { it.body != null }
        .flatMap { method ->
            val findings = mutableListOf<ValidatedValueExposeUncheckedReturn>()

            object : TreeScanner<Unit, Unit>() {
                override fun visitClass(node: ClassTree, unused: Unit?) {

```

```

        return
    }

    override fun visitReturn(node: ReturnTree, unused: Unit?) {
        if (isExposeUncheckedInvocation(node.expression)) {
            val lineNumber = sourceLine(compilationUnit, node, trees)
            findings += ValidatedValueExposeUncheckedReturn(
                lineNumber,
                method.name.toString(),
                exposeUncheckedReturnReason(sourceFile, lineNumber)
            )
        }
        super.visitReturn(node, unused)
    }
}.scan(method.body, Unit)

findings
}
}

fun isExposeUncheckedInvocation(expression: ExpressionTree?): Boolean {
    return when (expression) {
        is ParenthesizedTree -> isExposeUncheckedInvocation(expression.expression)
        is TypeCastTree -> isExposeUncheckedInvocation(expression.expression)
        is MethodInvocationTree -> expression.methodSelect.toString().substringAfterLast(".") == "exposeUnchecked"
        else -> false
    }
}

fun rawTypeName(typeText: String?): String? {
    if (typeText == null) {
        return null
    }

    return typeText.substringBefore("<").trim()
}

fun typeArguments(typeText: String?): List<String> {
    if (typeText == null) {
        return emptyList()
    }

    val start = typeText.indexOf('<')
    val end = typeText.lastIndexOf('>')

```

```
if (start < 0 || end <= start) {  
    return emptyList()  
}
```

```
val arguments = mutableListOf<String>()  
val current = StringBuilder()  
var depth = 0
```

```
for (character in typeText.substring(start + 1, end)) {  
    when (character) {  
        '<' -> {  
            depth += 1  
            current.append(character)  
        }  
        '>' -> {  
            depth -= 1  
            current.append(character)  
        }  
        ';' -> {  
            if (depth == 0) {  
                arguments += current.toString().trim()  
                current.clear()  
            } else {  
                current.append(character)  
            }  
        }  
        else -> current.append(character)  
    }  
}
```

```
val finalArgument = current.toString().trim()  
if (finalArgument.isNotEmpty()) {  
    arguments += finalArgument  
}
```

```
return arguments  
}
```

```
fun validatedValueTypeArgumentText(extendsText: String?): String? {  
    val superType = rawTypeName(extendsText)  
    if (superType != "ValidatedValue" && superType != "org.owasp.untrust.vv.ValidatedValue") {  
        return null  
    }  
}
```

```

    return typeArguments(extendsText).firstOrNull()
}

fun simpleTypeName(typeText: String?): String? {
    return rawTypeName(typeText)?.substringAfterLast(".")
}

fun isOptionalType(typeText: String?): Boolean {
    val rawType = rawTypeName(typeText)
    ?: return false

    return rawType == "Optional" || rawType == "java.util.Optional"
}

fun isCustomValidationForRareCasesTraits(classInfo: ValidatedValueClassInfo): Boolean {
    val superType = rawTypeName(classInfo.extendsText)
    ?: return false

    return superType == "CustomValidationForRareCasesTraits" ||
        superType == "org.owasp.untrust.vv.CustomValidationForRareCasesTraits"
}

fun isDirectValidatedValue(classInfo: ValidatedValueClassInfo): Boolean {
    val superType = rawTypeName(classInfo.extendsText)
    return superType == "ValidatedValue" || superType == "org.owasp.untrust.vv.ValidatedValue"
}

fun isDirectValidationTraits(typeText: String?): Boolean {
    val rawType = rawTypeName(typeText)
    ?: return false

    return rawType == "ValidationTraits" || rawType == "org.owasp.untrust.vv.ValidationTraits"
}

fun resolveSuperInfo(
    classInfo: ValidatedValueClassInfo,
    classesByQualifiedNames: Map<String, ValidatedValueClassInfo>,
    classesBySimpleName: Map<String, List<ValidatedValueClassInfo>>
): ValidatedValueClassInfo? {
    val superType = rawTypeName(classInfo.extendsText) ?: return null

    classesByQualifiedNames[superType]?.let { return it }
    classesByQualifiedNames["${classInfo.packageName}.$superType"]?.let { return it }
}

```



```

    val simpleName = superType.substringAfterLast(".")
    return classesBySimpleName[simpleName]?.singleOrNull()
}

fun inheritsFromValidatedValue(
    classInfo: ValidatedValueClassInfo,
    classesByQualifiedNames: Map<String, ValidatedValueClassInfo>,
    classesBySimpleName: Map<String, List<ValidatedValueClassInfo>>,
    visited: Set<String> = emptySet()
): Boolean {
    if (classInfo.qualifiedName in visited) {
        return false
    }

    if (isDirectValidatedValue(classInfo)) {
        return true
    }

    val superInfo = resolveSuperInfo(classInfo, classesByQualifiedNames, classesBySimpleName)
    ?: return false

    return inheritsFromValidatedValue(
        superInfo,
        classesByQualifiedNames,
        classesBySimpleName,
        visited + classInfo.qualifiedName
    )
}

fun implementsOrInheritsValidationTraits(
    classInfo: ValidatedValueClassInfo,
    classesByQualifiedNames: Map<String, ValidatedValueClassInfo>,
    classesBySimpleName: Map<String, List<ValidatedValueClassInfo>>,
    visited: Set<String> = emptySet()
): Boolean {
    if (classInfo.qualifiedName in visited) {
        return false
    }

    if (classInfo.implementsTexts.any(::isDirectValidationTraits)) {
        return true
    }

    val superInfo = resolveSuperInfo(classInfo, classesByQualifiedNames, classesBySimpleName)

```

?: return false

```
return implementsOrInheritsValidationTraits(  
    superInfo,  
    classesByQualifiedNames,  
    classesBySimpleNames,  
    visited + classInfo.qualifiedName  
)  
}
```

```
fun hasOwnedVvTraitsClass(  
    classInfo: ValidatedValueClassInfo,  
    vvTraitSimpleNames: Set<String>,  
    vvTraitQualifiedNames: Set<String>  
): Boolean {  
    val traitsClass = classInfo.nestedClasses.singleOrNull { it.simpleName == "Traits" }  
    ?: return false  
  
    val superType = rawTypeName(traitsClass.extendsText)  
    ?: return false
```

```
    return superType in vvTraitQualifiedNames || simpleTypeName(superType) in vvTraitSimpleNames  
}
```

```
fun allowReasonProblem(classInfo: ValidatedValueClassInfo): String? {  
    val reason = classInfo.allowReason ?: return null  
  
    return if (reason.length < MINIMUM_VALIDATED_VALUE_ALLOW_REASON_CHARACTERS) {  
        "ALLOW reason is ${reason.length} characters; minimum is $MINIMUM_VALIDATED_VALUE_ALLOW_REASON_CHA  
    } else {  
        null  
    }  
}
```

```
fun customValidationTraitsReasonProblem(classInfo: ValidatedValueClassInfo, config: ValidatedValueGuardConfig): Str  
if (!isCustomValidationForRareCasesTraits(classInfo)) {  
    return null  
}
```

```
val reason = classInfo.customValidationTraitsReason  
?: return "extends CustomValidationForRareCasesTraits without an immediately preceding $CUSTOM_VALIDATION  
  
return if (reason.length < config.minimumCustomValidationTraitsReasonCharacters) {  
    "$CUSTOM_VALIDATION_TRAITS_REASON_MARKER reason is ${reason.length} characters; minimum is ${config.mi
```

```

    } else {
        null
    }
}

```

```

fun exposeUncheckedReturnProblem(returned: ValidatedValueExposeUncheckedReturn, config: ValidatedValueGuardConfig) {
    val reason = returned.allowReason
    if (reason != null) {
        if (reason.length < config.minimumExposeUncheckedReasonCharacters) {
            "$INTENTIONALLY_EXPOSE_UNCHECKED_MARKER reason is ${reason.length} characters; minimum is ${config.minimumExposeUncheckedReasonCharacters}"
        } else {
            null
        }
    }
}

```

```

fun collectValidatedValueFindings(
    classInfos: List<ValidatedValueClassInfo>,
    config: ValidatedValueGuardConfig
): Map<String, List<ValidatedValueFinding>> {
    val classesByQualifiedNames = classInfos.associateBy { it.qualifiedName }
    val classesBySimpleName = classInfos.groupBy { it.simpleName }
    val vvTraitInfos = classInfos
        .filter {
            it.packageName == "org.owasp.untrust.vv" &&
            implementsOrInheritsValidationTraits(it, classesByQualifiedNames, classesBySimpleName)
        }
    val vvTraitSimpleNames = vvTraitInfos.map { it.simpleName }.toSet() + VV_TRAIT_BASE_SIMPLE_NAMES
    val vvTraitQualifiedNames = vvTraitInfos.map { it.qualifiedName }.toSet() + VV_TRAIT_BASE_QUALIFIED_NAMES

    return classInfos
        .mapNotNull { classInfo ->
            val customTraitsProblem = customValidationTraitsReasonProblem(classInfo, config)
            if (!inheritsFromValidatedValue(classInfo, classesByQualifiedNames, classesBySimpleName)) {
                return@mapNotNull customTraitsProblem?.let { problem ->
                    classInfo.sourcePath to listOf(ValidatedValueFinding(classInfo.lineNumber, classInfo.qualifiedName, problem))
                }
            }

            val findings = mutableListOf<ValidatedValueFinding>()

            customTraitsProblem?.let { problem ->
                findings += ValidatedValueFinding(classInfo.lineNumber, classInfo.qualifiedName, problem)
            }
        }
}

```

```

allowReasonProblem(classInfo)?.let { problem ->
    findings += ValidatedValueFinding(classInfo.lineNumber, classInfo.qualifiedName, problem)
}

val hasValidAllow = classInfo.allowReason != null && allowReasonProblem(classInfo) == null

if (!hasValidAllow) {
    if (!isDirectValidatedValue(classInfo)) {
        findings += ValidatedValueFinding(
            classInfo.lineNumber,
            classInfo.qualifiedName,
            "inherits from another ValidatedValue descendant instead of directly extending ValidatedValue."
        )
    }

    if (Modifier.FINAL !in classInfo.modifiers) {
        findings += ValidatedValueFinding(
            classInfo.lineNumber,
            classInfo.qualifiedName,
            "is a ValidatedValue descendant but is not final."
        )
    }

    if (isDirectValidatedValue(classInfo) && !hasOwnedVvTraitsClass(classInfo, vvTraitSimpleNames, vvTraitQualifi
        findings += ValidatedValueFinding(
            classInfo.lineNumber,
            classInfo.qualifiedName,
            "directly extends ValidatedValue but does not own a nested Traits class extending a traits class from pack
        )
    }

    if (isDirectValidatedValue(classInfo) && isOptionalType(classInfo.validatedValueTypeArgumentText)) {
        findings += ValidatedValueFinding(
            classInfo.lineNumber,
            classInfo.qualifiedName,
            "uses ValidatedValue<${classInfo.validatedValueTypeArgumentText}, ...>. ValidatedValue must wrap the a
        )
    }
}

classInfo.exposeUncheckedReturns.forEach { returned ->
    exposeUncheckedReturnProblem(returned, config)?.let { problem ->

```

```

        findings += ValidatedValueFinding(
            returned.lineNumber,
            classInfo.qualifiedName,
            problem
        )
    }
}

if (findings.isEmpty()) {
    null
} else {
    classInfo.sourcePath to findings
}
}

.groupBy({ it.first }, { it.second })
.mapValues { (_, nestedFindings) -> nestedFindings.flatten() }
}

fun inspectSourceSetForValidatedValueInheritance(
    sourceSet: SourceSet,
    rootDir: File,
    config: ValidatedValueGuardConfig
): Map<String, List<ValidatedValueFinding>> {
    val sourceFiles = sourceSet.allJava.files
        .filter { it.extension == "java" && it.exists() }
        .sortedBy { it.absolutePath }

    if (sourceFiles.isEmpty()) {
        return emptyMap()
    }

    val compiler = ToolProvider.getSystemJavaCompiler()
        ?: throw GradleException("A JDK is required to inspect Java source files. Gradle is not running with a JDK compiler")

    val diagnostics = DiagnosticCollector<JavaFileObject>()

    compiler.getStandardFileManager(diagnostics, Locale.ROOT, Charsets.UTF_8).use { fileManager ->
        val javaFiles = fileManager.getJavaFileObjectsFromFiles(sourceFiles)
        val task = compiler.getTask(
            StringWriter(),
            fileManager,
            diagnostics,
            compilerOptionsFor(sourceSet),
            emptyList<String>(),

```

```

        javaFiles
    ) as JacoTask

    val trees = Trees.instance(task)
    val parsedUnits = task.parse().toList()

    val errors = diagnostics.diagnostics
        .filter { it.kind == Diagnostic.Kind.ERROR }

    if (errors.isNotEmpty()) {
        val message = errors.joinToString("\n") { diagnostic ->
            val sourceName = diagnostic.source?.name ?: "<unknown source>"
            "- $sourceName:${diagnostic.lineNumber}: ${diagnostic.getMessage(Locale.ROOT)}"
        }

        throw GradleException("Failed to analyze Java source files before checking ValidatedValue inheritance:\n$message")
    }

    return collectValidatedValueFindings(parsedUnits.flatMap { compilationUnit ->
        collectValidatedValueClassInfo(compilationUnit, trees, rootDir)
    }, config)
}

fun validatedValueInheritanceFailureGuidance(config: ValidatedValueGuardConfig): String {
    return """

```

VALIDATED VALUE INHERITANCE POLICY FAILURE

ValidatedValue classes must be leaf values:

- Extend ValidatedValue directly.
- Mark the value **class final**.
- Define an owned nested Traits **class** that **extends** one of the traits classes in **package** `org.owasp.untrust.vv`.
- **Do not use** `Optional<T>` as the ValidatedValue value type. The value **class** should validate the present value, **while** absence belongs at the boundary as `Optional<ValueClass>` on a DTO field or route handler argument, **for** example:


```
record SomethingRequest(Optional<CommentId> commentId, Title title)
```
- **Do not** add methods that **return** `exposeUnchecked()`. That hides the safety boundary behind a harmless-looking accessor. Call `exposeUnchecked()` at the actual **use** site when raw access is intentionally needed.
- Avoid `CustomValidationForRareCasesTraits` unless the validation really cannot be expressed with the standard traits.

Do not create intermediate value base classes such as `RequiredTextValue` or `OptionalTextValue` or inherit an owned Traits **class from** outside **package** `org.owasp.untrust.vv` unless a human intentionally accepts that abstraction with an escape hatch comment.

The comment is **for** exceptional designs only: explain why direct `ValidatedValue` inheritance or the existing `vv` traits are not suitable, and how the deeper inheritance or non-`vv` trait-base risk is constrained.

```

${precedingEscapeHatchUsage(INTENTIONALLY_EXPOSE_UNCHECKED_MARKER, config.minimumExposeUncheckedRe

${precedingEscapeHatchUsage(CUSTOM_VALIDATION_TRAITS_REASON_MARKER, config.minimumCustomValidationTra

${precedingEscapeHatchUsage(VALIDATED_VALUE_ALLOW_MARKER, MINIMUM_VALIDATED_VALUE_ALLOW_REASON_C
"""").trimIndent()
}

```

```

val forbidValidatedValueInheritance by tasks.registering {
    group = "verification"
    description = "Fails the build when ValidatedValue descendants are not final direct values with owned vv traits."

    val javaExtension = project.extensions.getByType<JavaPluginExtension>()
    val sourceSets = javaExtension.sourceSets

    sourceSets.configureEach {
        inputs.files(allJava)
        inputs.files(compileClasspath)
    }
    inputs.file(rootProject.file("validated_value_guardrail.json"))

    doLast {
        val config = readValidatedValueGuardConfig(rootProject.file("validated_value_guardrail.json"))
        val findingsBySource = sourceSets
            .map { sourceSet -> inspectSourceSetForValidatedValueInheritance(sourceSet, rootProject.projectDir, config) }
            .fold(emptyMap<String, List<ValidatedValueFinding>>()) { acc, findings ->
                (acc.keys + findings.keys).associateWith { key ->
                    acc.getOrElse(key, emptyList()) + findings.getOrElse(key, emptyList())
                }
            }

        if (findingsBySource.isNotEmpty()) {
            val violations = findingsBySource
                .toSortedMap()
                .flatMap { (sourcePath, findings) ->
                    findings.sortedBy { it.lineNumber }.map { finding ->

```

```

        "$sourcePath:${finding.lineNumber} ${finding.className} ${finding.reason}"
    }
}

throw GradleException(
    "ValidatedValue inheritance violations found:\n" +
    violations.joinToString("\n") { " - $it" } +
    "\n\n" +
    validatedValueInheritanceFailureGuidance(config)
)
}
}
}

tasks.matching { it.name == "check" }.configureEach {
    dependsOn(forbidValidatedValueInheritance)
}

```

gradle/json_config_reader.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/gradle/json_config_reader.gradle.kts>


```

import groovy.json.JsonSlurper
import org.gradle.api.GradleException
import java.io.File

fun jsonConfigPath(rawPath: String, baseDir: File): File {
    val file = File(rawPath)
    return if (file.isAbsolute) file else File(baseDir, rawPath)
}

fun jsonConfigContext(file: File): String {
    return file.relativeToOrSelf(rootProject.rootDir).path.replace(File.separatorChar, '/')
}

fun deepMergeJsonConfig(base: Map<String, Any?>, override: Map<String, Any?>): Map<String, Any?> {
    val merged = base.toMutableMap()

    override.forEach { (key, value) ->
        val baseValue = merged[key]

        merged[key] = if (baseValue is Map<*, *> && value is Map<*, *>) {
            @Suppress("UNCHECKED_CAST")
            deepMergeJsonConfig(baseValue as Map<String, Any?>, value as Map<String, Any?>)
        } else {
            value
        }
    }

    return merged
}

fun readJsonConfigStrings(config: Map<String, Any?>, sourceFile: File): Map<String, String> {
    val rawStrings = config["strings"] ?: return emptyMap()

    if (rawStrings !is Map<*, *>) {
        throw GradleException("${jsonConfigContext(sourceFile)} field 'strings' must be an object when provided.")
    }

    return rawStrings.map { entry ->
        val key = entry.key
        val value = entry.value

        if (key !is String || key.isBlank()) {
            throw GradleException("${jsonConfigContext(sourceFile)} field 'strings' must only contain non-empty string keys")
        }
    }
}

```

```

    if (value !is String || value.isBlank()) {
        throw GradleException("${jsonConfigContext(sourceFile)} string '$key' must be a non-empty string.")
    }

    key to value
}.toMap()
}

fun resolveJsonConfigStringReference(
    value: String,
    strings: Map<String, String>,
    sourceFile: File,
    stringStack: List<String> = emptyList()
): String {
    if (!value.startsWith("$")) {
        return value
    }

    if (value.startsWith("$$$$")) {
        return value.drop(1)
    }

    val stringId = value.drop(1)

    if (stringId.isBlank()) {
        throw GradleException("${jsonConfigContext(sourceFile)} contains a blank string reference.")
    }

    if (stringId in stringStack) {
        val cycle = (stringStack + stringId).joinToString(" -> ")
        throw GradleException("${jsonConfigContext(sourceFile)} contains a string reference cycle: $cycle")
    }

    val referencedValue = strings[stringId]
    ?: throw GradleException("${jsonConfigContext(sourceFile)} references unknown string id '$stringId'.")

    return resolveJsonConfigStringReference(referencedValue, strings, sourceFile, stringStack + stringId)
}

fun resolveJsonConfigStringReferences(value: Any?, strings: Map<String, String>, sourceFile: File): Any? {
    return when (value) {
        is String -> resolveJsonConfigStringReference(value, strings, sourceFile)
        is Map<*, *> -> value.map { entry ->

```

```

        val key = entry.key

        if (key !is String || key.isBlank()) {
            throw GradleException("${jsonConfigContext(sourceFile)} JSON config object keys must be non-empty strings")
        }

        key to resolveJsonConfigStringReferences(entry.value, strings, sourceFile)
    }.toMap()
    is List<*> -> value.map { item -> resolveJsonConfigStringReferences(item, strings, sourceFile) }
    else -> value
}
}

fun normalizeJsonConfigValue(value: Any?, baseDir: File, includeStack: List<File>): Any? {
    return when (value) {
        is Map<*, *> -> resolveJsonConfigIncludes(value, baseDir, includeStack)
        is List<*> -> value.map { item -> normalizeJsonConfigValue(item, baseDir, includeStack) }
        else -> value
    }
}

fun resolveJsonConfigIncludes(rawMap: Map<*, *>, baseDir: File, includeStack: List<File>): Map<String, Any?> {
    val includeValue = rawMap["include"]
    val includedConfig = when (includeValue) {
        null -> emptyMap()
        is List<*> -> includeValue.fold(emptyMap<String, Any?>()) { mergedIncludes, rawIncludePath ->
            if (rawIncludePath !is String || rawIncludePath.isBlank()) {
                throw GradleException("JSON config field 'include' must contain only non-empty string paths.")
            }

            val includeFile = jsonConfigPath(rawIncludePath, baseDir)
            deepMergeJsonConfig(mergedIncludes, readJsonConfigObject(includeFile, includeStack))
        }
        else -> throw GradleException("JSON config field 'include' must be an array of paths.")
    }

    val currentConfig = rawMap
        .filterKeys { key -> key != "include" }
        .map { (key, value) ->
            if (key !is String || key.isBlank()) {
                throw GradleException("JSON config object keys must be non-empty strings.")
            }

            key to normalizeJsonConfigValue(value, baseDir, includeStack)
        }

```

```

    }
    .toMap()

    return deepMergeJsonConfig(includedConfig, currentConfig)
}

fun readJsonConfigObject(file: File, includeStack: List<File> = emptyList()): Map<String, Any?> {
    if (!file.exists()) {
        throw GradleException("${jsonConfigContext(file)} is required.")
    }

    val canonicalFile = file.canonicalFile

    if (canonicalFile in includeStack) {
        val cycle = (includeStack + canonicalFile)
            .joinToString(" -> ") { jsonConfigContext(it) }

        throw GradleException("JSON config include cycle detected: $cycle")
    }

    val parsed = JsonSlurper().parse(canonicalFile)

    if (parsed !is Map<*, *>) {
        throw GradleException("${jsonConfigContext(file)} must contain a JSON object.")
    }

    val mergedConfig = resolveJsonConfigIncludes(parsed, canonicalFile.parentFile, includeStack + canonicalFile)
    val strings = readJsonConfigStrings(mergedConfig, canonicalFile)

    @Suppress("UNCHECKED_CAST")
    return resolveJsonConfigStringReferences(mergedConfig, strings, canonicalFile) as Map<String, Any?>
}

rootProject.extensions.extraProperties["readJsonConfigObject"] = { file: File ->
    readJsonConfigObject(file)
}

```

gradle/preceding_comment_as_escape_hatch.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/gradle/preceding_comment_as_escape_hatch.gradle.kts>

```
import java.io.File
```

```
data class PrecedingEscapeComment(  
    val startLine: Long,  
    val endLine: Long,  
    val text: String  
)
```

```
fun normalizePrecedingEscapeCommentText(text: String): String {  
    return text  
        .lines()  
        .joinToString(" ") { line ->  
            line.trim()  
                .removePrefix("//")  
                .removePrefix("/*")  
                .removePrefix("*")  
                .removeSuffix("*/")  
                .trim()  
        }  
        .replace(Regex("\\s+"), " ")  
        .trim()  
}
```

```
fun precedingEscapeLineCommentText(rawLine: String): String {  
    return rawLine.substringAfter("//")  
}
```

```
fun extractPrecedingEscapeComments(sourceFile: File): List<PrecedingEscapeComment> {  
    val comments = mutableListOf<PrecedingEscapeComment>()  
    val lines = sourceFile.readlines()  
    var lineIndex = 0  
  
    while (lineIndex < lines.size) {  
        val rawLine = lines[lineIndex]  
        val trimmed = rawLine.trim()  
        val lineNumber = lineIndex + 1L  
  
        if (trimmed.startsWith("//")) {  
            val startLine = lineNumber  
            val text = StringBuilder(precedingEscapeLineCommentText(rawLine))  
            var endLine = lineNumber  
            lineIndex++  
  
            while (lineIndex < lines.size && lines[lineIndex].trim().startsWith("//")) {
```

```

        text.append("\n")
        text.append(precedingEscapeLineCommentText(lines[lineIndex]))
        endLine = lineIndex + 1L
        lineIndex++
    }

```

```

    comments += PrecedingEscapeComment(
        startLine = startLine,
        endLine = endLine,
        text = normalizePrecedingEscapeCommentText(text.toString())
    )
    continue
}

```

```

if (trimmed.startsWith("/*")) {
    val startLine = lineNumber
    val text = StringBuilder(rawLine.substringAfter("/*"))
    var endLine = lineNumber

```

```

    while (!lines[lineIndex].contains("*/") && lineIndex + 1 < lines.size) {
        lineIndex++
        text.append("\n")
        text.append(lines[lineIndex])
        endLine = lineIndex + 1L
    }

```

```

    comments += PrecedingEscapeComment(
        startLine = startLine,
        endLine = endLine,
        text = normalizePrecedingEscapeCommentText(text.toString())
    )
}

```

```

    lineIndex++
}

```

```

return comments
}

```

```

fun parenthesisDelta(text: String): Int {
    return text.count { it == '(' } - text.count { it == ')' }
}

```

```

fun annotationRanges(lines: List<String>): List<IntRange> {

```

```

val ranges = mutableListOf<IntRange>()
var lineIndex = 0

while (lineIndex < lines.size) {
    val trimmed = lines[lineIndex].trim()

    if (!trimmed.startsWith("@")) {
        lineIndex++
        continue
    }

    val start = lineIndex
    var end = lineIndex
    var depth = parenthesisDelta(trimmed)

    while (depth > 0 && end + 1 < lines.size) {
        end++
        depth += parenthesisDelta(lines[end])
    }

    ranges += start..end
    lineIndex = end + 1
}

return ranges
}

fun annotationRangeContaining(ranges: List<IntRange>, lineIndex: Int): IntRange? {
    return ranges.firstOrNull { lineIndex in it }
}

fun previousPrecedingEscapeLine(
    lines: List<String>,
    beforeLineNumber: Long,
    skipClosingBrace: Boolean
): Long? {
    val annotations = annotationRanges(lines)
    var index = beforeLineNumber.toInt() - 2

    while (index >= 0) {
        val trimmed = lines[index].trim()

        if (trimmed.isEmpty() || (skipClosingBrace && trimmed == "}") {
            index--
        }
    }
}

```

```

        continue
    }

    val annotationRange = annotationRangeContaining(annotations, index)
    if (annotationRange != null) {
        index = annotationRange.first - 1
        continue
    }

    return index + 1L
}

return null
}

fun precedingEscapeReasonBefore(
    sourceFile: File,
    targetLineNumber: Long,
    marker: String,
    skipClosingBrace: Boolean
): String? {
    val lines = sourceFile.readlines()
    val previousLine = previousPrecedingEscapeLine(lines, targetLineNumber, skipClosingBrace)
    ?: return null

    val comment = extractPrecedingEscapeComments(sourceFile).firstOrNull { it.endLine == previousLine }
    ?: return null

    if (!comment.text.contains(marker)) {
        return null
    }

    return comment.text.substringAfter(marker).trim()
}

fun precedingEscapeValidationProblem(
    sourceFile: File,
    targetLineNumber: Long,
    marker: String,
    minimumReasonCharacters: Int,
    gateName: String,
    skipClosingBrace: Boolean
): String? {
    val reason = precedingEscapeReasonBefore(sourceFile, targetLineNumber, marker, skipClosingBrace)

```



```
?: return "$gateName escape hatch comment containing $marker must immediately precede line $targetLineNum
```

```
return if (reason.length < minimumReasonCharacters) {  
    "$gateName escape hatch reason is ${reason.length} characters; minimum is $minimumReasonCharacters."  
} else {  
    null  
}  
}
```

```
fun precedingEscapeHatchUsage(  
    marker: String,  
    minimumReasonCharacters: Int,  
    escapedConstructExample: String
```

```
): String {  
    return ""
```

Escape hatch format:

```
// $marker  
// <specific justification of at least $minimumReasonCharacters characters>  
${escapedConstructExample.lines().joinToString("\n") { " $it" }}
```

Block comments are also accepted. Annotations may appear between the escape hatch comment and the escaped construct.

Important instruction for AI/code-generation tools:

Do not add or **use** an escape hatch comment unless the user has explicitly approved that specific exception. First **try** to satisfy the build gate by following the preferred design.

```
"".trimIndent()  
}
```

```
rootProject.extensions.extraProperties["precedingEscapeReasonBefore"] = {  
    sourceFile: File,  
    targetLineNumber: Long,  
    marker: String,  
    skipClosingBrace: Boolean ->  
    precedingEscapeReasonBefore(sourceFile, targetLineNumber, marker, skipClosingBrace)  
}
```

```
rootProject.extensions.extraProperties["precedingEscapeValidationProblem"] = {  
    sourceFile: File,  
    targetLineNumber: Long,  
    marker: String,  
    minimumReasonCharacters: Int,  
    gateName: String,
```

```

skipClosingBrace: Boolean ->
precedingEscapeValidationProblem(
    sourceFile,
    targetLineNumber,
    marker,
    minimumReasonCharacters,
    gateName,
    skipClosingBrace
)
}

rootProject.extensions.extraProperties["precedingEscapeHatchUsage"] = {
    marker: String,
    minimumReasonCharacters: Int,
    escapedConstructExample: String ->
    precedingEscapeHatchUsage(marker, minimumReasonCharacters, escapedConstructExample)
}

```

gradle/wrapper/gradle-wrapper.properties

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/gradle/wrapper/gradle-wrapper.properties>

```

distributionBase=GRADLE_USER_HOME
distributionPath=wrapper/dists
distributionUrl=https\://services.gradle.org/distributions/gradle-9.4.1-bin.zip
networkTimeout=10000
validateDistributionUrl=true
zipStoreBase=GRADLE_USER_HOME
zipStorePath=wrapper/dists

```

guidance.json

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/guidance.json>

```
{
  "strings": {
    "nullIsADefect": "Null must be made impossible by the framework at application boundaries, not handled defensively"
  }
}
```

local_catch_guardrail.json

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/local_catch_guardrail.json>

```
{
  "minimumReasonCharacters": 120
}
```

method_call_guardrail.json

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/method_call_guardrail.json>

```
{
  "include": [
    "guidance.json"
  ],
  "minimumAllowReasonCharacters": 120,
  "rules": [
    {
      "type": "java.lang.Exception",
      "includeSubclasses": true,
      "message": "Avoid exposing exception details as user-viewable output",
      "disallowedMethods": [
        "getMessage",
        "toString"
      ]
    },
    {
      "type": "java.util.Optional",
      "includeSubclasses": false,
      "message": "$nullIsADefect",
      "disallowedMethods": [
        "ofNullable"
      ]
    },
    {
      "type": "java.util.stream.Stream",
      "includeSubclasses": false,
      "message": "$nullIsADefect",
      "disallowedMethods": [
        "ofNullable"
      ]
    },
    {
      "type": "java.util.Objects",
      "includeSubclasses": false,
      "message": "$nullIsADefect",
      "disallowedMethods": [
        "isNull",
        "nonNull",
        "requireNonNull"
      ]
    },
    {
      "type": "org.jooq.DSLContext",
      "includeSubclasses": true,
```

```
"message": "Use DSL like api instead of direct sql strings",
"disallowedMethods": [
  "execute",
  "fetch",
  "fetchOptional"
]
}
]
}
```

saferprocess/README.md

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/saferprocess/README.md>

saferprocess

`saferprocess` is a small Java wrapper around `java.lang.ProcessBuilder`.

The goal is not to make process execution safe by magic. The goal is to make an important security distinction visible in

- the executable is a hardcoded value;
- or the executable is a path already constrained by `BoxedPath`;
- the later values are process arguments;
- callers **do** not construct one mixed command **string**;
- the low-level conversion to Java's platform API happens in one place.

Why **This** Exists

`java.lang.ProcessBuilder` is already better than `Runtime.exec(String)` because it accepts separate command argument

But the native API still puts the executable and all arguments in the same list:

```
new java.lang.ProcessBuilder( "/usr/bin/convert", inputPath.toString(), outputPath.toString()
).start();
```

Reviewers must remember that the first **string** is special. It chooses the program that runs. **If** that value becomes user

This library makes that distinction explicit:

```
ProcessBuilder.start( Hardcoded.of("/usr/bin/convert"), List.of(inputPath.toString(),
outputPath.toString()) );
```

It also accepts an executable selected through a `BoxedPath` filesystem boundary:

```
ProcessBuilder.start( boxedConvertExecutable, List.of(inputPath.toString(), outputPath.toString()) );
```

Usage

```
import java.io.IOException; import java.nio.file.Path; import java.util.List;
import org.owasp.untrust.valuedescriptors.Hardcoded; import
org.owasp.untrust.saferprocess.ProcessBuilder;
public final class ImageConverter { private static final Hardcoded CONVERT =
Hardcoded.of("/usr/bin/convert");
public Process convert(Path inputPath, Path outputPath) throws IOException { return
ProcessBuilder.start( CONVERT, List.of(inputPath.toString(), outputPath.toString()) ); }
```

When the executable path is not hardcoded but is still selected inside a trusted filesystem sandbox, [use](#) the `BoxedPath`

```
import java.io.IOException; import java.util.List;
import org.owasp.untrust.boxedpath.BoxedPath; import
org.owasp.untrust.saferprocess.ProcessBuilder;
public final class ToolRunner { private final BoxedPath executable;
public ToolRunner(BoxedPath executable) { this.executable = executable; }
public Process run(String input, String output) throws IOException { return ProcessBuilder.start(
executable, List.of(input, output) ); }
```

[If](#) you need to further configure the platform builder, request it explicitly:

```
java.lang.ProcessBuilder builder = ProcessBuilder.toJavaProcessBuilder(
Hardcoded.of("/usr/bin/convert"), List.of(inputPath.toString(), outputPath.toString()) );
builder.redirectErrorStream(true); Process process = builder.start();
```

What `Hardcoded` Means

`Hardcoded` marks that the executable value is authored in code rather than supplied by a request or other user-controlled data.

Good sources **for** a process executable include:

- hardcoded executable paths reviewed with the application;

What `BoxedPath` Means

`BoxedPath` marks that path construction has already passed through the OWASP Untrust boxed-path boundary.

Use this overload when the executable is selected **from** a trusted server-side directory, such as an application-owned directory.

Good `BoxedPath` executable sources include:

- a path resolved under a server-controlled tools directory;
- a path selected **from** a small application allow-list and then boxed;
- a deployment-managed executable path that should remain inside a configured sandbox.

Bad sources include:

- request parameters;
- request body fields;
- uploaded files;
- database rows written by users;
- environment variables treated as attacker-controllable in your deployment model.

What Not To **Do**

Do not pass user-controlled data as the executable:

```
ProcessBuilder.start( Hardcoded.of(request.converterName()), List.of(inputPath.toString(),  
outputPath.toString()) );
```

This example should be rejected in review: wrapping user input in `Hardcoded` lies about the value source.

Do not treat `BoxedPath` as an executable allow-list:

```
ProcessBuilder.start( toolsDirectory.resolve(request.toolName()), List.of(inputPath.toString(),  
outputPath.toString()) );
```

Even **if** the path stays inside the sandbox, the application still needs to decide which programs are approved.

Do not rebuild shell command strings:

```
ProcessBuilder.start( Hardcoded.of("/bin/sh"), List.of("-c", "convert " + inputPath + " " + outputPath)
);
```

Do not treat path conversion as a storage-boundary check. **If** paths are user-influenced, validate or box them before the

Do not make the wrapper accept a single `List<String>` where the first item is secretly the executable. That recreates the

Design Notes

This wrapper intentionally stays small:

- `Hardcoded` identifies the executable as an application-authored value.
- `BoxedPath` identifies an executable path already constrained by filesystem sandboxing.
- `ProcessBuilder.start(...)` starts a process with an explicit executable source and separate arguments.
- `ProcessBuilder.toJavaProcessBuilder(...)` exposes the native builder only after the command line has been assembled.

If you later need server-side configuration or allow-list based executable selection, add a separate explicit type **for** that

saferprocess/build.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/saferprocess/build.gradle.kts>


```
plugins {  
    `java-library`  
}  
  
group = "org.owasp.untrust"  
version = "0.1.0"  
  
java {  
    toolchain {  
        languageVersion = JavaLanguageVersion.of(21)  
    }  
}  
  
repositories {  
    mavenCentral()  
}  
  
dependencies {  
    api("org.owasp.untrust:valuedescriptors:0.1.0")  
    api("io.github.owasp-untrust:untrust-boxedpath:0.3")  
}
```

saferprocess/git_init_remote.txt

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/saferprocess/git_init_remote.txt>

```
remote add origin https://github.com/owasp-untrust/saferprocess\_java.git
```

saferprocess/settings.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/saferprocess/settings.gradle.kts>

```
rootProject.name = "saferprocess"  
  
includeBuild("../ValueDescriptors")
```

saferprocess/src/main/java/org/owasp/untrust/saferprocess/ProcessBuilder.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/saferprocess/src/main/java/org/owasp/untrust/saferprocess/ProcessBuilder.java>

```

package org.owasp.untrust.saferprocess;

import java.io.IOException;
import java.util.ArrayList;
import java.util.List;
import java.util.Objects;

import org.owasp.untrust.boxedpath.BoxedPath;
import org.owasp.untrust.valuedescriptors.Hardcoded;

public final class ProcessBuilder {
    private ProcessBuilder() {
    }

    public static Process start(Hardcoded executable, List<String> arguments) throws IOException {
        return toJavaProcessBuilder(executable, arguments).start();
    }

    public static Process start(BoxedPath executable, List<String> arguments) throws IOException {
        return toJavaProcessBuilder(executable, arguments).start();
    }

    public static java.lang.ProcessBuilder toJavaProcessBuilder(Hardcoded executable, List<String> arguments) {
        Objects.requireNonNull(executable);
        return toJavaProcessBuilder(executable.exposeUnchecked(), arguments);
    }

    public static java.lang.ProcessBuilder toJavaProcessBuilder(BoxedPath executable, List<String> arguments) {
        Objects.requireNonNull(executable);
        return toJavaProcessBuilder(executable.toString(), arguments);
    }

    private static java.lang.ProcessBuilder toJavaProcessBuilder(String executable, List<String> arguments) {
        Objects.requireNonNull(arguments);

        List<String> commandLine = new ArrayList<>();
        commandLine.add(executable);
        commandLine.addAll(arguments.stream()
            .map(Objects::requireNonNull)
            .toList());

        return new java.lang.ProcessBuilder(commandLine);
    }
}

```

```
}  
}
```

string_concat_guardrail.json

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/string_concat_guardrail.json>

```
{  
  "minimumReasonCharacters": 200  
}
```

tesseract_mock_java/README.md

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/tesseract_mock_java/README.md>

```
# tesseract_mock_java
```

Java command-line mock **for** Tesseract-compatible OCR tests.

It accepts the same minimal CLI shape BusyBee uses **for** synchronous OCR:

```
.\gradlew.bat installDist .\build\install\tesseract_mock_java\bin\tesseract_mock_java.bat image.png  
stdout -l eng --psm 3
```

The mock does not perform OCR. It chooses one of five hardcoded outputs based on the input file size, requested lang

Do not use this in production. Point BusyBee at the real Tesseract executable instead.

tesseract_mock_java/build.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/tesseract_mock_java/build.gradle.kts>

```
plugins {  
    application  
}  
  
group = "org.owasp.untrust"  
version = "0.1.0"  
  
java {  
    toolchain {  
        languageVersion = JavaLanguageVersion.of(21)  
    }  
}  
  
application {  
    mainClass = "org.owasp.untrust.tesseractmock.TesseractMock"  
}
```

tesseract_mock_java/settings.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/tesseract_mock_java/settings.gradle.kts>

```
rootProject.name = "tesseract_mock_java"
```

tesseract_mock_java/src/main/java/org/owasp/untrust/tesseractmock/TesseractMock.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/tesseract_mock_java/src/main/java/org/owasp/untrust/tesseractmock/TesseractMock.java>

```

package org.owasp.untrust.tesseractmock;

import java.nio.charset.StandardCharsets;
import java.nio.file.Files;
import java.nio.file.Path;
import java.util.List;

public final class TesseractMock {
    private static final List<String> OUTPUTS = List.of(
        "Buy printer paper",

        "Order two boxes of A4 printer paper for the office supply cabinet.",
        "Renew passport",

        "Fill the renewal form, attach a photo, and schedule the appointment.",
        "Pay utility bill",

        "Electricity bill is due this Friday. Confirm payment receipt afterward.",
        "Call project vendor",

        "Ask for the revised delivery estimate and request written confirmation.",
        "Prepare release notes",

        "Summarize fixed bugs, migration notes, and known limitations for review."
    );

    private TesseractMock() {}

    public static void main(String[] args) throws Exception {
        Arguments parsed = Arguments.parse(args);
        byte[] imageBytes = Files.readAllBytes(parsed.input());
        int outputIndex = Math.floorMod(imageBytes.length + parsed.language().hashCode() + parsed.pageSegmentationModel().hashCode(), OUTPUTS.size());
        System.out.print(OUTPUTS.get(outputIndex));
    }
}

```

```

}

private record Arguments(Path input, String outputBase, String language, int pageSegmentationMode) {
    private static Arguments parse(String[] args) {
        if (args.length != 6) {
            usage();
        }
        if (!"stdout".equals(args[1])) {
            usage();
        }
        if (!"-l".equals(args[2])) {
            usage();
        }
        if (args[3].isBlank()) {
            usage();
        }
        if (!"--psm".equals(args[4])) {
            usage();
        }
        return new Arguments(Path.of(args[0]), args[1], args[3], Integer.parseInt(args[5]));
    }

    private static void usage() {
        System.err.println("usage: tesseract <input-file> stdout -l <language> --psm <mode>");
        System.exit(2);
    }
}

```

validated_value_guardrail.json

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/validated_value_guardrail.json>

```

{
  "routeValidatedValueTypeName": "org.owasp.untrust.vv.foundation.SelfValidating",
  "minimumCustomValidationTraitsReasonCharacters": 120,
  "minimumExposeUncheckedReasonCharacters": 120
}

```

vv/README.md

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/README.md>

```
# vv_java
```

Validated value primitives **for** OWASP Untrust Java applications.

This library provides base classes, traits, exceptions, and helper types **for** turning raw strings into strongly typed validated values.

```
## Coordinates
```

Current Gradle metadata:

```
group = "org.owasp.untrust" version = "0.1.0"
```

Expected dependency once published:

```
dependencies { implementation("org.owasp.untrust:vv:0.1.0") }
```

Spring-specific adapters live in a separate optional module so the core validated-value library does not depend on Spring:

```
dependencies { implementation("org.owasp.untrust:vv_spring:0.1.0") }
```

For local development before all libraries are published:

```
// settings.gradle.kts includeBuild("../BuildGates") includeBuild("../ValueDescriptors")
```


Requirements

- JDK 21
- Gradle wrapper included in [this](#) repository
- ``buildgates_java``
- ``valuedescriptor_java``
- SLF4J API
- Spring Security core
- Jakarta Servlet API

Optional ``vv_spring`` module:

- Spring Vault Core

Prebuilt Values

The ``org.owasp.untrust.vv.prebuilt`` [package](#) contains reusable validated values that are production-ready enough to [use](#) directly when their policy matches your domain:

- ``PendingApiKey``
- ``ApiKey``
- ``CreditCard``

[Use](#) these instead of copying example code when the built-in validation and [public](#) rendering [match](#) the application requirement.

Secret Values And ``vv_spring``

The core ``vv`` module contains the generic secret-value model. The optional ``vv_spring`` module supplies a Spring Vault-backed ``SecretStore<String>`` adapter. [Use this](#) split when an application wants route-bound secret values without making the core validation library depend on Spring.

``PendingApiKey`` and ``ApiKey`` in ``org.owasp.untrust.vv.prebuilt`` provide the standard API-key shape:

- ``PendingApiKey`` represents the untrusted route/binding value.
- ``PendingApiKey.from(String)`` supports framework deserialization [from](#) a raw [string](#).
- ``PendingApiKey`` validates the raw API key and implements ``PendingSecret``.
- Calling the inherited pending-secret ``hide(...)`` method creates a ``SecretValueInitializer``.

- `ApiKey` extends `SecretValue<String>` and accepts that initializer in its constructor.
- `ApiKey` stores only the secret reference/display value pair as its `public` object state. Reading the actual secret goes through the configured `SecretStore<String>`.

With Spring Vault:

```
import org.owasp.untrust.vv.prebuilt.ApiKey; import org.owasp.untrust.vv.prebuilt.PendingApiKey;
import org.owasp.untrust.vv.visibility.secret.SecretReference; import
org.owasp.untrust.vv.visibility.secret.SecretValueInitializer; import
org.owasp.untrust.vv.visibility.secret.vault.SpringVaultStringSecretStore; import
org.springframework.vault.core.VaultOperations;

public final class AiCredentialService { private final SpringVaultStringSecretStore secretStore;

public AiCredentialService(VaultOperations vaultOperations) { this.secretStore = new
SpringVaultStringSecretStore( vaultOperations, "secret"); }

public ApiKey storeUserApiKey( SecretReference reference, PendingApiKey pendingApiKey, String
displayValue) {

SecretValueInitializer<String, ApiKey> initializer = pendingApiKey.hide( secretStore, reference,
displayValue);

return new ApiKey(initializer); } }
```

In application code, use `PendingApiKey` directly when all configured AI providers accept the same key character policy. If a provider needs stricter rules, create a small route-specific subclass or sibling pending type and keep the `final` stored value as `ApiKey`.

If the UI needs a masked suffix such as `*****1234`, compute it in the domain-specific pending type or the immediate route/service boundary before calling `hide(...)`; do not expose the raw secret again later merely to create a display value.

Avoid:

- returning the full API key from any HTTP response;
- logging the full API key;
- storing the raw API key in ordinary application DTOs;
- passing a primitive `String` API key through service layers after binding;
- putting Spring Vault dependencies in the core `vv` module.

Build:

.\gradlew.bat build

On Unix-like shells:

./gradlew build

Design Model

A validated value should be constructed **from** untrusted raw input exactly once, then passed through the rest of the ap

The basic validation pipeline is:

1. Check raw **string** constraints that are independent of parsing.
2. Parse the raw **string** into a typed value.
3. Normalize the parsed value.
4. Check constraints on the normalized value.
5. Store only the validated value inside the wrapper.

``ValidatedValue<T, Traits>`` implements **this** pipeline. A concrete value **class** supplies a ``ValidationTraits<T>`` implement

Core Types

``ValidatedValue<T, Traits>``

Base **class for** domain-specific validated values.

Typical subclass shape:

```
public final class TaskName extends ValidatedValue<String, TaskName.Traits> { public
TaskName(String raw) throws ValidationException { super(raw, new Traits()); }
public static final class Traits extends PrintableUnicodeStringTraits { @Override public Hardcoded
descriptionInErrors() { return Hardcoded.hardcoded("task name"); }
@Override public Bounds bounds() { return new Bounds(1, 100); }
@Override public String reformatString(String raw) { return raw.trim(); }
@Override public Optional<ValidationException>
findExtraValidationProblemInPrintableValue(String value) { return Optional.empty(); } }
```

Use a concrete validated value at the boundary:

```
TaskName taskName = new TaskName(request.name()); taskService.createTask(taskName);
```

Do not keep passing the original raw **string** deeper into the application.

`ValidationTraits<T>`

Defines validation behavior:

- `descriptionInErrors()`
- `parse(String raw)`
- `normalize(T parsed)`
- `findValidationProblemInRaw(String raw)`
- `findValidationProblemInNormalizedValue(T normalized)`

Raw validation should handle cheap, parser-independent checks such as length limits. **Let** parsing validate format whe

`ValidationException`

Thrown when input fails validation. It preserves:

- optional raw value
- optional parsed value
- validation error messages
- optional additional value such as a min/max bound

Use this for validation failures, not **for** authorization or persistence failures.

`ViewableUuidValue`

Validated wrapper **for** UUID strings.

Example:

```
ViewableUuidValue id = new ViewableUuidValue("123e4567-e89b-12d3-a456-426614174000");  
UUID raw = id.exposeUnchecked();
```

``ViewableUuidValue`` is intentionally non-**final** because UUID-backed identifiers may share parsing behavior.

Trait Helpers

``BoundedAnyContentStringTraits``

For strings with length limits but no content restrictions.

Use when any content is allowed and only size/normalization matters.

``PrintableUnicodeStringTraits``

For strings that must contain printable Unicode characters.

Use for names, labels, descriptions, comments, and other human-entered text where control characters should be rejected.

``RegexStringTraits``

For printable strings that must **match** a welcome-list regex.

Example:

```
public final class Username extends ValidatedValue<String, Username.Traits> { public
Username(String raw) throws ValidationException { super(raw, new Traits()); }

public static final class Traits extends RegexStringTraits { private static final Pattern USERNAME =
Pattern.compile("[A-Za-z0-9_]{3,32}");

@Override public Hardcoded descriptionInErrors() { return Hardcoded.hardcoded("username"); }
@Override public Bounds bounds() { return new Bounds(3, 32); }
@Override public Pattern welcomeListRegex() { return USERNAME; }
@Override public Optional<ValidationException> findExtraValidationProblem(String value) { return
Optional.empty(); } } }
```

Prefer welcome-list patterns over block-list patterns.

`BoundedValueTraits<T>`

For parsed values that have both raw **string** length bounds and typed value bounds.

Use for numbers, dates, times, and other ordered values.

`EnumValidationTraits<E>`

For case-insensitive parsing of **enum** constants.

Example:

```
enum DatabaseType { MYSQL, POSTGRES }
```

```
EnumValidationTraits<DatabaseType> traits = new EnumValidationTraits<>(DatabaseType.class,  
Hardcoded.hardcoded("database type"));
```

`RareTraitsCaseWhereParsingIsTheWholeValidation<T>`

For rare cases where parsing fully validates the domain.

Use sparingly. Supply a clear justification in code comments or annotations. UUID parsing is a typical example.

`CustomValidationForRareCasesTraits<T>`

Escape hatch for specialized validation that does not fit the common trait shapes.

Prefer the more specific traits when possible.

PII Rendering Interfaces

`Pii<T>`

Marker **interface for** wrapped PII values that must define **public** rendering.

`ErasedPii<T>`

Always renders as:

`MaskedPii<T>`

Renders a partial value **while** hiding the middle.

Examples:

a -> a* **abcd** -> a* abcdef -> ab***ef

Use masking only when revealing a prefix/suffix is acceptable **for** the data category.

Entity Authorization Helpers

`Entity<Snapshot>` and `DataAccess<Snapshot>` provide a small authorization pattern around entity IDs:

- parse and validate the entity UUID
- load a snapshot by ID
- apply an authorization justification **function**
- **return** `AuthorizedAccess<Snapshot>` only after authorization succeeds
- optionally hide existence with `DisclosurePolicy.HIDE\_EXISTENCE`

Example shape:

```
Entity<TaskSnapshot> entity = new Entity<>(taskIdFromRoute);
```

```
AuthorizedAccess<TaskSnapshot> access = entity.authorize( authentication,  
taskRepository::findSnapshotById, (auth, snapshot) -> snapshot.owner().equals(auth.getName()) ?  
Optional.of("task owner") : Optional.empty() );
```

Use the returned `AuthorizedAccess` as the proof that the entity was both found and authorized.

What Not To Do

- Do not pass raw request strings past the boundary where a validated value should be created.
- Do not call `exposeUnchecked()` unless the receiving API genuinely needs the raw typed value.
- Do not validate parser-specific format manually when the parser can do it more correctly.
- Do not use block lists for validation when a welcome-list trait is appropriate.
- Do not make a validated value non-final unless there is a documented security reason.
- Do not use `RareTraitsCaseWhereParsingIsTheWholeValidation` just because it is shorter to implement.
- Do not log raw PII values.
- Do not use `MaskedPii` when full erasure is required.
- Do not catch `ValidationException` and continue with the original raw value.
- Do not treat successful validation as authorization. Validation and authorization are separate decisions.

Bad:

```
String rawTaskId = request.getParameter("taskId"); taskRepository.load(rawTaskId);
```

Better:

```
ViewableUuidValue taskId = new ViewableUuidValue(request.getParameter("taskId"));  
taskRepository.load(taskId);
```

Bad:

```
LOGGER.info("User email is {}", email.exposeUnchecked());
```

Better:

```
LOGGER.info("User email is {}", email.toPublicString());
```

Repository Notes

This repository is intended to become:

https://github.com/owasp-untrust/vw_java

Dependency order:

1. ``buildgates_java``
2. ``valuedescriptor_java``
3. ``vv_java``

Publish ``buildgates_java`` and ``valuedescriptor_java`` before publishing [this](#) repository.

vv/build.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb38c70d2075a285c0507ee6e2/vv/build.gradle.kts>

```

plugins {
    `java-library`
    id("org.owasp.untrust.build-gates")
}

group = "org.owasp.untrust"
version = "0.1.0"

java {
    toolchain {
        languageVersion = JavaLanguageVersion.of(21)
    }
}

repositories {
    mavenCentral()
}

dependencies {
    api("org.owasp.untrust:valuedescriptors:0.1.0")
    implementation("org.owasp.untrust:buildmetadata:0.1.0")
    implementation("commons-validator:commons-validator:1.10.1")
    implementation("com.ibm.icu:icu4j:78.1")
    implementation("org.slf4j:slf4j-api:2.0.17")
    //implementation("org.springframework.security:spring-security-core:7.0.5")
    //implementation("jakarta.servlet:jakarta.servlet-api:6.1.0")
    api("com.fasterxml.jackson.core:jackson-annotations:2.20")
}

```

vv/settings.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/settings.gradle.kts>

```
pluginManagement {  
    includeBuild("../BuildGates")  
}  
  
rootProject.name = "vv"  
  
include("vv_spring")  
  
includeBuild("../BuildGates")  
includeBuild("../ValueDescriptors")
```

vv/src/main/java/org/owasp/untrust/vv/MultiLine.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/MultiLine.java>

```

package org.owasp.untrust.vv;

import com.fasterxml.jackson.annotation.JsonCreator;
import org.owasp.untrust.valuedescriptors.Hardcoded;
import org.owasp.untrust.valuedescriptors.foundation.PubliclyExposed;
import org.owasp.untrust.vv.foundation.ValidatedWrappedValue;
import org.owasp.untrust.vv.traits.LineTextTraits;

public final class MultiLine extends ValidatedWrappedValue<String> implements PubliclyExposed<String> {
    @JsonCreator(mode = JsonCreator.Mode.DELEGATING)
    public MultiLine(String raw) {
        super(raw, new Traits());
    }

    public static MultiLine from(String raw) {
        return new MultiLine(raw);
    }

    @Override
    public String exposeUnchecked() {
        return exposeUnchecked(EXPOSE_HALF_BAKED_VALUE_INTENDED_FOR_INTERNAL_LIBRARY_USE_ONLY_MARKER);
    }

    public static final class Traits extends LineTextTraits {
        @Override
        public Bounds rawBounds() {
            return new Bounds(0, 10_000);
        }

        @Override
        public boolean allowNewlines() {
            return true;
        }

        @Override
        public boolean allowEmoji() {
            return true;
        }

        @Override
        public boolean requirePathSafeText() {
            return false;
        }
    }
}

```

```
@Override
public Hardcoded descriptionInErrors() {
    return lineTextDescription("multi-line", "text with emoji");
}
}
```

vv/src/main/java/org/owasp/untrust/vv/MultiLineStrictText.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/MultiLineStrictText.java>

```
package org.owasp.untrust.vv;

import com.fasterxml.jackson.annotation.JsonCreator;
import org.owasp.untrust.valuedescriptors.Hardcoded;
import org.owasp.untrust.valuedescriptors.foundation.PubliclyExposed;
import org.owasp.untrust.vv.foundation.ValidatedWrappedValue;
import org.owasp.untrust.vv.traits.LineTextTraits;

public final class MultiLineStrictText extends ValidatedWrappedValue<String> implements PubliclyExposed<String> {
    @JsonCreator(mode = JsonCreator.Mode.DELEGATING)
    public MultiLineStrictText(String raw) {
        super(raw, new Traits());
    }

    public static MultiLineStrictText from(String raw) {
        return new MultiLineStrictText(raw);
    }

    @Override
    public String exposeUnchecked() {
        return exposeUnchecked(EXPOSE_HALF_BAKED_VALUE_INTENDED_FOR_INTERNAL_LIBRARY_USE_ONLY_MARKER);
    }

    public static final class Traits extends LineTextTraits {
        @Override
        public Bounds rawBounds() {
            return new Bounds(0, 10_000);
        }

        @Override
        public boolean allowNewlines() {
            return true;
        }

        @Override
        public boolean allowEmoji() {
            return false;
        }

        @Override
        public boolean requirePathSafeText() {
            return false;
        }
    }
}
```

```
@Override
public Hardcoded descriptionInErrors() {
    return lineTextDescription("multi-line", "strict text");
}
}
```

vv/src/main/java/org/owasp/untrust/vv/SingleLine.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/SingleLine.java>

```
package org.owasp.untrust.vv;

import com.fasterxml.jackson.annotation.JsonCreator;
import org.owasp.untrust.valuedescriptors.Hardcoded;
import org.owasp.untrust.valuedescriptors.foundation.PubliclyExposed;
import org.owasp.untrust.vv.foundation.ValidatedWrappedValue;
import org.owasp.untrust.vv.traits.LineTextTraits;

public final class SingleLine extends ValidatedWrappedValue<String> implements PubliclyExposed<String> {
    @JsonCreator(mode = JsonCreator.Mode.DELEGATING)
    public SingleLine(String raw) {
        super(raw, new Traits());
    }

    public static SingleLine from(String raw) {
        return new SingleLine(raw);
    }

    @Override
    public String exposeUnchecked() {
        return exposeUnchecked(EXPOSE_HALF_BAKED_VALUE_INTENDED_FOR_INTERNAL_LIBRARY_USE_ONLY_MARKER);
    }

    public static final class Traits extends LineTextTraits {
        @Override
        public Bounds rawBounds() {
            return new Bounds(0, 10_000);
        }

        @Override
        public boolean allowNewlines() {
            return false;
        }

        @Override
        public boolean allowEmoji() {
            return true;
        }

        @Override
        public boolean requirePathSafeText() {
            return false;
        }
    }
}
```



```
@Override
public Hardcoded descriptionInErrors() {
    return lineTextDescription("single-line", "text with emoji");
}
}
```

vv/src/main/java/org/owasp/untrust/vv/SingleLineStrictText.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/SingleLineStrictText.java>

```

package org.owasp.untrust.vv;

import com.fasterxml.jackson.annotation.JsonCreator;
import org.owasp.untrust.valuedescriptors.Hardcoded;
import org.owasp.untrust.valuedescriptors.foundation.PubliclyExposed;
import org.owasp.untrust.vv.foundation.ValidatedWrappedValue;
import org.owasp.untrust.vv.traits.LineTextTraits;

public final class SingleLineStrictText extends ValidatedWrappedValue<String> implements PubliclyExposed<String> {
    @JsonCreator(mode = JsonCreator.Mode.DELEGATING)
    public SingleLineStrictText(String raw) {
        super(raw, new Traits());
    }

    public static SingleLineStrictText from(String raw) {
        return new SingleLineStrictText(raw);
    }

    @Override
    public String exposeUnchecked() {
        return exposeUnchecked(EXPOSE_HALF_BAKED_VALUE_INTENDED_FOR_INTERNAL_LIBRARY_USE_ONLY_MARKER);
    }

    public static final class Traits extends LineTextTraits {
        @Override
        public Bounds rawBounds() {
            return new Bounds(0, 10_000);
        }

        @Override
        public boolean allowNewlines() {
            return false;
        }

        @Override
        public boolean allowEmoji() {
            return false;
        }

        @Override
        public boolean requirePathSafeText() {
            return false;
        }
    }
}

```

```
@Override
public Hardcoded descriptionInErrors() {
    return lineTextDescription("single-line", "strict text");
}
}
```

vv/src/main/java/org/owasp/untrust/vv/ViewableUuidValue.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/ViewableUuidValue.java>

```

package org.owasp.untrust.vv;

import java.util.UUID;

import com.fasterxml.jackson.annotation.JsonCreator;
import org.owasp.untrust.vv.exceptions.ValidationException;
import org.owasp.untrust.vv.foundation.ValidatedWrappedValue;
import org.owasp.untrust.vv.traits.RareTraitsCaseWhereParsingIsTheWholeValidation;
import org.owasp.untrust.buildmetadata.NonFinalValidatedValue;
import org.owasp.untrust.buildmetadata.StringConcatenationSafe;
import org.owasp.untrust.valuedescriptors.Hardcoded;
import org.owasp.untrust.valuedescriptors.foundation.PubliclyExposed;

import static org.owasp.untrust.valuedescriptors.Hardcoded.hardcoded;

// VALIDATED VALUE INHERITANCE REASON:
// All validated values that have a uuid type (basically all id types) can use
// a common ancestor since they all validate only to the extent of parsing -
// there should (normally) be no limitation on uuid range when used as an id.
@StringConcatenationSafe("UUIDs have a fixed format and length, so concatenation won't cause issues.")
@NonFinalValidatedValue("All validated values that have a uuid type (basically all id types) can use a common ancestor")
public class ViewableUuidValue extends ValidatedWrappedValue<UUID> implements PubliclyExposed<UUID> {
    @JsonCreator(mode = JsonCreator.Mode.DELEGATING)
    public ViewableUuidValue(String raw) throws ValidationException {
        super(raw, new Traits());
    }

    public static ViewableUuidValue from(String raw) {
        return new ViewableUuidValue(raw);
    }

    public static class Traits extends RareTraitsCaseWhereParsingIsTheWholeValidation<UUID> {
        @Override
        public Hardcoded descriptionInErrors() {
            return hardcoded("UUID");
        }

        @Override
        public UUID parse(String raw) throws IllegalArgumentException {
            return UUID.fromString(raw);
        }

        @Override
        public Bounds rawBounds() {

```

```

    // A UUID string has a fixed length of 36 characters (including hyphens)
    return new Bounds(36, 36);
}

@Override
public UUID normalize(UUID parsed) throws ValidationException {
    return parsed;
}
}

@Override
public UUID exposeUnchecked() {
    return exposeUnchecked(EXPOSE_HALF_BAKED_VALUE_INTENDED_FOR_INTERNAL_LIBRARY_USE_ONLY_MARKER);
}
}

```

vv/src/main/java/org/owasp/untrust/vv/examples/ExistingUsername.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/examples/ExistingUsername.java>

```

package org.owasp.untrust.vv.examples;

import java.util.function.Predicate;

import com.fasterxml.jackson.annotation.JsonCreator;
import org.owasp.untrust.vv.foundation.CrossValidationCandidate.FullyValidated;

public class ExistingUsername extends UsernameBase<ExistingUsername> {
    private ExistingUsername(FullyValidated<String, ExistingUsername> opaqueValue) {
        super(opaqueValue);
    }

    public static final class Candidate extends CandidateBase<ExistingUsername> {
        @JsonCreator(mode = JsonCreator.Mode.DELEGATING)
        public Candidate(String raw) {
            super(raw);
        }

        public static Candidate from(String raw) {
            return new Candidate(raw);
        }

        public FullyValidated<String, ExistingUsername> crossValidate(Predicate<String> exists) {
            return crossValidateExists(exists);
        }
    }
}

```

vv/src/main/java/org/owasp/untrust/vv/examples/RegisterUsername.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/examples/RegisterUsername.java>

```

package org.owasp.untrust.vv.examples;

import java.util.function.Predicate;

import com.fasterxml.jackson.annotation.JsonCreator;
import org.owasp.untrust.vv.foundation.CrossValidationCandidate.FullyValidated;

public class RegisterUsername extends UsernameBase<RegisterUsername> {
    private RegisterUsername(FullyValidated<String, RegisterUsername> opaqueValue) {
        super(opaqueValue);
    }

    public static final class Candidate extends CandidateBase<RegisterUsername> {
        @JsonCreator(mode = JsonCreator.Mode.DELEGATING)
        public Candidate(String raw) {
            super(raw);
        }

        public static Candidate from(String raw) {
            return new Candidate(raw);
        }

        public FullyValidated<String, RegisterUsername> crossValidate(Predicate<String> exists) {
            return crossValidateDoesNotExist(exists);
        }
    }
}

```

vv/src/main/java/org/owasp/untrust/vv/examples/TaskId.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/examples/TaskId.java>

```

package org.owasp.untrust.vv.examples;

import java.util.UUID;

import com.fasterxml.jackson.annotation.JsonCreator;
import org.owasp.untrust.valuedescriptors.foundation.ExposableValue;
import org.owasp.untrust.valuedescriptors.Hardcoded;
import org.owasp.untrust.vv.foundation.SelfValidating;
import org.owasp.untrust.vv.foundation.ValidatedWrappedValue;
import org.owasp.untrust.vv.traits.RareTraitsCaseWhereParsingIsTheWholeValidation;

import static org.owasp.untrust.valuedescriptors.Hardcoded.hardcoded;

public final class TaskId extends ValidatedWrappedValue<UUID>
    implements SelfValidating<UUID>, ExposableValue<UUID> {
    @JsonCreator(mode = JsonCreator.Mode.DELEGATING)
    public TaskId(String raw) {
        super(raw, new Traits());
    }

    public static TaskId from(String raw) {
        return new TaskId(raw);
    }

    @Override
    public UUID exposeUnchecked() {
        return exposeUnchecked(EXPOSE_HALF_BAKED_VALUE_INTENDED_FOR_INTERNAL_LIBRARY_USE_ONLY_MARKER)
    }

    private static final class Traits
        extends RareTraitsCaseWhereParsingIsTheWholeValidation<UUID> {
        @Override
        public Hardcoded descriptionInErrors() {
            return hardcoded("task id");
        }

        @Override
        public Bounds rawBounds() {
            return new Bounds(36, 36);
        }

        @Override
        public UUID parse(String raw) {
            return UUID.fromString(raw.trim());
        }
    }

```



```
}

@Override
public UUID normalize(UUID parsed) {
    return parsed;
}
}
```

vv/src/main/java/org/owasp/untrust/vv/examples/UsernameBase.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/examples/UsernameBase.java>

```

package org.owasp.untrust.vv.examples;

import java.util.function.Predicate;
import java.util.regex.Pattern;

import org.owasp.untrust.vv.foundation.CrossValidatedReceiver;
import org.owasp.untrust.vv.foundation.CrossValidationCandidate;
import org.owasp.untrust.vv.foundation.ValidatedWrappedValue;
import org.owasp.untrust.vv.foundation.CrossValidationCandidate.FullyValidated;
import org.owasp.untrust.vv.traits.RegexStringTraits;
import org.owasp.untrust.vv.visibility.MiddleMaskedValue;
import org.owasp.untrust.valuedescriptors.Hardcoded;

import static org.owasp.untrust.valuedescriptors.Hardcoded.hardcoded;

public abstract class UsernameBase<ReceiverOfValidated extends UsernameBase<ReceiverOfValidated>>
    extends CrossValidatedReceiver<String, ReceiverOfValidated>
    implements MiddleMaskedValue {
    // ExposableValue<String> {
    protected UsernameBase(FullyValidated<String, ReceiverOfValidated> opaqueValue) {
        super(opaqueValue);
    }

    public static class CandidateBase<ReceiverOfValidated extends UsernameBase<ReceiverOfValidated>>
        extends ValidatedWrappedValue<String>
        implements MiddleMaskedValue,
            CrossValidationCandidate<String, ReceiverOfValidated> {
        protected CandidateBase(String raw) {
            super(raw, new Traits());
        }

        protected FullyValidated<String, ReceiverOfValidated> crossValidateExists(Predicate<String> exists) {
            return crossValidate(exists, hardcoded("Username does not exist."));
        }

        protected FullyValidated<String, ReceiverOfValidated> crossValidateDoesNotExist(Predicate<String> exists) {
            return crossValidate((value) -> !exists.test(value), hardcoded("Username does not exist."));
        }

        @Override
        public String toString() {
            return toPublicString();
        }
    }
}

```

```

private static final class Traits extends RegexStringTraits {
    private static final Pattern USERNAME =
        Pattern.compile("[A-Za-z0-9._-]{1,80}");

    @Override
    public Hardcoded descriptionInErrors() {
        return hardcoded("username");
    }

    @Override
    public Bounds rawBounds() {
        return new Bounds(1, 80);
    }

    @Override
    public String reformatString(String raw) {
        return raw.trim();
    }

    @Override
    public Pattern welcomeListRegex() {
        return USERNAME;
    }
}

@Override
public String exposeUnchecked(ExposeHalfBakedValueIntendedForInternalLibraryUseOnlyMarker marker) {
    return exposeUnchecked(marker);
}
}

```

vv/src/main/java/org/owasp/untrust/vv/exceptions/EntityAccessForbiddenException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/exceptions/EntityAccessForbiddenException.java>

```

package org.owasp.untrust.vv.exceptions;

import org.owasp.untrust.buildmetadata.StringConcatenationSafe;
import org.owasp.untrust.vv.ViewableUuidValue;

@StringConcatenationSafe("This exception message deliberately includes a ViewableUuidValue identifier. The id type is ")
public class EntityAccessForbiddenException extends RuntimeException {
    public EntityAccessForbiddenException(ViewableUuidValue entityId) {
        super("Access to entity with ID " + entityId + " is forbidden.");
    }
}

```

vv/src/main/java/org/owasp/untrust/vv/exceptions/EntityNotFoundException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/exceptions/EntityNotFoundException.java>

```

package org.owasp.untrust.vv.exceptions;

import org.owasp.untrust.buildmetadata.StringConcatenationSafe;
import org.owasp.untrust.vv.ViewableUuidValue;

@StringConcatenationSafe("This exception message deliberately includes a ViewableUuidValue identifier. The id type is ")
public class EntityNotFoundException extends RuntimeException {
    public EntityNotFoundException(ViewableUuidValue entityId) {
        super("Entity with ID " + entityId + " not found.");
    }
}

```

vv/src/main/java/org/owasp/untrust/vv/exceptions/ValidationException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/exceptions/ValidationException.java>

```

package org.owasp.untrust.vv.exceptions;

import java.util.List;
import java.util.Optional;

public class ValidationException extends IllegalArgumentException {
    private final Optional<String> m_rawValue;
    private final Optional<?> m_parsedValue;
    private final List<String> m_validationErrors;
    private final Optional<Object> m_additionalValue;

    public ValidationException(String rawValue, String validationErrorDesc, Object additionalValue) {
        this(Optional.of(rawValue), Optional.empty(), List.of(validationErrorDesc), Optional.of(additionalValue));
    }

    public ValidationException(Object parsedValue, String validationErrorDesc, Object additionalValue) {
        this(Optional.empty(), Optional.of(parsedValue), List.of(validationErrorDesc), Optional.of(additionalValue));
    }

    public ValidationException(Optional<String> rawValue, Optional<?> parsedValue, List<String> validationErrors, Opti
        super("Validation failed.");
        m_rawValue = rawValue;
        m_parsedValue = parsedValue;
        m_validationErrors = List.copyOf(validationErrors);
        m_additionalValue = additionalValue;
    }

    public Optional<String> rawValue() {
        return m_rawValue;
    }

    public Optional<?> parsedValue() {
        return m_parsedValue;
    }

    public List<String> validationErrors() {
        return m_validationErrors;
    }
}

```

vv/src/main/java/org/owasp/untrust/vv/foundation/CrossValidate

dReceiver.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/foundation/CrossValidatedReceiver.java>

```
package org.owasp.untrust.vv.foundation;

import org.owasp.untrust.vv.foundation.CrossValidationCandidate.FullyValidated;
import org.owasp.untrust.vv.foundation.HalfBakedExposable.ExposeHalfBakedValueIntendedForInternalLibraryUseOr
import org.owasp.untrust.valuedescriptors.foundation.ExposableWrappedValue;

public abstract class CrossValidatedReceiver<T, ReceiverOfValidated extends CrossValidatedReceiver<T, ReceiverOfVa
    extends ExposableWrappedValue<T> {
    protected CrossValidatedReceiver(FullyValidated<T, ReceiverOfValidated> validated) {
        super(validated.m_validated);
    }
}
```

vv/src/main/java/org/owasp/untrust/vv/foundation/CrossValidationCandidate.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/foundation/CrossValidationCandidate.java>

```

package org.owasp.untrust.vv.foundation;

import java.util.Optional;
import java.util.function.BiFunction;
import java.util.function.Function;
import java.util.function.Predicate;

import org.owasp.untrust.valuedescriptors.foundation.PubliclyRepresentable;
import org.owasp.untrust.valuedescriptors.Hardcoded;

public interface CrossValidationCandidate<T, ReceiverOfValidated> extends CrossValidatedReceiver<T, ReceiverOfValidated>,
    extends HalfBakedExposable<T>, PubliclyRepresentable {
    //default <In> LinkT performLink(In in) {
    //}

    public static class FullyValidated<T, ReceiverOfValidated> {
        T m_validated; // package private

        private FullyValidated(T validated) {
            m_validated = validated;
        }
    }

    default <C> FullyValidated<T, ReceiverOfValidated> crossValidate(C context, BiFunction<T, C, Optional<String>>> findCrossValidationError) {
        T partiallyValidated = exposeUnchecked(EXPOSE_HALF_BAKED_VALUE_INTENDED_FOR_INTERNAL_LIBRARY_USE_ONLY, context, findCrossValidationError);
        Optional<String> crossValidationError = findCrossValidationError.apply(partiallyValidated, context);
        if (crossValidationError.isPresent()) {
            throw new IllegalArgumentException("Value does not satisfy cross-validation constraints.", new Exception(crossValidationError.get()));
        }
        return new FullyValidated<>(partiallyValidated);
    }

    default <C> FullyValidated<T, ReceiverOfValidated> crossValidate(Function<T, Optional<String>>> findCrossValidationError) {
        T partiallyValidated = exposeUnchecked(EXPOSE_HALF_BAKED_VALUE_INTENDED_FOR_INTERNAL_LIBRARY_USE_ONLY, findCrossValidationError);
        Optional<String> crossValidationError = findCrossValidationError.apply(partiallyValidated);
        if (crossValidationError.isPresent()) {
            throw new IllegalArgumentException("Value does not satisfy cross-validation constraints.", new Exception(crossValidationError.get()));
        }
        return new FullyValidated<>(partiallyValidated);
    }

    default <C> FullyValidated<T, ReceiverOfValidated> crossValidate(Predicate<T> isOk, Hardcoded errorDescription) {
        T partiallyValidated = exposeUnchecked(EXPOSE_HALF_BAKED_VALUE_INTENDED_FOR_INTERNAL_LIBRARY_USE_ONLY, isOk);
        if (isOk.test(partiallyValidated)) {
            return new FullyValidated<>(partiallyValidated);
        }
    }

```

```
    } else {  
        throw new IllegalArgumentException("Value does not satisfy cross-validation constraints.", new Exception(error  
    }  
}  
}
```

vv/src/main/java/org/owasp/untrust/vv/foundation/HalfBakedExposable.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/foundation/HalfBakedExposable.java>

```
package org.owasp.untrust.vv.foundation;  
  
public interface HalfBakedExposable<T> {  
    record ExposeHalfBakedValueIntendedForInternalLibraryUseOnlyMarker() {}  
    static final ExposeHalfBakedValueIntendedForInternalLibraryUseOnlyMarker EXPOSE_HALF_BAKED_VALUE_INTEND  
    T exposeUnchecked(ExposeHalfBakedValueIntendedForInternalLibraryUseOnlyMarker marker);  
}
```

vv/src/main/java/org/owasp/untrust/vv/foundation/PendingValidated.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/foundation/PendingValidated.java>


```
package org.owasp.untrust.vv.foundation;

import org.owasp.untrust.valuedescriptors.foundation.PubliclyRepresentable;

public interface PendingValidated<T> extends SelfValidating<T>, PubliclyRepresentable {
    T exposeForValidationOnly();

    @Override
    default String toPublicString() {
        return "[pending validation]";
    }
}
```

vv/src/main/java/org/owasp/untrust/vv/foundation/SelfValidating.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/foundation/SelfValidating.java>

```

package org.owasp.untrust.vv.foundation;

import java.util.Optional;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

import org.owasp.untrust.buildmetadata.StringConcatenationSafe;
import org.owasp.untrust.vv.exceptions.ValidationException;
import static org.owasp.untrust.valuedescriptors.Hardcoded.hardcoded;

@StringConcatenationSafe("Validation error messages in this interface are assembled only from developer-authored n
public interface SelfValidating<T> {
    //public static class AlreadyParsedMarker { };
    //static AlreadyParsedMarker ALREADY_PARSED = new AlreadyParsedMarker();
    public static final Logger LOGGER = LoggerFactory.getLogger(SelfValidating.class);

    default <V extends ValidationTraits<T>> T validate(
        String raw,
        V traits
    ){
        return prepareAndValidateValue(raw, traits);
    }

    default <V extends ValidationTraits<T>> T validateParsed(
        T unvalidated,
        V traits
    ){
        return prepareAndValidateValue(unvalidated, traits);
    }

    private static <T, Traits extends ValidationTraits<T>>
    T prepareAndValidateValue(String raw, Traits traits) throws ValidationException {
        Optional<ValidationException> validationProblem = traits.findValidationProblemInRaw(raw);
        if (validationProblem.isPresent()) {
            throw validationProblem.get();
        }

        T parsedValue = parse(raw, traits);
        return prepareAndValidateValue(parsedValue, traits);
    }

    private static <T, Traits extends ValidationTraits<T>>
    T parse(String raw, Traits traits) throws ValidationException {

```

```

try {
    return traits.parse(raw);
}

// LOCAL CATCH REASON:
// If parsing throws an exception, we wrap it in a ValidationException to provide more context.
// This also adds uniformity.
// This is the expected parsing failure exception: IllegalArgumentException, which is commonly used for parsing failures i
catch (IllegalArgumentException e) {
    throw new ValidationException(
        raw,
        hardcoded("Invalid format for ").concat(traits.descriptionInErrors()).value(),
        e);
}

// LOCAL CATCH REASON:
// If parsing throws an exception, we wrap it in a ValidationException to provide more context.
// This also adds uniformity.
// While parse() is suppose to only throw InvalidArgumentException, we catch all exceptions to be safe and to avoid unex
catch (Exception e) {
    LOGGER.warn(
        "Parsing in traits class {} failed with exception {} that is not IllegalArgumentException. This is unexpected an
        hardcoded(traits.getClass().getSimpleName()),
        hardcoded(e.getClass().getSimpleName()),
        e);
    throw new ValidationException(
        raw,
        hardcoded("Parsing failed for ").concat(traits.descriptionInErrors()).value(),
        e);
}
}

private static <T, Traits extends ValidationTraits<T>> T prepareAndValidateValue(T parsedValue, Traits traits) throws
    parsedValue = traits.normalize(parsedValue);

    Optional<ValidationException> validationProblem = traits.findValidationProblemInNormalizedValue(parsedValue);
    if (validationProblem.isPresent()) {
        throw validationProblem.get();
    }

    return parsedValue;
}
}

```

vv/src/main/java/org/owasp/untrust/vv/foundation/ValidatedWrappedValue.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/foundation/ValidatedWrappedValue.java>

```
package org.owasp.untrust.vv.foundation;

public class ValidatedWrappedValue<T> implements SelfValidating<T>, HalfBakedExposable<T> {
    private T m_validatedValue;

    protected ValidatedWrappedValue(String raw, ValidationTraits<T> traits) {
        m_validatedValue = validate(raw, traits);
    }

    @Override
    public T exposeUnchecked(ExposeHalfBakedValueIntendedForInternalLibraryUseOnlyMarker marker) {
        return m_validatedValue;
    }
}
```

vv/src/main/java/org/owasp/untrust/vv/foundation/ValidationTraits.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/foundation/ValidationTraits.java>

```
package org.owasp.untrust.vv.foundation;

import java.util.Optional;

import org.owasp.untrust.valuedescriptors.Hardcoded;
import org.owasp.untrust.vv.exceptions.ValidationException;

public interface ValidationTraits<T> {
    Hardcoded descriptionInErrors();

    Optional<ValidationException> findValidationProblemInRaw(String raw);

    T parse(String raw);

    T normalize(T parsed);

    Optional<ValidationException> findValidationProblemInNormalizedValue(T normalized);
}
```

vv/src/main/java/org/owasp/untrust/vv/prebuilt/ApiKey.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/prebuilt/ApiKey.java>

```

package org.owasp.untrust.vv.prebuilt;

import org.owasp.untrust.vv.visibility.secret.SecretReference;
import org.owasp.untrust.vv.visibility.secret.SecretStore;
import org.owasp.untrust.vv.visibility.secret.SecretValue;
import org.owasp.untrust.vv.visibility.secret.SecretValueInitializer;

public class ApiKey extends SecretValue<String> {
    public ApiKey(SecretValueInitializer<String, ApiKey> initializer) {
        super(initializer);
    }

    public ApiKey(
        SecretStore<String> store,
        SecretReference reference,
        String displayValue) {
        super(store, reference, displayValue);
    }

    @Override
    protected String revalidate(String value) {
        return validate(value, new PendingApiKey.Traits());
    }
}

```

vv/src/main/java/org/owasp/untrust/vv/prebuilt/CreditCard.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/prebuilt/CreditCard.java>

```

package org.owasp.untrust.vv.prebuilt;

import java.util.Optional;

import org.apache.commons.validator.routines.CreditCardValidator;
import com.fasterxml.jackson.annotation.JsonCreator;
import org.owasp.untrust.valuedescriptors.Hardcoded;
import org.owasp.untrust.vv.exceptions.ValidationException;
import org.owasp.untrust.vv.foundation.ValidatedWrappedValue;
import org.owasp.untrust.vv.foundation.ValidationTraits;
import org.owasp.untrust.vv.visibility.MaskedValue;

import static org.owasp.untrust.valuedescriptors.Hardcoded.hardcoded;

public final class CreditCard extends ValidatedWrappedValue<String> implements MaskedValue<String> {
    @JsonCreator(mode = JsonCreator.Mode.DELEGATING)
    public CreditCard(String raw) {
        super(raw, new Traits());
    }

    public static CreditCard from(String raw) {
        return new CreditCard(raw);
    }

    @Override
    public String mask(String value) {
        if (value.length() <= 4) {
            return "[sensitive]";
        }

        char[] masked = new char[value.length()];
        int hiddenLength = value.length() - 4;
        for (int i = 0; i < hiddenLength; i++) {
            masked[i] = '*';
        }
        value.getChars(hiddenLength, value.length(), masked, hiddenLength);
        return String.valueOf(masked);
    }

    @Override
    public String toString() {
        return toPublicString();
    }
}

```

```

public static final class Traits implements ValidationTraits<String> {
    private static final CreditCardValidator VALIDATOR = CreditCardValidator.genericCreditCardValidator(12, 19);

    @Override
    public Hardcoded descriptionInErrors() {
        return hardcoded("credit card number");
    }

    @Override
    public Optional<ValidationException> findValidationProblemInRaw(String raw) {
        if (VALIDATOR.isValid(raw)) {
            return Optional.empty();
        }

        return Optional.of(new ValidationException(
            "[sensitive]",
            "Invalid credit card number.",
            "Failed credit card syntax or check digit validation."));
    }

    @Override
    public String parse(String raw) {
        return raw;
    }

    @Override
    public String normalize(String parsed) {
        return parsed;
    }

    @Override
    public Optional<ValidationException> findValidationProblemInNormalizedValue(String normalized) {
        return Optional.empty();
    }
}

```

vv/src/main/java/org/owasp/untrust/vv/prebuilt/PendingApiKey.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp


```
_untrust/5e578e471fdb38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/  
prebuilt/PendingApiKey.java>
```

```

package org.owasp.untrust.vv.prebuilt;

import java.util.regex.Pattern;

import com.fasterxml.jackson.annotation.JsonCreator;
import org.owasp.untrust.valuedescriptors.Hardcoded;
import org.owasp.untrust.vv.foundation.SelfValidating;
import org.owasp.untrust.vv.traits.RegexStringTraits;
import org.owasp.untrust.vv.visibility.secret.PendingSecret;

import static org.owasp.untrust.valuedescriptors.Hardcoded.hardcoded;

public class PendingApiKey
    implements PendingSecret<String, ApiKey>, SelfValidating<String> {
    private static final int DISPLAY_SUFFIX_LENGTH = 4;

    private final String m_value;

    @JsonCreator(mode = JsonCreator.Mode.DELEGATING)
    protected PendingApiKey(String raw) {
        this.m_value = validate(raw, new Traits());
    }

    public static PendingApiKey from(String raw) {
        return new PendingApiKey(raw);
    }

    public String displayValue() {
        return "****" + suffix();
    }

    public String suffix() {
        int start = Math.max(0, m_value.length() - DISPLAY_SUFFIX_LENGTH);
        return m_value.substring(start);
    }

    public static class Traits extends RegexStringTraits {
        private static final Pattern API_KEY =
            Pattern.compile("[A-Za-z0-9_\\-]{20,512}");

        @Override
        public Hardcoded descriptionInErrors() {
            return hardcoded("AI API key");
        }
    }
}

```

```

@Override
public Bounds rawBounds() {
    return new Bounds(20, 512);
}

@Override
public Pattern welcomeListRegex() {
    return API_KEY;
}
}

@Override
public String exposeUnchecked(ExposeHalfBakedValueIntendedForInternalLibraryUseOnlyMarker marker) {
    return m_value;
}
}

```

vv/src/main/java/org/owasp/untrust/vv/traits/BoundedAnyContentStringTraits.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/traits/BoundedAnyContentStringTraits.java>

```
package org.owasp.untrust.vv.traits;

import java.util.Optional;

import org.owasp.untrust.vv.exceptions.ValidationException;

public abstract class BoundedAnyContentStringTraits extends BoundedValueTraits<String> {
    public String reformatString(String raw) {
        return raw;
    }

    @Override
    public String parse(String raw) {
        return raw;
    }

    @Override
    public String normalize(String parsed) {
        String reformatted = reformatString(parsed);
        return normalizeReformattedString(reformatted);
    }

    protected String normalizeReformattedString(String reformatted) {
        return reformatted;
    }

    @Override
    public Optional<Bounds> valueBounds() {
        return Optional.of(rawBounds());
    }

    @Override
    public int valueForBounds(String normalized) {
        return normalized.length();
    }

    @Override
    public Optional<ValidationException> findValidationProblemInBounded(String normalized) {
        return findExtraValidationProblem(normalized);
    }

    public Optional<ValidationException> findExtraValidationProblem(String normalized) {
        return Optional.empty();
    }
}
```

```
}  
}
```

vv/src/main/java/org/owasp/untrust/vv/traits/BoundedValueTraits.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/traits/BoundedValueTraits.java>

```

package org.owasp.untrust.vv.traits;

import java.util.Optional;

import org.owasp.untrust.buildmetadata.StringConcatenationSafe;
import org.owasp.untrust.vv.exceptions.ValidationException;
import org.owasp.untrust.vv.foundation.ValidationTraits;

import static org.owasp.untrust.valuedescriptors.Hardcoded.hardcoded;

@StringConcatenationSafe("Bounded trait messages are assembled from developer-authored validation labels and fix
public abstract class BoundedValueTraits<T> implements ValidationTraits<T> {
    public record Bounds(int minimum, int maximum) {
    }

    public abstract Bounds rawBounds();

    public Optional<Bounds> valueBounds() {
        return Optional.empty();
    }

    public int valueForBounds(T normalized) {
        throw new UnsupportedOperationException("valueForBounds must be implemented when valueBounds is prese

    }

    public Optional<ValidationException> findValidationProblemInBounded(T normalized) {
        return Optional.empty();
    }

    @Override
    public Optional<ValidationException> findValidationProblemInRaw(String raw) {
        Bounds bounds = rawBounds();
        int length = raw.length();
        if (length < bounds.minimum()) {
            return Optional.of(new ValidationException(raw, descriptionInErrors()
                .concat(hardcoded(": Value is too short.")).value(), bounds.minimum()));
        }

        if (length > bounds.maximum()) {
            return Optional.of(new ValidationException(raw, descriptionInErrors()
                .concat(hardcoded(": Value is too long.")).value(), bounds.maximum()));
        }

        return Optional.empty();
    }
}

```

```

}

@Override
public Optional<ValidationException> findValidationProblemInNormalizedValue(T normalized) {
    Optional<Bounds> maybeBounds = valueBounds();
    if (maybeBounds.isPresent()) {
        Bounds bounds = maybeBounds.get();
        int value = valueForBounds(normalized);
        if (value < bounds.minimum()) {
            return Optional.of(new ValidationException(normalized, descriptionInErrors()
                .concat(hardcoded(": Value is too small.")).value(), bounds.minimum()));
        }

        if (value > bounds.maximum()) {
            return Optional.of(new ValidationException(normalized, descriptionInErrors()
                .concat(hardcoded(": Value is too big.")).value(), bounds.maximum()));
        }
    }

    return findValidationProblemInBounded(normalized);
}
}

```

vv/src/main/java/org/owasp/untrust/vv/traits/CustomValidationForRareCasesTraits.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/traits/CustomValidationForRareCasesTraits.java>

```

package org.owasp.untrust.vv.traits;

import java.util.Optional;

import org.owasp.untrust.vv.exceptions.ValidationException;
import org.owasp.untrust.vv.foundation.ValidationTraits;

public abstract class CustomValidationForRareCasesTraits<T> implements ValidationTraits<T> {
    public record Bounds(int minimum, int maximum, String message) {
    }

    public abstract Bounds rawBounds();

    @Override
    public Optional<ValidationException> findValidationProblemInRaw(String raw) {
        Bounds bounds = rawBounds();
        if (raw.length() < bounds.minimum()) {
            return Optional.of(new ValidationException(raw, bounds.message(), bounds.minimum()));
        }

        if (raw.length() > bounds.maximum()) {
            return Optional.of(new ValidationException(raw, bounds.message(), bounds.maximum()));
        }

        return Optional.empty();
    }
}

```

vv/src/main/java/org/owasp/untrust/vv/traits/EnumValidationTraits.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/traits/EnumValidationTraits.java>


```

package org.owasp.untrust.vv.traits;

import java.util.Optional;
import java.util.stream.Stream;

import org.owasp.untrust.buildmetadata.StringConcatenationSafe;
import org.owasp.untrust.valuedescriptors.Hardcoded;
import org.owasp.untrust.vv.exceptions.ValidationException;
import org.owasp.untrust.vv.foundation.ValidationTraits;

import static org.owasp.untrust.valuedescriptors.Hardcoded.hardcoded;

@StringConcatenationSafe("Enum validation messages are assembled from developer-authored Hardcoded descriptions")
public class EnumValidationTraits<E> extends Enum<E> implements ValidationTraits<E> {
    private final Class<E> enumClass;
    private final Hardcoded description;
    private final int maxEnumLength;

    public EnumValidationTraits(Class<E> enumClass, Hardcoded description) {
        this.enumClass = enumClass;
        this.description = description;
        this.maxEnumLength = Stream.of(enumClass.getEnumConstants())
            .map(value -> value.toString().length())
            .max(Integer::compareTo)
            .orElse(0);
    }

    @Override
    public Hardcoded descriptionInErrors() {
        return description;
    }

    @Override
    public Optional<ValidationException> findValidationProblemInRaw(String raw) {
        if (raw.isEmpty()) {
            return Optional.of(new ValidationException(raw, hardcoded("Cannot have an empty ").concat(description).value));
        }

        if (raw.length() > maxEnumLength) {
            return Optional.of(new ValidationException(raw, hardcoded("Length is too long to be a valid ").concat(description).value));
        }

        return Optional.empty();
    }
}

```

```

@Override
public E parse(String raw) {
    for (E value : enumClass.getEnumConstants()) {
        if (value.toString().equalsIgnoreCase(raw.trim())) {
            return value;
        }
    }

    throw new IllegalArgumentException(hardcoded("Unsupported ").concat(description).concat(hardcoded(" ").value
}

@Override
public E normalize(E parsed) {
    return parsed;
}

@Override
public Optional<ValidationException> findValidationProblemInNormalizedValue(E normalized) {
    return Optional.empty();
}
}

```

vv/src/main/java/org/owasp/untrust/vv/traits/LineTextTraits.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/traits/LineTextTraits.java>

```

package org.owasp.untrust.vv.traits;

import java.text.Normalizer;
import java.util.Optional;

import com.ibm.icu.text.BreakIterator;
import com.ibm.icu.text.UnicodeSet;
import org.owasp.untrust.buildmetadata.StringConcatenationSafe;
import org.owasp.untrust.valuedescriptors.Hardcoded;
import org.owasp.untrust.vv.exceptions.ValidationException;

import static org.owasp.untrust.valuedescriptors.Hardcoded.hardcoded;

@StringConcatenationSafe("Line text validation messages and descriptions are assembled from fixed developer-authored")
public abstract class LineTextTraits extends BoundedAnyContentStringTraits {
    private static final UnicodeSet RGI_EMOJI = new UnicodeSet("[:RGI_Emoji:]").freeze();

    public abstract boolean allowNewlines();

    public abstract boolean allowEmoji();

    public abstract boolean requirePathSafeText();

    @Override
    protected final String normalizeReformattedString(String reformatted) {
        return Normalizer.normalize(reformatted, Normalizer.Form.NFC);
    }

    @Override
    public final Optional<ValidationException> findValidationProblemInNormalizedValue(String normalized) {
        Optional<ValidationException> boundedProblem = super.findValidationProblemInNormalizedValue(normalized);
        if (boundedProblem.isPresent()) {
            return boundedProblem;
        }

        if (requirePathSafeText()) {
            Optional<ValidationException> pathProblem = findPathSafeTextProblem(normalized);
            if (pathProblem.isPresent()) {
                return pathProblem;
            }
        }

        BreakIterator characterIterator = BreakIterator.getCharacterInstance();
        characterIterator.setText(normalized);
    }

```

```

int start = characterIterator.first();
for (int end = characterIterator.next(); end != BreakIterator.DONE; start = end, end = characterIterator.next()) {
    String cluster = normalized.substring(start, end);
    if (!isAllowedCluster(cluster)) {
        return Optional.of(new ValidationException(
            normalized,
            descriptionInErrors().concat(hardcoded(": Value contains a disallowed character or emoji sequence.")).val
            Optional.empty()));
    }
}

return Optional.empty();
}

private boolean isAllowedCluster(String cluster) {
    if (allowEmoji() && RGI_EMOJI.contains(cluster)) {
        return true;
    }

    return isAllowedTextualCluster(cluster);
}

private boolean isAllowedTextualCluster(String cluster) {
    boolean hasTextualBase = false;

    int[] codePoints = cluster.codePoints().toArray();
    for (int codePoint : codePoints) {
        if (isEmojiCombiningCodePoint(codePoint)) {
            return false;
        }

        if (requirePathSafeText() && isForbiddenPathCodePoint(codePoint)) {
            return false;
        }

        if (isCombiningMark(codePoint)) {
            continue;
        }

        if (!isAllowedTextualBaseCodePoint(codePoint)) {
            return false;
        }
    }
}

```

```

        hasTextualBase = true;
    }

    return hasTextualBase;
}

private boolean isAllowedTextualBaseCodePoint(int codePoint) {
    if (codePoint == ' '){
        return true;
    }

    if (allowNewlines() && codePoint == '\n') {
        return true;
    }

    int type = Character.getType(codePoint);
    return type == Character.UPPERCASE_LETTER
        || type == Character.LOWERCASE_LETTER
        || type == Character.TITLECASE_LETTER
        || type == Character.MODIFIER_LETTER
        || type == Character.OTHER_LETTER
        || type == Character.DECIMAL_DIGIT_NUMBER
        || type == Character.LETTER_NUMBER
        || type == Character.OTHER_NUMBER
        || type == Character.CONNECTOR_PUNCTUATION
        || type == Character.DASH_PUNCTUATION
        || type == Character.START_PUNCTUATION
        || type == Character.END_PUNCTUATION
        || type == Character.INITIAL_QUOTE_PUNCTUATION
        || type == Character.FINAL_QUOTE_PUNCTUATION
        || type == Character.OTHER_PUNCTUATION
        || isAllowedSymbol(type);
}

private boolean isAllowedSymbol(int type) {
    return type == Character.MATH_SYMBOL || type == Character.CURRENCY_SYMBOL;
}

private Optional<ValidationException> findPathSafeTextProblem(String normalized) {
    String[] pathSegments = normalized.split("[\\\\]", -1);
    for (String segment : pathSegments) {
        if (segment.equals(".") || segment.equals("..")) {
            return Optional.of(new ValidationException(
                normalized,

```

```

        descriptionInErrors().concat(hardcoded(": Path segments must not be '.' or '..").value(),
        Optional.empty());
    }
}

return Optional.empty();
}

private static boolean isForbiddenPathCodePoint(int codePoint) {
    return codePoint == '<'
        || codePoint == '>'
        || codePoint == ':'
        || codePoint == '"'
        || codePoint == '|'
        || codePoint == '?'
        || codePoint == '*';
}

private static boolean isCombiningMark(int codePoint) {
    int type = Character.getType(codePoint);
    return type == Character.NON_SPACING_MARK
        || type == Character.ENCLOSING_MARK
        || type == Character.COMBINING_SPACING_MARK;
}

private static boolean isEmojiCombiningCodePoint(int codePoint) {
    return codePoint == 0x20E3 || isVariationSelector(codePoint);
}

private static boolean isVariationSelector(int codePoint) {
    return (codePoint >= 0xFE00 && codePoint <= 0xFE0F)
        || (codePoint >= 0xE0100 && codePoint <= 0xE01EF);
}

protected static Hardcoded lineTextDescription(String lineMode, String emojiMode) {
    return Hardcoded.of(lineMode).concat(Hardcoded.of(" ")).concat(Hardcoded.of(emojiMode));
}
}

```

vv/src/main/java/org/owasp/untrust/vv/traits/RareTraitsCaseWhereParsingIsTheWholeValidation.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/traits/RareTraitsCaseWhereParsingIsTheWholeValidation.java>

```
package org.owasp.untrust.vv.traits;

public abstract class RareTraitsCaseWhereParsingIsTheWholeValidation<T>
    extends BoundedValueTraits<T> {
}
```

vv/src/main/java/org/owasp/untrust/vv/traits/RegexStringTraits.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/traits/RegexStringTraits.java>

```

package org.owasp.untrust.vv.traits;

import java.util.Optional;
import java.util.regex.Pattern;

import org.owasp.untrust.buildmetadata.StringConcatenationSafe;
import org.owasp.untrust.vv.exceptions.ValidationException;

import static org.owasp.untrust.valuedescriptors.Hardcoded.hardcoded;

@StringConcatenationSafe("Regex validation messages are assembled from fixed developer-authored fragments and k
public abstract class RegexStringTraits extends BoundedAnyContentStringTraits {
    public abstract Pattern welcomeListRegex();

    public Optional<ValidationException> findExtraValidationProblem(String normalized) {
        return Optional.empty();
    }

    @Override
    public Optional<ValidationException> findValidationProblemInNormalizedValue(String normalized) {
        Optional<ValidationException> boundedProblem = super.findValidationProblemInNormalizedValue(normalized);
        if (boundedProblem.isPresent()) {
            return boundedProblem;
        }

        if (!welcomeListRegex().matcher(normalized).matches()) {
            return Optional.of(new ValidationException(
                normalized,
                hardcoded("Invalid ").concat(descriptionInErrors()).concat(hardcoded(".")).value(),
                welcomeListRegex()));
        }

        return findExtraValidationProblem(normalized);
    }
}

```

vv/src/main/java/org/owasp/untrust/vv/visibility/MaskedValue.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp

_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/MaskedValue.java>

```
package org.owasp.untrust.vv.visibility;

import org.owasp.untrust.valuedescriptors.foundation.PubliclyRepresentable;
import org.owasp.untrust.valuedescriptors.foundation.ExposableValue;
import org.owasp.untrust.vv.foundation.HalfBakedExposable;

public interface MaskedValue<T> extends
    Sensitive<T>,
    HalfBakedExposable<T>,
    PubliclyRepresentable {
    @Override
    default String toPublicString() {
        return mask(exposeUnchecked(EXPOSE_HALF_BAKED_VALUE_INTENDED_FOR_INTERNAL_LIBRARY_USE_ONLY_MA
    }

    String mask(T value);
}
```

vv/src/main/java/org/owasp/untrust/vv/visibility/MiddleMaskedValue.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/MiddleMaskedValue.java>

```
package org.owasp.untrust.vv.visibility;

public interface MiddleMaskedValue extends MaskedValue<String> {
    @Override
    default String mask(String value) {
        if (value.length() < 2) {
            return "aa****aa";
        }

        int hiddenLength = Math.max(4, value.length() - 4);
        char[] masked = new char[4 + hiddenLength];
        value.getChars(0, 2, masked, 0);
        for (int i = 2; i < 2 + hiddenLength; i++) {
            masked[i] = '*';
        }
        value.getChars(value.length() - 2, value.length(), masked, 2 + hiddenLength);
        return String.valueOf(masked);
    }
}
```

vv/src/main/java/org/owasp/untrust/vv/visibility/PiiValue.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/PiiValue.java>

```

package org.owasp.untrust.vv.visibility;

import org.owasp.untrust.vv.foundation.HalfBakedExposable;
import org.owasp.untrust.vv.foundation.SelfValidating;

public abstract class PiiValue<T, V extends SelfValidating<T> & HalfBakedExposable<T>>
    extends VisibleValue<T, V>
    implements Sensitive<T> {
    protected PiiValue(V value) {
        super(value.exposeUnchecked(EXPOSE_HALF_BAKED_VALUE_INTENDED_FOR_INTERNAL_LIBRARY_USE_ONLY_MARKER));
    }

    protected final T exposedValue() {
        return exposeUnchecked(EXPOSE_HALF_BAKED_VALUE_INTENDED_FOR_INTERNAL_LIBRARY_USE_ONLY_MARKER);
    }

    @Override
    public String toPublicString() {
        return "[sensitive]";
    }
}

```

vv/src/main/java/org/owasp/untrust/vv/visibility/RedactedValue.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/RedactedValue.java>

```

package org.owasp.untrust.vv.visibility;

public interface RedactedValue<T> extends Sensitive<T> {
    public static final String PUBLIC_REPLACEMENT = "[sensitive]";

    @Override
    default String toPublicString() {
        return PUBLIC_REPLACEMENT;
    }
}

```

vv/src/main/java/org/owasp/untrust/vv/visibility/Sensitive.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/Sensitive.java>

```
package org.owasp.untrust.vv.visibility;

import org.owasp.untrust.valuedescriptors.foundation.PubliclyRepresentable;

// marker interface for values that are sensitive and should not be exposed publicly
public interface Sensitive<T> extends PubliclyRepresentable {
}
```

vv/src/main/java/org/owasp/untrust/vv/visibility/SensitiveConcat.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/SensitiveConcat.java>

```
package org.owasp.untrust.vv.visibility;

import java.util.ArrayList;
import java.util.List;
import java.util.Optional;
import java.util.function.Function;

import org.owasp.untrust.valuedescriptors.foundation.ExposableValue;
import org.owasp.untrust.valuedescriptors.foundation.PubliclyRepresentable;

public class SensitiveConcat implements CharSequence, Appendable, ExposableValue<String>, PubliclyRepresentable {
    private final List<String> exposedParts = new ArrayList<>();
    private final List<String> publicParts = new ArrayList<>();
    private final Optional<Function<String, String>> elementTransform;

    public SensitiveConcat() {
        this.elementTransform = Optional.empty();
    }

    public SensitiveConcat(int capacity) {
        this();
    }

    public SensitiveConcat(String str) {
        this();
        append(str);
    }

    public SensitiveConcat(CharSequence seq) {
        this();
        append(seq);
    }

    public SensitiveConcat(Function<String, String> elementTransform) {
        this.elementTransform = Optional.of(elementTransform);
    }

    public SensitiveConcat(Function<String, String> elementTransform, int capacity) {
        this.elementTransform = Optional.of(elementTransform);
    }

    public SensitiveConcat(Function<String, String> elementTransform, String str) {
        this(elementTransform);
        append(str);
    }
}
```

```

}

public SensitiveConcat append(Object obj) {
    if (obj instanceof PubliclyRepresentable publiclyRepresentable) {
        return append(publiclyRepresentable);
    }

    return append(String.valueOf(obj));
}

public SensitiveConcat append(PubliclyRepresentable value) {
    String publicValue = value.toPublicString();
    publicParts.add(transform(publicValue));

    if (value instanceof ExposableValue<?> exposableValue) {
        exposedParts.add(transform(String.valueOf(exposableValue.exposeUnchecked())));
    } else {
        exposedParts.add(transform(publicValue));
    }

    return this;
}

@Override
public SensitiveConcat append(CharSequence seq) {
    return append(String.valueOf(seq));
}

@Override
public SensitiveConcat append(CharSequence seq, int start, int end) {
    return append(String.valueOf(seq).subSequence(start, end));
}

public SensitiveConcat append(String value) {
    String transformed = transform(value);
    exposedParts.add(transformed);
    publicParts.add(transformed);
    return this;
}

@Override
public SensitiveConcat append(char c) {
    return append(String.valueOf(c));
}

```

```

public SensitiveConcat appendCodePoint(int codePoint) {
    return append(new String(Character.toChars(codePoint)));
}

public SensitiveConcat pushPrefix(String prefix) {
    String transformedPrefix = transform(prefix);
    exposedParts.add(0, transformedPrefix);
    publicParts.add(0, transformedPrefix);
    return this;
}

@Override
public String exposeUnchecked() {
    return combine(exposedParts);
}

@Override
public String toPublicString() {
    return combine(publicParts);
}

@Override
public String toString() {
    return toPublicString();
}

@Override
public int length() {
    return exposeUnchecked().length();
}

@Override
public char charAt(int index) {
    return exposeUnchecked().charAt(index);
}

@Override
public CharSequence subSequence(int start, int end) {
    return exposeUnchecked().subSequence(start, end);
}

private String transform(String value) {
    return elementTransform.map(transform -> transform.apply(value)).orElse(value);
}

```

```

}

private static String combine(List<String> parts) {
    int length = 0;
    for (String part : parts) {
        length += part.length();
    }

    char[] combined = new char[length];
    int offset = 0;
    for (String part : parts) {
        part.getChars(0, part.length(), combined, offset);
        offset += part.length();
    }
    return String.valueOf(combined);
}
}

```

vv/src/main/java/org/owasp/untrust/vv/visibility/TokenizedValue.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/TokenizedValue.java>


```

package org.owasp.untrust.vv.visibility;

import org.owasp.untrust.valuedescriptors.foundation.ExposableValue;

public final class TokenizedValue<T> implements Sensitive<T>, ExposableValue<T> {
    private final T m_value;
    private final String m_replacement;

    public TokenizedValue(T value, String replacement) {
        this.m_value = value;
        this.m_replacement = replacement;
    }

    @Override
    public T exposeUnchecked() {
        return m_value;
    }

    @Override
    public String toPublicString() {
        return m_replacement;
    }

    @Override
    public String toString() {
        return toPublicString();
    }
}

```

vv/src/main/java/org/owasp/untrust/vv/visibility/VisibleValue.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/VisibleValue.java>

```

package org.owasp.untrust.vv.visibility;

import org.owasp.untrust.valuedescriptors.foundation.PubliclyRepresentable;
import org.owasp.untrust.vv.foundation.HalfBakedExposable;
import org.owasp.untrust.vv.foundation.SelfValidating;

public abstract class VisibleValue<T, V extends SelfValidating<T>>
    implements HalfBakedExposable<T>, PubliclyRepresentable {
    private final T m_value;

    protected VisibleValue(T value) {
        this.m_value = value;
    }

    public final T exposeUnchecked(ExposeHalfBakedValueIntendedForInternalLibraryUseOnlyMarker marker) {
        return m_value;
    }

    @Override
    public final String toString() {
        return toPublicString();
    }
}

```

vv/src/main/java/org/owasp/untrust/vv/visibility/encryption/EncryptedOnlyValue.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/encryption/EncryptedOnlyValue.java>

```

package org.owasp.untrust.vv.visibility.encrypted;

import java.util.Arrays;
import java.util.HexFormat;

import org.owasp.untrust.valuedescriptors.foundation.ExposableValue;
import org.owasp.untrust.vv.foundation.HalfBakedExposable;
import org.owasp.untrust.vv.foundation.SelfValidating;
import org.owasp.untrust.vv.visibility.PiiValue;

public final class EncryptedOnlyValue<T, V extends SelfValidating<T> & HalfBakedExposable<T>>
    extends PiiValue<T, V>
    implements ExposableValue<T> {
    private final byte[] ciphertext;
    private final byte[] iv;

    public EncryptedOnlyValue(V value, byte[] ciphertext, byte[] iv) {
        super(value);
        this.ciphertext = ciphertext.clone();
        this.iv = iv.clone();
    }

    @Override
    public T exposeUnchecked() {
        throw new UnsupportedOperationException("Encrypted-only value cannot expose the original value.");
    }

    public byte[] ciphertext() {
        return ciphertext.clone();
    }

    public byte[] iv() {
        return iv.clone();
    }

    @Override
    public String toPublicString() {
        return HexFormat.of().formatHex(ciphertext);
    }

    @Override
    public boolean equals(Object other) {
        if (this == other) {
            return true;
        }
    }

```

```

    }
    if (!(other instanceof EncryptedOnlyValue<?, ?> that)) {
        return false;
    }
    return Arrays.equals(ciphertext, that.ciphertext)
        && Arrays.equals(iv, that.iv);
}

@Override
public int hashCode() {
    return 31 * Arrays.hashCode(ciphertext) + Arrays.hashCode(iv);
}
}

```

vv/src/main/java/org/owasp/untrust/vv/visibility/encryption/PendingEncryption.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/encryption/PendingEncryption.java>

```

package org.owasp.untrust.vv.visibility.encryption;

import org.owasp.untrust.valuedescriptors.foundation.PubliclyRepresentable;
import org.owasp.untrust.vv.foundation.HalfBakedExposable;
import org.owasp.untrust.vv.foundation.SelfValidating;

public final class PendingEncryption<T, V extends SelfValidating<T> & HalfBakedExposable<T>, K>
    implements PubliclyRepresentable {
    private final V value;
    private final PiiEncryptor<T, K> encryptor;

    public PendingEncryption(V value, PiiEncryptor<T, K> encryptor) {
        this.value = value;
        this.encryptor = encryptor;
    }

    public RetainedEncryptedValue<T, V> retainRaw(K key, byte[] iv) {
        return new RetainedEncryptedValue<>{
            value,
            encryptor.encrypt(
                value.exposeUnchecked(HalfBakedExposable.EXPOSE_HALF_BAKED_VALUE_INTENDED_FOR_INTERNAL_L
                key,
                iv),
            iv);
        }
    }

    public EncryptedOnlyValue<T, V> encryptedOnly(K key, byte[] iv) {
        return new EncryptedOnlyValue<>{
            value,
            encryptor.encrypt(
                value.exposeUnchecked(HalfBakedExposable.EXPOSE_HALF_BAKED_VALUE_INTENDED_FOR_INTERNAL_L
                key,
                iv),
            iv);
        }
    }

    @Override
    public String toPublicString() {
        return "[pending pii]";
    }

    @Override
    public String toString() {
        return toPublicString();
    }

```

```
}  
}
```

vv/src/main/java/org/owasp/untrust/vv/visibility/encryption/PiiEncryptor.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/encryption/PiiEncryptor.java>

```
package org.owasp.untrust.vv.visibility.encryption;  
  
@FunctionalInterface  
public interface PiiEncryptor<T, K> {  
    byte[] encrypt(T value, K key, byte[] iv);  
}
```

vv/src/main/java/org/owasp/untrust/vv/visibility/encryption/RetainedEncryptedValue.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/encryption/RetainedEncryptedValue.java>

```

package org.owasp.untrust.vw.visibility.encryption;

import java.util.Arrays;
import java.util.HexFormat;

import org.owasp.untrust.valuedescriptors.foundation.ExposableValue;
import org.owasp.untrust.vw.foundation.HalfBakedExposable;
import org.owasp.untrust.vw.foundation.SelfValidating;
import org.owasp.untrust.vw.visibility.PiiValue;

public final class RetainedEncryptedValue<T, V extends SelfValidating<T> & HalfBakedExposable<T>>
    extends PiiValue<T, V>
    implements ExposableValue<T> {
    private final byte[] ciphertext;
    private final byte[] iv;

    public RetainedEncryptedValue(V value, byte[] ciphertext, byte[] iv) {
        super(value);
        this.ciphertext = ciphertext.clone();
        this.iv = iv.clone();
    }

    @Override
    public T exposeUnchecked() {
        return exposedValue();
    }

    public byte[] ciphertext() {
        return ciphertext.clone();
    }

    public byte[] iv() {
        return iv.clone();
    }

    @Override
    public String toPublicString() {
        return HexFormat.of().formatHex(ciphertext);
    }

    @Override
    public boolean equals(Object other) {
        if (this == other) {
            return true;
        }
    }

```

```

    }
    if (!(other instanceof RetainedEncryptedValue<?, ?> that)) {
        return false;
    }
    return Arrays.equals(ciphertext, that.ciphertext)
        && Arrays.equals(iv, that.iv);
}

@Override
public int hashCode() {
    return 31 * Arrays.hashCode(ciphertext) + Arrays.hashCode(iv);
}
}

```

vv/src/main/java/org/owasp/untrust/vv/visibility/hash/HashOnlyValue.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/hash/HashOnlyValue.java>


```

package org.owasp.untrust.vv.visibility.hash;

import java.util.Arrays;
import java.util.HexFormat;

import org.owasp.untrust.valuedescriptors.foundation.ExposableValue;
import org.owasp.untrust.vv.foundation.HalfBakedExposable;
import org.owasp.untrust.vv.foundation.SelfValidating;
import org.owasp.untrust.vv.visibility.PiiValue;

public final class HashOnlyValue<T, V extends SelfValidating<T> & HalfBakedExposable<T>>
    extends PiiValue<T, V>
    implements ExposableValue<T> {
    private final byte[] hash;

    public HashOnlyValue(V value, byte[] hash) {
        super(value);
        this.hash = hash.clone();
    }

    @Override
    public T exposeUnchecked() {
        throw new UnsupportedOperationException("Hash-only value cannot expose the original value.");
    }

    public byte[] hash() {
        return hash.clone();
    }

    @Override
    public String toPublicString() {
        return HexFormat.of().formatHex(hash);
    }

    @Override
    public boolean equals(Object other) {
        if (this == other) {
            return true;
        }
        if (!(other instanceof HashOnlyValue<?, ?> that)) {
            return false;
        }
        return Arrays.equals(hash, that.hash);
    }
}

```

```
@Override
public int hashCode() {
    return Arrays.hashCode(hash);
}
}
```

vv/src/main/java/org/owasp/untrust/vv/visibility/hash/PendingHash.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/hash/PendingHash.java>

```
package org.owasp.untrust.vv.visibility.hash;

import org.owasp.untrust.valuedescriptors.foundation.PubliclyRepresentable;
import org.owasp.untrust.vv.foundation.HalfBakedExposable;
import org.owasp.untrust.vv.foundation.SelfValidating;

public final class PendingHash<V extends SelfValidating<T> & HalfBakedExposable<T>, T>
    implements PubliclyRepresentable {
    private final V value;

    public PendingHash(V value) {
        this.value = value;
    }

    public V valueObjectForValidationFlow() {
        return value;
    }

    public RetainedHashedValue<T, V> retainRaw(PiiHasher<T> hasher) {
        return new RetainedHashedValue<>{
            value,
            hasher.hash(value.exposeUnchecked(HalfBakedExposable.EXPOSE_HALF_BAKED_VALUE_INTENDED_FOR_INT
        }

    public HashOnlyValue<T, V> hashOnly(PiiHasher<T> hasher) {
        return new HashOnlyValue<>{
            value,
            hasher.hash(value.exposeUnchecked(HalfBakedExposable.EXPOSE_HALF_BAKED_VALUE_INTENDED_FOR_INT
        }

    @Override
    public String toPublicString() {
        return "[pending pii]";
    }

    @Override
    public String toString() {
        return toPublicString();
    }
}
```

vv/src/main/java/org/owasp/untrust/vv/visibility/hash/PiiHasher.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/hash/PiiHasher.java>

```
package org.owasp.untrust.vv.visibility.hash;
```

```
@FunctionalInterface
```

```
public interface PiiHasher<T> {
```

```
    byte[] hash(T value);
```

```
}
```

vv/src/main/java/org/owasp/untrust/vv/visibility/hash/RetainedHashedValue.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/hash/RetainedHashedValue.java>

```

package org.owasp.untrust.vv.visibility.hash;

import java.util.Arrays;
import java.util.HexFormat;

import org.owasp.untrust.valuedescriptors.foundation.ExposableValue;
import org.owasp.untrust.vv.foundation.HalfBakedExposable;
import org.owasp.untrust.vv.foundation.SelfValidating;
import org.owasp.untrust.vv.visibility.PiiValue;

public final class RetainedHashedValue<T, V extends SelfValidating<T> & HalfBakedExposable<T>>
    extends PiiValue<T, V>
    implements ExposableValue<T> {
    private final byte[] hash;

    public RetainedHashedValue(V value, byte[] hash) {
        super(value);
        this.hash = hash.clone();
    }

    @Override
    public T exposeUnchecked() {
        return exposedValue();
    }

    public byte[] hash() {
        return hash.clone();
    }

    @Override
    public String toPublicString() {
        return HexFormat.of().formatHex(hash);
    }

    @Override
    public boolean equals(Object other) {
        if (this == other) {
            return true;
        }
        if (!(other instanceof RetainedHashedValue<?, ?> that)) {
            return false;
        }
        return Arrays.equals(hash, that.hash);
    }
}

```

```

@Override
public int hashCode() {
    return Arrays.hashCode(hash);
}
}

```

vv/src/main/java/org/owasp/untrust/vv/visibility/secret/PendingSecret.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/secret/PendingSecret.java>

```

package org.owasp.untrust.vv.visibility.secret;

import org.owasp.untrust.valuedescriptors.foundation.PubliclyRepresentable;
import org.owasp.untrust.vv.foundation.HalfBakedExposable;

public interface PendingSecret<T, ReceiverOfInitializer>
    extends HalfBakedExposable<T>, PubliclyRepresentable {
    default SecretValueInitializer<T, ReceiverOfInitializer> hide(
        SecretStore<T> store,
        SecretReference reference,
        String displayValue) {
        store.write(reference, exposeUnchecked(EXPOSE_HALF_BAKED_VALUE_INTENDED_FOR_INTERNAL_LIBRARY_USE_C
        return new SecretValueInitializer<>(store, reference, displayValue);
    }

    @Override
    default String toPublicString() {
        return "[pending secret]";
    }
}

```

vv/src/main/java/org/owasp/untrust/vv/visibility/secret/SecretReference.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/secret/SecretReference.java>

[_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/secret/SecretReference.java](https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/secret/SecretReference.java)>

```
package org.owasp.untrust.vv.visibility.secret;

public record SecretReference(String path) {
    public SecretReference {
        if (path.isBlank()) {
            throw new IllegalArgumentException("Secret reference path must not be blank.");
        }
        if (path.startsWith("/") || path.contains("..")) {
            throw new IllegalArgumentException("Secret reference path must be relative and bounded.");
        }
    }
}
```

vv/src/main/java/org/owasp/untrust/vv/visibility/secret/SecretStore.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/secret/SecretStore.java>

```
package org.owasp.untrust.vv.visibility.secret;

public interface SecretStore<T> {
    void write(SecretReference reference, T value);

    T read(SecretReference reference);
}
```

vv/src/main/java/org/owasp/untrust/vv/visibility/secret/SecretValue.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/secret/SecretValue.java>

```

package org.owasp.untrust.vv.visibility.secret;

import org.owasp.untrust.valuedescriptors.foundation.ExposableValue;
import org.owasp.untrust.valuedescriptors.foundation.PubliclyRepresentable;
import org.owasp.untrust.vv.foundation.SelfValidating;

public abstract class SecretValue<T>
    implements PubliclyRepresentable, ExposableValue<T>, SelfValidating<T> {
    private final SecretStore<T> store;
    private final SecretReference reference;
    private final String displayValue;

    protected <Derived extends SecretValue<T>> SecretValue(SecretValueInitializer<T, Derived> initializer) {
        this.store = initializer.store();
        this.reference = initializer.reference();
        this.displayValue = initializer.displayValue();
    }

    protected SecretValue(
        SecretStore<T> store,
        SecretReference reference,
        String displayValue) {
        this.store = store;
        this.reference = reference;
        this.displayValue = displayValue;
    }

    public SecretReference reference() {
        return reference;
    }

    @Override
    public T exposeUnchecked() {
        return revalidate(store.read(reference));
    }

    @Override
    public String toPublicString() {
        return displayValue;
    }

    @Override
    public String toString() {
        return toPublicString();
    }

```



```
}  
  
protected abstract T revalidate(T value);  
  
}
```

vv/src/main/java/org/owasp/untrust/vv/visibility/secret/SecretValueInitializer.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/secret/SecretValueInitializer.java>

```

package org.owasp.untrust.vv.visibility.secret;

import org.owasp.untrust.vv.foundation.SelfValidating;

// ReceiverOfInitializer ensures that an initializer emitted by a PendingX is used on
// a SecretX and not on a SecretY.
// This is a compile-time check that prevents accidental misuse of initializers.
public final class SecretValueInitializer<T, ReceiverOfInitializer> {
    private final SecretStore<T> store;
    private final SecretReference reference;
    private final String displayValue;

    SecretValueInitializer(
        SecretStore<T> store,
        SecretReference reference,
        String displayValue) {
        this.store = store;
        this.reference = reference;
        this.displayValue = displayValue;
    }

    SecretStore<T> store() {
        return store;
    }

    SecretReference reference() {
        return reference;
    }

    String displayValue() {
        return displayValue;
    }
}

```

vv/src/main/java/org/owasp/untrust/vv/visibility/token/PendingTokenization.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/token/PendingTokenization.java>

```

package org.owasp.untrust.vv.visibility.token;

import org.owasp.untrust.valuedescriptors.foundation.PubliclyRepresentable;
import org.owasp.untrust.vv.foundation.HalfBakedExposable;
import org.owasp.untrust.vv.foundation.SelfValidating;

public final class PendingTokenization<V extends SelfValidating<T> & HalfBakedExposable<T>, T>
    implements PubliclyRepresentable {
    private final V value;

    public PendingTokenization(V value) {
        this.value = value;
    }

    public RetainedTokenizedValue<T, V> retainRaw(String token) {
        return new RetainedTokenizedValue<>(value, token);
    }

    public TokenOnlyValue<T, V> tokenOnly(String token) {
        return new TokenOnlyValue<>(value, token);
    }

    @Override
    public String toPublicString() {
        return "[pending pii]";
    }

    @Override
    public String toString() {
        return toPublicString();
    }
}

```

vv/src/main/java/org/owasp/untrust/vv/visibility/token/RetainedTokenizedValue.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/token/RetainedTokenizedValue.java>

```

package org.owasp.untrust.vv.visibility.token;

import org.owasp.untrust.valuedescriptors.foundation.ExposableValue;
import org.owasp.untrust.vv.foundation.HalfBakedExposable;
import org.owasp.untrust.vv.foundation.SelfValidating;
import org.owasp.untrust.vv.visibility.PiiValue;

public final class RetainedTokenizedValue<T, V extends SelfValidating<T> & HalfBakedExposable<T>>
    extends PiiValue<T, V>
    implements ExposableValue<T> {
    private final String token;

    public RetainedTokenizedValue(V value, String token) {
        super(value);
        this.token = token;
    }

    @Override
    public T exposeUnchecked() {
        return exposedValue();
    }

    public String token() {
        return token;
    }

    @Override
    public String toPublicString() {
        return token;
    }
}

```

vv/src/main/java/org/owasp/untrust/vv/visibility/token/TokenOnlyValue.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_owasp_untrust/5e578e471fdb9db38c70d2075a285c0507ee6e2/vv/src/main/java/org/owasp/untrust/vv/visibility/token/TokenOnlyValue.java>

```

package org.owasp.untrust.vv.visibility.token;

import org.owasp.untrust.valuedescriptors.foundation.ExposableValue;
import org.owasp.untrust.vv.foundation.HalfBakedExposable;
import org.owasp.untrust.vv.foundation.SelfValidating;
import org.owasp.untrust.vv.visibility.PiiValue;

public final class TokenOnlyValue<T, V extends SelfValidating<T> & HalfBakedExposable<T>>
    extends PiiValue<T, V>
    implements ExposableValue<T> {
    private final String token;

    public TokenOnlyValue(V value, String token) {
        super(value);
        this.token = token;
    }

    @Override
    public T exposeUnchecked() {
        throw new UnsupportedOperationException("Token-only value cannot expose the original value.");
    }

    public String token() {
        return token;
    }

    @Override
    public String toPublicString() {
        return token;
    }
}

```